the more traditional mission, shielding or baffling are typically much lower cost than most other spacecraft components, we're in a very benign radiation environment and more uniform thermal environment, and we're in a regime where it is at least possible to get at the spacecraft in the future if something goes wrong. I don't want to imply that all scientific spacecraft should be in LEO, but that option should be a part of the cost reduction trade for many missions.

Finally, we should consider the use of "service provided" systems (i.e., existing, commercial ground stations) to reduce the overall mission cost, as discussed in Sec. 13.5.9 below. This typically provides better coverage, lower operating cost, and little or no up-front ground system development cost. The central issue here is that reducing mission cost and schedule is not just a low-cost payload, spacecraft bus, and operations, but starts with the entire mission design as discussed in Chaps. 1–5.

13.5.7 Launch

As discussed in Sec. 13.1.1, although launch is typically not the largest element of cost in most missions, it nonetheless drives mission cost. It simply isn't worth launching a $2 million spacecraft on a $10 or $20 million launch vehicle. In addition, the lack of launch-on-demand, which the Soviets/Russians have had for decades and the Chinese are developing, prevents us from creating responsive, low-cost systems that would be capable of taking some of the workload off large, very expensive satellites and could prevent them from having to cover all the world all the time.

The major approaches to reducing launch cost are shown in Table 13-16. The single most effective approach to reducing both cost and schedule is not to launch to orbit at all. Depending on the goals of the experiment, test, or mission, there are multiple alternatives to a dedicated orbital launch. These are shown in Table 13-17. Balloon flights can provide hours or days at high altitude at very low cost. If 0-g is important, drop towers and drop tubes can provide excellent 0-g conditions for 5–10 sec by dropping a payload of up to 1,000 kg inside a vacuum tube. (See Fig. 13-6.) The data and payload are available essentially immediately and the experiment can typically be repeated twice per day. Periods of 0-g up to about 20–25 seconds (and even longer periods of lunar gravity or Mars gravity) are available from aircraft parabolic flights. Up to 40 parabolas a day can be flown, but perhaps the major benefit is that the experimenter and a few others can fly along, watch what happens, and make adjustments and corrections in real time. (See Fig. 13-7.)

The next step up from parabolic flights are suborbital flights on a sounding rocket. These can provide up to 12 minutes of excellent 0-g and an altitude of up to 1,200 km. This means you can get to LEO altitudes and above with vacuum and a full view of the Earth and space, just as you would in LEO. The only thing missing is the orbital velocity and a large chunk of the price tag.



Fig. 13-6. Drop Tower at ZARM in Bremin, Germany. Drop towers provide 5–10 sec of very good 0-g and can typically do 2 "flights" per day with instant access to the payload after the test. (Photo courtesy ZARM.)



Fig. 13-7. Parabolic Flight. While the quality of 0-g is not as good as drop towers or sounding rockets, a major advantage is the experimenter can ride along, see what is happening, and make corrections on the next parabola only a few minutes away. (Photo courtesy 0-g Corp.)

For going all the way to orbit at lower cost for small payloads, the principal options are rides as secondary payloads or shared launches. The ASAP (*Ariane Structure for Auxiliary Payloads*) Ring on the Ariane V provides accommodations for up to 6 spacecraft of 100 kg each and multiple slots can be used. The ASAP ring has been in use for many years and has provided the ride to orbit for many low-cost satellites. More recently, the ESPA (*EELV Secondary Payload Adapter*) ring has been developed which provides similar services for the Atlas and Delta vehicles (see Fig. 13-8). The ESPA ring is qualified to mount a 6,800 kg primary satellite using the EELV standard interface plane (SIP) bolt pattern, and six 180 kg secondary satellites on a Delta IV or Atlas V vehicle. The standard secondary interface is a 0.38 m diameter bolt circle with 24 fasteners. The ring is about 1.5 m in diameter and 0.61 m tall, it is made of half-inch thick

Table 13-16. Summary of Approaches to Reducing Launch Cost and Schedule. While launch is typically not the principal element of space mission cost, it largely drives the mission cost. (For a more extended discussion of launch costs, see London [1994, 1996]. The Reinventing Space website [2011] also includes multiple papers on this topic.)

LAUNCH

Reducing Space Mission Cost—Launch		© 2010, Microcosm, Inc.
Technique or Action	Mechanism	Comment
L1. *Use Alternatives to Orbital Missions*	Dramatically reduces "launch" cost	Use balloons, drop towers, parabolic flights, or suborbital flights as low-cost testing alternatives.
L2. *Design for Multiple Launch Vehicles*	Increases competition; reduces schedule risk	Used by all of the commercial constellations. Serves to protect schedule as well as reduce cost.
L3. *Use ASAP or ESPA Rings*	Significantly lower cost	Does not allow selection of orbit or launch time. Not applicable for operational missions.
L4. *Rideshare*	Shares launch cost	Only works if systems have compatible orbits and schedules.
L5. *Low-Cost Small Launch*	Allows operational smallsats plus tests of technology for larger missions	Low-cost small launchers (several sizes) are critical to a long-term cost reduction program. Payloads are getting smaller and launchers are getting bigger.
L6. *Build Small Launchers to Inventory*	Reduces mission risk and permits higher system risk	Can dramatically reduce mission risk by allowing satellite replacement. Less of a target for enemy attack. Mission becomes less susceptible to system failures or orbital debris.

Table 13-17.Space Principal Alternatives to a Dedicated Launch to Orbit. The main way to reduce launch costs is to use one of these alternatives to a dedicated orbital launch. (Updated from London [1994, 1996].)

Option	Characteristics	Weight Limits	Principal Constraints	Approximate Cost	Sources
Balloon Flights	Hours to days at ≈ 30 km altitude	Up to 70 kg for low-cost flights	Not in space, not 0-g, weather concerns	$5K to $15K	U. of Wyoming, USAFA, NSBF
Drop Towers	1 to 10 sec of 0-g with immediate payload recovery	Up to 1,000 kg	Brief "flight," 5 to 50 g landing acceleration, entire experiment package dropped	~ $10K per experiment	ZARM, JAMIC, NASA LeRC and MSFC, Vanderbilt U.
Drop Tubes	1 to 5 sec of 0-g with immediate sample retrieval	<0.01 kg	Brief "flight," 20 to 50 g landing acceleration, instrumentation not dropped with sample	~ $0.02K per experiment	ZARM, JAMIC, NASA LeRC and MSFC, Vanderbilt U.
Aircraft Parabolic Flights	Fair 0-g environment, repeated 0-g cycles	Effectively unlimited	Low gravity is only 10^{-2} g	$6.5K to $9K per hour	NASA LeRC and JSC, Novespace, Zero-g, ESA
Sounding Rockets	Good 0-g environment, altitude to 1,200 km, duration of 4 to 12 minutes	Up to 600 kg	Much less than orbital velocities	$1M to $2M	NASA GSFC, NRL, ESA/Sweden, OSC, EER, Bristol Aerosp.
Secondary Payloads	Capacity that is available in excess of primary's requirements	Up to ~ 1,000 kg	Subject to primary's mission timeline and orbit	<$10M	Ariane, OSC, MDA, Russia
Shared Launches	Flights with other payloads having similar orbital requirements	Up to ~ 5,000 kg	Integration challenges launch timing and orbit must be compatible	Up to ~ $60M	Ariane, OSC, Russia

aluminum and weighs 125 kg. Modified versions of the adapter are available depending on payload and launch vehicle requirements. ESPA rings may also be stacked to provide additional capacity.

ESPA provides the required thermal, mechanical, and electrical interfaces to the secondary payloads (SPL). SPL's can be deployed from the ESPA at a predetermined point along the primary mission trajectory or after primary spacecraft separation. Once it is separated from the launch vehicle, ESPA can be used as a satellite bus structure, as it was done for the NASA LCROSS mission [Maly, et al. 2009], for which ESPA was the structural hub of a free-flyer satellite, with interior mounting of a large hydrazine tank.

An ESPA ring is being developed to radially deploy CubeSats from an ESPA port after the SPL has separated from the port. [Maly, et al. 2009]. This structure houses a 3U P-POD. The "P-POD shooter" uses the volume on



Fig. 13-8. EELV Secondary Payload Adapter (ESPA) Ring (Courtesy of MOOG CSA Engineering).

the interior of the ring. It mounts to the cylinder wall inside an ESPA port, and the port remains available for SPL mounting.

The *Small Launch ESPA* is a scaled version of the EELV ESPA. [Maly, et al. 2009]. It accommodates radial mounting for up to 6 low-profile small satellites on standard 0.2 m or 0.38 m diameter interfaces. It is suited for CubeSats and satellites below 180 kg. It can accommodate secondary payloads up to 100 kg on Minotaur IV and Delta II.

Depending on the specific mission needs, there are quite a few alternatives to a dedicated launch to orbit. Of course, each approach has both strengths and limitations, but all of them can provide potentially large reductions in both cost and schedule.

For larger spacecraft there are fewer options for reducing cost and schedule, although the use of some of the alternatives above for testing elements of the system may be able to find problems early in the program and, therefore, avoid more expensive fixes later. Perhaps the best option for reducing both cost and schedule, or at least helping to prevent overruns, is to design the spacecraft for multiple launch vehicles. The cost of launch is typically negotiated between whoever is buying the launch and the launch provider. Clearly, there is more potential for negotiation if more than one launch provider is possible. Designing for multiple launch vehicles is usually not hard or expensive because the payload environments of all of the launch vehicles are typically similar, except for the Minotaur that provides up to 13 g's of axial acceleration because it is made from decommissioned ICBMs for which the loads were not a principal design consideration.

An equally important reason for designing for multiple launch vehicles is to protect the schedule. Recall that launch systems have approximately a 90% success rate. When a launch failure occurs, there is a significant downtime until the next launch of that system (see Sec. 26.1.6). In addition, if your payload was the next in line at the time of the failure, it may have been moved further back by higher priority launches when the launch system resumes operations. For this reason, nearly all of the constellation builders use multiple launch providers. This also provides a continuing negotiating position. Thus, if a constellation needs to launch 50 satellites, they may choose Launch Provider A for 15, Launch Provider B for 15, and reserve the last 20, depending on the performance of the first ones. Note that constellations may also use launch vehicles of different sizes by launching multiple satellites on a larger launcher. This can work out well or badly. Iridium launched its entire constellation without a launch failure. Unfortunately, GlobalStar lost 12 satellites on a single Zenit 2 launch failure [Harland and Lorenz, 2005].

As discussed in Sec. 13.1.1, a key to reducing both cost and schedule for systems of all sizes is the development of a low-cost, small, responsive launch vehicle. (See also Wertz [2010].) This is needed for both operational smallsats and for rapid testing of both technology and processes applicable to larger systems. It also provides for the rapid introduction of new technology, which is evolving particularly quickly in small spacecraft. Building launch vehicles to inventory, as needed for launch-on-demand, is primarily an issue of whether it is worth the interest cost on the money required to build the vehicle for the time period from when it is completed until it is launched [Wertz, 2010]. Thus, at 10% interest, holding the vehicle in inventory for 6 months would increase the build cost by 5% and the total launch cost by less than that, say 4%, plus an incremental cost for storage and maintenance.

This book does not address the design and development of launch systems. However, London [1994] provides an excellent overview of why launch systems cost as much as they do and ways to reduce launch system cost. Wertz [2000] provides a cost model intended to compare reusable vs. expendable launch vehicles, which has been updated to model the added cost of launch-on-demand systems [Wertz, 2004]. A number of papers on low-cost launch systems are available at the Reinventing Space website [Reinventing Space, 2011].

13.5.8 Spacecraft Technology

As shown in Table 13-18, reducing the cost of spacecraft is largely a matter of finding lower cost, lighter, or more competent components. One of the better ways to do this is to have the spacecraft do more of the functions in software and less in hardware. (See Chap. 20.) This has multiple advantages, such as:

- Lower mass
- Lower recurring cost
- Much higher functionality
- **Can be changed, upgraded, and fixed on orbit**

and also disadvantages:

- High non-recurring development cost
- Development process is difficult to manage

Table 13-18. Summary of Approaches for Reducing Spacecraft Cost and Schedule. Note that all of the systems engineering approaches of Sec. 13.5.5 are key to reducing spacecraft cost as well. (See Table 13-8. For additional discussion, see Secs. 25.3 and 25.4 and Fleeter [1996, 2000]. The Reinventing Space website [2011] also has many papers on this topic. For a detailed discussion of building low-cost communications satellites, see Davidoff [1998, now out of print] or Ford [2010].)

Reducing Space Mission Cost—Spacecraft Technology		© 2010, Microcosm, Inc.
Technique or Action	**Mechanism**	**Comment**
Sp1. Use Plug-and-Play Technology	Dramatically reduces cost and schedule for I&T	Has not been used much to date because it isn't minimum mass. Substantial work currently ongoing.
Sp2. More Extensive Use of Software	Minimizes mass and often allows use of a general-purpose processor	Can update and revise on-orbit as needed. Allows spacecraft to become a general purpose unit with specialized functions implemented in software.
Sp3. Use COTS Software	Immediately available; dramatically lower cost; tested through use	Possible drawbacks: May need modification and thorough testing. Typically not optimal for the application.
Sp4. Use COTS Hardware	Immediately available; dramatically lower cost; tested through use; less need for spares	Reduces both cost and risk when combined with large margins.
Sp5. Use CubeSat Hardware	All of the above plus built for space use	A specific example of COTS hardware for space applications. Most CubeSat hardware is in stock and available for immediate delivery.
Sp6. More Microelectronics	Lighter weight and lower cost than either mechanical parts or analog electronics	Takes advantage of dramatic recent growth in microelectronics. Key issue will be radiation tolerance—not a problem for low Earth orbit, but could be for higher orbits.
Sp7. Use Commercial Battery Technology	Both much lower cost and higher power density	Takes advantage of continuing advances in battery technology due to widespread use personal electronics of all sorts.
Sp8. Use More Composite Materials	Can be lighter, stronger, and lower cost	Can build much lighter, stronger structures with shorter schedules than metal tanks and structures. Potential problems include low thermal conductivity and near-zero coefficient of thermal expansion—very different than metals.
Sp9. Use Non-space Equipment	Takes advantage of existing designs, testing through use, and mass production	Typically not optimal. Often must be space qualified or put through major test program. Takes advantage of advances in design and extensive testing through use. Example: Carpenter tape antennas and hinges.
Sp10. Standardized Components and Interfaces	Reduces both cost and schedule. Avoids reinventing the wheel.	Has been remarkably unsuccessful in older space applications because it is sub-optimal in terms of weight and power. May be able to use these more in the future.
Sp11. Avoid Large Engines for In-Space Applications	Reduces cost, mass, and need for additional control components	Large engines in space often require separate control system and represent largest threat to the on-orbit spacecraft. Small thrusters can be controlled by spacecraft control system and recovery from errors may be possible.
Sp12. Hosted Payloads	Shares spacecraft bus and launch	Potential to increase cost and delay schedule if added payload creates conflicting requirements or forces a larger launch vehicle.

SPACECRAFT

- Subsystem interfaces that are all in the spacecraft computer are difficult to control

The ability to fix the software on orbit is a key consideration for reducing cost and increasing reliability. This implies the need to ensure that mission operations has procedures and processes in place to change out the on-orbit software. Doing more in software also suggests that there is a major advantage to being able to fly the latest computer available. (Item SE5 above.) In effect, the spacecraft becomes a general purpose processor with most of the work being done in software. Because both software and on-board processors are evolving very rapidly, this also implies a large advantage to lower cost, short-lived spacecraft. It is likely that you have much more processing capability in your cell phone than many traditional on-orbit spacecraft. This means that newer spacecraft will typically be more competent than older spacecraft, such that the value of an on-orbit asset continues to decline. Some of the features that we can reasonably expect from future software controlled spacecraft, include:

- Software-defined radio
- Pre-processing of images such that only the needed information is sent to the ground (i.e., FireSat II sends the fire location, size, and intensity direct to the end user in the field)

- More responsive systems, such that the spacecraft can send more detailed data if and when it is requested by the end user
- Autonomous on-board control of both orbit and attitude such that the spacecraft always knows where it is and where it's looking
- Precise control of spacecraft motion based on dynamic analysis such that all motions are both rapid and nearly jitter free

These features don't reduce cost directly, but rather allow low-cost small spacecraft to be much more capable, such that they can do the job of older, larger, much more expensive systems.

Another processor-related function intended specifically to reduce cost is the use of more plug-and-play electronics. Here the goal is to make an interface between the various spacecraft components and subsystems that will be essentially similar to the USB port on your computer in which multiple different items can be plugged in and begin to function immediately. This greatly reduces the time and cost associated with spacecraft integration and test. In addition, this allows the potential, for example, of a new more capable or more relevant payload to be put into a spacecraft that is in storage waiting for a need to be launched.

Using CubeSat hardware components, even if the satellite itself is not a CubeSat, is also a good way to drive down both cost and schedule. (For a discussion of CubeSats, see Secs. 2.1.8 and 25.4.) The two main advantages here are that CubeSat components are very low-cost and are maintained in stock and bought off-the-shelf, such that you can have them available with days. This not only reduces the cost and schedule of the component themselves, but also effectively eliminates, or greatly reduces, the need for spares because components can be obtained from inventory at any time. CubeSat components also make use of more modern technology, such as lithium-ion batteries, and that technology is tested on orbit much more rapidly because CubeSats themselves are launched more often and more quickly than traditional satellites.

Note that essentially all of the systems engineering approaches discussed in Sec. 13.5.5 serve to drive down the spacecraft bus cost. In addition, the effect of these processes is often multiplied when they are used together. For example, assume there is an existing optical instrument that is light weight and low cost. If we can use an existing instrument, it can drive down cost, risk, and schedule by a large amount. Of course, this instrument may or may not meet all of the mission requirements when flown in a traditional spacecraft. However, we can use a combination of trading on requirements (Item SE1); mission approaches, such as flying low (Item M2); and spacecraft approaches, such as the use of CubeSat components (Sp5) to create very capable, but also very low-cost space systems. This is the approach used for developing NanoEye, which can provide high-resolution imagery in a spacecraft with a projected recurring cost of less than $2 million [Van Allen, et al. 2010]. Typically, reducing the cost of a single component, i.e., buying lower cost reaction wheels, will have only a very small effect on the cost of the spacecraft and even less effect on the cost of the mission. Significantly reducing the cost and schedule of the mission as a whole typically requires that we use multiple, synergistic approaches to create a system that can meet our end objectives at very low cost and risk in a short period of time. For a much more extensive discussion of reducing spacecraft cost and schedule see Wertz and Larson [1996]. A large number of papers on this topic are also available on the Reinventing Space website [Reinventing Space, 2011].

There is at least one area in which low-cost spacecraft builders and traditional manufacturers would agree—test as much as possible and, if possible, test it like it flies. As discussed in Secs. 23.3 and 23.4, testing is critical to finding both integration flaws and design flaws, often associated with the interaction between various components. Low cost satellite manufacturers will try to find low-cost ways to conduct testing and will orient the testing toward specific questions that must be answered—i.e., will the spacecraft survive the launch loads, will it overheat in the vacuum environment, will it work in the radiation environment? As much as possible, low cost tests will be designed to verify these characteristics. Traditional systems will tend to use a more formal approach and have a "required" series of tests. (Note that thermal vacuum testing has the merit of driving out gasses that have been trapped in the material such that the test article will outgas less on orbit.) Testing is an important part of the regime for both low-cost and high-cost systems.

13.5.9 Operations

Traditionally, mission operations have been an expensive and complex activity; run from a mission operations center; requiring multiple people; and 24 hour coverage, 7 days a week. This, in turn, implies either 4 or 5 operations teams and, the management and communications needed to make them work smoothly together as discussed in Sec. 29.2. As summarized in Table 13-19, there are a number of approaches to reduce this cost.

The most direct approach to reducing operations costs is to reduce the operations crew to a single shift of 40 hours/week. This reduces the number of people, overhead, management, and communications costs. It also requires that the spacecraft be capable of "taking care of itself" for an extended period, including probably long weekends. Ordinarily, this is much easier with small spacecraft that have large design margins and are capable of at least maintaining themselves in nearly any orientation. Many small spacecraft are operated by one person on a very part-time basis often by using a safe mode combined with paging from the ground station. [Marshall, et al. 1996.]

Another approach for reducing operations cost is the use of service-provided ground stations. Here we are making use of existing ground stations located around the world that are both manned and maintained in order

Table 13-18◀, Fig. 13-8◀, Eq. 13-1◀

Table 13-19. Summary of Approaches for Reducing Mission Operations Cost and Schedule. Multiple approaches are now available for reducing the cost of mission operations. (For a more extended discussion, see Marshall, et al. [1996] and many papers from the Reinventing Space website [2011].)

OPERATIONS

Reducing Space Mission Cost—Operations		© 2010, Microcosm, Inc.
Technique or Action	Mechanism	Comment
O1. Use Service-Provided Ground Systems	Lower cost with little or no non-recurring cost	Substantial redundancy and large area of coverage. Disadvantage is that your mission may have to share priority with others.
O2. Share Ground System Across Programs	Shares cost among two or more user organizations	Critical to compromise on requirements rather than simply combine all requirements from multiple programs.
O3. Use Iridium or GlobalStar Modem	Very low cost, continuous communications link	Low data rate, but nearly continuous coverage. Only applicable to LEO.
O4. Use Autonomous Orbit Control	Reduces personnel requirement on the ground	Also reduces propellant cost and provides precise timing for future coverage. Allows planned coverage with precision.
O5. Fully Autonomous Systems, On-board and in Operations	May allow one-shift coverage and less-frequent commanding	Autonomous systems may allow ground operations by a single shift or even one person maintaining the ground system, computers, and software. May increase non-recurring cost.
O6. Fly the Spacecraft Over the Internet	Simplifies operations by making spacecraft just another InterNet site	Can use secure Internet or encrypted data to protect data and commanding. Means spacecraft can be controlled from virtually anywhere.
O7. Use AMSAT Resources for Science Data Return	Lower cost by having an unpaid network	Has worked successfully in astronomy for decades. Can reduce both cost and provide high reliability by having multiply redundant ground segments.
O8. Common Software for Test and Operations	Reduces both cost and schedule. Avoids reinventing the wheel	May be less efficient and less user friendly than the operations group would prefer.

to communicate with multiple spacecraft. This also provides a high level of redundancy and excess coverage. The main disadvantage is that you have to share priority with others. However, this can be overcome by complementing the service-provided system with dedicated remote antennas built specifically for your system. When used in conjunction with a service-provided system, these remote sites are not required to have near-100% reliability, because the other ground stations provide back-up and coverage in areas beyond the reach of 1 or a few dedicated remote antennas. Generally, the cost of the service-provided system is in the range of several hundred dollars per data pass, which is usually a great deal less than maintaining a dedicated ground system.

In conjunction with service-provided systems, there is the potential for simply flying the spacecraft over the Internet. This is done by using the service-provided system for communications between the satellite and the ground. The ground station then puts the data on the Internet, which is then downloaded by as many end users as need it. (A variety of encryption techniques are available that keep the data secure, if needed.) Commands are sent to the spacecraft via the same process and, again, can be encrypted to avoid others intentionally or inadvertently taking over the spacecraft. In this way, the spacecraft becomes effectively just another node on the Internet that you can talk to, get data from, and control from any location where Internet access is available.

One of the more cumbersome and critical ground station functions is maintaining the spacecraft orbit, particularly in low LEO where atmospheric drag is high. This can be accommodated by using a GPS receiver on board for navigation and autonomous on-board orbit control. [Conger, et al. 2002; Wertz, 2003; Plam, et al. 2008.] A secondary advantage of this approach is that you will know in advance (years in advance, if desired) just where your spacecraft is located at any given time to about 1 km in-track and more precisely in cross-track and radial.

Finally, another approach is to use AMSAT resources for science data return. This approach of making use of the amateur community has worked in astronomy for decades as amateur astronomers make most of the observations of variable stars for which it is simply too expensive to tie up the manpower and resources of professional astronomers. This would not only provide data return at much lower cost, it would also create a high level of interest in multiple communities where amateurs were collecting useful science data and genuinely helping in the exploration of space.

13.5.10 Summary

In summary, there are several broad lessons from looking at the multiple approaches for dramatically reducing space mission cost and schedule:

- Significantly reducing overall mission cost and schedule typically requires using multiple techniques that complement each other. Unless there is a single large cost or schedule driver, making a change in only one approach or one part of the system is unlikely to have a major impact on the system as a whole

- Reducing cost and schedule is not just a matter of finding a low-cost spacecraft bus or payload. It is a mission problem involving the full range of mission engineering issues in order to provide the end user the data they need, when they need it, at low cost and with high reliability
- Truly reducing overall space mission cost significantly will require at least some investment in and development of both low-cost small spacecraft and low-cost, small, responsive launch systems
- The greatest impact comes from mission diversity in which small spacecraft are used for some operational activities and also as a test-bed to rapidly and economically develop both processes and technology for reducing cost, schedule, and workload for larger missions

The issue of mission diversity is important. Using a naval analogy, having even a few less battleships and a lot more PT boats changes the nature of the game. Diversity will allow us to do things we simply couldn't do before with a flexibility and responsiveness that isn't possible today in space. The right combination of assets will make us much less susceptible to enemy attack or the vagaries of launch failures, system failures, or collisions with orbital debris. It also makes the overall space mission enterprise much lower cost which gives us the potential of undertaking projects that today we simply cannot afford to undertake.

Our ultimate objective is to find ways to reverse the Space Spiral as shown in Fig. 13-6. What we would like to do is make use of the vast knowledge and experience base on reducing mission cost from throughout the world to begin to create much lower cost missions which will allow both shorter schedules and more missions, which will allow a reduced demand for 0 failures (that was unrealistic in the first place), which will allow further reductions in mission cost, and so on. In a real sense, we want to reinvent space—to return to the drive and excitement of the early space program, but with the advantages of modern technology and the experience base that 50 years of space exploration have brought.

13.6 Avoiding Cost and Schedule Overruns

This topic is particularly challenging because, in some respects, overruns in cost and schedule are one of the biggest problems in the space program and, in another respect, they aren't a bad thing at all. As we will see, they erode credibility and rob other programs of the funding to keep them going. On the other hand, in R&D programs, they are a necessary consequence of attempts to dramatically reduce cost and schedule, and these attempts are critical to the future of the space program.

An obvious solution to avoiding cost and schedule overruns is to never undertake anything challenging, to never push on reducing cost, reducing schedule, or trying to achieve more with less. I call this approach *Objectives Erosion*—it is basically the bureaucratic response to perceived "Requirements Creep." In the Objectives Erosion approach, if there are consistent cost and schedule overruns, then the obvious solution is to ask for less performance, for a higher price, in a longer time, until eventually we have programs that do nothing, or at least nothing new, for dramatically high cost, in a time frame such that the program is no longer useful.

While this might, or might not, be the right solution for the space community, it is not one that I'm willing to recommend. It is my belief that we need to dramatically reduce cost and schedule on space programs and to try to do much more, faster, and, most likely, with fewer resources. If we set a goal of major reductions in cost and schedule, we will not always get there. But we will never get there if we don't try.

As an example, let's assume that, using the cost models of Chap. 11, we estimate that a particular program will take 5 years and cost $100M. We apply as many of the cost reduction techniques from Sec. 13.5 as are appropriate and believe that we can accomplish the mission in 2 years and $20M or, if we add margin, 2.5 years and $25 million. In fact, the program ends up taking 3 years and $40M. Have we done well, or have we done badly? From the point of view of the contracting officer, we have done badly and our award fee should be cut to zero. We overran the schedule by 25% and the cost by 60%. In this view, we are either cheating or incompetent and the program manager should be hung in effigy and never allowed to manage another program. But, from a different perspective, we have saved the government 2 years and $60M. We should get a thank-you letter from the Secretary of Defense and twice the normal award fee. It is likely that if we had not set aggressive goals and had not pushed hard to get the cost and schedule down as far as possible, the program would have taken longer and cost even more. After 50 years of experience in space, we have very good cost models that project what it would cost if we built the system using the traditional approach. In my view, a new approach to reduce cost and schedule should be judged in part on whether it can accomplish what it set out to do (i.e., meet the stated cost and schedule goals) and in part on how it does relative to a traditional approach based on standard cost models.

Because there is a difference of opinion across the community and no guaranteed answers, I will provide both a "traditional approach" to avoiding cost and schedule overruns and an "alternative approach" (i.e., mine). Cost and schedule overruns are important issues that are worth serious discussion of alternative approaches to see what is best for any given program.

In 2007, the 10 largest DoD space programs were collectively overrun by $32 billion. Also in 2007, major weapon systems programs were over budget by $295 billion (26%), up from $42 billion in 2000 [Deloitte Consulting, 2008]. The schedule delays are not as well documented, but certainly average many years.

NASA programs suffer similar problems and often cause delays in follow-on programs. The Chandra X-Ray

Observatory was most likely delayed 10 to 15 years by cost and schedule overruns on Hubble Space Telescope. The James Webb Space Telescope started at $2 billion and is currently at $6.5 billion, if it can meet its delayed launch date of 2015, which seems unlikely.

This history of cost and schedule overruns is dramatically damaging to follow-on programs and to the credibility of both government and contractor personnel. This situation is particularly hard if you're trying to reduce cost, because it is difficult for people to believe any estimates that are below traditional values, much less any projections that are well below. But if we don't have at least some cost and schedule overruns, then we're not trying hard enough to reduce cost and contain the schedule. As in other elements of reducing mission cost, we need to find a reasonable balance—with R&D and development programs that have some cost and schedule overruns (and some underruns) and operational programs that are built on time and within budget.

In spite of different approaches to overruns, there are some aspects that I believe most observers would agree on. Cost and schedule overruns are much too widespread to be a result of bad program management, contractors that intentionally mislead the government, or poor individual decisions. In general, program managers are very experienced and very good at what they do. Everybody is working hard to meet cost and schedule, yet somehow we have trouble actually doing it. It may be that our estimates are too low or our expectations too high, but overruns are a fact of life on most programs, particularly in R&D activities.

The overrun problem is something much more fundamental than bad estimating or bad management and, most likely, has multiple causes that may well feed off each other. For example, in space systems there is a culture of "all is forgiven" if things work right in the end. Therefore, if there are cost and schedule overruns, there is more pressure to be sure you get it done "right," which, of course, leads to even more cost and schedule overruns. The problem is—What about the science you didn't get or the soldiers who died because the system wasn't in place when it was needed. Cost and schedule overruns are systematic problems that are somehow "built in" to the process or the culture. This is what we need to find and change.

13.6.1 Traditional Causes and Solutions

As we have pointed out previously, the Apollo program went from a challenge to Congress by President Kennedy to landing a man on the Moon in 8 years. Similarly, the Manhattan Project to create the atomic bomb was completed in 30 months. The CORONA project to create the first photographic reconnaissance satellite went from contractor selection to launch in less than 12 months [Nowinski and Kohler, 2006]. Historically, complex programs went much quicker than they do today.

Despite that, some observers see a strong correlation between system technical complexity and cost and schedule overruns. In 2007, technical complexity produced an average 8.7% cost overrun relative to the baseline budget, as shown in Fig. 13-9. Many programs rely on leading edge technologies that are still maturing. Sophisticated software and other advanced technologies deliver greater functionality, but require a much higher level of integration and interoperability. When large weapons systems were less complex, the development cycle was shorter. Modern jet fighter programs take in excess of 13 years and satellite systems routinely require 10 years or more [Deloitte Consulting, 2008].

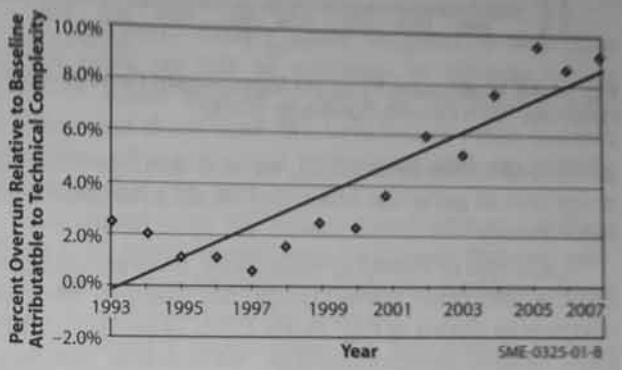


Fig. 13-9. Technical Complexity as a Cause of Increasing Cost Overruns (from Deloitte Consulting [2008]).

Recommended solutions to the problem of technical complexity include the following [Deloitte Consulting, 2008]:

- Divide programs into less complex work packages with shorter durations. Time is a clear predictor of budget and schedule risk. The longer the program, the greater the risk
- Don't approve contracts for system development and demonstration until the underlying technology is proven. Technology maturity is critical to program success, and should, therefore, be required prior to committing billions of dollars to development contracts
- Require the DoD and armed services to set a program's critical design requirements and priorities. Defense contractors cannot be expected to make the appropriate affordability trade-offs and "stand-in" for the customer
- Improve defense contractor capabilities in systems engineering, integration, and testing
- Provide more slack time in the schedule to react to unanticipated technical challenges
- Create budgets that reflect technical complexity risk and realistic assumptions

While many of these recommendations have merit, I believe the idea of "technical complexity" is rather subjective in this context. It is difficult to imagine tasks inherently more technically complex than the Manhattan Project or Apollo. In addition, the recommendations aren't consistent with the lessons learned from Apollo. If we required all of the underlying technology to be fully

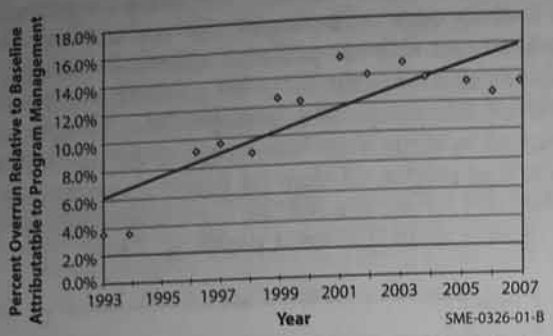


Fig. 13-10. Inadequate Program Management Contributes to Overruns (from Deloitte Consulting [2008]).

developed before proceeding, we would still be thinking about how to get to the Moon and would have spent a lot more money.

In addition to technical complexity, program management is also often given "credit" for many of the cost overruns as shown in Fig. 13-10. Other causes frequently cited include frequent program management turnover, unstable program requirements, lack of empowerment, funding constraints, and funding breaks as discussed in Sec. 13.2. This leads to a number of additional recommendations for reducing cost and schedule overruns [Deloitte Consulting, 2008]:

- Defense acquisition authorities need to understand and guard against "low bid" scenarios, where due to immature requirements specifications or lack of skill and talent, source selection awards turn out to be "change order" problems
- Establishing clear performance goals prior to program startup and applying appropriate monitoring processes to report effectiveness and efficiency
- Establishing an enterprise-wide definition of risk. Designing, implementing and maintaining an effective risk management program. Top executives set the tone, design, direction, and metrics; however, risk management should permeate all layers of the organization
- Defining clear roles and responsibilities, accountability, and authority for managing program costs and schedules. Inside the organization, establishing close coordination between multiple supporting functions, executive management, and the board
- Providing management and governing bodies with improved transparency into the organization's risk management practices. Executives and the Board should keep risk on the agenda and require timely information from operations and other supporting functions for critical decisions
- Ensuring the overall risk program has adequate support. Certain functions (e.g., legal, IT, HR, finance) should not just own risk, but also help other parts of the business manage risk effectively
- Budgets and schedules need to contain and be certified as "risk tolerant;" i.e., enough slack and contingency time is included to mitigate financial, operational, funding and human resource risks

Note that the first bullet above means that any attempts to dramatically reduce cost will be systematically and forcefully eliminated, which means that it may prove counterproductive. Similarly, there are a number of other traditional approaches that are more likely to make the problem worse rather than better. One of these is called "Back-to-Basics" in which the general idea is to take smaller steps in order to reduce the potential for overruns. This is primarily a bureaucratic approach to slowing down and stretching out procurement programs.

What's the problem with a logical progression of (a) develop technology and then (b) build a system? The first problem is that it inherently stretches the schedule and unfortunately, the government works on a 1-year budget cycle and a 4-year (or shorter) major review cycle. Typically, the government stops a program while it thinks, and it thinks very slowly. No manager likes to be told that stopping a program almost guarantees significant increases in cost and schedule, and nobody wants to take "credit" for these increases. The second problem is that good mission engineering requires working the technology and the mission simultaneously. (See Chap. 3 to 5.) To do good mission engineering we need to trade among the requirements, and, hopefully, between requirements and cost, and the only way to do that is if they are all moving forward together. It seems very unlikely that Apollo would have been faster, better, or cheaper if we had first developed the Saturn V to a largely arbitrary set of requirements and then thought about how to use it to get to the Moon.

13.6.2 An Alternative View of Causes and Solutions

Because I don't entirely agree with the more traditional approaches to reduce cost and schedule overruns, the list below provides what I believe are the major causes and some recommended solutions.

C1. Decision-Making Issues. There is no incentive for the Government to make decisions promptly, particularly if there is a possible risk. As we have discussed above, natural funding and decision cycles are at 1-year and 4-year intervals and many programs now extend over multiple cycles. This means they will be reviewed, re-evaluated, re-assessed, and most likely re-engineered many times from concept to implementation. This adds cost and delay, and delay adds more cost. Everybody can come up with a new thing to test, but nobody has the power to say "No, we don't need that test." As a practical matter, it is virtually impossible for the US to do a 6-month, $1 million program. Our decision making process simply wouldn't support that time frame or that cost.

C2. Contracting Issues. Contracting delays at every phase cause multiple breaks in execution that, in turn, create both cost and schedule overruns for the reasons we

have discussed. Another problem is that cost margin isn't allowed in the contracting process and will nearly always be removed by the contract negotiation. This builds in the probability of cost overruns. The third principal contracting issue is that contracting is designed as an adversarial process. Many authors have pointed out the need for the government and industry to work together as a team (see, for example, Nowinski and Kohler [2006]), but the contracting process, and often the culture within both government and industry, strongly discourages that.

C3. Mission Design Issues. Unfortunately, mission and system engineering is often short changed on space programs, frequently because of the need to create requirements and then buy a system that meets those requirements at minimum cost. This means that trades are rarely done between major cost elements (Sec. 5.3) and Figures of Merit are rarely computed or used (Sec. 5.4). System design teams typically won't trade on cost as a parameter, relying on the idea that "It costs what it costs."

As an example of this, different orbits have a cost—but often not to the mission design community. Sec. 10.3.2 creates and evaluates an "Orbit Cost Function," but orbit cost is almost never considered as an element of mission design. (See also Sec. 12.3.2 of Wertz [2009].) Two examples of poor choices in this arena are the use of highly elliptical orbits for communications (See Sec. 10.4.2.3) or Sun synchronous orbits to avoid eclipses.

C4. Programmatic Issues. Perhaps the most substantial issue here is that there is no advantage to under-running a contract in either cost or schedule. As discussed in Sec. 13.5.2, there should be some positive reward for achieving what you would like to get done and this is rarely done in government contracting. External reviewers with "no skin in the game" have no motivation to reduce cost or schedule, but might be blamed if something goes wrong. Therefore, there is no recognition of "reasonable risk," and the push is always toward reducing technical risk, even at the expense of schedule or cost risk, for which the external reviewer won't get blamed. Finally, forcing risk onto the contractor, while not necessarily causing cost and schedule overruns, forces out small, low-cost contractors, which leads to an environment of high cost and long schedules.

C5. Cultural Issues. As discussed in Sec. 13.2, the biggest impediments to both reducing cost and schedule and constraining cost and schedule overruns are typically cultural issues. People don't like to talk about cost or reducing cost and discussing it is often discouraged, both within the government and industry. In addition, it is often the case that the issue isn't cost, but cost overruns, so this encourages the contractor to bid high. That may help the overrun issue, but doesn't help in our long-term goal of trying to reduce cost and schedule. Also, the government and others have a strong inclination to not believe people who say they can do things more economically and are afraid they will simply introduce cost overruns.

Often the introduction of TRLs (Sec. 1.3.4) will cause more harm than good. There is a tendency to spend lots of time deciding whether something can be used in space based on heritage (i.e., TRL), rather than either engineering analysis or common sense. Also, there is a general understanding in the space community that both cost and schedule overruns are acceptable, so long as the system performs well when it's on orbit. Unfortunately, few people think about the other program that was canceled to cover that cost overrun, the disaster monitoring data that missed because the mission was late, or the soldier that died because the system wasn't yet in place.

In summary, today's space program is too often driven more by fear of unknown or adverse consequences, rather than by a desire to get the job done quickly, and at low cost. We have lost in many respects the sense of urgency and the professional engineering judgment that goes with it.

Associated with the causes listed above, I believe there is a commensurate set of solutions that can mitigate, though not entirely eliminate the problem of cost and schedule overruns. These are:

S1. Tiered Expectations. The key here is to have different expectations, and a different set of responses to cost and schedule overruns in a multi-billion dollar operational program than in a multi-million dollar R&D program. There should be FFP contracts for GEO communications satellites or other well-established technologies. (This is in place today.) Here, cost and schedule overruns should be a major problem area. On the other hand, there should be CPFF contracts for programs that include R&D or first unit development with large margins for small programs, new technology, or programs aimed at dramatically reducing cost. The key is how much we are trying to save and what the projected cost would be relative to traditional cost models. This is the example described at the opening of this section. If a program would traditionally cost $100M and we are trying to do it for $10M, an overrun of $10M, while not desirable, is still good performance and a major benefit. On the other hand, a cost overrun of $5B on a $5B operational system should not be acceptable.

S2. Create a Risk Mitigation Plan. The goal here is to identify and attack key risk areas in advance of their becoming major cost or schedule overruns. We would like to monitor progress closely in areas of perceived risk and have back-ups available—**including being willing to back off on performance requirements in order to contain cost and schedule.** This is the fundamental characteristic of trading on requirements. This implies having a goal-oriented team, not one that is process-oriented.

S3. Cultural Approaches. Perhaps the most important element here is to make cost data known and make cost and schedule matter. We want to create a true team activity where we work together for the common good, including cost and schedule. We want to create a plan of attack—design how to attack the problem of excessive cost and schedule and work together toward reducing them. We want to think about the opportunities that are missed—not because the system wasn't perfect, but

because we didn't get it there when it was needed or couldn't afford it because of excessive cost overruns.

S4. Programmatic Approaches. The most important approach here is to avoid schedule breaks. Schedule breaks may well be the single largest cause of cost and schedule overruns. There are a number of approaches for avoiding them and it's very important to do so. If schedule breaks can't be avoided, then the next best approach is to recognize that cost and schedule will both increase significantly and make the revised values the new baseline. Also avoid the tyranny of TRLs and the need to do technology development in series. It is much better to develop processes and equipment in parallel with the system because this allows for trades on what the process or equipment needs to do and allows us to either reassign requirements to other parts of the system or, reduce the requirements to be consistent with the capabilities. During this development, we should also have back-ups and alternative approaches. We need better systems and mission engineering throughout the program, including the creation of large margins so that the system is less susceptible to small variations in performance.

Another programmatic approach is to reward cost and schedule underruns. On the whole, rewards are much more effective than punishment. For example, if a contractor is able to drive down cost, we might choose to split the cost savings by using 40% of the savings to reduce the government's cost, 25% to new cost reduction studies or technology, 25% to the contractor, and 10% to the individuals who made it happen. Of course, rewards don't need to be money. We could give preference points toward future programs and contracts or a "thank you meeting" with the Administrator of NASA or the Secretary of the Air Force.

S5. Contracting Approaches. The major method here to use a "should cost" approach based on traditional cost models. Set the budget based on splitting "should cost" from the cost model and "bid cost" from the contractor and provide a tiered reward based on how much cost can be reduced. Similarly, a design-to-cost approach can be used that provides the best capability available for the assigned cost. Allow and reward creativity by specifying **what** and not **how**. For example, it is much better to award a contract for a system to achieve attitude determination to 0.01 deg, 3 sigma, rather than for a blue star sensor with an 8 deg square FOV and sensitivity to magnitude 4.5 in visible light.

We should encourage creativity in contracting by using methods such as the Cassini Resource Exchange in cost, mass, and power developed by Wessen, Porter, and others at JPL [Wessen and Porter, 1998]. Planetary missions are particularly susceptible to cost overruns, in part because there is no motivation for PIs to not go back and ask the program office for more money, more mass, or more power. The basic idea of the resource exchange is to create a "stock market" in cost, mass, and power. At experiment selection, the program office divided all the cost, mass, and power among the PIs, with no margin. If one of the PIs needed more mass, they get it by trading cost or power to one of the other PIs; similarly for cost and power. The program office tracked (and posted on their website), the value of mass and power. The decision-making process was now in the hands of the PIs to determine the best way to use the limited resources available to accomplish the best mission they could within the constrains of power, mass, and cost. The bottom line—the program came in on budget in cost, mass, and power. As this end result became apparent, the value of mass and power went down—i.e., you got more for your excess power (or mass) if you traded early.

S6. Use Multiple SmallSats. Sets of small satellites can do many of the things that larger satellites are more traditionally used for. Smallsats have inherently much shorter schedules, which implies far fewer schedule breaks. It also reduces the cost of individual failures. Cost and schedule overruns are likely to be much smaller dollar values, although, of course, this may not matter to the rule-oriented bureaucracy. Teamwork is easier in small programs and individual responsibility is greater. There are two approaches for using sets of smallsats. The satellites can each be individual missions that work together for better coverage or timeliness, such as the SSTL Disaster Monitoring Constellation. Alternatively, the mission functions can be broken down and split among multiple satellites, as in the DARPA F6 program.

In summary, there are many potential solutions. What they all have in common is that cost and schedule have to matter from the outset.

If our specific goal is to reduce, or contain the schedule, then the answer is to determine the value of doing so and reward that behavior. Make it worthwhile to come in ahead of schedule. The classic example of this is the Northridge earthquake in Los Angeles on Jan. 17, 1994, which caused freeway overpasses to collapse on the I10, at the time the single most heavily used highway in the US. Unfortunately, the freeway doesn't work as well when the bridges have collapsed. The state of California offered a bonus of $250,000/day for every day ahead of schedule that the bridge reopened. The I10 reopened 88 days after the earthquake, 66 days ahead of schedule, the contractor earned a bonus of $16 million, and millions of commuters got back on the freeway 2 months earlier than they otherwise would have, and probably years ahead of what would be the case with normal bidding and contracting. Returning to space missions, if we can constrain the schedule for both decision-making and contracting, we will have gone a long way toward constraining the cost as well.

As with nearly all aspects of space mission engineering, it's a matter of finding the right balance. Some cost and schedule overruns are acceptable in R&D or development programs that are pushing dramatic improvements. Continuing overruns on operational programs are dramatically bad. The hard part is being able to appreciate the difference in a rule-based system.

It seems clear that the right answer is returning to some of the approaches used at the beginning of the space program—being aggressive in what we attempt;

Table 13-19◀, Fig. 13-10◀, Eq. 13-1◀

forcing rapid, parallel decisions making; and doing good up-front mission engineering, but doing it quickly. We need dramatic reductions in both mission cost and schedule. The most certain way to fail is not to make an attempt by concentrating only on overruns and not on what has been attempted or achieved. We need to use what we have learned over 50 years of space exploration along with modern technology to change the way we do business in space. We need to Reinvent Space.

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14 Overview of Spacecraft Design

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14.1 The Spacecraft Design Process

14.2 Spacecraft System Design Drivers
Mass, Power, Cost, and Schedule; Lifetime and Reliability; Total Delta V; Orbit; Payload Accommodation

14.3 Spacecraft Configuration Alternatives
Attitude Control Technique; Need for Propulsion and Its Impact; Solar Array Configuration; Communication Antenna Placement; Spacecraft Autonomy

14.4 Partitioning Spacecraft into Subsystems
Propulsion; Attitude Determination and Control; Position and Orbit Determination and Control; On-board Processing; Telemetry, Tracking, and Command; Power; Structures and Mechanisms; Thermal Control; Other Specialities (Cross-cutting Disciplines)

14.5 Creating Preliminary Spacecraft Budgets
Mass; Propellant; Power; Pointing and Alignment; Data System

14.6 Design Evolution
LRO; WISE; WIRE

14.7 Examples
SCS; FireSat II

14.8 Future of Spacecraft Design

14.1 The Spacecraft Design Process

Having defined the overall mission objectives and instrumentation, it is now time to develop the initial spacecraft bus design. The spacecraft bus's purpose is the support of the payloads. The mass, power consumption, data rate, pointing requirements, and other aspect of the payloads will directly impact the overall configuration of the spacecraft. The bus must take care of final orbit adjustments; orbit maintenance; pointing; commanding; data storage and downlink; power generation, storage, and delivery; structural support; and thermal control. Exactly how these functions are accomplished will greatly impact the overall cost of the space mission.

In Chap. 3, you learned about establishing the requirements and constraints for the overall mission through discussions with your customer. As you go through this chapter, you may find the design running into trouble with some of those early decisions. Do not be afraid to discuss these issues with your customer—you may find some flexibility you did not anticipate. The Conops from Chap. 4, the environments from Chap. 7, and the orbit as described in Chaps. 8–10 will all drive the basic configuration of the spacecraft, as you will see in Sec. 14.3. This chapter is the next step toward refining your mission architecture that you started in Chap. 5. Sec. 14.7 will expand on the FireSat II and SCS examples of Sec. 5.6. Throughout this chapter, you will need to use the techniques of Chap. 6 as you capture many spacecraft and subsystem requirements and refine your mission requirements. Depending on your mission, you will usually improve your payload definition by working through Chaps. 15–17 first, but you may want to work through the spacecraft design in this chapter to figure out the resources available for the payload. The FireSat II mission takes the first approach and SCS takes the second. To complete your spacecraft design, you will need to iterate with the subsystem designs as described in Chaps. 18–22.

It cannot be said too often: Space Mission Engineering is a complex process with many intertwined steps. You need to hold the big picture objectives in your head while you delve down into the details. You will need to use good engineering judgment as you move back and forth between levels, iterating the design. There is no single, correct solution and "optimum" is a matter of opinion (with your customer's opinion being the most important). Decisions you make will involve millions of dollars, and a single mistake can end a mission. Section 14.6 describes the challenges of three real missions as they moved from preliminary design through operations. As you will see, the space business is one of incremental gain punctuated by catastrophic failures. This isn't a business for the timid, so let's quit wasting time and get started.

Table 14-1 describes the basic steps used to design a spacecraft. The items in Table 14-2 impact the system configuration and must be well understood by everyone working on the design. Note that payload configuration, including its mass, power, data rate, and pointing requirements are key drivers. Chapters 15–17 will guide you through establishing payload parameters. Docu-

Table 14-1. Spacecraft Design Steps.

Step	Reference
1. Document design drivers and requirements, including payload parameters	Sec. 14.2, Chaps. 15–17
2. Determine overall configuration and key trades	Sec. 14.3
3. Partition into subsystems	Sec. 14.4
4. Establish budgets and subsystem requirements	Sec. 14.5
5. Develop subsystem designs	Chaps. 18–22
6. Evaluate overall system configuration against requirements and constraints	Chap. 5
7. Explore options	Perspective of Others
8. Document and Iterate	Steps 1 to 7

menting the drivers creates a clear agreement between the builders and the stakeholders, so the design proceeds in the correct direction. The design drivers will lead to an approximate size and general layout for the system, with the attitude control technique, the propulsion configuration, and the solar array configuration (or other power generation method) defined through some key trade studies. We can then partition the spacecraft into subsystems and establish budgets and requirements for each of those subsystems. With the top-level subsystem requirements in place, we can now hand off pieces of the system to discipline experts who will further define details. As the subsystem designs begin to take shape, we must track the mass and power totals and ensure that the overall system continues to fulfill the mission requirements and constraints. Our team must document their design decisions along the way, leaving a clear path for later reviewers to follow. Inevitably, details of the subsystems will conflict with each other or the mass or power will exceed the original allocations. The systems engineer must remain constantly vigilant for these issues. Sometimes subsystem leads simply misunderstand the requirements, and sometimes the requirements just cannot be met within the budget; but when these obstacles show up, the systems engineer must quickly resolve the issue so effort is not being wasted on the wrong direction. Be sure to talk with other experts and get different perspectives. Ask your subsystem leads for options, not just one solution. Sometimes you will need to revisit the original spacecraft system drivers, and the builders must renegotiate the agreement with the stakeholders. There is no cookbook formula for this process other than simply "iterate."

14.2 Spacecraft System Design Drivers

Table 14-2 summarizes the primary drivers of spacecraft design. These drivers are covered in the rest of this section.

Table 14-2. Spacecraft Design Drivers. The key parameters that drive a spacecraft design come from mission-level decisions and the payload configuration.

Spacecraft Design Drivers	Mission Source	Impact	Thresholds	Ref
Mass	Payload mass	Launch vehicle	<250 kg, <1,000 kg, <7,000 kg	Sec. 14.2.1
Power Consumption	Payload design	Power sys, solar array config	beyond Mars or < 2 weeks	Sec. 14.2.1
Cost	Funding organization	Ripples throughout system	<$5M, <$100M, <$500 M	Sec. 14.2.1
Schedule	Funding organization	Development process	<3 yrs, <5 yrs, >5 yrs	Sec. 14.2.1
Lifetime	Mission design	Redundancy, quality of parts	<1 yr, <3 yrs, <5 yrs, >5 yrs	Sec. 14.2.2
Reliability	Mission design	Redundancy, quality of parts, margin	Experimental, operational, human rated	Sec. 14.2.2, Chap. 24
Delta V	Mission design	Propellant load	0, 100, 1,000, 2,000	Sec. 14.2.3, Sec. 10.3
Orbit	Mission design	Solar array, thermal, radiation reg, launch vehicle	<1,000 km, high-Earth, planetary	Sec. 14.2.4, Ch. 8–10
Payload Accommodation — Data Rate, Vol, Latency	Mission purpose	Comm, data storage, ground	Thresholds changing fast	Sec. 14.2.5
— Pointing Requirements	Res or antenna beam	Attitude control system	<5 deg, <0.5 deg, <20 arcsec	Sec. 14.2.5
— Mass, Volume, FOVs	Payload constraints	Mechanical design	Diameter: <1 m, <5 m	Sec. 14.2.5
— Other	Timing accuracy, contamination sensitivity	Data system, master oscillator, cost of ground processing	Mission specific	Sec. 14.2.5

14.2.1 Mass, Power, Cost, and Schedule

Mass is the fundamental resource for the spacecraft designer. The cost of space missions is ultimately driven by the high cost of delivery to orbit. Because the launch vehicle is expensive, and because the spacecraft cannot be retrieved without the cost of a second launch, developers spend significant effort (and money), in the form of extensive analysis and testing, ensuring that the spacecraft will meet the mission objectives. The more it costs to get to orbit, the more the stakeholders are willing to spend to analyze and test the system before launch. Launch vehicles are expensive because it requires a tremendous amount of energy to get to orbit. A 1,000 kg spacecraft in a 600 km circular orbit has a total energy of 34 GJ (kinetic and potential energy) relative to the Earth's surface! That's 9,570 kWhr—enough energy to operate a house for almost half a year. To put this much energy into a spacecraft requires what amounts to a controlled explosion, with all of the force of the chemical energy going out the nozzle of the rocket. This is a very difficult problem, with little margin for error, resulting in an expensive system. More spacecraft mass means a larger and more expensive rocket, so mass is a fundamental cost driver in space systems, with distinct breakpoints when the mass reaches the limit of a particular vehicle.

Because it is the major cost driver, system mass must be estimated early, and the designer must keep careful track of the mass throughout the development. Mass estimation comes in 3 steps:

1. A rough order of magnitude based on payload mass (details in this section)
2. An estimate based on functions and configuration (details in Sec. 14.5)
3. A bottoms-up budget (using Sec. 14.5 guidelines combined with Chaps. 18–22).

Based on historical data and the type of mission (see App. A and Table 14-18), the spacecraft dry mass (including payload) will be between 2.5 and 6.5 times the payload mass. This dry mass, along with the propellant mass (see Sec. 14.5.2) will establish the total mass for the system. From this mass and the target orbit, the spacecraft designer can quickly determine what class launch vehicle is required (see Table 26-7) and, from that, determine the order of magnitude for the cost of the system.

We will further refine this mass estimate in Sec. 14.5, followed by a more detailed estimate based on subsystem conceptual designs. The subsystem roll-up should get the mass estimate within about 30% of the final number at launch (see Sec. 14.6).

The other key resource for the spacecraft designer is power. The power consumption of the electronic components in the system must be generated and delivered to the loads. We establish the power generating capability early in the design process, because power systems considerations, such as solar array size and configuration, will ripple throughout the system design (see Sec. 14.3.3). Once the power system is sized, it becomes another constraint on the design of the rest of the system. Large payload power demands, for payloads such as RF communications or radar, will result in large power systems, with power being a major driver of the overall system design.

Besides mass, power, and cost, schedule tends to be the other dominating constraint driving all other decisions in the development of a spacecraft. The people funding the spacecraft expect a return from the system within a certain amount of time in order to make the funding worthwhile. The spacecraft designer only has a limited time to consider options and choose solutions. The builders have a limited time to assemble the system and check it out. If the time available is unusually short, it can drive all aspects of the system development and potentially jeopardize mission success (see Sec. 14.6.1). If the time is long, it gives the designers time to optimize the system and develop new technologies to reduce the overall mass. But generally, time is money; so a longer development time will cost more money. When establishing the top-level architecture, the system designer should work with the stakeholders to establish a schedule that strikes a balance between driving cost and driving technical decisions and performance.

14.2.2 Lifetime and Reliability

Spacecraft lifetime and the probability that the mission achieves that lifetime will drive decisions about redundancy (duplicate components) and reliability of components (see Table 14-3). The importance of the system will also drive reliability, with experimental, operational, and human-rated providing general classes of increasing reliability (see Table 14-4). Redundant components add to the overall mass and cost of the system. Without a failure, these components are, as the name indicates, unnecessary. Short missions will generally be *single-string*, with only a single copy of each component. Any failure in a single-string spacecraft will generally lead to the loss of the mission; although some areas of the design may lend themselves to *graceful degradation*, where a failure leads to loss of performance but not loss of the entire mission. With thorough testing and careful part selection, single-string spacecraft can be very reliable, as demonstrated by NASA's Small Explorers and by AMSAT spacecraft which continue to operate 10 to 20 years after launch. Testing is easier without the need to verify redundant paths, so testers are more likely to find design flaws, but single-string spacecraft are vulnerable to the failure of a single part or the error in assembly of a single component.

As the desired mission lifetime is increased beyond one year, the designer should start to consider adding redundancy for key components. Various analytic techniques are available to help identify weak links in the design (see Chap. 24). Items that cannot be easily tested (such as components associated with deployments, actu-

Table 14-3. Conservative Redundancy Strategy for Different Lifetimes. This table is based on common practice for NASA missions. It is intended primarily to demonstrate that longer lifetimes will tend to drive more back-up hardware into the design. The actual application of redundancy for your mission should be based on your mission objectives and tolerance for failure.

Lifetime	Redundancy
<1 year	Single-string
1–3 years	Selected redundancy
3–5 years	Block redundancy
>5 years	Fully-cross strapped

Table 14-4. Classes of Reliability.

Class	Description
Experimental	Generally short mission to demonstrate capability or gather data to answer a specific question. Missions in this class are often secondary payloads on a launch vehicle
Operational	Mission to gather critical data or to provide critical services
Human-Rated	Vehicles that must support humans

ators for solar arrays and antennas, and propulsion valves and thrusters) should be the first items considered for redundancy. Complex circuits involving analog circuitry (such as power converters) are the next areas which reliability analysis will generally identify. Beyond 3 years, the spacecraft will frequently carry a second copy of each critical component in a *block-redundant* configuration, where use of a redundant component requires switching all components to the alternate side. In a fully *cross-strapped* system, each redundant component can be separately switched into operation, without changing the state of the other components. We use this configuration for long duration missions or missions where reliability is particularly important (such as human space flight). See Fig. 14-1.

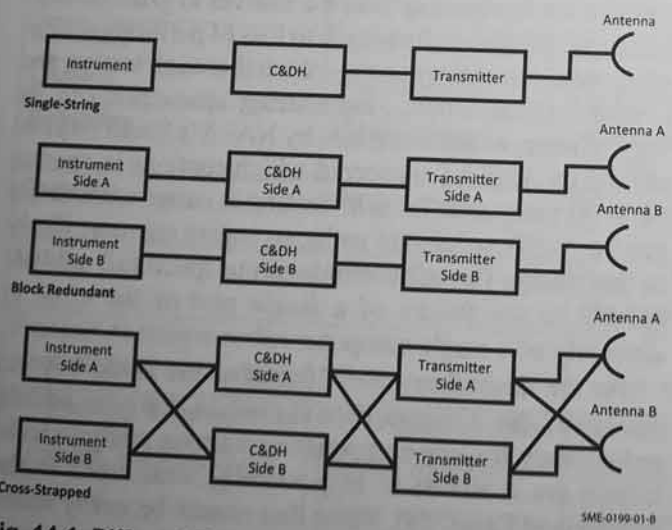


Fig. 14-1. Different Types of Redundancy.

One thing to keep in mind with redundancy is that it will not protect you from design flaws—if side A fails because of a poor circuit design, side B will probably experience the same failure. Consider extra testing, better parts, and more thorough analysis of your single-string design before you add the expensive mass of redundant components. Also consider *functional redundancy*, where completely different components provide the same function, in order to eliminate risks associated with design and manufacturing flaws. Watch carefully the circuitry involved with switching between redundant components. I worked on a spacecraft that included two power-control boxes with a relay to switch between them. It turned out that the relay was less reliable than the box, so we eliminated the relay and one of the boxes and ended up with a **more** reliable system.

Hurley and Purdy [2010] points out that the full reliability of a system includes delays in deploying the system. If you need the system operating but it is still on the ground, then your reliability is zero during that period. One of the biggest challenges in the space business is balancing the costs and delays associated with ensuring reliable performance against the costs and delays associated with a failure after launch. Dealing with rare, catastrophic events is challenging—there are not enough samples for reliable statistics, and our intuition frequently steers us in the wrong direction. Chapter 24 contains a more thorough discussion of risk and reliability.

14.2.3 Total Delta V

The total change in velocity, or *delta V*, required throughout the lifetime of a mission will drive propulsion requirements (see Sec. 14.3.2) and propellant mass. Section 10.3 covers the development of a delta V budget for a mission. The total delta V combined with the *specific impulse* of the thrusters will define the overall mass of propellant required. Specific impulse, I_{sp}, is a measure of thruster efficiency related to the exit velocity of the propellant, with typical units of seconds (see Sec. 18.1). As a first cut at total propellant mass, use the rocket equation with an I_{sp} of 220 sec for a mono-propellant hydrazine system, 300 sec for a bi-propellant system, or whatever I_{sp} is appropriate for the design's propulsion system. For example, a spacecraft with dry mass of 1,000 kg, requiring 1,300 m/s of delta V from a mono-propellant hydrazine system, uses this form of the *rocket equation*:

$$M_p = M_f\left(e^{\Delta V/(I_{sp}g_0)} - 1\right) \qquad (14\text{-}1)$$
$$= 1000\cdot\left(e^{1300/(9.8\cdot 220)} - 1\right) = 828 \text{ kg}$$

where M_f is the dry mass of the spacecraft and g is the acceleration due to Earth's gravity. With a bi-propellant system, the total propellant mass is only 556 kg (replace 220 sec with 300 sec). This method will provide a rough estimate of required propellant, with Sec. 14.5.2 describing the details of a propellant budget.

Table 14-4, Fig. 14-1, Eq. 14-1

Different types of propulsion system are commonly used for different delta V totals. If a system has no delta V requirements, propulsion is probably not needed (the exception may be for attitude control). For a small delta V, cold gas is usually a good choice, since pressurized nitrogen is non-toxic and non-flammable and so requires no extra valves for safety. As the delta V gets above about 100 m/s, mono-propellant hydrazine becomes the fuel of choice, since it does not require the complexity of a separate oxidizer. Above 1,000 m/s, the higher efficiency of a bi-propellant system will compensate for the expense of a heavier and more complex propulsion system. For extremely high delta V's, above 2,000 m/s, an ion engine, with its very high I_{sp}, is the only practical choice at this time. These thresholds, based on mass efficiency, are only general rules of thumb, so you should take a careful look at the specifics of your system, including a consideration of solid rockets, before choosing a propulsion system. Section 14.3.2 and Chap. 18 provide more information on the selection of a propulsion system.

14.2.4 Orbit

The mission orbit design, described in Chap. 10, drives many aspects of the spacecraft. The space environment will vary based on where the spacecraft goes (see Chap. 7). The orbit and the mission targets will define where the Sun will be with respect to the spacecraft body, impacting the solar array design (see Sec. 14.3.3) and the thermal design. Spacecraft will frequently keep one part of the body away from the Sun, where radiators are placed to dump heat. The altitude of the orbit will impact the thermal influence of the orbited body on the spacecraft. For low-Earth orbit, the Earth fills almost half the sky, with a temperature near 300 K. For a low lunar orbit, the Moon will fill almost half the sky, but its temperature varies from 150 to 400 K! For a geosynchronous orbit, the Earth is only 17 deg wide, so its thermal influence is minimal, and the spacecraft sees mostly deep space at 3 K. The orbit defines how far away the ground stations will be, impacting the radio link design (see Sec. 21.1). Gravity gradient and atmospheric drag are altitude dependent (see Sec. 19.1.2). These effects, along with Sun location and obstruction of stars by the orbited body will impact the attitude control design.

14.2.5 Payload Accommodation

Spacecraft design is all about payload accommodation. The spacecraft exists to provide for all of the payload's needs. The payload command and data generation requirements will impact the data system design. The instantaneous data rate of the payload interface will impact exactly how data is transferred to the spacecraft bus, including not only the electrical data standards but also the software involved with the transfer. The average data rate over time combined with the frequency of downlinks will impact data storage requirements (see Sec. 14.5.5). The latency requirements of the mission will determine how often the spacecraft must transmit data, impacting the communications system and the overall power consumption. The payloads define how accurately the spacecraft must point, driving the design of the attitude control system. Structurally, the spacecraft must support the payload through launch, and it must provide a stable platform for measurements. Fields of view of multiple instruments can be a challenge, with each instrument needing to stay out of the view of the others, while all mounting to a small panel facing in the same direction. Some payloads are physically large, such as payloads with large antennas or optical apertures. Optical payloads will require a clean environment to protect the optics, and ultra-violet instruments are particularly sensitive to organic contamination. Contamination control considerations can drive the selection of materials and the handling of flight equipment, causing ripple effects throughout the design. Precise timing of measurements can drive onboard clock requirements, resulting in significant mass and power impacts in order to maintain a thermally stable oscillator. The system designer must look carefully at the payload, paying particular attention to unique features, in order to fully understand impacts across the system.

Pointing accuracy typically drives sensor accuracy. For pointing requirements on the order of several degrees, a set of coarse Sun sensors and a magnetometer can provide sufficient pointing accuracy. Horizon sensors and fine Sun sensors provide accuracies around a half a degree or so. Below a half a degree down to tens of arcseconds, a star tracker combined with an inertial measurement unit (gyro) will provide sufficient knowledge. Below this level, special equipment (both sensors and actuators) and extra care in the mounting of components will be necessary. See Sec. 14.3.1 and Sec. 19.1 for more details on attitude control techniques.

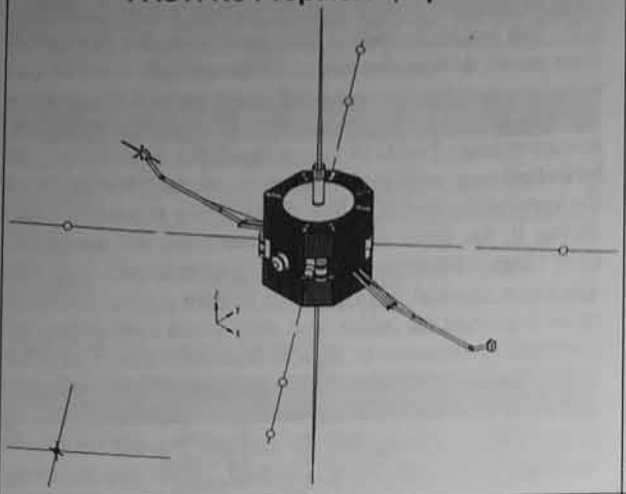
14.3 Spacecraft Configuration Alternatives

Having defined the key drivers of the system, it is now time to look at major architectural decisions that will affect all other aspects of the spacecraft design. These key decisions include choosing an attitude control technique, determining whether propulsion is needed and, if so, how it will be implemented, determining the solar array and antenna configurations, and determining an overall autonomy approach. Figure 14-2 provides some examples of spacecraft configurations to give you a feel for the options and decisions ahead.

The overall size of the spacecraft will tend to push aspects of the design in particular directions. Table 14-5 provides some general rules of thumb to help you scale up or scale down from previous designs. These general trends will always have exceptions. Connectors, fittings for propulsion or cryogenics, electronics parts, and other standard products come in fixed sizes, independent of the size of the rest of the system. If you are trying to scale significantly smaller, these items can cause your mass estimates to be unrealistically low.

Table 14-4◀, Fig. 14-1◀, Eq. 14-1◀

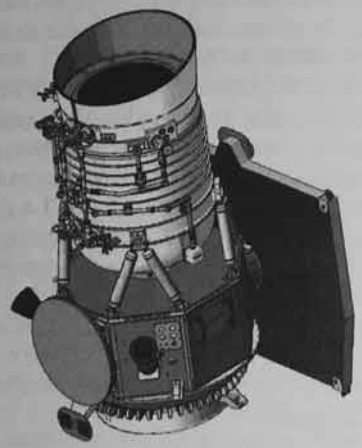
FAST: No Propulsion; Spinner



Mission: Particle and field measurements in Earth's auroral acceleration region
Orbit: 350 × 4,200 km, 83 deg
Payload: Electron and ion instruments, magnetometers, electric-field instruments
Configuration Features:
- ***Unique features:*** Single deck for avionics in spin plane, with solar array substrate and a top panel providing radiation shielding
- ***Solar array:*** Wrap-around with conductive surface (minimize impact on measurements) and thick (1.3 mm) coverglass (radiation protection)
- ***Appendage:*** 30 m wire booms, 3 m axial booms, 1 m magnetometer booms
- ***Attitude control:*** 5 RPM orbit-normal spinner, 1 deg pointing accuracy; wire coils used for magnetic torquing; Sun and horizon sensors
- ***Redundancy:*** Single-string

Mass: 191 kg
Average Power: 52 W

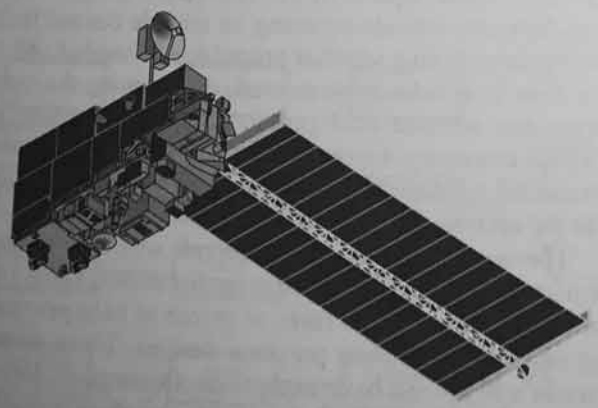
WISE: No Propulsion; 3-Axis; Fixed Array



Mission: All-sky infrared survey
Orbit: 525 km, 97.5 deg, 6 pm ascending node
Payload: Solid-hydrogen cryostat cooling infrared telescope
Configuration Features:
- ***Unique features:*** Fixed, passive, phased-array Ku-band antenna (circular, on anti-sun side); struts support and thermally isolate cryostat from spacecraft bus; load isolation system near separation ring; zenith pointing and dawn/dusk orbit allow fixed array
- ***Solar array:*** 3 m^2, fixed
- ***Appendage:*** None
- ***Attitude control:*** 3-axis, constant slew rate for sky survey, 75 arcsec pointing accuracy, 4 wheels; 3 torque bars
- ***Redundancy:*** Single-string

Mass: 661 kg
Average Power: 301 W

TERRA: LEO with Propulsion; Nadir Pointer

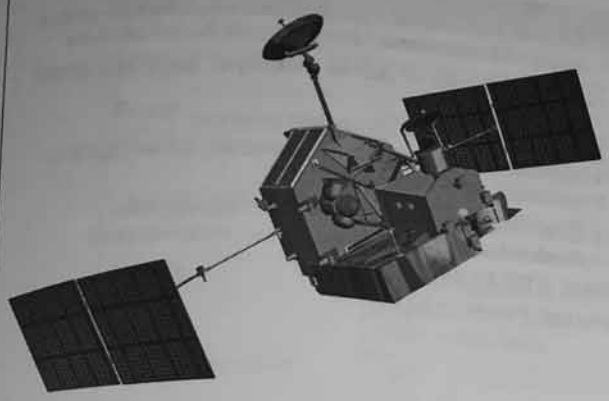


Mission: Earth observation
Orbit: 705 km circular, 98.2 deg, 10:30 pm ascending node
Payload: IR radiometer, high-resolution imager, visible and IR spectrometers
Configuration Features:
- ***Unique features:*** Radiators on anti-sun side; numerous ground-observing apertures
- ***Solar array:*** 45 m^2, single-axis gimbal
- ***Appendage:*** High-gain antenna (HGA)
- ***Attitude control:*** 3-axis, nadir pointer, 2.5 arcminute pointing accuracy
- ***Propulsion:*** 65 kg hydrazine and pressurant
- ***Redundancy:*** Redundant and cross-strapped

Mass: 4,761 kg wet, 4,396 kg dry
Average Power: 2,159 W

Fig. 14-2. Spacecraft Configuration.

TRMM: LEO with Propulsion; Nadir Pointer



Mission: Tropical rainfall measurement
Orbit: 350 km circular, 35 deg
Payload: Precipitation radar, microwave radiometer, visible and IR radiometers, and lightning imager
Configuration Features:

- ***Unique features:*** Low orbit to support radar measurements; symmetric arrays to balance drag torques; wheels in tetrahedron; large flat-panel radar antenna; yaw flip to keep Sun on same side
- ***Solar array:*** 2 deployed panels, single-axis gimbal; >3,000 W beginning of life
- ***Appendage:*** HGA for communication with TDRSS; antenna for spinning microwave radiometer
- ***Attitude control:*** 3-axis, nadir pointer, 0.2 deg pointing accuracy, reaction wheels
- ***Propulsion:*** 899 kg hydrazine and pressurant; large propellant load to compensate for drag at low altitude
- ***Redundancy:*** Fully redundant and cross-strapped

Mass: 3,504 kg wet, 2,605 kg dry
Average Power: 1,019 W

MAP: High Earth (L2)

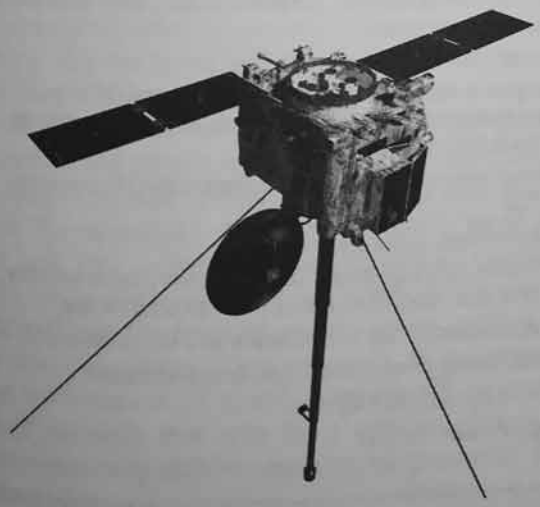


Mission: Measure variations in microwave background radiation
Orbit: Earth Libration, L2
Payload: Microwave receivers
Configuration Features:

- ***Unique features:*** Sun shield shadows spacecraft avionics and instruments; instrument antennas perpendicular to sun line
- ***Solar array:*** Deployed in petal arrangement (hidden by Sun shield in this view)
- ***Appendage:*** Sun shield
- ***Attitude control:*** 3-axis, scanning at 2.8 deg/s, 0.25 deg pointing accuracy, 3 wheels, thrusters
- ***Propulsion:*** 194 kg of hydrazine and pressurant
- ***Redundancy:*** Selective redundancy

Mass: 812 kg wet, 724 kg dry
Average Power: 371 W

STEREO: High Earth (Fly Away)



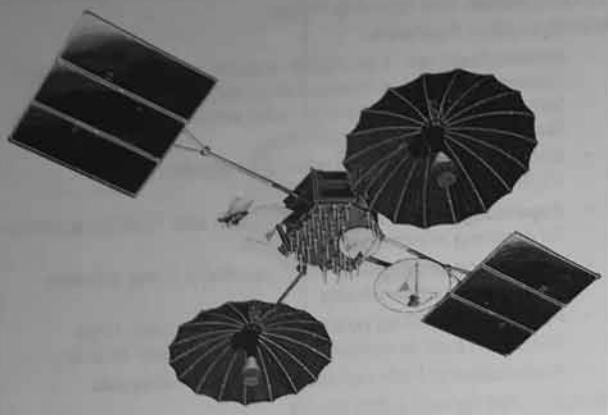
Mission: Solar Observation from 2 spacecraft
Orbit: Solar orbit, Earth fly away
Payload: Extreme UV imager, coronographs, visible imagers, particle and field instruments
Configuration Features:

- ***Unique features:*** Booms keep sensitive instruments away from the influence of the spacecraft electronics
- ***Solar array:*** 2 deployed and stationary wings of 2 panels each
- ***Appendage:*** Particle and field instruments, HGA
- ***Attitude control:*** 3-axis, 7 arcsec pointing accuracy, wheels, thrusters
- ***Propulsion:*** 61 kg of hydrazine and pressurant
- ***Redundancy:*** Selective redundancy

Mass: 619 kg wet, 558 kg dry
Average Power: 514 W

Fig. 14-2. Spacecraft Configuration. (Continued)

TDRS: High Earth (GEO)

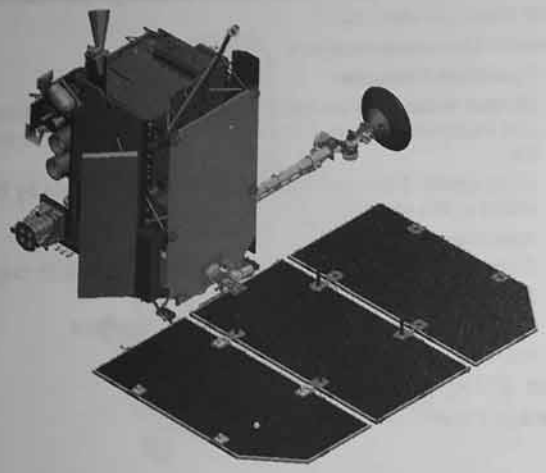


Mission: Communications relay for spacecraft
Orbit: Geosynchronous
Payload: S- and K_u-band radio relays
Configuration Features:

- ***Unique features:*** 2 high-gain antennas for single-access users, S-band phased array for multiple-access users
- ***Solar array:*** 30 m^2 across 2 deployed, single-axis-gimbal wings
- ***Appendage:*** Solar arrays and antennas
- ***Attitude control:*** 3-axis, nadir pointer, 0.2 deg pointing accuracy
- ***Propulsion:*** 552 kg hydrazine and pressurant
- ***Redundancy:*** Fully redundant and cross-strapped

Mass: 2,197 kg wet, 1,523 kg dry
Average Power: 1,621 W

LRO: Planetary (Moon)

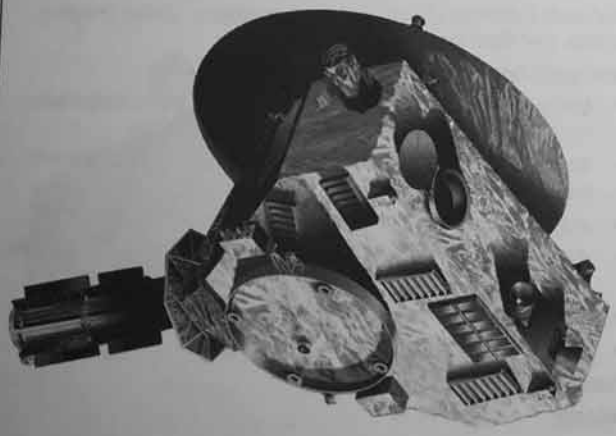


Mission: Lunar mapping
Orbit: Lunar, 50 km circular, 90 deg inclination
Payload: Wide- and narrow-angle cameras, laser altimeter, UV spectrometer, IR sensor, radiation monitor, neutron detector, synthetic aperture radar
Configuration Features:

- ***Unique features:*** Avionics on honeycomb deck with heat pipes; modular assembly
- ***Solar array:*** 10 m^2, 2-axis gimbaled
- ***Appendage:*** 2-axis gimbaled HGA
- ***Attitude control:*** 3-axis using 4 wheels, nadir pointer
- ***Propulsion:*** 897 kg hydrazine and He pressurant
- ***Redundancy:*** Single-string

Mass: 1,915 kg
Average Power: 620 W

New Horizons*: Planetary (Pluto)



Mission: Pluto exploration
Orbit: Fly-by
Payload: UV spectrometer, IR imaging spectrometer, visible camera, solar wind monitor, particle spectrometer, dust counter, radio science instrument
Configuration Features:

- ***Unique features:*** large high-gain antenna (HGA) for communicating over long distance; small spacecraft to achieve high velocity
- ***Power source:*** Radioisotope thermoelectric generator (RTG)
- ***Appendage:*** RTG cooling fins
- ***Attitude control:*** 3-axis using thrusters for encounter, 5 RPM spin stabilized about HGA axis for cruise
- ***Propulsion:*** 77 kg of hydrazine and He pressurant
- ***Redundancy:*** redundant and cross-strapped

Mass: 478 kg wet, 401 kg dry
Average Power: 200 W

*Fountain et. al. [2008]

Fig. 14-2. Spacecraft Configuration. (Continued)

Table 14-5. Scaling of Spacecraft Features. These are general trends assuming relative proportions stay the same (all dimensions grow or shrink equally). Any one dimension is considered r. The lateral frequency is important for launch. Place heavier (denser) items closer to the launch vehicle interface to keep the height down and the frequency up (see Sec. 22.1.3 for more details).

Feature	Scaling
Surface Area	Increases slower than volume (r^2 vs. r^3)
Mass	Increases with volume
Power	Increases with volume (more components)
Thermal Control	Surface area for radiators becomes more scarce on larger spacecraft, since power goes with volume
Solar arrays	Small spacecraft, with lower power, are more likely to have enough surface area for body-mounted arrays
Stability	Torque required to rotate a spacecraft goes up with moment of inertial. MOI is proportional product of mass and r^2 so torque goes as r^5. Larger spacecraft are much more stable.
Lateral Frequency	Larger structures tend to have lower frequencies, with an interplay among MOI of base, height, and mass—goes as $1/r$ for proportional growth, $1/(\text{c.m. height})^{1.5}$ for constant base and mass, as center of mass (c.m.) moves.

14.3.1 Attitude Control Technique

Depending on the pointing requirements of the system, various options exist for stabilizing and pointing a spacecraft. All spacecraft must have a method to counteract external torques (see Sec. 19.1.2 for more on external torques) and orient the system in order to keep the instruments pointed in the correct direction. Many early spacecraft were spin stabilized, using the momentum of the entire system to maintain the needed pointing accuracy. But the spinning can add complexity to the instrumentation if accurate measurements are required in all three directions. With modern computational power, three-axis stabilized spacecraft have become the norm, with only special circumstances driving the design toward a spinner.

Spinner

Spin stabilization is an ancient technique. A good discus throw relies on spin stabilization, as does a modern plastic disc thrown in a game of ultimate, and rifling in guns spins projectiles to improve accuracy. Older launch vehicles, like the Scout and the Delta II, used spinning upper stages to ensure the thrust went in the desired direction. Spin stabilization is still used today for some kick motors and certain special spacecraft applications.

A spin stabilized system intentionally puts momentum in the spacecraft body. The body's momentum vector will stay oriented in inertial space, reacting at right angles to any torque, just as a tilted, spinning top on a table will turn the force of gravity trying to tip it over into a slow precession of the spin axis. In the near-zero friction environment of space, a spacecraft set to spinning will maintain that spin for a very long time. The system is passively stable, requiring no power to maintain orientation.

But a system is only unconditionally stable if it is spinning around its maximum moment of inertia, like the flying disc. A system can spin for a short period of time around its minimum moment of inertia, but any energy damping in the system will tend to re-orient the spin axis so that the body is spinning around its maximum moment of inertia. A pencil-shaped object can spin around its long axis, but eventually it will settle into a flat spin. Spacecraft designers take advantage of this property by intentionally spinning around the maximum moment of inertia and installing a *nutation damper*. The nutation damper is generally a fluid-filled structure, where any asymmetry in the spin of the system will cause the fluid to flow, with the viscosity of the fluid dissipating energy.

Practical spinning spacecraft will need some method to control the spin rate and some method to adjust the orientation of the spin axis (spin precession control). The spin rate can generally be set once and left for a long time, but an orbit-normal spinner in low-Earth orbit will need to keep its spin axis moving at the rate the orbit is precessing around the Earth (see Secs. 9.5.2 and 9.5.3 for an explanation of orbit nodal regression). Both the spin rate and the precession can be managed from the ground, without a sophisticated control system on board the spacecraft. Spin rate can be measured with a sensor that produces a blip as its field of view crosses the horizon or moves past the Sun, or with a sensor (magnetometer) that measures the changing magnetic field (for a low-Earth orbiter). Sun sensors and magnetometers can also be used to measure the spin axis orientation.

Spin-stabilized spacecraft sometimes have a *de-spun platform*, in which part of the spacecraft remains inertially fixed while the rest of the system spins. A motor rotates between the spun and de-spun sections at the spacecraft spin rate, and slip rings carry signals across the interface. The Geostationary Earth Observing Satellites (GOES) monitored weather in the United States from de-spun platforms for many years.

Instead of spinning the entire spacecraft, some systems utilize a *momentum wheel*, which is a motor-driven wheel which spins at a constant rate, putting a fixed

momentum into the system, but leaving the rest of the system unspinning—a configuration known as *momentum bias*. This configuration requires some method of external torquing to counteract environmental torques. One example is a *B-dot algorithm*, where a spacecraft in low-Earth orbit uses the rate of change of the measured magnetic field to set the current in an electromagnet, and that magnet torques the spacecraft opposite to the body's spin rate. This stabilization technique, with a momentum bias, B-dot for rate damping, and a bang-bang controller to precess the momentum axis toward the Sun, was used by the early Small Explorer missions as an analog *safe hold*. A safe hold mode provides a very simple contingency control mode when the spacecraft processor is unavailable due to some fault condition.

Spinning spacecraft frequently utilize body-mounted solar arrays for power generation (see Sec. 14.3.3). Because of the constantly-changing orientation, more solar cells must be carried to generate the same amount of power. Spinning spacecraft tend to be power limited. The 191 kg FAST spacecraft shown in Fig. 14-3 operated on less than 60 W orbit average power.

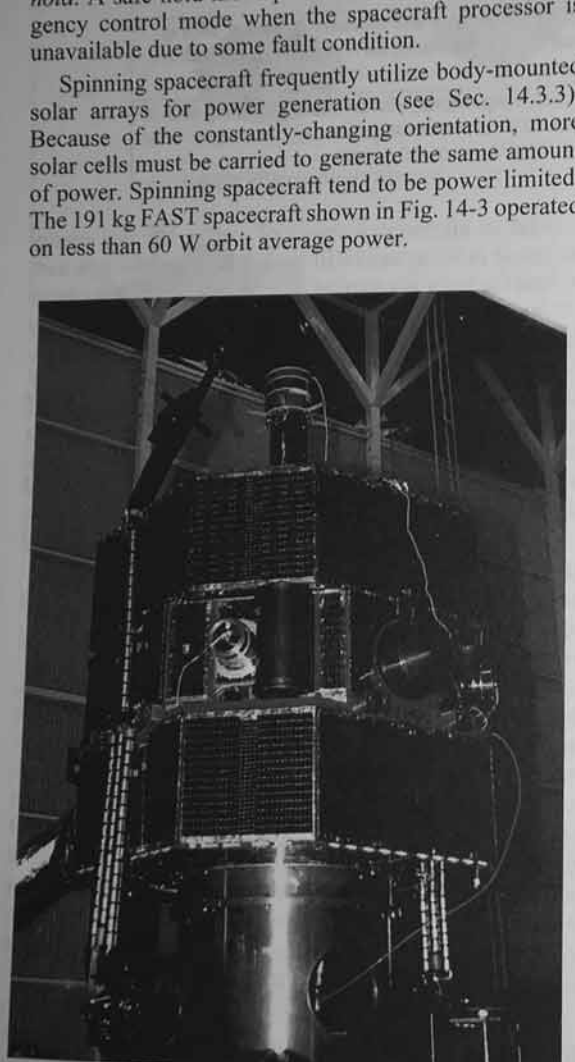


Fig. 14-3. Fast Auroral Snapshot Explorer (FAST) was a Spin-stabilized Spacecraft. (Photo courtesy of NASA GSFC)

If the upper stage of the launch vehicle or a kick stage spins for stability, the spacecraft will often need to de-spin after the burn. Thrusters can provide the torque necessary to stop the spin, but another option is a *yo-yo despin*. A yo-yo de-spin uses weights which release from the spacecraft and unwind a cable, similar to a toy yo-yo. As the weights move out, they reduce the spin rate of the system by increasing the moment of inertia, just like a figure skater extending her arms to reduce her spin rate. The spin momentum of the spacecraft body is transferred to the weights, and then the weights are released when the cable is fully unwound. Unfortunately, this leaves at least two weights with cables free-flying, creating a collision hazard for other spacecraft, so this technique is not used near operational orbits anymore. The SAMPEX spacecraft used a yo-yo despin after launching in 1992 (see Fig. 14-4).



Fig. 14-4. SAMPEX Spacecraft. SAMPEX Launched on a Scout Rocket, with a Spinning Upper Stage. The yo-yo mechanism for the despin wrapped around the spacecraft when the arrays were folded. You can see the attach band going horizontally across the middle of the spacecraft. (Photo courtesy of NASA GSFC)

Three-Axis Stabilized

Most modern spacecraft utilize three-axis stabilization. Attitude sensors provide feedback to a control system where actuators such as reaction wheels or thrusters rotate the system to its proper orientation. Processors now have enough computational capability to perform the necessary calculations to operate a 3-axis control algorithm while simultaneously handling the instrument commanding and data collection. Although the control algorithm is more complicated, other aspects of the system become less complex with 3-axis control. There is none of the complexity associated with a de-spun platform, the solar arrays can generate more power with a fixed orientation to the Sun, and the instruments do not need to compensate for spin motion in their measurements.

The highest precision pointing control is achieved on spacecraft using *reaction wheels* for attitude control. The spacecraft controls the speed of spinning wheels in order to adjust the torque applied to the body. External torques on the spacecraft are counteracted by torquing the reaction wheels, with the net result being a gradual increase in the wheel speed without changing the attitude of the spacecraft. Eventually, the system must use some external force, such as that introduced by thrusters or electromagnetic torquers, to remove the momentum

Table 14-5, Fig. 14-4, Eq. 14-1◀

from the system. Low-Earth orbiting spacecraft which do not otherwise require propulsion typically will use *electromagnetic torquers* for momentum unloading, using the torque produced by the electromagnets attempting to align with the Earth's magnetic field.

Three-axis stabilization takes on different appearances depending on where the spacecraft is pointed. Systems typically operate in an inertial reference frame, but the target is not always inertial. Many systems look at the body that is being orbited, and some systems use the Sun as the orienting target.

Inertial Pointer—Inertially-pointed spacecraft typically use the stars as reference. This configuration is most convenient for spacecraft observing the sky. The solar array configuration of an inertial pointer will vary, depending on the orientation of the orbit with respect to the Sun and any precession of the orbit. Engineers typically talk about X, Y, and Z axes for an inertially-pointed spacecraft.

Nadir Pointer—Nadir pointing is most useful for a spacecraft observing the body being orbited. As in an aircraft, axes are frequently described as roll, pitch, and yaw; with *roll axis* oriented about the direction of travel, *pitch axis* oriented normal to the orbit plane, and *yaw axis* oriented around the nadir vector. In order to remain nadir pointing, the spacecraft must rotate once per orbit around the pitch axis. This rotation rate must vary if the orbit is not perfectly circular.

Although horizon sensors were frequently used in the past to control pointing with a nadir pointer, modern systems usually use star sensors and on-board computation to calculate the correct direction to point in order to maintain a nadir orientation. Momentum bias in the pitch axis can help stabilize the spacecraft—in one quarter of an orbit, the roll axis direction rotates about the pitch axis by 90 deg, assuming the alignment that the yaw axis had in inertial space. So errors about the yaw axis become errors about roll, and the spacecraft can operate with sensors and actuators in only two axes.

Nadir-pointed spacecraft will generally keep one side of the pitch axis oriented away from the Sun so that radiators can dump heat from the electronics into space. As the Sun moves with respect to the orbit plane, the spacecraft may need to perform a *yaw flip*. In a yaw flip, the spacecraft is rotated 180 deg about the yaw axis, usually when the Sun is near the plane of the orbit.

Solar Pointer—Solar pointing is frequently used as a safe attitude, since spacecraft frequently use solar power and simple sensors and algorithms can quickly find the Sun. Also, any mission observing the Sun will be a Sun pointer. A sun sensor will define two of the three axes, and some other sensor (like a star sensor or magnetometer) must determine the orientation about the Sun vector.

Other

A small set of simple spacecraft require no active control. They are naturally stable by design. Most common in this category are *gravity-gradient* spacecraft, which use the gradient of the gravitational field in orbit to align the minimum moment of inertia axis (usually the longest axis) with the nadir vector (remember that the MOI is calculated around the axis). Spacecraft have been designed with *passive magnetic dampers*, where the Earth's magnetic field induces currents in the rotating spacecraft, and those currents dissipate power through resistive losses, reducing the tumble rate. A low-altitude spacecraft can use the atmosphere to provide orientation by placing the center of mass ahead of the center of pressure from the atmospheric drag—like a weathervane. Spacecraft which use *aerobraking*, where atmospheric drag intentionally changes the orbit instead of using thrusters, typically use this technique.

14.3.2 Need for Propulsion and Its Impact

Propulsion configuration is another major architectural decision that must be made early in the spacecraft design. Propulsion systems can be used to move the spacecraft to its final orbit, maintain that orbit or change it as necessary, maintain attitude control, and perform a final change of orbit for disposal of the system. The spacecraft designer should first consider whether propulsion is needed at all. Is there enough capability in the launch vehicle to put the spacecraft in its final orbit? Can the system operate with the imperfect performance of a real launch vehicle, where the final orbit is slightly different from the optimal? Can the system tolerate the change in orbit due to atmospheric drag? Is there any need for propulsion to dispose of the system? Can magnetic torquers or some passive system such as gravity gradient be used to counter environmental forces and dump accumulated momentum? If the answer to any of these questions is "no," then propulsion of some type is required. Any system going beyond low-Earth orbit will need some sort of propulsion system.

Achieving Final Orbit

If the launch vehicle does not take the spacecraft to its final orbit, the spacecraft designer can include a kick stage which is jettisoned at the end of the orbit adjust. A *kick stage* provides a large delta V over a short period of time in order to make a large change to the orbit of a spacecraft. A kick stage is generally a solid rocket motor. Jettisoning the stage eliminates the mass of the spent motor from the system, reducing the amount of fuel that is needed for later orbit adjusts and maintenance. As an example of a kick stage, geosynchronous spacecraft frequently use an *apogee kick motor* fired at the apogee of a geosynchronous transfer orbit (see Sec. 9.6.1) to raise the perigee of the orbit to near geosynchronous altitude. The designer must trade the fuel mass saved in later maneuvers if the stage is jettisoned against the mass and complexity associated with separating part of the spacecraft. Integral propulsion systems enable the builders to concentrate on a single system, with all margin carried in one set of fuel tanks. Any unused margin from the orbit adjustment can be used for orbit maintenance. For orbit adjustments requiring high precision, such as those used for planetary trajectory maneuvers or for the final orbit adjustments, the precision of a liquid system is generally

Table 14-5◀, Fig. 14-4◀, Eq. 14-1◀

preferred over a solid kick stage. A liquid propulsion system can be shut off when the appropriate delta V has been achieved, but a solid burns until the fuel is expended. The integral propulsion system also allows the operators to change the orbit of the spacecraft multiple times, perhaps adjusting to changing mission demands. A final consideration with staging is the disposal of the spent stage—does the kick stage end up in an orbit that can jeopardize operational assets?

The high thrust of a large delta V maneuver will require either spin stabilization or active thruster control to maintain the proper attitude during the burn. Spin stabilization requires precision balancing of the system, and it frequently requires some sort of nutation control system to keep the spin straight while the mass properties of the system change with expulsion of the propellant. Active thruster control may require a gimbaled nozzle on the main rocket or simply sufficient torque authority from the attitude control thrusters. For a liquid burn using multiple thrusters, the delta V thrusters can sometimes be individually turned off briefly (known as *off-pulsing* or *off-modulation*) to maintain control. Alternatively, attitude thrusters can be on-pulsed briefly while the main engines burn continuously.

Station Keeping

Once in the final orbit, many missions must maintain that orbit to a high precision. *Station-keeping maneuvers* are small adjustments to the velocity of the spacecraft in order to keep the system within some orbit tolerance. Perturbations to the orbit can be caused by atmospheric drag or gravity effects. Adjustments are generally a few meters per second or less. Most spacecraft use a liquid propulsion system for these high-precision maneuvers, but geostationary communication spacecraft, which have large power generation capability, frequently use electric thrusters. Electric thrusters have a much larger specific impulse, resulting in less fuel required for a long operational life. Because the maneuvers are small, liquid systems generally use smaller thrusters for station keeping. These smaller thrusters can also be used for attitude control and momentum management.

Attitude Control and Momentum Management

Many spacecraft use reaction wheels (see Sec. 19.1.4) for attitude control. These wheels can provide very high precision pointing, and they can temporarily compensate for external torques, but eventually an external force is needed to compensate for the momentum which builds up in the system (the wheels cannot keep spinning faster forever; they become *saturated*, or spinning at their maximum rate). In the absence of the Earth's magnetic field, or for systems which already have propulsion for orbit adjustment, a propulsion system provides a good method to provide an external torque on the system. Controllers will perform a *delta H maneuver*, changing the momentum (mathematical symbol of H) of the system without changing the attitude or the orbit. The wheels are commanded to a new speed and the thrusters fire to hold the attitude, or the thrusters fire and the wheels hold attitude by spinning down.

Table 14-5◀, Fig. 14-4◀, Eq. 14-1◀

Some spacecraft utilize propulsion exclusively for attitude control. Any time the attitude drifts outside a *deadband*, a thruster will fire, pushing the attitude back into its control box. This technique works well for systems that do not require precision pointing and can handle the fuel mass used. Since fuel is expended for every excursion outside the deadband, the system cannot take advantage of long-term averaging of external torques, so attitude control using thrusters will generally consume more fuel than attitude control with wheels and thrusters for momentum dumping. Any errors in the control algorithm (such as an overly tight deadband) can quickly consume the mission's fuel budget. The system designer must trade the mass, complexity, and precision of the wheels against the extra fuel mass and risk of a propulsion-only system.

Disposal

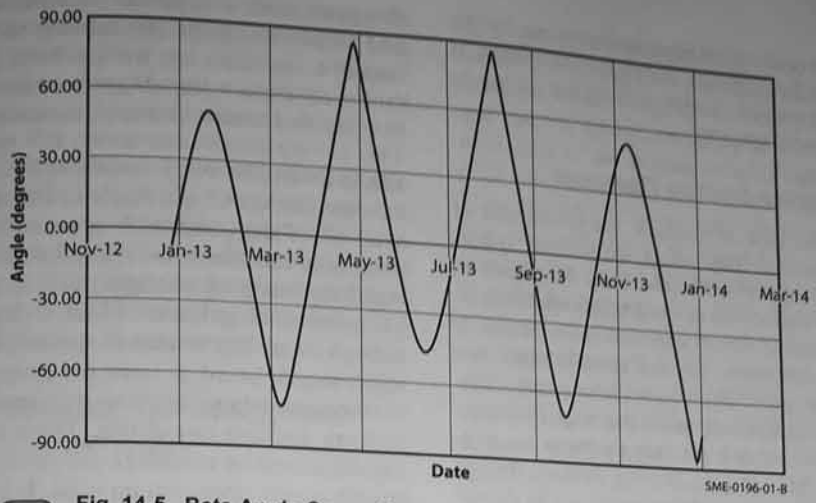
Spacecraft designers must consider the final disposition of the system, to prevent damage to other spacecraft and loss of life on the ground. Chapter 30 provides a full discussion of disposal. If the spacecraft will be moved to a different orbit at the end of the mission, then the propulsion system must include this capability. A controlled de-orbit of the spacecraft, where the impact point is targeted for an unpopulated part of the ocean, will require significant delta V. For a mono-propellant hydrazine system, the mass of the fuel required for a controlled de-orbit from low-Earth orbit is about 7% of the total mass of the system.

14.3.3 Solar Array Configuration

Most spacecraft use photovoltaic solar arrays for generating power. The solar array configuration will constrain the overall structural design and impact the orientation of the spacecraft in flight. Large solar arrays must be folded and firmly locked for launch. Deployed arrays can impact the stability of the spacecraft and interfere with instrument fields of view. The arrays must be positioned to control shadowing (see Sec. 21.2.2).

After determining the solar array area required (see Table 21-12), we must determine a suitable layout for the array. The array design must operate over the full range of the solar *beta angle*, the angle between the Sun and the plane of the orbit. For an Earth-orbiting spacecraft, the beta angle varies as the orbit precesses and as the Earth goes around the Sun (see Fig. 14-5). Depending on where the spacecraft points and how the Sun moves with respect to that pointing, the solar array configuration can be body-mounted, deployed and fixed, or deployed and articulated.

Body-mounted arrays provide limited power, since their area is constrained by the surface area of the spacecraft, which is constrained by the launch vehicle fairing volume. This may not be an issue for small spacecraft, which have limited volume available for power-consuming components. For spinning spacecraft (see Sec. 14.3.1), wrap-around body-mounted solar arrays are generally the required configuration in order to generate power throughout the spin. Spin-stabilized spacecraft tend to be



Live Calc

Fig. 14-5. Beta Angle Over a Year for a 400 km, 70 deg Inclination Orbit.

power constrained, although the simple attitude control system helps reduce the power demands of the bus. The COBE spacecraft used an alternate configuration for additional power, with deployed, double-sided arrays placed 120 deg apart and radial to the spin axis (see Fig. 14-6). This configuration doubles the number of solar cells, and it requires deployment mechanisms, but it generated for COBE over 2 times more power for the same size spacecraft with body-mounted arrays. Special circumstances may allow a fixed solar array on a non-spinning spacecraft. The Wide-Field Infrared Survey Explorer (WISE) was a zenith pointer flying in a Sun-synchronous orbit, with significant room in the launch-vehicle fairing. WISE flew a panel array that was fixed to the body, with no deployment necessary (see Fig. 14-2). Body mounted arrays on multiple spacecraft faces can provide an economical solution, eliminating the need for array pointing or even spacecraft pointing.

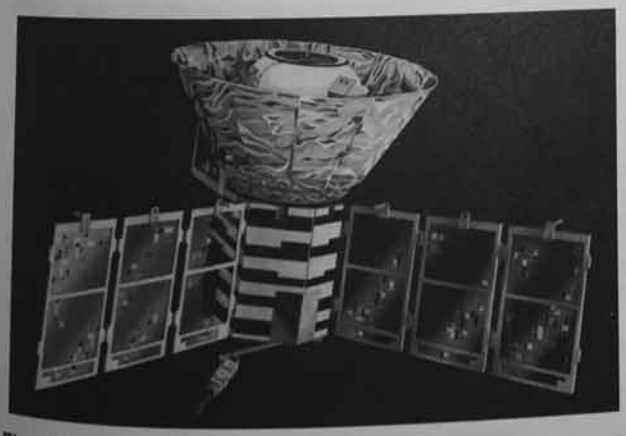


Fig. 14-6. COBE.

Most non-spinning spacecraft use deployed solar panels. If you can design the mission to constrain the Sun to within about 30 deg of a particular body vector, the array can probably remain rigidly fixed to the spacecraft body after deployment. If the direction to the Sun cannot be this constrained, then the spacecraft will need articulated arrays. For low inclination orbits, the Sun will remain near the plane of the orbit. A single axis of rotation (on the pitch axis for a nadir pointer) will probably be sufficient to keep the arrays pointed toward the Sun. Higher inclination orbits will have a large variation in beta angle, resulting in the need for two-axis gimbals on the array. A significant exception to this rule is the *Sun-synchronous orbit*, where the orbit precesses at the same rate that the Earth orbits the Sun. In this case, the beta angle variation is again small. Gimbals on solar arrays add significant complexity to the system, so the designer should avoid them if possible. In addition to the complexity of a mechanism in space which must operate over thousands of cycles, gimbals are a source of jitter, they require significant hardware and software for control, they complicate safe mode, and the thermal design, including blankets on moving parts, is challenging. The problems have certainly been solved many times, but it does add to the overall cost of the system.

The final aspect of solar array design involves deciding whether to place arrays on one side of the spacecraft or symmetrically on two sides. At first glance, it would appear that a single panel would always be the way to go—fewer mechanisms and deployments to worry about and no shadowing from the spacecraft body for high beta angles. But as a panel gets longer, the moments of inertia (MOI) go as the length **squared**, requiring more torque from the attitude control system. Balancing the array on two sides of the spacecraft cuts the contribution from the array to the MOI in half. Longer arrays will also be less stiff, so they will be more likely to create jitter and couple into the control bandwidth of the attitude control system (see Sec. 19.1.5). In low-Earth orbit, solar arrays produce significant drag area, so symmetric arrays will keep the center of pressure of the system near the center of mass. An offset of the two centers will produce a

Table 14-5◀, Fig. 14-6, Eq. 14-1◀

torque on the spacecraft which must be overcome by the control system (see Sec. 19.1.2). Away from the drag of an atmosphere, asymmetric arrays still create torque in the system due to solar pressure.

14.3.4 Communication Antenna Placement

Depending on the type of mission, the placement of communication antennas may affect the overall spacecraft configuration (see TDRSS and New Horizons in Fig. 14-2). For low-Earth orbit science missions communicating with the ground, omni antennas will generally support the needed data rates. You can usually place two antennas, each viewing half the sky, without much difficulty, although solar arrays may get in the way. For more distant missions, medium or high-gain antennas are generally required (see Sec. 21.1.5). If your mission design can afford to stop collecting measurements while downlinking data, you can fix a high-gain antenna to the spacecraft body and use the spacecraft to point it at the ground station. For continuous operation, the antenna will mount on gimbals, and a boom will keep the antenna from striking the spacecraft body. A dual-axis-gimbaled antenna hanging out on a boom will affect your attitude control stability, although not as much as a gimbaled solar array, since the solar array mass is generally much greater than the mass of even a large high-gain antenna.

14.3.5 Spacecraft Autonomy

Although it does not have a large impact on the general shape of the spacecraft, the level of autonomy in the system can greatly impact the mission costs and the system design. All spacecraft must perform some functions without interaction from the ground, but the amount of autonomy designed into the system will vary greatly with application. More autonomous spacecraft will require more complex software, faster processors, and more memory. The hardware resources will demand more power and mass, and the entire system will require more testing. There are three main reasons for autonomy:

1. Reduce the operations costs
2. Execute mission critical functions when the ground is unavailable
3. Protect the spacecraft from unexpected events or failures

Early in the design process, the mission designer should decide the general approach for operation of the spacecraft. For a short mission (a few months), it probably makes sense for the ground operators to monitor the spacecraft continuously, with very little action taking place without explicit commands from the ground. It is generally prudent to have the spacecraft builders present for the early check-out of the spacecraft, so having them stick around for another month or two will not add much more to the cost, certainly less than adding a bunch of autonomous functions. For more extended missions, the designers trade complexity on board against manpower and autonomy on the ground to control the system. For instance, operators can work reduced hours if the spacecraft can store a time-tagged command sequence and execute that sequence over the course of a day or two. This is very simple autonomy, still requiring the operators to determine every command that the spacecraft will execute, but allowing a single uplink where the operators send all of the commands at once. More sophisticated autonomy can enable the spacecraft to respond to incoming data, changing configurations in order to maximize the information gathered. Modern flight computers have enough capability to enable spacecraft to autonomously select targets based on some pre-programmed criteria, or to autonomously rendezvous with another spacecraft and perform docking operations. These sorts of algorithms require extensive modeling and testing on the ground to ensure proper operation on orbit. It is up to the mission designer to work with the spacecraft and ground system teams to determine the optimum balance between autonomy and operators.

Some missions, especially planetary missions, must execute critical functions, such as an orbit insertion burn, without contact from the ground. In planetary missions, the time delay in radio signal from the ground to the spacecraft does not allow operators to send real-time commands in reaction to events. The timing of most orbit maneuvers is such that on-board time-tagged commands must be used to ensure proper execution.

The long communications delays in planetary missions also require the spacecraft to autonomously recover from failures during critical events. If the spacecraft only has one opportunity to perform a burn in order to achieve orbit, on-board autonomy must handle failures and switch over to redundant components—the spacecraft must be *fail operational*. But even for missions that do not have critical maneuvers, system designers usually add autonomy to preserve the spacecraft after failures or unexpected anomalies. In these situations, it is simpler to design the spacecraft to *fail safe*, going to a mode which preserves the system until the ground can diagnose the situation and hopefully correct it. A spacecraft *safe mode* is the safe state to which the spacecraft transitions when a failure is detected. Safe is generally defined as:

1. *Power Positive*—more power being generated by the solar arrays than being consumed by the system, so that the battery recharges. If the battery becomes fully discharged, there is no longer any way to communicate with the spacecraft, and the attitude control system will no longer function, preventing a change in attitude that would allow recharging of the battery. In other words, a spacecraft with a dead battery is dead.
2. *Thermally Benign*—all temperatures must stay within survival limits
3. *Commandable*—the orientation of antennas and the configuration of the receiver must be such that

Table 14-5◀, Fig. 14-6◀, Eq. 14-1◀

the ground can send commands in order to correct problems. The transmitter is frequently off in safe mode to save power, but the mode must support output of housekeeping telemetry when the ground commands transmit, to enable troubleshooting.

Safe mode generally requires some control of the attitude of the spacecraft. Safe mode considerations can drive antenna placement and avionics hardware features. Many spacecraft have hardware-decoded commands to allow resets and transitions to redundant hardware even when the software is not functioning (see Sec. 29.1 Step 6 for more discussion of autonomy).

14.4 Partitioning Spacecraft into Subsystems

After completing the top-level configuration trades, it is time to go to the next level of detail in the spacecraft design. This extra level of detail will require partitioning the spacecraft into subsystems, with a different subsystem expert responsible for each of the partitions. Significant work can then occur in parallel, with the systems engineer ensuring that each subsystem lead stays connected with the overall system concept. By dividing the system in a way that minimizes interfaces between subsystems, the systems engineer can minimize the effort required to coordinate among the subsystem engineers. Ideally, each spacecraft would have a unique subsystem partitioning designed for that particular mission. Practically, our spacecraft development organizations are already divided into particular disciplines (see Table 14-6), so the subsystems we use for spacecraft development are generally the same for each spacecraft, reducing the required coordination between organizations. So we may end up with the attitude determination and control group designing a box that communicates with each of the attitude sensors and actuators, separate from the command and data handling (C&DH) box. Then the C&DH group only has to worry about a single interface with the ADC, instead of each of the sensor and actuator interfaces. We may face a small increase in mass in order to provide clear lines of responsibility, but the overall cost of the system will be less if the organization operates more efficiently. Over the long term, we may see re-organizations within the industry, as software takes over many functions, digital logic operates at higher speeds, and the distinctions between TT&C, C&DH, ADC, and Power electronics blur.

A classic partitioning of functionality in a spacecraft is the hardware vs. software trade. Functions performed in hardware are deterministic and static, and they can be done independently of other tasks on the spacecraft. If you have a simple function that must always operate the same way, a hardware solution will ensure that ground operators cannot disturb the critical functionality (such as a hardware-decoded command for resetting the system to a known state). If you have a function which requires significant computation, hardware can remove that burden from the flight processor (encoding of downlink telemetry is an example). Software provides the flexibility to finish the details of a function late in the development, or even change functionality after launch. Software does not add to the mass of the spacecraft, but

Table 14-6. List of Common Spacecraft Subsystems.

Subsystem Name	Function	Reference
Propulsion	Spacecraft thrust, including fuel storage and plumbing	Chap. 18
Attitude Determination and Control (ADC) or Attitude Control System (ACS)	Sensors, actuators, and software necessary to control the spacecraft pointing	Sec. 19.1
Position and Orbit Determination and Control	Sensors and software necessary to control the spacecraft orbit	Sec. 19.2
On Board Processing or Command and Data Handling (C&DH)	Electronics and software used to receive and distribute commands and to store and forward payload data and spacecraft telemetry	Chap. 20
Telemetry, Tracking, and Command (TT&C) or RF Communications	Radio and associated hardware, such as cabling and antennas, used to communicate with the ground or other spacecraft	Sec. 21.1
Power	Electronics, power generation, and power storage devices, as well as harnessing for power distribution	Sec. 21.2
Structures and Mechanisms	All the hardware that supports the spacecraft, including the primary structural components, brackets, fasteners, and the actuators and mechanisms associated with deployed or movable structures	Sec. 22.1
Thermal Control	All of the hardware necessary to control the temperature of the spacecraft	Sec. 22.2

it is not always cheaper than hardware, because more complex code requires more complex testing. How much can be done in software is generally limited by the speed of the processors available. As flight processors have become faster, more functionality has moved to software, and this has generally made our spacecraft smaller, more capable, and more adaptable. Programmable logic, such as Field-Programmable Gate Arrays (FPGA's), blur the lines between hardware and software, since the logic can be treated as a piece of hardware, but it can have its functionality changed via programming late in the design cycle. Some programmable logic includes embedded micro-controllers, with the ability to change the code on the fly, further blurring the distinction.

The rest of the section will discuss the subsystems most frequently encountered in the spacecraft industry, include driving requirements for each subsystem and an introduction to the perspective of each—what do the subsystem leads worry about, and what does the systems engineer need to know to communicate with each. The tables of driving requirements are based on the experience of dozens of preliminary mission designs; they provide a good starting point, but each mission will have some unique features.

14.4.1 Propulsion

The propulsion subsystem provides the necessary thrust to change the orbit of the spacecraft and dump momentum. The total change in velocity (delta V) dominates the propellant budget. The three parameters in Table 14-7, combined with the configuration selected through the trades discussed in Sec. 14.3.2 will define the overall propulsion configuration. The propulsion lead must work with the attitude control and orbit control leads to ensure proper thruster sizing; with the mechanical lead to ensure proper support of the tank mass and proper placement of thrusters; and with the thermal lead to ensure proper temperature control (hydrazine has a freezing point near that of water). Electronics to operate the propulsion can either be part of the propulsion subsystem or part of the attitude control system. The mass summaries in App. A place the control electronics for propulsion in the attitude control system.

Table 14-7. Propulsion Design Drivers.

Major Design Drivers	Driven By	Impact
Delta V	Mission design	Fuel required
Spacecraft Dry Mass	Spacecraft design	Fuel required
Thrust for Maneuvers and Control	Trajectory and ADC design	Thruster sizing

14.4.2 Attitude Determination and Control

The attitude determination and control (ADC) subsystem, also known as the attitude control system (ACS), determines the orientation of the spacecraft in space, it reorients the spacecraft toward commanded or pre-programmed targets, and it holds that attitude. It takes four parameters to describe the attitude of the spacecraft—three define a vector in inertial space and a fourth describes the rotation about that vector. Attitude is frequently represented using *quaternions*, which are an extension to the complex number system. Quaternions are easily manipulated with matrix mathematics and are well suited for computer computation. For a 3-axis-controlled spacecraft, the ADC is a major piece of the overall spacecraft bus. The sensors and actuators add significant mass to the system, and the software required to control a spacecraft's attitude can consume a major portion of the total processor bandwidth. Table 14-8 shows the major drivers of the ADC design.

The fundamental driving requirement for ADC is the pointing control, typically driven by how accurately the payload must be pointed. Observational payloads around the Earth or any other body typically need position information relative to the body in addition to attitude information in order to locate the observation with respect to a map of the body. Communications payloads and outward looking payloads generally do not need high-precision position information. Communications payloads and space weather instrumentation (measuring electrons, ions, and fields) generally have looser pointing requirements than other payloads.

How accurately the spacecraft must point will impact the accuracy of the sensors and the precision of the actuators. Typically, more accurate sensors are heavier and more expensive, consume more power, and have lower reliability—more complexity is required for more accuracy. The on-board knowledge is typically better than the pointing because of the errors introduced by the actuators. The ground knowledge of pointing is generally better than the on-board solution. Errors introduced by predicted orbit position (rather than definitive orbit solutions after the fact) and by less accurate on-board algorithms (as compared with what can be done on the ground) will reduce the spacecraft's instantaneous estimate of attitude. The stability of the payload pointing will be affected by imbalance of reaction wheels and motion of spacecraft appendages as well as structural stiffness and stability over temperature, both for the payload mounting and the ADC sensor and actuator mounting. The stiffness of appendages must typically put the first resonance at a frequency that is at least a factor of 10 above the controller bandwidth to prevent amplification of oscillations. If this is not practical, a structural filter can be incorporated in the control system, but this approach requires significant testing and analysis (see Sec. 19.1.5).

The actuators provide the torque to re-point the spacecraft. The actuators must provide more torque for faster attitude changes or larger spacecraft moments of inertia. Reaction wheel systems also provide momentum storage, so that external torques from effects such as solar pressure, atmospheric drag, and gravity gradient can be temporarily compensated without thrusters or electromagnetic torquers. (See Sec. 19.1.2 for more details on external torques.) The wheels will spin faster and faster as the torques are

Table 14-8. Attitude Determination and Control Design Drivers.

Major Design Drivers	Driven By	Impact
Pointing Control Accuracy	Payload requirements	Sensors and actuators
Pointing Knowledge	Payload requirements	Sensors
Stability	Payload requirements	Actuators, appendages
Maneuvers	Payload requirements	Actuators
Max Slew Rate	Payload requirements	Actuators
Spacecraft Moment of Inertia	Spacecraft config	Actuators
Time Between Momentum Unloads	Operational availability	Momentum storage
External Torques	Spacecraft config, orbit	Momentum storage
Sun, Earth, Moon Avoidance	Payload requirements	Algorithms, sensors
Safing Requirements	Reliability, lifetime	Algorithms, sens & act
Solar Array Configuration	Orbit	Stability, external torq
Antenna Pointing Requirements	Data system design	Pointing ctrl, algorithm

reacted, storing momentum, until the spacecraft uses thrusters or electromagnets to remove the excess momentum. The dumping of momentum will typically add some instability into the system, so we would like to go as long as possible without dumping momentum.

For many payloads, such as those with sensitive or cold detectors, pointing at the Sun can be fatal. Some payloads must avoid the Earth or even the Moon. These considerations, along with the safing considerations described in Sec. 14.3.5, will impact the required sensors and on-board algorithms. For instance, most spacecraft carry coarse Sun sensors which face in all directions on the system, allowing it to autonomously determine where to find the Sun.

The other major function of the ADC subsystem is the control of solar array and antenna pointing. The ADC subsystem has the on-board knowledge of attitude, so it is fairly easy to add information about where to find the Earth and where to find the Sun, then generate the proper commands to the hardware to point the antenna and solar panel. The ground typically uploads an *ephemeris table* which describes in inertial coordinates the direction of the Sun or the direction of the Earth vs. time from the expected position of the spacecraft. Alternatively, the spacecraft can use an on-board algorithm to calculate the Sun and Earth directions, with a modest increase in processing power. For Earth-orbiting missions, a Global Positioning Satellite (GPS) receiver can provide very accurate orbit information, possibly eliminating the need for additional position determination algorithms.

14.4.3 Position and Orbit Determination and Control

Historically, spacecraft orbit determination was a function of the ground system, with the spacecraft providing some hardware to enable the necessary measurements. During ground contacts, coherent transponders would provide range and range-rate information, which the operators then processed to determine the spacecraft orbit. The controllers would then send the appropriate command to the propulsion system to adjust the orbit as necessary. While this method is still quite common, modern computational capability has enabled autonomous orbit control by spacecraft. An on-board GPS receiver determines the spacecraft position and velocity, and then the spacecraft processor can compute an orbit for the system. This orbit is refined over multiple measurements, with algorithmic filtering ensuring that a bad sample will not cause a bad final solution. The spacecraft can then compare this orbit solution to the desired orbit and calculate and execute the propulsive maneuvers necessary to adjust the orbit. This technique will allow one spacecraft to get close enough to another spacecraft so that more specialized sensors, such as a scanning lidar or optical sensors with image processing, can calculate the maneuvers necessary for autonomous rendezvous and even docking between spacecraft. More information about orbit determination and control is in Sec. 19.2.

14.4.4 On-board Processing

The On-board Processing subsystem is frequently known as the Command and Data Handling subsystem (C&DH). It receives commands from the ground via the communications system, passes them to the appropriate components and payloads, collects telemetry from across the system, stores that telemetry, collects and stores science data, and forwards the real-time and stored data to the ground via the communications system. The C&DH contains the flight processor, where the software runs, and it contains the non-volatile memory where the software is stored. The primary drivers for the C&DH are in Table 14-9.

The C&DH is the primary point of contact for the system's interaction with the payloads. The hardware and software interfaces for commanding the payloads and reading the data impact the complexity of the C&DH. For data rates below 100 kbps, a slower-speed serial interface like the MIL-STD-1553 can be used for communication with the payloads. Using a common, robust data standard will improve reliability and reduce integration and testing costs. Higher data rates demand more complex interfaces, such as Spacewire (standard ECSS-E-ST-50-12C based on IEEE 1355) or custom solutions. As the data rates increase, more work must be done by the C&DH hardware, with less data handling being done in software.





Table 14-9. On-board Processing Design Drivers.

Major Design Drivers	Driven By	Impact
Instrument Data interface	Payload requirements	Hardware design
Processing requirements	Ops concept, algorithms	Processor, power
Data Storage volume	Payload, ops concept	Hardware design
Timing accuracy	Payload	Oscillator, hardware design

Hardware solutions must be fixed earlier in the build cycle, since software can generally be changed after the hardware is complete, through reprogrammable memory. The designer should pay particular attention to the *handshaking*, the interaction between the payload and the C&DH, on the data interface which regulates the data flow. For most applications, it is critical that the handshaking allows the C&DH to pause, with data held on the instrument side of the interface, during high-speed transfers. This allows time for the C&DH to execute other critical functions without dropping payload data.

The processing requirements define the speed of the processor, the dynamic memory (typically Random Access Memory or RAM), and the non-volatile memory (usually Programmable Read Only Memory, PROM, or Electrically Erasable Programmable Read Only Memory, EEPROM). Table 14-10 provides a reference level of processor utilization for different speed processors. All three missions used a 3-axis attitude control system, and all three had a similar level of autonomy. The instrument managing software in the flight computer is part of the C&DH total. The increase in ACS utilization with the Rad6000 processor was due to an increase in complexity which outpaced the increase in speed—Triana ran extra filters and produced more telemetry. Also, the Rad6000 is a Reduced Instruction Set Computer (RISC) processor, so it did not have the floating point math built in like the 386/387 combination used on SWAS and WIRE. With LRO, the large increase in processor speed allowed substantially more instrument support on the C&DH side, with the ACS code approaching "insignificant." LRO could handle a lower ACS cycle due to its much larger size (larger spacecraft change pointing much slower). You can use this table as a starting point to estimate, for your processor, how much processing bandwidth will be required for these functions. Assuming about 25% utilization at 20 MHz and 10 Hz ACS cycle, increasing the speed to 132 MHz is a factor of 6.6, which cuts utilization to 3.8%, and running at 5 Hz instead of 10 Hz cuts utilization in half again, resulting in 1.9%—very close to the LRO measured value of 1.7%. For lower processing demands, consider operating the processor at a slower speed to save power. Faster processors generally consume more power.

The non-volatile memory stores the software code, with PROM used for code that is unchangeable in flight and EEPROM used for code that can be altered during

Table 14-10. Processor Utilization vs. Processor Type and Speed.

Year	Mission	Processor	ACS Rate	Utilization	
				ACS	C&DH
1995	SWAS/ WIRE	386/387 @ 16 MHz	10 Hz	23%	66%
2000	Triana	Rad6000 @ 20 MHz	10 Hz	28%	63%
2007	LRO	Rad750 @ 132 MHz	5 Hz	1.7%	48%

flight. The *boot code* copies the software from non-volatile memory into RAM where it executes (the software pulls itself up by the bootstraps). The RAM is used for temporary storage during software execution, and it can also be used for storage of some key telemetry information. The C&DH also contains a larger bank of RAM or perhaps even hard disks for storage of payload data. The volume of data generated between ground contacts determines how much memory the C&DH must carry. The LRO Rad750 processor card, selected in 2005, contained 64 kbytes of PROM, 4 Mbytes of EEPROM, and 36 Mbytes of radiation-hard RAM for the processor and another 8 Mbytes of RAM for the high-speed data interface. The larger data storage on LRO used 1800 Mbytes of Dynamic RAM on each of 4 cards. This RAM memory included 600 Mbytes for coding to correct errors caused by radiation.

The C&DH generally maintains the master clock for the spacecraft. An oscillator provides the reference, with a counter, called the *mission elapsed timer* (MET), keeping track of the number of cycles in the oscillator since applying power to the system. The count represents seconds and subseconds. The software then adds an offset to the MET to determine the seconds and subseconds since some *epoch*. The epoch is the reference date and time to which all spacecraft time is referenced. A typical epoch is midnight starting January 1, 2001. The sum of the MET and offset is the number of seconds and fractional seconds since that epoch. The required time-tagging accuracy of the payload determines the precision and accuracy of the onboard oscillator and the frequency and methodology of maintaining the clock. A high-precision oscillator will have a built-in oven to maintain the temperature of the oscillator. Without an oven, over the typical temperature range of a spacecraft, the system will be able to maintain the clock to within 0.5 seconds over a day. The absolute accuracy of an oscillator's frequency is usually poorer than its precision (its variation from the initial value). This means that the clock will generally drift somewhat from the ideal, but the drift will be about the same every day. More C&DH details can be found in Chap. 20.

14.4.5 Telemetry, Tracking, and Command

The Telemetry, Tracking, and Command (TT&C) subsystem provides the radio link between the spacecraft and the ground, and it provides signaling necessary to

determine the spacecraft's distance and velocity over time. The distance, or *ranging*, information is determined by measuring the delay of ranging signals that are sent from the ground to the spacecraft, received by the TT&C, then retransmitted to the ground. The velocity information is determined by the ground measuring the *Doppler shift* of the spacecraft's signal at the station over time. Because of the fixed velocity of radio waves, the frequency of the signal will shift up with an approaching spacecraft and down with a receding spacecraft as the waves either bunch up or spread out. The equation for the shifted frequency is:

$$f = f_{xmit} \cdot \left(1 - \frac{V_{rel}}{c}\right) \qquad (14\text{-}2)$$

Where f_{xmit} is the transmit frequency on board the spacecraft, V_{rel} is the relative velocity receeding along the line of sight, ($V_{rel} < 0$ for an approaching spacecraft) and c is the speed of light. This same shift occurs for the signal sent from the ground to the spacecraft. Most Doppler tracking is done with a TT&C system that transmits at a frequency which is proportional to the received frequency, called a *coherent transponder*, so that the ground knows exactly what frequency to expect from the spacecraft. The difference between the expected frequency and the actual frequency is the *two-way Doppler shift*, since the signal is shifted on the way both to and from the spacecraft. If the spacecraft does not transmit a coherent signal, then its frequency is set by an internal oscillator, and the ground will only expect a *one-way Doppler shift* from the oscillator-derived frequency. Table 14-11 contains the major drivers for the TT&C design.

Table 14-11. TT&C Design Drivers.

Major Design Drivers	Driven By	Impact
Data Rate & Distance	Payload, ops concept	RF power, antenna
Frequency	Data rate, regulations	Hardware design
Ground Station EIRP	Ops concept	Recvr sensitivity, ant
Ground Station G/T	Ops concept	Xmit pwr, antenna
Duty Cycle	Ops concept	Orbit-average power

Radio Frequency (RF) engineers use link budgets to describe the performance of the radio links in the TT&C. A full discussion of link budgets is in Sec. 21.1.4. The most important things for the systems engineer to know about link budgets are:

- Links are generally described in decibels (dB). This is a logarithmic scale, so an increase in 3 dB means a doubling of the power
- RF power required is proportional to the data rate, so a doubling of the data rate will require a doubling of the transmitted power
- RF power is inversely proportional to the distance squared. So a doubling of the distance results four times more transmitter power
- Antennas can provide significant gain, meaning that you can reduce the transmitted power by orders of magnitude if you use a high-gain antenna
- The gain of an antenna is inversely proportional to its beamwidth. If the antenna focuses all of its power in a narrow beam, the link is more efficient, so the antenna gain is much greater than 1. But the spacecraft must then point that beam
- The gain of a high-gain antenna is proportional to the area of the antenna, so doubling the diameter of an antenna increases the gain by roughly 6 dB
- The antenna size is proportional to the frequency
- It is generally a good idea to maintain 3 dB of margin in the link. This covers unexpected factors and later degradation

The frequency selection is usually dictated by international regulations and planned data rates. See Table 21-3 for more information.

The main performance parameters for a ground station are the *Effective Isotropic Radiated Power (EIRP)*, which is a combination of the transmitter power and the antenna gain, and the *G/T ratio*, which is the antenna gain divided by the noise temperature of the receiver. The G/T is a convenient figure of merit which comes out of the link equation (see Eq. (21-4)). With a higher EIRP from the ground, the spacecraft antenna gain can be reduced. With a higher G/T on the ground, the spacecraft transmitter power or antenna gain can be reduced.

The spacecraft power consumed by the TT&C transmitter is substantially greater than the RF power transmitted. A 5 W transmitter will consume about 30–35 W. For higher transmitter powers (>20 W), alternate amplifier technology is used, so transmitters become more efficient. A 50 W transmitter will consume about 120–160 W. The receiver in the TT&C should be continuously powered, so that the ground can always get a command to the spacecraft. The transmitter can be turned off when not in view of a ground station, reducing the average power consumption of the TT&C.

Some general rules of thumb for omni-directional antennas and a typical S-band TT&C:

- For spacecraft in low-Earth orbit, omni antennas will support about 2 kbps command link from TDRSS
- Those same omni antennas will support about 8 kbps link to TDRSS and 4,000 kbps to a 13 m ground station using a 5 W transmitter

More details on the TT&C system can be found in Sec. 21.1.

Table 14-11, Fig. 14-6◀, Eq. 14-2

The international community has developed standards for spacecraft communications. The Consultative Committee for Space Data Systems (CCSDS) maintains the standards documents. These standards include coding on the radio links as well as command and telemetry packet and transfer frame formats.

14.4.6 Power

The power subsystem provides all of the power generation, storage, and distribution for the spacecraft. The design drivers are in Table 14-12.

Table 14-12. Power Subsystem Design Drivers.

Major Design Drivers	Driven By	Impact
Power Consumption	Payload Req	Solar array, battery
Power Distribution	Spacecraft design	Power electronics, wiring
Eclipse Duration	Orbit	Battery
Bus Voltage	Spacecraft design	Power electronics, wiring
Payload Duty Cycle	Ops concept	Solar array, battery

The needs of the payload will drive the overall power consumption of the spacecraft. Something will need to generate the power—usually a photovoltaic solar array, although deep-space or high-power missions may use other techniques as described in Sec. 21.2.6. Very short missions may simply use stored energy from a battery. When array power is not available, a battery will keep the system operating. The power electronics controls the charging of the battery, and it controls the switching of power to the spacecraft loads. Lithium ion batteries are becoming the first choice for most spacecraft today.

Most spacecraft use a 28 V bus. Since there is power lost in the conversion of 28 V to the lower voltages typically used by electronics, very small spacecraft use lower-voltage buses. If the spacecraft has large power demands and long harness runs, it may make sense to use a larger voltage. The International Space Station operates around 160 V on its power bus. The total power consumed by a component is:

$$P_{comp} = V \times I \tag{14-3}$$

where P_{comp} is the component power and I is the current into the component. Higher voltages will reduce the current used by components, and it will therefore reduce the power lost in harnessing, since the power dissipation is:

$$P_{dis} = I^2 \times R \tag{14-4}$$

where P_{dis} is the power dissipation, I is the current in the harness, and R is the resistance of the harness. Most components on a 28 V bus will draw constant power, because of the design of the internal electronics. This means that a **drop** in bus voltage will cause components to draw **more** current. Be careful during the testing of such components, because laboratory power supplies typically reduce their voltage if a component's current exceeds the supply's current limit—if the supply goes into current limiting mode, the supply voltage drops, causing the component under test to draw even more current, eventually taking the voltage down below the normal operating range. For this reason, you should require all components on the spacecraft bus to survive indefinitely without damage any voltage below the operating range all the way to 0 V.

Transistors have largely replaced relays for spacecraft power switching. Modern circuitry allows for resettable circuit-breaker functionality to protect harnessing and the power bus, rather than using fuses for over-current conditions. Latching relays are still used in cases where a system must maintain a switch in a given configuration even after a power outage. Relays also find some application in safety circuits, since they completely isolate the source voltage from the potentially hazardous actuator.

The solar arrays typically require a long development cycle. The systems engineer must define the power requirements early, so that the power engineer can size the array. The mechanical team then must design and fabricate the array substrate, before the array manufacturer can populate the array. The systems engineer must then carefully monitor the power consumption of the spacecraft to prevent exceeding the array's capability.

One way to delay the sizing of the array without delaying the delivery of the array is through the use of a *modular design* for the solar array. In this approach many individual solar array modules are assembled onto a frame to create the full-sized array. The solar array builder can assemble each of the modules while the spacecraft team is still sizing the power system, with the exact array configuration coming later. The spacecraft builder can keep an inventory of modules for rapid solar array development. The WIRE spacecraft solar array was a modular design. The mass impact of this approach was small [Everett, et al. 2000]. Figure 14-7 shows the WIRE engineering unit solar array structure, with one frame populated with modules and the other unpopulated. Section 21.2 provides more details on power systems.

14.4.7 Structures and Mechanisms

The spacecraft structure supports all the spacecraft components and payloads throughout the launch, and it provides a stable platform for on-orbit operations. Table 14-13 describes the design drivers for the structure.

The acceleration of the launch vehicle is the dominant load for the structure, with the maximum load generally occurring near the end of a stage, since the mass of the fuel is almost gone but the rocket still has the same thrust as it did early in the flight. Some liquid launch systems can reduce the thrust toward the end of the burn to limit this effect. Expect loads of about 6 to 10 g's, depending

Table 14-13. Mechanical Subsystem Design Drivers.

Major Design Drivers	Driven By	Impact
Launch Vehicle	Launch mass	Structure strength, stiffness
Payload Accommodation	Payload config	Structure shape
Solar Array Area	Power system design	Appendage, overall shape
Structural Stability	Pointing requirements	Stiffness, joint design, material selection
Center of Mass Offset	Launch vehicle, ACS design	Layout, balance mass
Thermal Radiator Area	Power dissipation	Structure layout

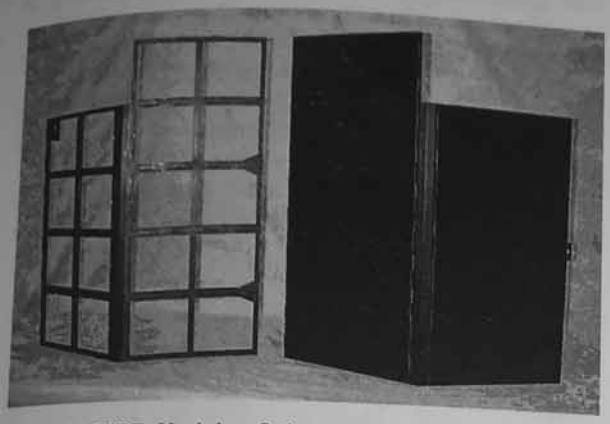


Fig. 14-7. WIRE Modular Solar Array. (Photo courtesy of NASA GSFC)

on the rocket, with smaller systems creating higher loads. Lateral loads, although generally smaller, can also have significant impact on the design. Most vehicles produce less than 3 g's laterally, but the drop transient of the Pegasus can be significantly more, especially near the end of the spacecraft, since there is a rotational component to the transient. Some vehicles have a sine transient during motor ignition, especially as one stage burns out and the next starts. The spacecraft's natural frequencies must stay away from the launch vehicle driving frequencies in order to avoid coupling excessive energy into the system. *Load isolation systems* have become more common in recent years. Composed of flexures and dampers and functioning like the springs and shock absorbers on a car, a load isolation system reduces the high-frequency components of loads transmitted from the launch vehicle into the spacecraft. Launch vehicles typically impose requirements on spacecraft frequency and the position of the center of mass.

In addition to the static and transient loads, a launch vehicle produces vibro-acoustic loads. The noise from the launch vehicle couples into the spacecraft through the air around the vehicle during the early part of the flight. Blankets inside the fairing can help reduce this environment. Large structures, like the solar array panels, will absorb energy from the acoustic environment and couple it into the structure. Additionally, the launch vehicle will transmit vibration directly through the mounting to the spacecraft. The vibration loads tend to dominate the structural design of smaller components.

Any devices in the system which release energy suddenly will produce a shock load. The separation interface with the launch vehicle is generally the biggest source of shock, but other releases, such as the deployment of a solar array, can produce a shock transient. Shock loads tend to cover a large frequency range, with much of the higher frequencies attenuating rapidly across interfaces.

Mechanical engineers must take all of the above-described loads into consideration when designing a structure that will also provide the necessary views for all of the payloads and spacecraft components. Additionally we ask them to minimize the mass, so we can maximize the payload. Structural modeling is an important aspect of the design process, with modern computer modeling providing accurate predictions of the structural frequency and the stress on individual items. Despite the success of structural modeling, it is still important to conduct some tests to confirm the validity of assumptions built into the model.

Payloads which require high-precision pointing place additional demands on the spacecraft structure. We need to ensure that the movement of the structure, as it expands and contracts with temperature changes, does not excessively move the payload or spacecraft attitude sensors. *Flexures* bend in one direction while remaining stiff in another, preventing the buildup of stress which can distort a surface. We can use materials like graphite-composite, which do not change size much over temperature. An effective optical bench structure uses graphite-composite face-sheets over an aluminum honeycomb core. This arrangement produces a very stiff plate that is extremely light weight and does not change much in size over temperature. Section 22.1 provides more details on spacecraft structures, including some strategies for estimating mass.

Spacecraft mechanisms, which must operate in the micro-gravity environment of orbit, can be difficult to test on the ground in a 1 g environment. A motor with plenty of torque margin to move a solar array around in space may not have any hope of moving that array when it is loaded by gravity. During the design phase, it is important to carefully consider the need for each mechanism on the spacecraft, since each will require detailed analysis and each will be potentially difficult to test, creating a risk that it will not operate correctly on orbit. Simple springs and hinges generally provide a more reliable one-time operation, such as a solar-array deployment, than a motor. Any deployment must have a method to hold the component in place during launch, generally under compression, then release it at the appropriate time. Release mechanisms are generally electrically

Table 14-13, Fig. 14-7, Eq. 14-4◀

Table 14-14. Thermal Subsystem Design Drivers.

Major Design Drivers	Driven By	Impact
Thermal Environment	Orbit	Entire thermal design
Power Dissipation	Spacecraft design	Radiator area
Op and Survival Temps	Payload, components	Entire thermal design
Structural Configuration	Mechanical drivers	Restricted radiators
Structural Stability	Pointing requirements	Tight temperature limits

actuated. *Electrically-actuated explosive devices* (EED) can cut a cable or bolt, release the threads on a bolt, or provide some other such release by detonating a small explosive charge with heat generated by electrical current. EED's generally create a large shock, since they release suddenly. Other types of non-explosive actuators are available today, some of which can be reset, allowing multiple tests with the device that will actually fly.

Kick-off springs generally get a structure moving after its release, overcoming any static friction that might be present. Section 22.1.5 provides more details on mechanisms.

14.4.8 Thermal Control

The thermal subsystem keeps all components and payloads within operating temperature ranges during normal operations and it keeps everything within survival limits under all circumstances. Table 14-14 lists the design drivers for the thermal subsystem.

The power dissipation of the system combined with the orbit's thermal environment will drive the thermal design. The system must conduct internally-generated heat to the outside of the spacecraft where it can be radiated to space. Ideally, the radiators will face deep space, with its 3 K background temperature. For low-Earth orbit, the Earth is about 300 K and fills half the sky, making it more difficult to find room on the spacecraft for radiators that need to run below this temperature. Since the Sun is only a half-degree across from Earth distance and it is surrounded by deep space, radiators can face it, but they need white or shiny coatings which reflect optical energies and still radiate in the infrared. Smaller spacecraft generally have plenty of surface area to radiate the heat that can be generated by the few components that fit in their small volume. On larger spacecraft, it can be a challenge to move heat from its source out to a radiator. *Heat pipes* provide an effective means to transport heat within a spacecraft. Heat pipes use the evaporation and condensation of a working fluid like ammonia to transport the heat. The pipes can be designed to take advantage of capillary action in the micro-gravity environment to move the fluid to the source of the heat while the vapor moves to the radiator.

The biggest thermal challenges come when temperatures must be kept within a very narrow temperature range. The thermal engineer can solve temperature range issues by adding more radiator area, enabling the system to dissipate more heat. But that extra heat dissipation will be a liability during cold cases, resulting in additional heater power. In the extreme, we can run the radiator so cold that we always need some level of heater power to maintain the desired temperature. One of the biggest system-level trades related to the thermal system involves this tradeoff between thermal system performance, radiator area, and power consumption.

Section 22.2 contains more information about the thermal subsystem.

14.4.9 Other Specialties (Cross-cutting Disciplines)

The topics below; contamination control, risk & reliability, parts control, mission assurance, and safety; are typically handled by discipline experts who operate at the spacecraft system level. These are not subsystems, but they are part of the partitioning of work necessary to build a spacecraft. These considerations will impact the cost of the system.

Contamination Control

We have all seen the dust that accumulates on surfaces over time. Many of us have also experienced the film that forms on the inside of car windshields—a hard-to-remove scum that requires window cleaner or vinegar to remove, leaving the rag black. There are many sources for the dust, not the least of which is the surface layer of our skin flaking off as new layers grow below. The film on the windshield is due to *outgassing* of the plastics in the car—molecules in the plastic escape the material, float around in the car, and then deposit on the glass. The "new car smell" is actually your nose detecting those molecules floating around in the car's air. Clearly, we do not want to cover spacecraft optics with dust or a film of molecules, so we must be very careful about the environment in which we assemble the spacecraft, and we must choose materials for the spacecraft that will not excessively outgas. Controlling these contaminants is a specialty in the spacecraft business, with engineers focusing on processes and procedures for maintaining cleanliness levels.

There are some general principles and rules of thumb that the systems engineer needs to know. The spacecraft requires some top-level requirements for particles and molecular contamination. Typically, different surfaces will require different cleanliness levels. The top-level contamination requirements will also drive the selection of materials for use in the spacecraft. Those material requirements flow out to everything installed on the spacecraft. Some materials are never flown in space, because they will simply evaporate in a vacuum, leaving a film over everything. The payload is the chief driver of top-level contamination requirements, although star trackers, solar arrays, and sensitive thermal surfaces

(where we don't want a change in optical properties, since we are rejecting heat in the sunlight) can also be drivers. Ultraviolet instruments are particularly susceptible to molecular contamination, since the molecules generally have resonances at ultraviolet wavelengths, absorbing a significant portion of incoming photons. Molecular contamination will tend to discolor at optical wavelengths as ultraviolet radiation from the sunlight changes chemical bonds. These changes also make it difficult to remove the contaminants through heating.

IEST-STD-1246 is the standard for surface cleanliness, replacing FED-STD-1246 in the United States. A number and a letter, such as 450 A, are used to represent the particle and molecular contamination levels, respectively. The number corresponds to the largest single particle in microns allowed within a square foot, although there are also limits on the number of smaller particles. The letter is simply a non-volatile residue limit as defined in the standard.

Outgassing products tend to accumulate on cold surfaces. For surfaces below freezing, any water that evaporates out of the spacecraft, such as out of graphite epoxy structure, will accumulate. Critical components, such as telescopes, will typically use *decontamination heaters*, which keep critical surfaces warmer than the surrounding area, driving contaminants away. These decontamination heaters are powered during the early part of the mission, when the spacecraft is first experiencing the hard vacuum of space and components are outgassing at the highest rate. Sometimes decontamination heaters are operated after propulsive maneuvers, since thrusters may leave products behind. On top of the material and assembly constraints described above, the spacecraft designer must account for the power loads associated with these heaters.

Based on the cleanliness requirements, the contamination control plan will call out the *clean room* environment in which the spacecraft and its components must be assembled. A clean room is a facility where filters and air motion maintain particle count and organic and aromatic molecules below a certain level. ISO 14644 is the international standard for clean rooms and it replaces FED-STD-209 in the United States. The FED-STD-209 designations referred to the maximum number of particles greater than 0.5 μm in a cubic foot of air. The ISO 14644 standard uses 1 cubic meter as the standard volume with 0.1 μm as the standard particle for class 6 and cleaner and 0.5 μm still used for the dirtier classes. Table 14-15 describes common practices for the most common clean room classifications. The cleaner the room, the more care and monitoring required in order to maintain the classification, driving the cost up.

Risk and Reliability

Once a spacecraft is launched, it generally cannot be retrieved (the Space Shuttle provided some limited exceptions). A single failure in the wrong place can end a mission. The designers perform analyses to investigate what failures might be fatal and how likely it is that they might occur. A *risk* is a bad event that might occur. The criticality of the risk is the combination of likelihood that the bad event occurs and the consequences of that occurrence. The reliability of components is the likelihood that the components will fail. Designers can add redundancy or use more reliable parts to increase the reliability of the overall system. The system architecture will affect the criticality of each part. The designer must be careful to avoid an architecture that, despite redundancy, has *single points of failure*, where one failure can take out all of the redundancy. A full description of reliability and risk management is covered in Chap. 24.

Parts Control

The space industry has developed a set of processes and procedures to ensure that electronic parts will operate over the life of the mission. Modern electronic parts include features smaller than 1 μm! These features are susceptible to manufacturing flaws, particles, packaging issues, moisture, electrostatic discharge (during ground handling or in flight), and even individual protons ejected by the Sun. Screening procedures test the potential flight parts and destructively test parts made in the same batch or *lot*, before they are considered qualified for flight. This screening detects flaws in individual parts, flaws in the manufacturing process, and weaknesses induced by a radiation environment. Obviously, the more screening, the more expensive the parts will be. Good quality parts are expensive, but they reduce the expense of failures later in the development or in flight. Modern part manufacturing techniques have greatly improved the consistency and reliability of commercial electronic parts, but some level of screening, such as a 500-hour high-temperature burn in, perhaps conducted at the circuit-card level, is prudent even for the most inexpensive of space missions. Do not underestimate the impact of the radiation environment. Part response due to single-event effects and total ionizing dose is not always predictable (see Sec. 7.3 for more details). NASA's Electronic Parts and Packaging Program provides information on electronic parts on their website, including the Instructions for EEE Parts Selection, Screening, Qualification, and Derating (EEE-INST-002). Although these instructions are intended to provide direction to NASA developers, they provide good general information about a variety of parts and part qualities. See Sec. 24.1.3 for more information on parts.

Mission Assurance

Mission assurance, which includes *quality assurance*, refers to the discipline that monitors the spacecraft build and ensures that the builders follow proper processes and procedures. The mission assurance team reviews the plans for the assembly during the design phase. During build, they review and approve procedures, monitor the execution of those procedures, and inspect the flight hardware after build steps. Quality assurance typically refers to the monitoring and inspections. For any spacecraft development, it is essential that everyone involved with the

Table 14-15. Most Common Cleanroom Classes.

ISO 14644 Designation	FED-STD-209 Designation	Typical Practices
Class 6	***Class 1000***	Full-cover garments with clear plastic face masks; clean garments for each use; daily surface cleaning; precision cleaning of new items; items carefully restricted
Class 7	***Class 10,000***	Full-cover garments with cloth face masks; weekly cleaning of garments; weekly cleaning of surfaces; precision cleaning of new items; items restricted
Class 8	***Class 100,000***	Smocks, shoe covers, gloves and hair nets; regular cleaning of surfaces; wipe-down cleaning of new items; some items restricted

system pay attention to quality (since a single failure can take down a mission), but mission assurance provides that extra set of eyes to help spot those unavoidable human errors that are bound to happen. It is important to get the mission assurance team on board early in the development, to work with the builders so that everyone has the same expectations for later processes and procedures.

Safety

Spacecraft and especially launch vehicles store large amounts of energy. It is essential that the spacecraft designer consider what might happen to that energy if something goes wrong. The spacecraft manufacturer, the customer, and the launch site range will all have safety requirements that will impact the spacecraft design and handling. Table 14-16 shows some examples of the common safety considerations.

The propulsion hazard is pretty obvious. Once a spacecraft is fueled, the flammable and toxic fuel is generally under pressure. The tanks and lines are critical safety items. Any heaters on the system must be monitored to prevent a runaway heater from building pressure to the burst point or damaging insulation on wires. (The Apollo 13 mishap was caused by a runaway heater which melted insulation during ground testing. During flight, the exposed wiring caused a spark which started a fire, which heated the oxygen tank until it exploded, taking out the redundant oxygen tank.) The general strategy for safety of people is to ensure that no two failures or individual operations will cause a hazardous event. So the propulsion system generally has three valves in series, and any electrical operations require three switches in series, with each switch separately controlled. This strategy, with three *inhibits* (independent features that stop an event), is known as *two-fault tolerant*.

Batteries store a large amount of energy in a small volume. Overcharging can cause an explosion. A rapid discharge can lead to overheating and a fire or explosion. Batteries are generally designed to prevent some hazards, such as thermal switches in each cell to prevent thermal runaway. Other hazards must be controlled through procedures and protective circuits on the spacecraft. Frequently, the voltage of each individual cell must be monitored during charging or discharging before launch. The wiring for this monitoring must be within the spacecraft's mass budget, even if the monitoring circuits stay on the ground.

Deployables, such as fold-up solar arrays, generally use some electrically-activated device to release. That device may be hazardous in itself, like an EED, which uses a small explosive charge to cut a cable or bolt, or it may simply release the energy needed to unfold the deployable. Either way, three inhibits are needed to prevent electrical actuation. Transmitters on the spacecraft may also require three inhibits if the signal levels exceed the levels to which the launch vehicle has been designed. Cryogenic solids and liquids also require special care, since the simple warming of the system will result in an explosive buildup of pressure if it is not allowed to vent.

Table 14-16. Safety Impacts on Spacecraft Design.

Component	Hazard	Design impact
Propulsion System	Fire, explosion, toxic gas	Pressure vessel design guidelines, extra valves, extra electrical switches
Battery	Fire, explosion, toxic materials	Battery design rules, charging restrictions, cell monitoring
Deployables	Impact with personnel, interference with launch vehicle operation	Extra electrical switches
Transmitters	Interference with launch vehicle operation	Extra electrical switches
Cryogenics	Explosion	Relief valves, burst disks

Table 14-16, Fig. 14-7◀, Eq. 14-4◀

14.5 Creating Preliminary Spacecraft Budgets

After determining what subsystems are appropriate for the particular design, the next step is the allocation of resources and requirements to those subsystems. In other words, we must decide exactly what we expect each subsystem to do (the requirements), and we must decide how much of the spacecraft resources will be necessary to do that job. Obviously, this will be an iterative process. We start off knowing pretty well the subsystem driving requirements, and we discussed those requirements in the previous section, but our initial estimates of the resources to achieve those requirements is merely an educated guess based on experience. But we need to give our subsystem designers a starting point, so we make those guesses.

The fundamental resource of a spacecraft is mass. Because of the high cost of launch vehicles and the step function in cost when you outgrow a vehicle, the system design must stay within established mass limits. Power becomes another limited resource once you choose the solar array size—a choice that must generally occur early in the development. The long, multi-step assembly process for an array, combined with the array's impact on the attitude control and thermal systems will drive you to decide early the array configuration and size (see Sec. 14.3.3). Table 14-17 lists technical resource budgets that are typically tracked by spacecraft systems engineers. You may not need to track all of them for your system, and some of them may be tracked at the subsystem level, but it is a good idea to think through your resources early and decide what is important. The subsections below will walk you through some of the key budgets.

In order to ensure that the system stays within the mass capability of the launch vehicle, we set aside *margin*, which is extra mass that is not assigned to any particular subsystem. We also carry margin for all the other spacecraft resources, like power, signal levels on the radio link, fuel, processor utilization, and memory. One of the biggest challenges for a spacecraft systems engineer is properly managing margin. If you hold too much margin, the subsystem designs become overly constrained and the cost of the system rises, or you lose out on capability you could have had in your system. If you hold too little margin, you will find yourself with an assembled spacecraft that doesn't work—it is too heavy for the launch vehicle, its solar arrays don't generate enough power to support its function, there is insufficient fuel, or some other such crisis. The consequence of exceeding capability is severe—you don't want to get to the end and have something that doesn't work. Margin covers the uncertainty of the design, so you can reduce it as the design matures.

Figure 14-8 shows the typical management of margin for resources like mass and power. We allocate to a subsystem a mass total that includes the *current best estimate* and a *contingency* related to the maturity of the

Table 14-17. Typical Budgets Tracked by the Spacecraft Systems Engineer.

Parameter	Reference
Mass	Sec. 14.5.1
Power	Sec. 14.5.3
Pointing and Alignment	Sec. 14.5.4
Propellant or Delta V	Secs.14.5.2, 10.3
TT&C Link	Sec. 21.1.4
Data System	Secs. 14.5.5, 20.1, 20.2, 28.2
• ***CPU utilization***	
• ***Non-volatile memory***	
• ***Volatile processor memory***	
• ***Processor data bus utilization***	
• ***Peripheral data bus (i.e., 1553)***	
• ***Timing***	
• ***Data latency***	
• ***Operational availability***	

subsystem design. We want every subsystem designer to be able to meet their requirements within the total we have allocated, but we also want that designer to have some flexibility in the exact implementation. The contingency gives them the room to adjust, without impacting the rest of the system. On top of the allocation, we, as systems designers, carry additional *system margin* to handle unforeseen situations. Perhaps we need to add another function to our design after exploring the details—for instance, on LRO we added an extra box late in the design phase to provide launch-pad lightning suppression. Or perhaps a subsystem cannot achieve their requirements within the resource allocations. Or perhaps a design error requires compensation elsewhere in the system. The WISE mission added a load-isolation system to compensate for an undersized support within an already-built instrument. These sorts of problems are part of the normal design process, so we carry system margin to handle them. Our *total resource margin* is the sum of the contingencies for each subsystem plus the system margin. A *lien* is a potential impact to the system margin. We carry a lien when there is a significant risk that a subsystem will exceed an allocation (see Sec. 24.2 for more on risk management). The lien is still system margin, but it is margin that is only available for solving a specific problem. If we later retire the risk without it becoming a problem, we can release the lien back into the general pool of system margin.

A final note on margin: it is appropriate to launch with performance margin. Cutting the design to the edge may seem the most efficient, but some events are **unforeseeable**. Having extra processor capability at launch will allow you to add additional processing after launch. To do that, you will need more memory for code space. Extra power can solve mistakes in the thermal design that you couldn't properly test on the ground, or it can allow you to run the transmitter longer to downlink more data or compensate for a failed high-gain antenna. Fuel

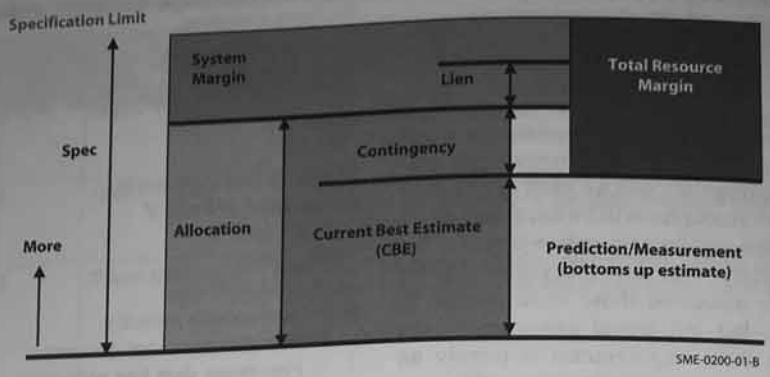


Fig. 14-8. Graphical Presentation of Margin.

margin will give you options to salvage a mission after a failed maneuver. Spacecraft are generally irretrievable for repair, so margin is your insurance. The systems engineer's biggest hope on launch day is that there is enough margin to cover the things we have forgotten. Buy your insurance wisely, though. The best margin can be consumed to increase return from your system if everything goes well. Extra fuel will extend the life of a spacecraft that requires orbit maintenance. Extra power can be used to boost communications bandwidth beyond requirements early in the mission or run extra instruments, with the system dropping back to required performance as the arrays degrade toward the end of life.

14.5.1 Mass

Appendix A and Table 14-18 will help you make a first cut at a mass budget. Start with your target launch mass, then subtract your propellant mass (see Sec. 14.5.2). Using App. A and Chaps. 15–17, determine a reasonable allocation for the payload. Remember that this allocation includes contingency above your current best estimate. For a new payload design, this contingency should be on the order of 30% of the CBE (see Sec. 14.6). Then set aside system margin. A good starting point is 25% of the remaining mass, based on historical trends. Use App. A and Table 14-18 and some judgment about subsystem complexity to take a first stab at the target CBE's for the subsystems. For some applications, this cut may be enough. The next iteration of mass requires quite a bit more work, with the details of Chaps.18–22 getting you a better estimate of the mass for each subsystem. With those better estimates in hand, take a look at your total mass margin, including the payload contingency, and decide whether your initial cut at payload mass allocation was good. Adjust as appropriate, take some of your system margin and allocate it as contingency, and document your allocations. If you run into trouble with this iterative process, go back and revisit your starting assumptions. Perhaps your target mass is too low. The SCS example in Sec. 14.7.1 will walk you through this process.

Table 14-18. Average Mass by Subsystem as a Percentage of Dry Mass for 4 Types of Spacecraft. Types include those with no propulsion, those in Low-Earth Orbit with propulsion, those in high-Earth orbit, and planetary missions. See App. A for more information.

Subsystem (% of Dry Mass)	No Prop	LEO with Prop	High Earth	Planetary
Payload	41%	31%	32%	15%
Structure and Mechanisms	20%	27%	24%	25%
Thermal Control	2%	2%	4%	6%
Power (incl. harness)	19%	21%	17%	21%
TT&C	2%	2%	4%	7%
On-Board Processing	5%	5%	3%	4%
Attitude Determination and Control	8%	6%	6%	6%
Propulsion	0%	3%	7%	13%
Other (balance + launch)	3%	3%	3%	3%
Total	100%	100%	100%	100%
Propellant	0%	27%	72%	110%

Table 14-18, Fig. 14-8, Eq. 14-4◀

14.5.2 Propellant

The propellant budget follows directly from the delta V budget. Sec.14.2.3 shows how you use the rocket Eq. (14-1), to determine mass of the fuel. In reality, there is not a single I_{sp} for all maneuvers. A large orbital maneuver may use a more efficient engine than smaller housekeeping maneuvers. You need to itemize each of the maneuvers, keeping track of the fuel mass used and the remaining mass of the system after each. That remaining mass is then used with the next maneuver's delta V and I_{sp} to calculate the fuel use.

Table 14-19 shows the Lunar Reconnaissance Orbiter fuel budget at launch. The total launch mass is the system dry mass (1,017.7 kg) plus: *residuals* (5.2 kg), which is fuel that cannot be expelled from the system; the *pressurant* (3.2 kg), which is a gas used to pressurize the system (helium in this case); and usable propellant (889.2 kg). The table itemizes specific uses for the fuel: the mid-course correction (MCC), thruster checks, lunar-orbit insertion (LOI), mission orbit insertion (MOI), regular station keeping, momentum unloading, extended mission, and additional margin. The critical MCC and LOI-1 burns were preceded by engineering burns to confirm the performance of the attitude control system during thruster maneuvers. The MCC burns were done in a *blow-down* configuration, without the pressurant valve open, resulting in a lower I_{sp} than later maneuvers. All other maneuvers were done in a *regulated* configuration, with the pressurant flowing through a regulator and providing constant pressure to the fuel. The LOI-1 maneuver used large thrusters with higher efficiency, while all other maneuvers used smaller thrusters for more accuracy. The station keeping maneuvers have a somewhat lower efficiency than MOI-1 to account for the fact that short maneuvers don't give the thrusters enough time to get to full efficiency. Finally, there is a fuel mass associated with delta H, which are momentum dumping maneuvers. LRO had a requirement to carry fuel for an extended mission. The final, as-built system came in a bit under the target mass, which translated to additional margin.

Built into this budget are some conservative assumptions about the delta V required and the I_{sp} performance—we used the worst-case predictions, but our actual performance ended up somewhat better. It is important for the systems engineer to be aware of this hidden margin so that we don't add margin on top of the margin.

14.5.3 Power

The initial estimate for the solar array size should be based on an estimate of the power consumed by the spacecraft. Just like with mass, you should itemize the subsystems and eventually the individual components in a subsystem and add up the total power. Table 14-20 provides some general guidance on power consumption based on type of spacecraft, based on historical data. Some rules of thumb on power consumption:

- The faster a data system operates, the more power it will consume. The digital logic circuits that run our flight computers consume drastically more power when they are switching than when they are at a steady state. So faster processors consume more power.

Live Calc

Table 14-19. LRO Fuel Budget at Launch. (Courtesy of Charles Zakrzwski)

Total Launch Mass (kg)	1915.27					
Maneuvers	Delta V (m/s)	Effective I_{SP} (s)	Total Propellant Used for Maneuver (kg)	Delta H Propellant (kg)	Total Propellant Used (kg)	S/C Mass at End of Burn (kg)
MCC Eng	2	202	1.93	0	1.93	1,913
MCC Burn(s)	28	202	26.85	0	26.85	1,886
Thruster Check Out				0	0.23	1,886
LOI Eng	8	223.6	6.87	0	6.87	1,879
LOI – 1	583	223.6	438.64	0	438.64	1,441
LOI – 2-5	362	220	222.48	0	222.48	1,218
MOI – 1	48	220	26.80	0	26.80	1,191
MOI – Others	8	217	4.47	0	4.47	1,187
Station Keeping	162	217	86.98	0	86.98	1,100
Momentum Unloading		N/A		17	17.00	1,083
Extended Mission	69	217	34.54		34.54	1,048
Additional Margin	46	217	22.41	0	22.41	1,026
Totals	1316				889.20	

Table 14-19, Fig. 14-8◀, Eq. 14-4◀

Table 14-20. Average Power by Subsystem for 4 Types of Spacecraft. Types include those with no propulsion, those in Low-Earth Orbit with propulsion, those in high-Earth orbit, and planetary missions. See App. A for more information.

Subsystem (% of Total Power)	No Prop	LEO Prop	High Earth	Planetary
Payload	43%	46%	35%	22%
Structure and Mechanisms	0%	1%	0%	1%
Thermal Control	5%	10%	14%	15%
Power (incl. harness)	10%	9%	7%	10%
TT&C	11%	12%	16%	18%
On-board Processing	13%	12%	10%	11%
Attitude Determination and Control	18%	10%	16%	12%
Propulsion	0%	0%	2%	11%
Average Power (W)	299	794	691	749

- Communications systems trade power for antenna size. This is an important system-level trade. You may be trading a smaller antenna for a larger solar array.
- You will generally need more heater power if the spacecraft environment has large variations. There are ways around this problem, but generally you will need radiator area for the hot case and you will need heater power to make up for the heat lost by the radiator area during the cold case. Extra heater power available makes the thermal design easier.
- Power consumption estimates tend to peak around the time of Preliminary Design Review. The major uncertainty at this point is heater power consumption. Estimates for other components are generally pretty good by this point.

After you have determined the spacecraft power consumption, you need to size the solar array as described in Sec. 21.2.2. The solar array size will impact the attitude control system (primarily the system moment of inertia) and the structural design.

You need to think a bit about the safe mode power configuration. You want the spacecraft to consume as little power as possible in safe mode, so you can give the control system time to acquire the Sun on the solar array before depleting the battery. One of the power budgets to track will be the safe mode power consumption. Depending on your design, you may have several other driving cases, where you will have to track the power consumption for each of those modes.

14.5.4 Pointing and Alignment

The pointing and alignment budget encompasses not only the attitude control system pointing budget, but also structural effects on pointing. For payloads that demand accurate pointing, you will need to account for the alignment errors such as:

- Internal alignments within the payload
- The alignment of the payload to the structure
- The alignment of the attitude control sensors to the structure
- The movement of the sensors and payload with respect to each other due to temperature changes, one-time changes from the launch environment, and changes due to the structure no longer experiencing its own weight (also known as *1 g release*).

For the highest accuracy pointing, you should locate attitude sensors on the same structural support as the payload. On LRO, our instrument module supported the three instruments with tight pointing requirements along with our two star trackers. We made this module with graphite-composite facesheets to minimize changes due to thermal expansion, since graphite-composite has a much lower coefficient of thermal expansion than aluminum.

In addition to alignments, actuator performance will also affect pointing. Reaction wheels will provide continuously variable torque for smooth control, but imbalances in the wheels can create jitter. Thrusters and torque bars generally provide on-off, or *bang-bang*, control, so your pointing budget will need to account for a *dead-band*—angles over which the control system will allow an error before activating the actuator. You can improve pointing performance with control algorithms which include filtering, such as a Kalman filter. You can also include structural filtering within the algorithm to reduce interaction with flexible modes.

Since the errors associated with pointing and alignment are generally random and uncorrelated, we can combine terms as a root-sum-square rather than as an arithmetic sum. For high-performance pointing, you should plan to perform a *structural-thermal-optical performance* (*STOP*) analysis around the time of the preliminary design review, with updates as you finalize the design. A STOP analysis

Table 14-21. System-Level LRO Pointing Budget for Narrow Angle Camera. Each axis is allocated separately. The Attitude Determination and Control (ADC) errors are allocated to the knowledge (based on the sensors' performance and their alignment) and control (based on the algorithm and wheel performance). The jitter of solar array motion causes high-frequency errors that cannot be corrected. The thermal distortion is movement of the star trackers due to changing structural dimensions over temperature. The static bias accounts for errors that are fixed over time. The three separate items of static bias are uncorrelated, so they can be combined as a root sum of the squares. The budget also accounts for errors within the instrument itself.

		Arc-Sec		
		Rx	Ry	Rz
ADC	ADC Knowledge	30	30	30
	ADC Controller	15	15	15
Jitter	Jitter	30	30	30
Thermal	Thermal Distortion	60	60	60
Static Bias Contributors	1-g Release	75	75	75
	Launch Shift	20	20	20
	Ground Alignment Accuracy	300	300	300
Static Bias RSS		310	310	310
Instrument		21	21	21
	Total	**466**	**466**	**466**
	Requirement	**516**	**516**	**516**
	Unallocated Margin	**50**	**50**	**50**

uses a structural model along with temperature predictions to determine how much optical alignments will change during the course of mission operations. This analysis will highlight changes that might be necessary to meet pointing requirements. On LRO after performing the STOP analysis, we added flexures under the star tracker mount to prevent tracker alignment shifts caused by the mismatch between the tracker's aluminum base and the graphite-composite instrument module. Table 14-21 is an example pointing budget.

14.5.5 Data System

The data system includes several different budgets which are tracked by various engineers on the team. The interaction among processing capability, on-board data links, data storage, data latency, downlink data rate, RF link budgets, contact time, and ground station availability affects the overall architecture of the system. For the conceptual design of a spacecraft, you need to look for key drivers: is the collection rate high? Are there limited contact opportunities? Does the on-board data require a lot of processing before downlink? Based on your Concept of Operations (see Sec. 4.3), you will have a target number of passes and a duration for those passes. Taking the total data collected, divided by the contact time, you get your required downlink data rate. From there, use Sec. 21.1.4 to calculate your RF link budget. System-level considerations related to RF link budgets can be found in Sec. 14.4.5. You will size your data storage based on your data collection rate and the maximum time between passes. As your design matures, you will need to account not only for data collected by the payload, but also for system performance data that is generated by each component, called *housekeeping data*.

Table 14-22. LRO Data Volume Budget. LRO used a file system to collect data. The total files per day was used to determine ground handling overhead associated with number of files.

Type	Data per Orbit (Mbits)	Data per Day (Gbits)	Files per Orbit	File per Day
CRaTER Instrument	610.20	7.78	2.00	25.49
Diviner Instrument	180.52	2.30	22.60	288.00
LAMP Instrument	168.22	2.14	2.00	25.49
LEND Instrument	20.52	0.26	2.00	25.49
LOLA Instrument	225.96	2.88	2.00	25.49
LROC Instrument	34,659.16	441.67	28.00	356.81
Spacecraft Housekeeping	1,242.30	15.83	2.00	25.49
Total (Gbits):	37.11	472.87	61	772

Table 14-22 and Table 14-23 provide examples from LRO. We allocated to each instrument a certain amount of data collection per orbit based on the operations concept for that instrument. That allocation set the total data volume per day. Then we sized the recorder based on how much of that data would accumulate before we could start downlinking. LRO downlinked science data from the Moon to a single ground station, so transmission only occurred when the Moon was visible and high

enough in the sky at that ground station. Gaps between transmissions could be as much as 1,101 minutes (18.35 hours) when we accounted for a possible missed contact. The 412 Gbit recorder was sized to store the mission data over this period with margin. LRO's 100 Mbps downlink data rate enabled the transfer of all of this data within two 45-minute contacts (see Sec. 28.3 for more information on recorder sizing).

Table 14-23. LRO Recorder Usage Budget. The recorder was divided into partitions for ease of data management. The data collection was dominated by the LRO Camera (LROC) images. The current storage is based on the orbit data collection rate from Table 14-22 times the orbits between supports.

Max Time Between Supports:	1,101	Minutes	
Max Orbits Between Supports:	9.74		
Recorder Size:	412	Gbits	
	Partition 1 LROC	**Partition 2 Other Science**	**Partition 3 Spacecraft House-keeping**
Partition Size (Gbits):	376.32	14.00	22.00
Current Storage CBE (Gbits):	337.70	11.74	12.10
Remaining Space (Gbits):	38.62	2.26	9.90
Remaining Margin (%):	10.26%	16.11%	44.98%
Partition Usage (%):	89.74%	83.89%	55.02%
Additional Images/Orbits:	30.57	1.87	7.97

14.6 Design Evolution

This section describes how some real spacecraft evolved from the initial concept to the final product. The intent of the section is to provide some real-life lessons about where the design can go wrong. Each new design is unique, and it is impossible to forecast what problems will occur along the way, but you can be sure that there will be problems! Go into the design looking for places where problems can be mission ending. Be prepared for the unexpected. Tackle problems, or potential problems, as soon as you spot them. With all of that effort, and a little luck, you may have enough margin to overcome anything you have missed.

Figure 14-9 shows the growth in payload mass for 9 different NASA missions. Except for two outliers, all of the mass growth is between 20% and 37% of the value at the System Requirements Review. All of these payloads were based on some previous designs, but various mission-unique challenges led to similar growth. Despite this pattern in the payloads, Fig. 14-10 shows drastically different growths in the total system dry mass. It seems that the dry mass growth tends to be driven more by the available margin (see Fig. 14-11)—spacecraft designers will tend to expend mass margin to solve problems. On missions that had tight mass margin, like WIRE, the designers limited mass growth or even reduced mass in the system. On missions with large margin, the designers consumed much of the margin. Also notice in these graphs that the increase after the Critical Design Review (CDR) is limited to the 10–15% range, and even WIRE, which decreased after the Preliminary Design Review (PDR), still grew after CDR. (Milestones such as PDR and CDR are described in Sec. 3.4.2.)

Figure 14-12 shows the power trend for the payloads of 6 of the above missions (data from the other 3 missions was not readily available). Power does not show the same trend as mass. In fact, power frequently trends downward, especially after CDR. Fig. 14-13 and Fig. 14-14 shows a similar pattern for the overall power of the spacecraft. One of the main factors affecting the final drop in power consumption is the final update of the thermal model after vacuum testing. Spacecraft builders can frequently adjust blankets on radiators, based on actual performance, and reduce the overall heater power consumption. Margins also tend to trend upward at the end, with the solar array performance coming in a bit better than specification and the overall system power consumption dropping a bit. Because the solar array size must be fixed early in the process, it is important to be conservative with power estimates. As a spacecraft designer, you can keep this trend in mind, designing your mission to be able to take advantage of extra power margin to enhance performance—perhaps with more transmitter time or more payload operation. But don't cut the margin too tightly, or you will have no room to fix any late problems with the thermal design (larger radiators means more heater power) or other subsystem.

14.6.1 LRO

Launched on June 18, 2009, the Lunar Reconnaissance Orbiter (LRO), shown in Figs. 14-15 and 14-16, provided detailed mapping of the Moon, particularly in the polar regions. Because of the need to capture into and maintain a low (50 km), polar, lunar orbit, this 1916 kg spacecraft was almost half fuel. The conceptual design placed the system on a Delta II launch vehicle, with a spinning upper stage. A Presidential mandate tightly constrained the development schedule for LRO. The spacecraft preliminary design to support the 7 payloads included a custom fuel tank to carry the mono-propellant hydrazine. As the preliminary design review approached, the propulsion lead realized that the spinning upper stage of the Delta II would cause motion in the fuel that would interact with the Delta's guidance system. He could not guarantee that we would be able to achieve the nutation time constant that the launch vehicle required. Rather than continue with a design that might not work, the systems team considered other architectures, including a solid rocket motor kick stage for lunar orbit insertion (see Sec. 14.3.2). In the end, the customer decided to switch

Table 14-23, Fig. 14-8◀, Eq. 14-4◀

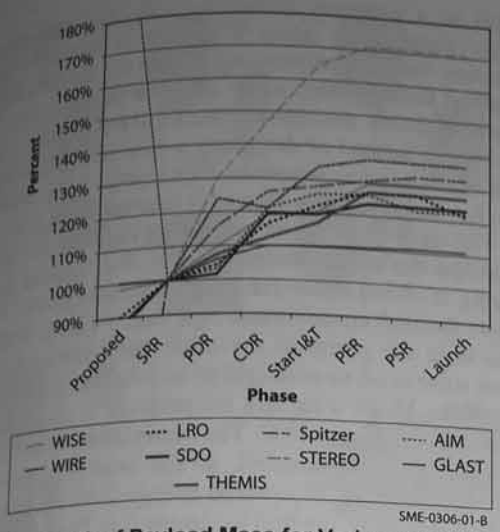


Fig. 14-9. Growth of Payload Mass for Various Spacecraft.

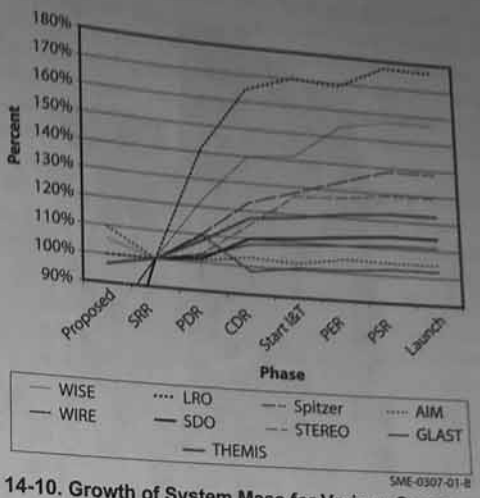


Fig. 14-10. Growth of System Mass for Various Spacecraft.

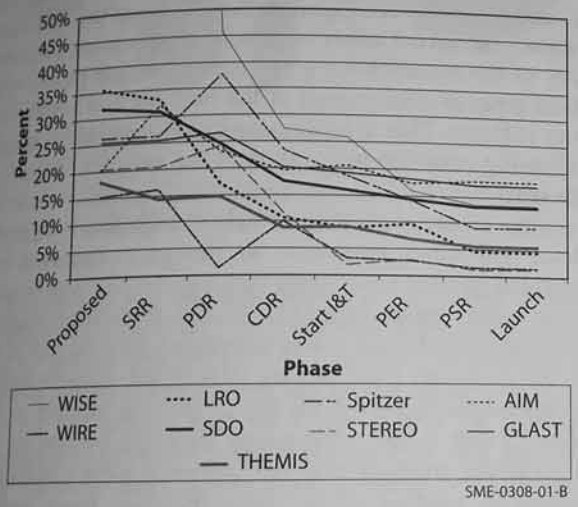


Fig. 14-11. Dry Mass Margin Depletion for Various Spacecraft.

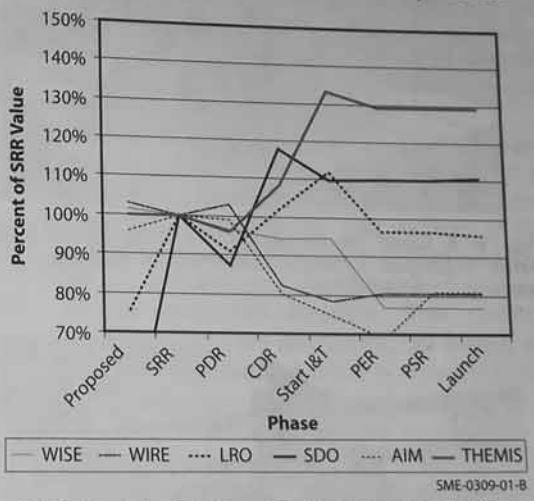


Fig. 14-12. Growth of Payload for Various Spacecraft.

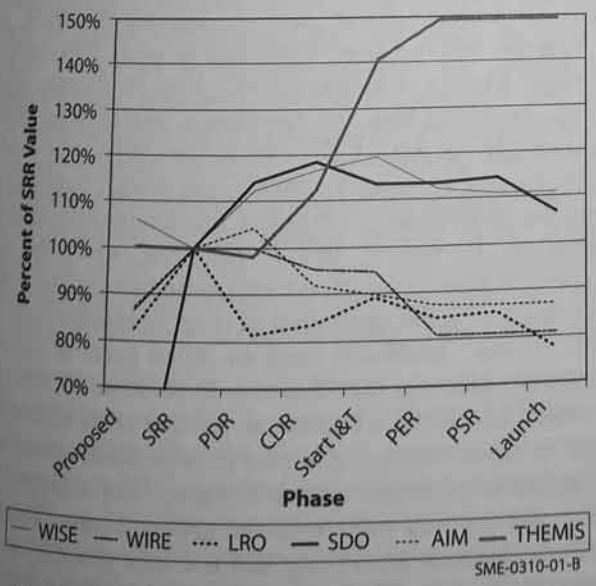


Fig. 14-13. Growth of System Power for Various Spacecraft.

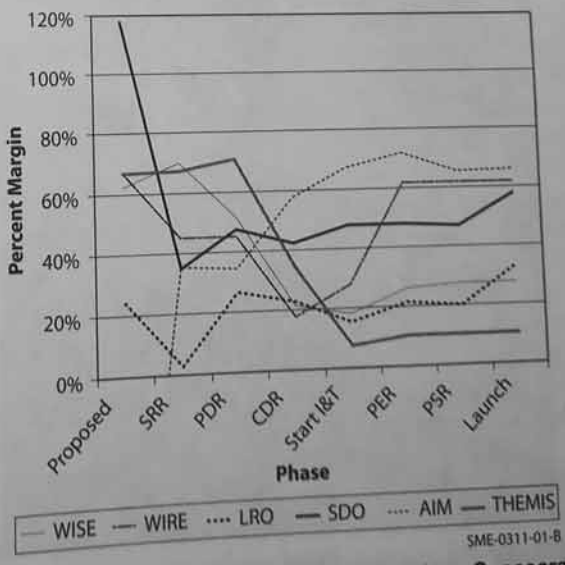


Fig. 14-14. Power Margin Depletion for Various Spacecraft.

Table 14-23◀, Fig. 14-14, Eq. 14-4◀



Fig. 14-15. Lunar Reconnaissance Orbiter Showing Solar Array, High-gain Antenna, and Thermal Radiator. (Photo courtesy of NASA GSFC)

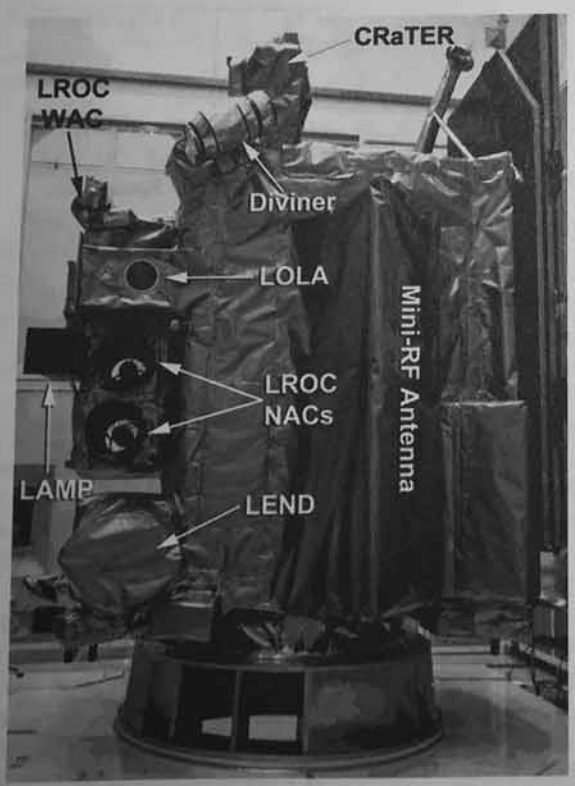


Fig. 14-16. Lunar Reconnaissance Orbiter with Instruments Labeled. (Photo courtesy of NASA GSFC)

LRO to a different launch vehicle. The non-spinning upper stage of the Atlas V eliminated issues related to nutation time constants, and the extra mass capability allowed the customer to fly an extra payload. The extra mass capability also allowed the LRO team to use existing fuel tanks instead of designing a new tank, reducing the risk of not meeting schedule. This change is the major reason for the increase in Fig. 14-9 between SRR and CDR.

The new vehicle and new tanks required the team to redesign the structure and, with it, the thermal system. Flying at 50 km above the Moon's surface presents a difficult thermal environment (see Sec. 14.2.4). When the Sun is in the plane of the orbit, the spacecraft travels from solar noon to midnight in less than an hour. At solar noon, the Moon's surface temperature is around 140 C. At midnight, it is –140 C. The Moon takes up half the sky as seen by the spacecraft, so the heat load goes from about 1300 W/m^2 of infrared energy from the Moon and 1300 W/m^2 from the Sun to almost zero from each in less than an hour. The thermal engineer had to couple the mass of the system together thermally, so that the electronics would not experience wild swings in temperature. All of the avionics were placed on a single panel, with heat pipes conducting all of the heat out to a zenith-facing radiator. This arrangement had the added benefit of allowing table-top integration of the avionics on the isothermal panel.

The mechanical team worked with the integration lead and other engineers to design a structure that enabled parallel processing (see Fig. 14-17). The propulsion team began assembly of the propulsion module almost a year before it was integrated with the rest of the system. At one point during integration, blanket technicians completed work on the propulsion module in one clean room while technicians installed heaters on the instrument module and the avionics were electrically tested in another clean room. Without these parallel operations, the LRO development would have taken at least another 4 to 6 months.

The launch vehicle change put the mechanical and thermal teams behind, and it delayed the PDR by 3 months. But the team continued to press forward to a Critical Design Review (CDR) just 10 months after PDR, by which time the mechanical and thermal teams had caught up. After CDR should theoretically be a calm time for the systems engineer—the design is complete and the vendors are all building the components to specification. In reality, this is the time when the real problems start to occur. As lead systems engineer, I was involved with the resolution of over 50 problems during this period—problems such as failed parts or components not meeting specification. In some cases, the team worked with NASA experts on parts or other specialties, and in some cases, I adjusted requirements across subsystems to maintain system integrity. For example, we changed the idle pattern in the ground system because the transponder sometimes did not lock properly to the baseline pattern. The transponder vendor was able to isolate the problem to a software design flaw, and he could

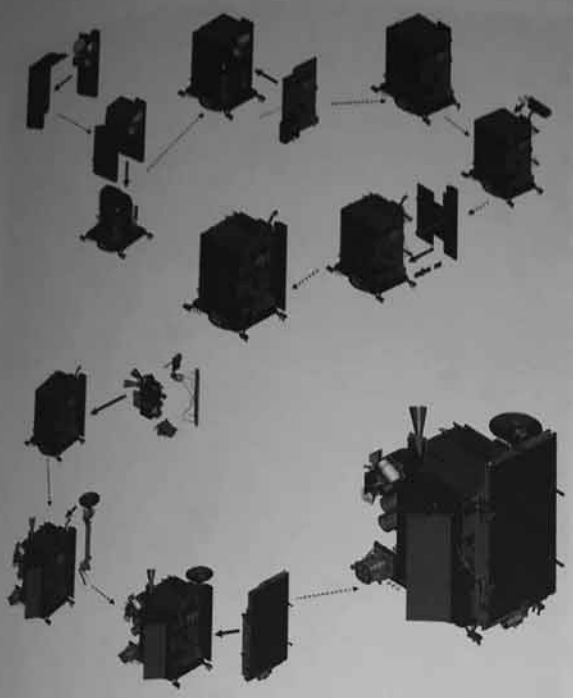


Fig. 14-17. LRO Modular Assembly.

prove that only our baseline pattern was a problem, so we changed the baseline rather than updating the transponder software.

Because of the interface tests we conducted during development, the team encountered very few issues during system integration. At our Pre-Environmental Review (PER), a member of the review team identified that we had missed running a test during the thermal vacuum testing of the solar array gimbal—we had not sent full solar array power through the system. We redesigned our system-level test to include this power-flow path. Within hours of the start of system-level thermal balance, we discovered a serious error in the gimbal thermal design—we were off by a factor of 14 in our estimate of power dissipation. The team came up with a simple design for an extended radiator and opened up some blankets, alleviating most of the problem. Had we not conducted this test, the spacecraft would have been seriously crippled in flight by overheating solar array gimbals. We encountered several other relatively minor problems during thermal vacuum as well as one major issue with an instrument. We were able to correct the problem with the instrument on the flight spare unit, and we installed the spare just before going to the launch site.

The mass chart (Fig. 14-9) shows how the mass settled out pretty well at CDR, except for the final 5% bump during assembly. Despite efforts to account for everything added to LRO, the final sum of components is still about 40 kg less than the actual measured mass. This extra mass is probably due to extra blanketing, tape, harnessing, harness shielding, brackets, and other such items. The lesson is clear—some bump up in mass after CDR is inevitable, despite our best efforts—there are always things we don't think of. Power consumption, on the other hand, dropped. The actual measured power of each component generally came in under specification. The thermal team, using actual power dissipation numbers, could reduce the size of the radiator, which reduced the heater loads under all conditions.

The mandated schedule was a driving factor throughout the spacecraft development. The team needed to make decisions quickly, always based on engineering judgment and experience and frequently with little analysis, and then we moved forward with the design. We had to decide what shortcuts to take to speed the development. But throughout the process, we were very aware of the possibility that a mistake could lead to mission failure. We fostered an environment where engineers were comfortable raising concerns with the project, and we explored those concerns. We cut corners and took risks, if the problems would manifest before launch. We relied on a solid test program to wring out any errors we may have made along the way. The lesson here actually came to the team in the form of a fortune cookie, "People forget how fast you did a job—but they remember how well you did it."

LRO achieved lunar orbit on June 23, 2009. All systems operated as planned (high praise for a spacecraft!), except for the laser altimeter, which experienced some mis-alignment when targeting the cold surfaces of the Moon. We eventually attributed this mis-alignment to a blanket that was a bit too tight, shrinking when cold and pulling on the instrument structure. This problem reduced some data from the instrument, but overall we were able to meet payload and mission objectives. We finished commissioning in mid-September 2009. As of this writing, LRO continues to operate successfully in lunar orbit, producing detailed images like Fig. 14-18 with enough resolution to spot trails left behind by the Apollo astronauts.

14.6.2 WISE

The Wide-field Infrared Survey Explorer (WISE), launched December 14, 2009, and shown in Fig. 14-19, used frozen hydrogen to cool infrared detectors so that it could produce an all-sky survey at 23, 12, 4.6, and 3.6 μm greater than 500 times more sensitive than IRAS. This mission built on the instrument developed for WIRE (see Sec. 14.6.3). It was competitively selected into the Explorer program in 2003, which limits the total mission cost. The target launch vehicle was initially a Taurus 2210. By the time of the System Requirements Review in December 2004, NASA was considering a dual manifest on the Delta II, but did not decide until March 2005. WISE became a sole payload on the Delta II when its co-manifested payload was cancelled.

The budget environment at NASA during the WISE development restricted the funds available, so the project delayed some parts of the build, such as the spacecraft bus. These delays are not unusual with NASA missions

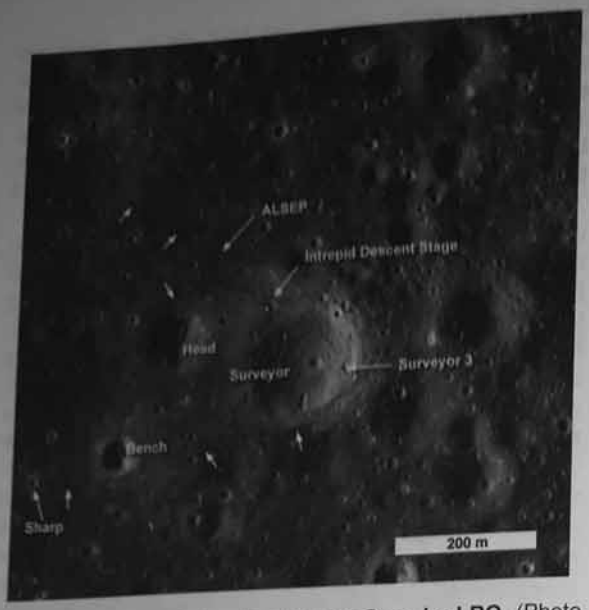


Fig. 14-18. Apollo 12 Landing Site as Seen by LRO. (Photo courtesy of NASA GSFC and ASU)

because of the political nature of the funding, but the project management was able to successfully schedule work to stay within each year's available funding and launch within a year and a half of the originally-targeted date. These types of constraints can be frustrating to spacecraft designers, but it is part of the business. As a designer, you must be aware of your constraints, and you must be flexible to handle delays. For instance, if you design for a low orbit, taking advantage of the smaller atmosphere at solar minimum (see Sec. 9.4.4), you may have a broken design if the launch is delayed.

The biggest challenge for WISE came when an instrument engineering unit failed during mechanical testing. The team discovered a mistake in the design of the structure that supported the telescope inside the *cryostat*. (The cryostat is an insulated container that stores the frozen hydrogen and uses it to cool the telescope and detectors below 20 K.) By this time, the cryostat assembly had already begun, and it was going to be a significant cost and schedule impact to replace the structure. The team considered a number of options, finally selecting incorporation of a *load-isolation system* into the design. A load isolation system functions like the suspension system in a car, reducing the higher frequency loads from the launch vehicle—smoothing the bumps in the road with springs and dampers. This load isolation system was able to reduce the dynamic environment enough to keep the cryostat support structure within the limits it could support. The extra mass margin of the Delta II enabled this decision.

The other major challenge to WISE was detector performance. The long-wavelength detectors did not have quite the sensitivity originally expected. The team tracked this performance measure throughout development, and concentrated extra effort on the detectors and the optical elements in the path when the potential problem was spotted. This effort did improve the situation, but in the end, the performance did not quite meet the original requirements. Upon consulting with the customer and the science team, everyone agreed that the mission objectives would still be achieved with slightly less performance at one wavelength, so they changed the requirement. The lesson here is to not be afraid to talk with your customer if your system is having trouble achieving a requirement.

Early images from WISE show excellent performance (see Fig. 14-20). The final all-sky survey is a new standard reference for astronomers. And the mission provided a big advance in asteroid science.

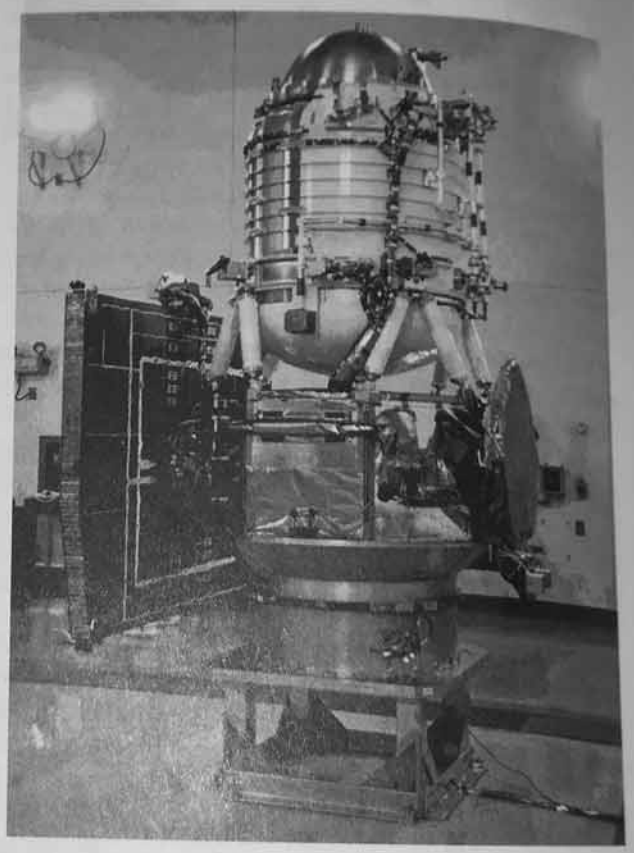


Fig. 14-19. WISE During Final Preparations for Launch. (Photo courtesy of NASA/JPL-Caltech)

14.6.3 WIRE

NASA selected the Wide-Field Infrared Explorer (WIRE) as the fifth Small Explorer mission in 1993. WIRE, shown in Fig. 14-21, used frozen hydrogen to cool infrared detectors to 7.5 K in order to study galaxies with high rates of star formation. The original proposal called for a 25 cm aperture telescope, but this aperture grew during the preliminary design phase to 30 cm. Although this growth greatly increased the science potential of the instrument, it also increased the size and mass of the instrument, but the team appeared to have sufficient mass margin to handle the change. As the detailed design of the system began, the cryostat team

Table 14-23◀, Fig. 14-19, Eq. 14-4◀



Fig. 14-20. IC342, also Known as the Hidden Galaxy, as Seen by WISE. (Photo courtesy of NASA/JPL-Caltech/UCLA)

realized that they were overly optimistic when scaling the mass from a much larger mission. Even though the mass of the overall structure reduced proportionally, the valves and fittings on the cryostat did not scale with size. With the small WIRE cryostat, the valves and fittings were a significant contributor to the total mass. The other big impact to mass margin at this time was a change in mission orbit. The preliminary designers did not take into consideration the dispersions in launch vehicle performance. The final, solid-rocket stage of the Pegasus vehicle might underperform enough to leave one apse of the orbit 90 km low (see "Achieving Final Orbit" in Sec. 14.3.2 for more information about solid vs. liquid propellant). The original baseline orbit was 400 km, but the mission could not meet its 4-month lifetime with a perigee at 310 km altitude. The orbit baseline was changed to be no lower than 450 km after launch vehicle dispersions, allowing for a launch at solar maximum, but cutting into the mass margin significantly.

WIRE had the advantage of well-established mass for the avionics, since the spacecraft bus was based on the already-built SWAS. The one area where mass could be reduced was the spacecraft structure. The mechanical team decided to use a fully-bonded graphite-composite structure in order to reduce mass. Major brackets, such as those supporting the reaction wheels and star tracker, were built into the structure, further reducing the mass. The mechanical team produced a structure that was only 11% of the total system mass, allowing the system design to absorb more instrument mass growth instead of implementing costly mass reduction of the payload.

The WIRE avionics were built immediately after the avionics for the fourth SMEX mission, Transition Region and Coronal Explorer (TRACE). The WIRE team made deliberate design decisions in the system so that the avionics would remain **identical** to TRACE's avionics. Boxes for one mission were tested back-to-back with the other. Much of the software was identical, except for the attitude control software. Overall, this combined-build strategy reduced the cost of the spacecraft bus for both TRACE and WIRE by about 20% over the originally anticipated cost.

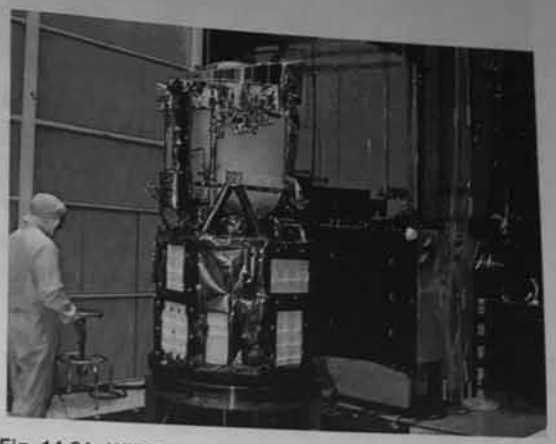


Fig. 14-21. WIRE During Testing. (Photo courtesy of NASA GSFC)

Throughout the entire mission development, the cryostat remained a major concern. There are layers and layers of insulation, there are multiple support structures to produce a thermally-isolated path, there is a labyrinth of plumbing, and the entire structure was bonded together, making rework difficult if not impossible. Besides the normal safety considerations of cryogenics (see "Safety" in Sec. 14.4.9), the cryogen is highly flammable in air. Launching on a Pegasus requires dropping it from an aircraft with people aboard, so the launch system included a safety vent for the cryogen to handle the case of the cryostat suddenly warming. Launch site operations of a cryostat requires around-the-clock monitoring and delicate close-out procedures (no small feat on a runway with 20+ MPH winds blowing, see Fig. 14-22 for a picture of the runway operations).

The WIRE team got everything right, except for one little glitch, lasting about 12 ms. A flaw in the pyro electronics box (a mundane part of the instrument electronics design), resulted in a start-up transient that prematurely released the instrument cover. The heat from the Earth caused cryogen to vent at an abnormally high rate. That venting caused the spacecraft to tumble faster, resulting in more exposure to the heat of the Earth. Within 24 hours, all of the mission cryogen was vented and WIRE was spinning at 60 revolutions per minute. It took us a week to use the magnetic torquers to despin the spacecraft. Although the instrument was useless, we did manage to get a little science out of our star tracker, but the primary mission was a total loss. There are a few lessons from the WIRE failure:



Fig. 14-22. WIRE Servicing While on Pegasus Under Carrier Aircraft. Dewars contain liquid helium. Access to the spacecraft is through a fairing door inside the tent to the left of the dewars. The safety vent is visible running up the side of the aircraft. (Photo courtesy of NASA GSFC)

- The space business is very unforgiving. A single tiny flaw can take down the whole mission, so the team must be constantly vigilant even with the routine parts of the design.
- Peer reviews of detailed designs are critical. Because of a combination of scheduling and poor judgment, no one outside the electronics designer's organization took a detailed look at the pyro electronics box circuit design. The flaw in the design was due to a feature of the programmable logic that was unknown to the designer. An outside reviewer may have caught the problem.
- The systems engineer must always consider the coupling of spacecraft mass loss to attitude control capability. I had considered scenarios with high cryogen vent rates from the perspective of instrument life, and I had considered the torques on the system from nominal vent rates, but I never put the two together. The placement of the hydrogen vent caused torques in the worst-case scenario that were only a factor of 2 greater than the torque authority of the magnetic torquers. Had we considered this, we may have placed the vent closer to the spacecraft center of mass and had enough authority.
- Spacecraft testing needs to continually look for unexpected events. We were continuously monitoring the pyro electronics for spurious signals, but the test equipment was overly sensitive to tiny glitches, and we had no way to capture the waveform of those glitches. When we first integrated the electronics, we connected an oscilloscope to capture the waveforms of the transients, but all we saw were tiny glitches. It turns out that the electronics only put out a large glitch, big enough to fire pyros, when it had been off for over 1.5 hours! We never actually captured a waveform after leaving the box off for a long time. With modern equipment, it should be standard procedure to continuously monitor waveforms, enabling engineers to review the signals after test equipment raises an alarm.
- It is generally a series of events that lead to a spacecraft failure—bad design, mistake during build, inadequate test, and a failure to anticipate something. The industry uses checks and balances along the way to try to catch flaws—design reviews, testing, mission assurance, etc. It is impossible to be perfect, and we cannot afford to be excessively conservative. There are plenty of examples where costs can be reduced, by cutting out some redundancy within the process, but the system designer should only deviate from standard practice with his eyes wide open.

14.7 Examples

14.7.1 SCS Example

Picking up the Supplemental Communications System design from Sec. 5.6, we are targeting 200 kg for each spacecraft, and we are assuming that all three spacecraft are launched on a single rocket, with the rocket taking all three to the 20,500 km altitude. (Note that this assumption implies a 6-hour mission for the launch vehicle—an assumption that should be revisited with a launch-vehicle expert, since this is much longer than a typical mission.) Based on the total delta V budget in Table 10-7, the fuel budget is in Table 14-24. The 2 kg for momentum management is an allocation without much analysis. Use Chap. 19 to estimate the external torques, and determine

Table 14-24. SCS Fuel Budget.

	delta V (m/s)	Effective I_{sp} (s)	Propellant for Maneuver (kg)	S/C Mass after Burn (kg)
1st burn	18	220	1.7	198.3
2nd burn	17	220	1.6	196.8
Rephasing	5	220	0.5	196.3
Momentum Management	0		2.0	194.3
Disposal	20	220	1.8	192.5
Residuals (kg)			0.5	
Total	60		8.0	
Total Launch Mass: 200 kg				

Table 14-24, Fig. 14-22, Eq. 14-4◀

Table 14-25. Driving Requirements for SCS and FireSat II Missions.

Parameter	SCS Value	FireSat II Value
Mass	Maximum of 200 kg	To be determined by first iteration
Power Consumption	To be determined by first iteration	To be determined by first iteration
Cost	Low-cost	Low-cost
Schedule	Not a major driver	Not a major driver
Lifetime	10 yrs	8 years
Reliability	Assume this is not a driver, since it provides supplemental support	Want high reliability, can rely on multiple spacecraft for redundancy
Delta V	60 m/s + ADC	764 m/s
Orbit	21,000 km circular, 0 deg inclination	700 km circular, 55 deg inclination
Payload Accommodation		
— *Data Rate, Volume, and Latency*	Maximize data rate through payload, which translates to maximizing power.	10 Mbps, 48 × 22 × 18 cm^3, 90 minute latency
— *Pointing Requirements*	Pointing of 1 degree accuracy to support cross-links. See text for more info.	0.1 pixel, 3 sigma; pixel is 500 m: 15 arcsec
— *Mass, Volume, Fields of View*		20 kg, 50 W without contingency
— *Other*		0.65 deg along track, 58 deg cross track
		Sun avoidance; Sun can dwell no more than 30 seconds within 1 degree of instrument boresight

if this number overly constrains the offset between center of mass and center of solar pressure. The residuals round the total to an even 8 kg.

Since the primary design objective is the maximization of communications bandwidth, we don't know exactly what the size of the payload will be. We know the final mass of the system, and we know the delta V, so we will use the design process to determine the power and mass available for the payload. Going down the list of drivers from Table 14-2 we create Table 14-25.

As you can see, "maximizing bandwidth" doesn't tell us much about the payload until we do some additional work. Chapter 16 explains how to design a communications payload. Since the antenna for our payload must cover 26.8 deg (horizon to horizon), we cannot drastically improve the gain of the payload antenna in order to improve data rate—all we can do is maximize the power available for the payload transmitters. Our first objective is estimation of the power available for the payload, so our payload designer can move forward. Our first cut at this won't be very accurate, so we will need to iterate. We can reduce the power required for cross links by increasing the gain on the cross-link antennas, which requires an increase in pointing accuracy. But these antennas can remain fixed to the spacecraft body, since the relative geometry between spacecraft remains fixed. In order to keep the system within the category of a "coarse" pointer (see Sec. 14.2.5), we will assume a 1 deg accuracy for pointing of the cross-link antennas.

Because this is a small, low-cost, supplemental system, we will assume a single-string system, even though the lifetime is 10 years. For this multi-spacecraft system, it probably makes more sense to carry extra spacecraft in the constellation to provide redundancy, rather than adding extra components to a single spacecraft. We will look for options of graceful degradation as appropriate.

The next step is defining the overall configuration of the spacecraft. We know we need to point one antenna at the Earth and point two cross-link antennas at two other spacecraft, so we must have a 3-axis-stabilized nadir pointing spacecraft, with 3 fixed payload antennas. The spacecraft flies too high for use of a magnetometer for attitude determination, and it will be in eclipse for part of the orbit, so Sun sensors won't always give a solution. The pointing requirement is loose enough to allow the use of a horizon sensor, but a low-cost star tracker could work if the ground regularly sends up orbit ephemeris information. Another possibility is the use of the cross-links themselves to provide pointing feedback to the spacecraft, adjusting the attitude to maximize the signal. This option will need further investigation on a later iteration. We will need some sort of gyro for measuring rate during attitude maneuvers, since star trackers generally cannot perform this function. The gyro-sun sensor combination can provide enough information for safe mode. Assume 4 reaction wheels in order to provide graceful degradation in the event that a wheel is lost during the 10 year lifetime.

We know we need a propulsion system. Based on the delta V, assume a hydrazine system. If we were to use cold gas, the 220 I_{sp} becomes 50 sec, resulting in a fuel mass of 25 kg. The practical considerations of a hydrazine system (see Sec. 18.8), will make its dry mass heavier than cold gas, but probably not 17 kg heavier. The 8 kg in the fuel budget leaves us with 192 kg for the dry mass of the spacecraft, including the payload.

Since the spacecraft operates with a 0 deg inclination, we know that the beta angle of the Sun will vary between 23.5 deg and –23.5 deg over the year. If we only use a single-axis gimbal on our solar array, tracking the pitch-axis motion, then the solar array output may drop as much as cos(23.5 deg) or 92% of its peak. Since the variation is small, a second gimbal on the array is not necessary. In order to minimize the solar pressure disturbance on attitude, we will put a solar array on each side of the spacecraft.

Table 14-25, Fig. 14-22◀, Eq. 14-4◀

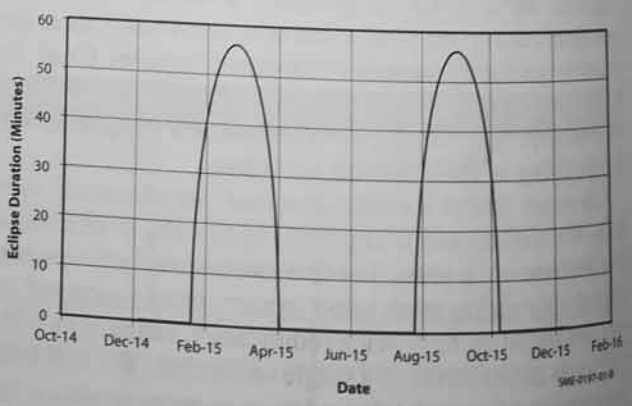
Table 14-26. SCS Initial Mass Estimates. The second column follows directly from Table 14-18 with a dry mass of 192 kg. The target mass is based on engineering judgment as described in the rationale column. Note that the propellant mass for this mission (4%) is a much lower percentage than the high-Earth mission average (72%), justifying a reduction in propulsion system dry mass. This lower percentage also reduced the target for the structure mass, since the structure does not need to support a large propellant load. The low propellant load followed directly from our assumption that the launch vehicle will take the spacecraft to nearly the mission orbit; this is frequently not the case for high Earth missions.

Subsystem	Mass Based on Table 14-18 (kg)	Target Mass (kg)	Rationale
Payload	61	47	Allows 30% growth contingency for new instrument
Structure	46	40	10% contingency, assuming straightforward structure design, 2 kg reduction for light propulsion
Thermal	8	7	15% contingency
Power	33	28	15% contingency
Communications	8	7	10% contingency, available parts
C&DH	6	5	10% contingency, available parts
Attitude Control	12	10	15% contingency to handle uncertainty in strategy
Propulsion Dry Mass	13	9	20% contingency for unique design, reduced by 2 kg because of small propellant load
Other/Margin	5	39	Total margin set at 25%
Total Dry Mass	**192**	**192**	
Propellant	8	8	Fuel budget
Launch Mass	200	200	

We will want the spacecraft to require little interaction with controllers most of the time, but during peak loading, we can expect controllers to be available continuously. Commands to the spacecraft will primarily be occasional configurations of the payload, even during peak loading. Other commands will be necessary to control the thruster firings associated with orbit adjustment and momentum dumping. The spacecraft must only store housekeeping data, since the payload is only acting as a bent pipe.

We will partition the system into the standard set of subsystems, but for a spacecraft this small, we should be able to combine some of the power electronics with the C&DH. Based on Table 14-18 and some guidance from App. A, we will start with estimates associated with a spacecraft in high-Earth orbit and apply some engineering judgment to come up with target masses for each subsystem. These estimates are shown in Table 14-26. The second column shows the mass derived by multiplying the percentages in the high-Earth column of Table 14-18 by the dry mass. We'll use the "other" total as margin. As described in Sec. 14.5.1, we should carry about 25% total mass margin at this point, so we assume some growth contingencies and reduce our target masses accordingly. The spacecraft's communication system is separate from the payload, providing the simple housekeeping and control functions using a single transponder and an omni antenna system.

Focusing on the power system, we need to know the maximum eclipse period. From Table 9-4, we know that we have a 12 hour orbit. Figure 14-23 shows that we have a maximum eclipse of 58 minutes every orbit, and two eclipse seasons of 80 days each per year, giving a total of 320 cycles per year, or 1,600 cycles in 5 years. These eclipses and the system power draw will define the size of the battery. The solar arrays will need to generate about 12/11 of the average usage since they will be in sunlight for only about 11 of the 12 hours. Using Fig. 21-6, the mass of the array will be about 34% of the 28 kg target or 9.5 kg. Using Table 21-15 and the GaAs improved MJ cells, we get 195 W/m^2 EOL and 2.8 kg/m^2, for a power output of 70 W/kg from the array, or about 665 W from the array. At a beta angle of 23.5 deg, we will get 92% of this, or about 612 W. During the maximum eclipse, the Sun angle is minimum, so the average power available is 11/12 of 665 W or about 610 W. Using Table 14-20, the power available for the payload is 35% of the total, or about 214 W. Assuming a 50% depth-of-discharge for the battery, we will need about 1,330 Whr for the hour-long eclipse. Based on



Live Calc

Fig. 14-23. Eclipse Duration vs. Date for SCS Spacecraft Over One Year.

Table 14-26, Fig. 14-23, Eq. 14-4◀

Table 21-18 with a lithium-ion battery at 125 Whr/kg, we need about 10.6 kg for the battery. Assuming 4.8 kg for electronics and harnessing (from Fig. 21-6, 17% of 28 kg power system), we have only consumed about 9.5 + 10.6 + 4.8 = 25 kg, which is reasonably close to the 28 kg target, given the roughness of the calculations. We will allocate 214 W to the payload as a starting point.

We now have a 47 kg, 214 W estimate for the SCS payload. This is our target design point at this time. We are carrying 30% margin, or 14.1 kg at the system level for our payload, so we need to make sure the payload designer does not carry excess margin on top of this. We need a power system that supports 610 W orbit-average. This starting point will be used in Chap. 16 to calculate a reasonable payload configuration, then we will feed the results of that back into an updated set of allocations and requirements.

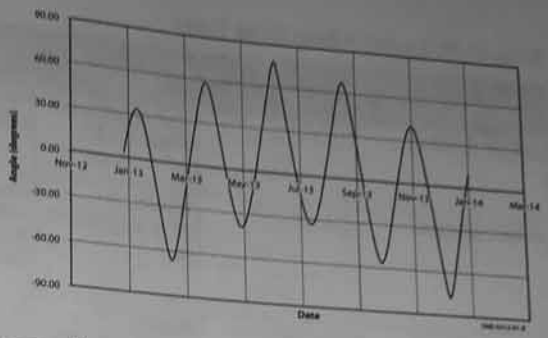
14.7.2 FireSat II Example

In the SCS example, we started with a target mass for the entire system, and we determined the mass and power available for the payload. In this example, we will start with the FireSat II payload and determine the mass and overall configuration for the spacecraft. The flow ends up looking a bit different, since we have some flexibility in our final system mass. Following the process defined in Sec. 14.1, we need to first document the design drivers and requirements for this system. Table 14-25 lists the requirements we know at this time. Since we need multiple spacecraft to achieve the coverage, we will plan to launch two spacecraft on a single vehicle, utilizing on-board fuel for the orbit adjusts.

The FireSat II constellation needs to reliably deliver the fire images, but it is important to get the system launched so that it can start to deliver the data. In order to achieve this balance, we will go with a single-string spacecraft bus, with back-ups coming through multiple spacecraft. Then we end up with an overall system that exhibits graceful degradation—we get very good coverage from multiple spacecraft, but if one fails, we still get coverage, just less frequently. We do not want to overly rush the build of the spacecraft, though, since we are building multiple copies—a single mistake will get multiplied.

The FireSat II instrument requires a nadir-pointing spacecraft. The orbit has a period of 99 minutes, with a maximum eclipse time of 35 minutes. Figure 14-24 shows the beta angle across a year of operations. The Sun gets as far as about 75 deg from the plane of the orbit, and it crosses to both sides of the plane. Since the instrument detectors must run cold, we will plan to yaw-flip the spacecraft at 0 deg beta angle, about 10 times per year, in order to keep the instrument radiator on the shaded side of the spacecraft. This yaw flip could be performed with reaction wheels, but it may take too long. We will assume a thruster assist as described in Sec. 19.1.4. Table 19-12 shows 49 g per maneuver, for a total of 3.9 kg over the 8 year mission.

With the large variation in Sun angle, we will need to conduct a trade on the solar array configuration.



Live Calc

Fig. 14-24. FireSat II Solar Beta Angle Over a Year of Operations.

Table 14-27 shows the options for the solar array trade, with advantages and disadvantages. A two-axis gimbal provides the smallest array area, but it is the most expensive gimbal. For 141 W orbit-average power consumption (see Table 14-28) and a 98% efficient lithium-ion battery (from Table 21-18), the power needed from the array is 220 W. Using Table 21-15 and the GaAs improved MJ cells, we get 197 W/m^2 EOL, so the required array area using a 2-axis gimbal is 1.1 m^2. With a single-axis gimbal, we arrange the array so it splits the 75 deg angle in half, and the total area is increased to compensate for the cosine loss. With a body-mounted array, the average projected area of the array must be at least 1.1 m^2. (When laying out body-mounted arrays, the designer needs to avoid very shallow angles, since the sunlight will tend to skip off the coverglass, despite anti-reflective coatings.) With a body-mounted array, the volume of the spacecraft will need to provide enough surface area for the array. With the yaw flip, we can limit the solar array covering to 4 sides of the spacecraft. With a cylindrical fairing on the launch vehicle, the smallest possible fairing that will provide 1.1 m^2 of surface area on the ends is 1.18 m in diameter, and that diameter requires a minimum length of 1.07 m. These dimensions are close to the Pegasus fairing dimensions, with other vehicles providing more room. Note also that a body-mounted solar array on a non-spinning spacecraft will generally run hotter than a deployed array, since it cannot radiate out the back.

For our initial conceptual design, we will choose the body-mounted array. We can avoid the cost of a deployment system, a gimbal system, and the software to drive the array. Our customer is very interested in simplicity. As we get into a more detailed design phase, we should re-visit this trade. We may be able to package a gimbaled array in a way that allows two spacecraft to fit in a smaller launch vehicle, or perhaps the mass of the gimbal and drive electronics will be less than the mass of solar cells on multiple faces. The detailed trade will require some drawings of the spacecraft and some careful subsystem analysis.

FireSat II will need a propulsion system. With a delta V as high as 764 m/s, we should compare the mass

Table 14-26◀, Fig. 14-24, Eq. 14-4◀

Table 14-27. FireSat II Solar Array Trade.

Array Configuration	Array Area	Rationale	Advantages	Disadvantages
2-Axis Gimbal	1.1 m^2	197 W/m^2 from Table 21-15	Smallest total array area	Expensive powered mechanism Deployment mechanism needed Software needed to drive gimbals
1-Axis Gimbal	1.4 m^2	Based on 37.5 deg cosine loss	Significantly less expensive that 2-axis gimbal	Expensive powered mechanism Deployment mechanism needed Larger array area Software needed to drive gimbals
Body Mounted	Avg projected area, >1.1 m^2	Higher temperature due to mounting	No expensive mechanisms	Very large total array area May increase volume of spacecraft Array will tend to run hotter, requiring more array area

of a mono-propellant system to a bi-propellant system. We should carry 30% margin on the payload estimate of 20 kg, resulting in 26 kg allocation. Using Table 14-18, LEO with propulsion, the payload is 31% of the dry mass, so dry mass is 84 kg. With an I_{sp} of 220 sec, the propellant mass is 36 kg for a monopropellant system. A bi-prop system with 300 sec I_{sp} requires 25 kg of propellant, so there is only an 11 kg savings, which will be eaten up by the extra complexity. We will go with a mono-propellant propulsion system, even if the dry mass grows some.

From LEO, we are in view of ground stations for about 10 minutes out of the 99 minute orbit. With a 10 Mbps continuous data collection rate, we will need to downlink every orbit at 100 Mbps to get all of the data down. We should investigate data compression and on-board processing to reduce the total data volume. If the on-board algorithm can identify fires, we can limit our downlinks to data immediately surrounding a fire, which will drastically reduce the total volume. At a 100 Mbps downlink rate, we will need an X-band transmitter, but reducing the data rate to 5 Mbps would allow us to use a less expensive S-band system (see Table 21-3).

Using typical subsystems and Tables 14-18 and 14-20, the mass and power allocations are shown in Table 14-28. Contingency of 30% has been added to both the mass and power numbers to allow for growth during development. This contingency has been added before applying the Sec. 14.5 tables, so the 30% has effectively been added to all of the subsystems. Given that this is a very small spacecraft, not all subsystems may scale per the table averages, so we should carry additional mass at the system level. We will assume a target launch mass of 150 kg for each spacecraft, leaving us some margin. This assumes that one spacecraft will support the other for launch. The 150 kg number will be used by mechanical, propulsion, and ADC engineers in their subsystem analysis. Table 14-29 shows the fuel budget for this wet mass.

As a sanity check on our power system mass, we can take a look at the weight of our body-mounted array. Assuming an octagonal spacecraft 1.2 m across the octagon and 1.1 m tall, with array on both ends and 5 of the side panels, the total surface area is about 4.6 m^2. At 2.8 kg/m^2 (from Table 21-15), the total array mass is 12.9 kg. 141 W for 35 minutes with a 30% depth of discharge requires 274 Whr. At 125 Whr/kg, per Table 21-18, the battery will be 2.2 kg. That only leaves 2.9 kg for the power electronics and the spacecraft harness. Our body-mounted array will probably drive our power allocation a bit above 18 kg, but we will wait for our detailed subsystem analysis to reallocate, using some of our 14 kg of system-level margin if necessary.

Table 14-28. FireSat II Mass and Power Using Tables from Sec. 14.5. Payload mass and power numbers include 30% contingency to allow for growth during development.

	Mass Based on Table 14-18 (kg)	Power Based on Table 14-20 (W)
Payload Total	26	65
Structure	23	1
Thermal	2	14
Power (incl. harness)	18	13
TT&C	2	17
On-Board Processing	4	17
ADC	5	14
Propulsion	3	0
Other	3	
Total Dry	86	141
Propellant mass (kg)	50	
Total Wet Mass (kg)	136	
Target Launch Mass (kg)	150	
System-level Margin (kg)	14	

At this point, we need to go into more detailed analysis of each of the subsystems, which will be covered in later chapters. To complete the conceptual design, we will need to rack up the mass and power totals from the

Table 14-28, Fig. 14-24◀, Eq. 14-4◀

Table 14-29. FireSat II Fuel Budget.

	delta V (m/s)	Effective I_{sp} (s)	Propellant for Maneuver (kg)	S/C Mass after Burn (kg)
1st Burn	280	220	18.3	131.7
2nd Burn	280	220	16.0	115.7
Maintenance	21	215	1.1	114.5
Rephasing	0	220	0.0	114.4
Momentum Management	0		3.9	110.6
Disposal	183	220	9.0	101.6
Residuals, Margin (kg)			1.6	100.0
Total	764		50.0	
Total Launch Mass: 150 kg				

subsystem analyses and compare that to the analysis we have done here, reconciling any discrepancies by considering options and performing trades. Once we have a consistent design, including mass and power totals that accommodate uncertainty in the development, we will document our design and deliver it to our customer.

14.8 Future of Spacecraft Design

Over the last 20 years, there have been some clear trends in spacecraft design. The most prominent change is the increasing role of software. With more powerful radiation-hard processors has come the ability to put more functionality into those processors. Functions in software can be changed after launch, improving the flexibility and versatility of today's spacecraft. With the increased software complexity has also come increased difficulty with software during development. Today's software subsystem lead is really a spacecraft systems engineer, working across all of the subsystems except structure. The industry is starting to realize this fact and incorporate within the systems team engineers who are familiar with the details of software design. Software is becoming more abstract, isolating most developers from the details of the hardware. Efficiency of the code itself is being traded for efficiency in the writing of the code—as we get faster processors and larger memory, we can afford code that requires more steps and more memory to perform the same task. Features such as file systems allow manipulation of large volumes of payload data with few steps on the part of the programmer and the operations team, while underneath the top-level code resides a very complex operating system which utilizes complex device drivers. As we look ahead at the next generation of spacecraft, we can expect to see very sophisticated software, with complex attitude control algorithms and filters; autonomous orbit adjustments and maintenance; file system transfers with the ground that allow autonomous, reliable transfer; sophisticated on-board signal processing; and autonomous target selection. In short, our spacecraft will be more capable of taking care of themselves, with a reduction in operations costs. Software development costs will go up because of the increasing complexity, but the functionality will increase faster than the costs, saving money in other areas. The biggest challenge with software, which will only get worse, is ensuring that the inevitable flaws in the code do not prematurely end the mission.

Along with improved software capability has come an increased use of data standards. Spacecraft can now afford the overhead associated with packaging data inside standard headers, which allow easier handling of the data on-board and on the ground. The work of the Consultative Committee for Space Data Systems [CCSDS, 2011] since the 1980s has taken spacecraft toward standard data interfaces, but the explosion of the Internet over the last 15 years has accelerated work on commercial data standards. Billions of dollars spent by commercial industry on Internet Protocol (IP), mobile IP, IP Security (IPsec), Ethernet, and Universal Serial Bus (USB) will enable the space industry to simplify spacecraft interfaces, spacecraft testing, and flight operations. Over the last 20 years, the MIL-STD-1553 [DoD, 1978] data bus has replaced more custom serial interfaces. Many spacecraft components now use this standard, greatly reducing the documentation and errors associated with the development of signaling between two electronics boxes. Over the last 10 years, Spacewire (standard ECSS-E-ST-50-12C [ESA, 2003] based on IEEE 1355) has started to claim the high-speed serial interfaces. But, over the same time, the commercial industry has solved many of the issues spacecraft face, using IP, Ethernet, and USB. The challenges of mobile IP, where users have limited power in hand-held devices, is very similar to the challenges of a spacecraft in Earth orbit with limited power. Command link security is becoming a bigger issue as electronics needed for spacecraft communication become cheaper, so the advantages of IPsec will become greater. Protocols used for bank transactions will be used to ensure command link security, rather than the current practice of custom hardware.



Field-Programmable Gate Arrays (FPGA) have drastically changed digital design over the last 25 years. Most digital circuits are now FPGA's with some support parts around them. These parts reduce the design cycle time, since mistakes or changes during development can be implemented by altering the programming of the FPGA, rather than changing physical wiring on the circuit boards. Radiation-hard versions of these parts are available for flight. FPGA's can even include simple microprocessors, and some FPGA's can initialize using external memory, so the ground can re-program circuit functionality. FPGA's blur the line between software and hardware, making the hardware much more flexible.

On the payload side, detectors used in imagers continue to grow. Larger detectors mean higher resolution and more data collected. Spacecraft will need to support higher data rates from the payloads and to the ground. Data storage requirements will grow. High data rate communication is already moving to higher frequency radio bands, such as the *K_a-band* which covers 26.5–40 GHz, because of crowding and bandwidth limitations at the lower frequencies. On-board processing is already used to compress measurement data; future missions will also use software to perform other processing on the data. For instance, with a mission like FireSat II, the on-board processing will identify regions in an image which contain fires, then downlink those portions of the image with higher resolution. Our society in general is demanding higher bandwidths, so there will be constant pressure on communications payloads, too, with similar impacts on the spacecraft design—higher power, larger antennas, and higher frequencies. Some applications, especially space-to-space links, will use optical communication.

Significant reductions in the cost of space transportation could drastically change the industry. It doesn't really make sense to launch a $20 million spacecraft on a $200 million rocket, but if the cost to lift a spacecraft into space came down by an order of magnitude, then it would make sense to reduce analysis and testing and risk additional failures. This cost reduction will require either a drastic increase in the number of launches, creating more of an assembly-line operation, with simpler rockets; or we will need some completely new technology to overcome the basic-physics challenge of accelerating mass to more than 7.5 km/s. There are some commercial companies experimenting with simpler rockets, so perhaps there will be some decrease in launch costs. The change should increase opportunities to fly spacecraft, but it will probably not be enough to drastically change how we design, build, and test space systems.

The design process itself has been changing with the increased capability of our computers. For over 20 years, we have been using computers for mechanical drafting and electrical schematic drawing. The process of going from computer to hardware has become more and more automated. The ongoing revolution now is in the integration of computer models, using standard data formats like those in ISO 10303. We will soon be able to create a model of the entire spacecraft, with mechanical, thermal, electrical, and dynamic properties all shared across the team. Preliminary designs will come together faster, with designers analyzing more options. Rather than configuration management with text documents, we will use the data model representation of the spacecraft to capture the current configuration. But we will still need the creativity and common sense of human designers. We will still need designers who understand the basics of spacecraft and who can exercise good judgment, quickly disregarding the absurd or foolish solutions and exploring the possibilities of what can go wrong. These creative and knowledgeable individuals are essential for our continued exploration and utilization of space.

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15 Overview of Payload Design

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15.1 Types of Space Payloads
15.2 Mission System Concept or Subject Trades—What is the System Measuring or Working With?
Subject Trades
15.3 Payload Design
15.4 The Electromagnetic Spectrum
15.5 Examples

The *Payload* is defined as the total complement of equipment carried by a spacecraft that interacts with the *subject* in performance of a particular mission. The subject of the mission is what exists outside the spacecraft that is sensed by or interacts with the payload, as discussed in Sec. 4.1 and Sec. 14.2. *Passive subjects* are ones that the mission design cannot control, for example, the fire used in the FireSat II example. *Active subjects* are ones which the design and basic parameters can be controlled by mission design, such as a GPS receiver on the Earth. Mission subjects will fall into at least one, and sometimes both of these types. Although typically unique to each mission, payloads are the main reason missions fly, and what drives the mission size, cost and risk. So, consequently, a critical part of mission analysis and design is to understand and recognize what drives a particular set of space payloads so that those elements can become an integral part of the overall system trade process designed to meet *mission objectives* at minimum cost and risk.

In this chapter, we will summarize the overall process of payload design and sizing, in Sec. 15.1 and Sec. 15.2 we discuss the different types of payloads, select the subject and discuss undertaking subject trades (i.e., how to best go about achieving the mission objective). Section 15.3 lays out the details of the payload design process, using the background information on the electromagnetic spectrum provided in Sec. 15.4, which is the primary way of observing and communicating from space. Finally, we will review some examples of this process in Sec. 15.5. *Communication Payloads* (Chap. 16) and *Observation Payloads* (discussed in Chap. 17) are the two most common types.

15.1 Types of Space Payloads

As shown in Table 2-1, there are many space communities with various objectives and goals, and Table 1-15 lists over 70 space mission applications. These communities and applications focus on **what** we are trying to achieve and **why** we are trying to achieve it. In contrast, the mission payload focuses on **how** to achieve it. Putting the payload in space and obtaining results is what the mission is all about. How the design meets the needs of the end users (or the people that pay for the mission) determines the why.

Mission payloads may be unique in the details of their implementation and objectives, but share many of the same design characteristics. There are, in fact, only a few broad types of payloads (defined in Table 15-1), however, each can be used for a wide variety of applications. For example, the Hubble Space Telescope is used for astronomical observation, but basically the same instrument, turned upside down, can be used for military reconnaissance or surveillance, and in GEO may be used for continuous environmental monitoring.

If the subject or payload is active, it will involve a more complex (and therefore, likely more expensive) design process, but also adds flexibility that can greatly improve performance. For example, in a communications system where the subject is active, we can put more performance on the spacecraft and make the ground system simpler (such as direct broadcast radio or TV), or we can ask the ground unit to perform much of the work or have the relevant data. (The GPS receiver in your car calculates where it is based on the relatively simple signals from the GPS satellites. The ground system also contains all of the maps on which the information is displayed.)

A similar relationship holds for the payload. A *passive payload* is one that is just observing, watching, or taking in data, such as a space telescope. While these payloads can be complex and expensive, they are simply dealing with what the outside world provides. In contrast, an *active payload* is one that interacts directly with the subject, such as a radar that provides its own illumination, essentially the equivalent of a flash photograph in visible light. This gives us a great deal of flexibility because we can control the illumination to provide just the characteristics that we need. Because we have brought our own illumination, the space radar works equally well

Table 15-1. Types of Space Payloads.

Payload Type	Sample Function	Active or Passive Payload	Active or Passive Subject	Examples	References
Observation	Remote sensing Space-based Earth-based Space weather	Passive or Active	Passive	Hubble Telescope LandSat Active: TerraSar X Weather Doppler Meteorology Wind Lidar	Chap. 17
Communications/ Navigation	One Way comm. 2-way comm. Navigation Tracking	Active	Active	Direct Broadcast TV Telephone GPS/GLONASS/GNSS Galileo Navstar Wireless Internet (WiFi)	Chap. 16
In-Situ	Rovers Space manufacturing In-situ measurements Exploration	Active	Passive	Mars or Lunar rovers Viking Mars lander Magnetic field meas. Gravity measurements	Squyres, et al. [2003] Gabriel [2007] Young [2007] Ball et al. [2007]
Action at a Distance	Space weapons Power transmission National security Spacecraft tracking	Active	Passive or Active	Space-based laser Space-based radar Satellite solar power Guidance Military data	Zachen [2007] Glaser et al. [1997] Klein [2006] Aviv [2006] Pillai et al. [2008]
People (Human spaceflight)	Exploration Space tourism Construction and repair Space experiments	Passive or Active	Passive	Apollo Space tourism Lunar colony ISS	Eckart [2006] Seedhouse [2008] Harland and Harvey [2008] Pelton and Marshall [2009]

at night and, because we also control the wavelength, we can pick one that allows us to penetrate clouds or light foliage. The demerit is that we need a really big "flashlight" for this task and both the power required and the thermal control to get rid of lots of excess heat are major problems and a big expense.

Payloads meeting these different objectives will be different in many of their details (such as exposure time or wavelength band), but will share many of the same design characteristics.

Communications Payloads. By far the largest and most "popular" group of spacecraft, the main goal of communication payloads is to transfer information. Communications missions, traditionally dominated by large geosynchronous spacecraft, range from one-way broadcast of television signals to wideband full-duplex telecommunication connectivity. The advances of technologies in these payloads have been astounding, from smaller spacecraft in LEO used as alternative architectures for global coverage to CubeSats as discussed in Sec. 25.4. In fact, there have been so many advances, that we have devoted all of Chap. 16 to a detailed discussion of this type of payload.

Observation Payloads. Observation payloads cover a wide and diverse range of missions and applications using remote sensing. Any observation that a spacecraft makes without directly contacting the object in question is considered remote sensing. Imaging the Earth's surface, sounding the Earth's atmosphere, providing early warning of a ballistic missile launch, tsunami, or hurricane, or seeking out new worlds by observing the chemical spectra of distant galaxies, are all remote sensing missions. We focus mostly on the electromagnetic spectrum to determine the nature, state, or features of physical objects or phenomenon.

As shown in Fig. 15-4, we can evaluate different aspects of the electromagnetic radiation to exploit different characteristics, with respect to spatial, spectral, and intensity information content, depending on the particular mission. We can also evaluate this information in the temporal context that supports comparisons and cause-and-effect relationships. Depending on the source of the electromagnetic radiation being sensed, we can establish the activity of the sensor. If the instrument measures direct or reflected solar radiation in the environment, we distinguish that it is a passive sensor. Active sensors, such as *radar* and *lidar*, emit radiation that generate a reflected return which the instrument measures. The physical principles of remote sensing and the categories of the sensors are the same, regardless of whether the payload is studying space weather, mapping the topography of Earth, or observing electromagnetic phenomena elsewhere in the universe.

One of the best known observation systems, the Hubble Space Telescope, has been repaired and kept up to date with the latest available remote sensing technology by a series of five servicing missions between 1993 and 2008. This remarkable capability and the unprecedented legacy of scientific discovery that it enabled was made possible by the skill of designers of the instruments and the US Space Shuttle astronauts. With the imminent retirement of the US Space Shuttle, future payload and spacecraft servicing may occur with remotely controlled or automated robotic systems. For the extended version of observation payloads, see Chap. 17.

Navigation. GPS, GLONASS, Galileo, and other international navigation systems have demonstrated a wealth of applications for military, civilian, academic, and recreational users. GPS provides information for real-time position, velocity, and time determination. It is available worldwide on a broad range of platforms, including pedestrians, cars, ships, commercial and military aircraft, and spacecraft. The heart of GPS is a spread-spectrum broadcast communication message that can be exploited using relatively low-cost receivers.

In Situ Science. Sample collection and evaluation serves an important role in planetary and space science. Perhaps the most elaborate instance of sample collection took place in the Apollo missions when approximately 300 kg of samples from the Moon were returned to Earth for analysis. Other examples of sample collection and analysis include planetary landers (such as Viking and Mars Exploration Rovers, Spirit and Opportunity).

Action at a Distance. While communication, navigation and remote sensing applications are quite mature and dominate the use of space, action-at-a-distance concepts such as space-based lasers, space-based radar and space-based solar power still occupy a relatively small niche in the realm of space mission design. In particular, concepts for weapons in space became a topic of intense study and debate as part of the Strategic Defense Initiative and space-based strategic missile defense. Development of certain operational space weapons has been prohibited under the Anti-Ballistic Missile Treaty of 1972. Although some experts view widespread weaponization of space as inevitable, it has not become a stated objective of US national policy. Of course, space has been used to support military objectives since the dawn of artificial spacecraft, but the vast majority of military space applications fall into the categories of remote sensing and communications. Extensive studies have been done on satellite solar power, i.e., generating solar power in space for use on Earth. Several commercial ventures in the US, Japan, and Europe have announced plans to develop space-based solar power systems.

Human Spaceflight. Human spaceflight involves a human payload consisting of a flight crew and possibly passengers. Human spaceflight has been an ongoing international activity by the US, Russia, and more recently, China since 1961 with the first human spaceflight by the Soviet cosmonaut Yuri Gagarin. Human spaceflight occurs through the Russian Soyuz program, the US Space Shuttle program, and the Chinese Shenzhou program. However, the US government is expected to lose human spaceflight launch capability [Kraemer, 2006] with retirement of the Space Shuttle. Recently, to help fill the gap between the Space Shuttle Program and a possible next US government human spaceflight program, there has been a growing trend in the US toward commercial human spaceflight to help supply the International Space Station and to stimulate commercial spaceflight and other applications through the Commercial Crew Development (CCDev) and Commercial Orbital Transportation Services programs. The 21st century will see Space Tourism become a reality, with pioneers like Sir Richard Branson of Virgin Galactic and John Carmack of Armadillo Aerospace leading the way; already they are building space ports in New Mexico and pre-selling seats on their upcoming launches due to begin in 2012.

Exploitation of physical resources in space, either from the Moon or asteroids, has sparked innovative and imaginative concepts for augmenting Earth's limited resources or enabling human exploration of the solar system. Many authors (e.g., Eckart [2006]) have created designs for lunar bases, but all require a dramatic reduction in launch cost. In the nearer term, however, space-based materials processing and manufacturing are more likely to mature and exploit the characteristics of the microgravity environment (see Chap. 7).

15.2 Mission System Concept or Subject Trade—What Is the System Measuring or Working With?

Space mission requirements and constraints have been defined in Sec. 3.5. Mission requirements dictate the technical performance of the payload while operational implementation is determined by the mission concepts and constraints. Frequently, the technical specification and operations concept for payloads are interrelated. For example, increasing temporal resolution, that is, reducing revisit time, may reduce the requirement for spatial resolution in an optical sensor system. We must ensure the mission requirements capture fundamental needs of the users without constraining the designer's ability to satisfy these requirements through alternative technical means.

As an example, the FireSat II overall mission requirements are to detect, locate, and identify forest fires, then define the level of detail needed to satisfy the overall mission. Based on sensor data, several questions need to be answered and decisions need to be made:

- What is the timeframe for detecting a fire? 20 minutes? 1 hour?
- What is the geographic extent of the fire?
- What are the primary combustibles?
- In what direction is the fire spreading? How quickly?
- Where is the fire burning hottest?

Table 15-1◀, Fig. 15-0, Eq. 15-0

- Can flaming fires be distinguished from smoldering fires?
- Can a distinction be made between petroleum, chemical, and organic material-fueled fires?
- Where to strategically place firefighting efforts to contain and suppress the fire most effectively.
- How much smoke and ash is the fire generating?
- Are any other sources of information available that might also be used (e.g., air, ground, or space-based)?

Mission priorities and objectives addressed by these questions will determine the specific observables linking payload performance with mission performance. To choose a remote sensing payload, the key steps to a disciplined and repeatable design begin with determining the elements of information that we need to address the problem. We must specify the physically observable quantities that contribute to elements of information about the problem in sufficient detail to ensure they can be detected by a spacecraft payload with sufficient resolution to provide meaningful insight into the subject.

Establishing different performance levels provides a framework for trading off performance across a number of different design features. For all missions, payload performance evaluation categories include physical performance constraints and operational constraints. Examples of physical performance constraints include limits on spatial, spectral, radiometric, and temporal resolution. Operational constraints include sensor duty cycle limits, tasking and scheduling limits on sensor time, and resource contention such as inability of the sensor to view two targets of interest simultaneously.

Within each sensor constraint category, we should establish an absolute minimum threshold such that any performance that does not meet this capability is unacceptable. At the other extreme, we should specify the performance goal in order to establish the performance that will fully satisfy the requirement. Table 15-2 illustrates a sample of performance levels for the FireSat II mission payload across the functional areas of resolution, quantity, timeliness, periodicity, geolocation accuracy, and completeness. These distinctions can be critical in determining the viability of a mission concept. In commercial remote sensing, for example, the range of performance requirements from threshold to goal is typically determined through market analysis and business development case studies. These studies frequently identify a minimum resolution (or other performance parameter) below which a remote sensing spacecraft concept will not be profitable. Some of the payload constraints identified in Table 15-2 could be used to constrain the payload or the higher level FireSat II satellite constellation or both. For example, based on the understanding of astrodynamics and satellite constellation design we developed in Chaps. 9 and 10, respectively, we know the timeliness threshold of 2 hours and the revisit interval threshold of 90 min cannot be achieved with a single payload in low Earth orbit unless that payload is on board a highly maneuverable satellite. On the other hand, if we were able to relax the completeness requirement to encompass only one specific part of the world such as the continental US, then this mission could be carried out with a payload on board a single geosynchronous satellite and the timeliness and revisit rate requirements would drive a variety of payload characteristics such as Earth scan rate and detector array size.

Mission parameters, such as identifying and locating forest fires, need to be established in such a way that we can evaluate, size, and design candidate sensors. This parameterization involves a process of requirements analysis that focuses on matching the tasks involved in the mission with categories of discipline capabilities. If we match mission requirements with existing or probable capabilities, the result is a set of potential information requirements. We then try to identify the characteristics of the subject (i.e., signatures) that correspond to information requirements through a set of rules. For example, for FireSat II, these rules yield a set of mission observables necessary to detect fires, such as specific wavelength bands and spectral sensitivities that our sensor needs. These observables provide the basis for the payload characteristics that comprise the baseline design to satisfy the mission. The basic sensor types and corresponding infor-

Table 15-2. Sample Threshold and Performance Goal Requirements for the FireSat II Payload. Performance goal represents the maximum meaningful level of performance across all design features. Surpassing performance goal does not increase the value of the system. Likewise, performance falling below threshold prevents the system from meeting mission requirements, which would likely lead to program termination.

	Acceptable	Desired
Subject Characteristics	Identify, locate, and track progress of fires	Determine thermal conditions within fires and products of combustion
Quantity	Simultaneously measure and track 7 fires	Simultaneously measure and track 20 fires
Timeliness	Report detection of fire within 2 hours	Report detection of fire within 20 min
Revisit Interval	Update status of fire every 90 min	Update status of fire every 45 min
Geolocation Accuracy	Determine location and extent of fire within ±1 km	Determine location and extent of fire within ±100 m
Completeness	Map fires in North America and one other selectable region (e.g., Persian Gulf)	Map fires globally

mation that might satisfy the FireSat II mission are shown in Table 15-3. It may also be possible to satisfy mission requirements by observing different phenomenologies, such as visible signatures associated with flame and smoke, thermal infrared signatures from fire, spectral analysis of the products of combustion, or an algorithm combining all of these.

The selection of a spacecraft payload represents the fundamental leap in determining how to satisfy mission requirements with a space sensor. In the previous section we introduced a top-down framework for considering the general problem of spacecraft design. Here we turn our attention to the payload; in particular, a methodology for determining the type of payload to employ and the physical quantities to measure.

Table 15-3. Simplified Subject Trades for FireSat II Mission. The information type allows for subject trades to be made among the different signatures that can be exploited to satisfy FireSat II mission requirements.

Sensor Type	Information Type
Electro-optical Imager	Visible return from light or smoke cloud produced by the fire
Spectrometer	Spectral signatures from products of combustion
Radiometer	Thermal intensity

The process for linking mission requirements to payload design, begins with a task (or mission requirement) and ends with a spacecraft payload design. Figure 15-1 divides this process into intermediate steps to focus on the requirements analysis, subject trades and payload analysis. In this section, we will focus on describing this process. Section 17.2 provides some of the specific techniques that are employed in this process for visible and IR systems. This process can be extended to consider cost as an independent variable (CAIV) and evaluate performance of different payload designs with respect to threshold and goal requirements as a function of cost to try to determine the best value for specific missions.

For the FireSat II mission design, we need to identify specific signatures that would allow candidate sensors to provide viable solutions to the mission requirement. We observe physical phenomena through signatures, and we must choose which signature will provide the desired information. The specific signatures that a payload senses must be evaluated in light of the particular focus of the mission. For example, a spectrometer that is sensitive enough to detect all fires, but which cannot be used to differentiate campfires from forest fires could generate a large false alarm rate and render it operationally useless. Defining the key signatures and observables that support the information content needed to satisfy the mission determines the performance limits for the payload design.

15.2.1 Subject Trades

The objective of a space mission is typically to detect, communicate, or interact. The subject, as an element of

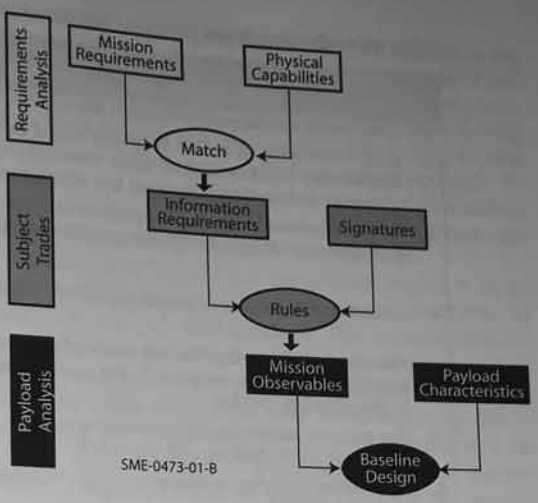


Fig. 15-1. Process for Linking Mission Requirements to Payload Design. The process moves from mission requirements to a payload design in three steps: requirements analysis, subject trades, and payload analysis.

the space mission, is the specific thing that the spacecraft will detect, communicate, or interact with. For GPS, the subject is the GPS receiver. For FireSat II, we could assume that the subject is the heat generated by the forest fire, but other subjects are possible: light, smoke, or changes in atmospheric composition.

What we choose as the subject will dramatically affect performance, cost, and the mission concept. Thus, we must do this trade carefully and review it from time to time to ensure it is consistent with mission objectives and our goal of minimizing cost and risk.

Table 15-4 summarizes the subject-trade process. We begin by looking at the basic mission objectives and then ask what subjects could meet these objectives. To do this, we should look at what we are trying to achieve, the properties of space we intend to exploit, and the characteristics of what we are looking at or interacting with. Table 15-5 shows examples of subject trades for four representative missions. As the missions change, the nature of the subject trades will also change. For FireSat II, we are looking for a well-defined subject (the forest fire), and we want to do this at minimum cost and risk. With the Space Telescope, we must ask, "What am I looking for? What am I trying to detect and how can I detect it?" For any of the science missions, we would ask, "Is the subject some distant and unknown object, and is it detectable in the part of the electromagnetic spectrum I am able to explore?"

For a space system intended to detect airplanes, the main subject trades would concern mission goals. Are the targets cooperative or noncooperative? Do we need to track over the poles? Should we track in high-density areas around airports or over the open oceans? The answers to these questions will determine the nature of the subject trades.

Table 15-4. Subject Trade Process. See discussion earlier in this section.

Step	FireSat II Example	Where Discussed
1. ***Determine fundamental mission objectives***	Detect and monitor forest fires	Sec. 3.3
2. ***Determine what possible subjects could be used to meet these objectives (i.e., what could the system detect or interact with to meet the objectives)***	Heat, fire, smoke, atmospheric composition	Sec. 15.2
3. ***Determine broad class of ways that the spacecraft can detect or interact with the possible subjects***	Heat –> IR flame, smoke –> visual atmospheric composition –> lidar	Sec. 15.2
4. ***Determine if subject is passive or controllable***	Initially assume passive fire detection	Secs 4.1, 15.2
5a. ***For controllable subjects, do trade of putting functionality at the subject, in the space system, or in the ground system***	N/A	Secs. 5.3, 15.2
5b. ***For passive subjects, determine general characteristics that can be detected***	Forest fire temperature range and total heat output	Sec. 15.2
6. ***Determine whether multiple subjects and payloads should be used***	Not initially	Sec. 15.2
7. ***Define and document initial subject selection***	IR detection of heat	Sec. 4.3
8. ***Review selection frequently for alternative methods and possible use of ancillary subjects***		Sec. 15.2

Table 15-5. Subject Trades Examples.

Mission	Property of Spacecraft Used	General Object of Study or Interaction	Alternative Mission Subjects	Key Subject Trades	Comments
FireSat II	Global perspective	Forest fires	Fire (visible or IR) Smoke (visible or IR) Increased CO_2 Decreased vegetation	None—IR detection probably best choice	See low-cost alternative in Chap. 13
Airplane Detection	Global perspective	Airplanes	Skin (radar, visible) Plume (IR) Radio emissions (RF)	Radar vs. IR vs. active RF	Need to examine goals; cooperative vs. noncooperative targets; high density vs. ocean tracking
Truck Communications System	Global perspective	Portable telecommunication centers	Current radio Current CB Standard TV New telecommunication center Cellular relay	Complexity of truck element vs. complexity of space & ground station	
Space Telescope	Above the atmosphere	Distant galaxies unknown phenomena	Quasars Galaxies Planets Visible spectrum Unknown objects	Is the subject known or unknown?	

Perhaps the easiest subject trades are those in which the system will be interacting with a ground element that is a part of the system, such as direct broadcast television or a truck communication system. In this case, the subject trade becomes simply an issue of how much capacity should go on the spacecraft vs. how much should go in the unit on the ground.

Determining whether a subject is active or passive is our next task. Active subjects can be controlled by the designer, such as ground stations, antennas, or receivers; transmitters used for ground communications, direct broadcast television, or data sources would qualify as well. Because we can control the subject, we can add more or less capability to it. Thus, we might choose to have a simple receiver on the ground with a high power, accurately pointed, narrow-beam transmitter on the spacecraft. Or we could place a sophisticated, sensitive receiver on the ground with a small, lower-cost system in

space. Usually, the solution will depend on the number of ground stations we wish to interact with. If there are many ground stations, as in direct-broadcast television, we will put as much capability as possible into the satellite to drive down the cost and complexity of the ground stations. On the other hand, if there are only a few ground stations, we can save money by giving these stations substantial processing and pointing capability and using a simpler, lighter-weight, and lower-cost satellite.

Passive subjects are those in which the characteristics may be known but cannot be altered. This includes phenomena such as weather, quasars, or forest fires. Even though we cannot control the object under examination, we can choose the subject from various characteristics. We could detect forest fires by observing either the fire itself or the smoke in the visible or infrared spectrum. We could detect atmospheric composition changes or, in principle, reductions in vegetation. Thus, even for passive subjects, the subject is part of the system trades.

We do not always know whether a given mission has passive or active subjects; in some cases, we can choose either type. For example, we could detect airplanes passively with an IR sensor or radar, or actively by listening for or interrogating a transponder on the airplane. Satellites that monitor the weather or environment could do complex observations or simply collect and relay data from sensors on the ground.

Using *multiple subjects* to meet our mission objectives can be advantageous, providing much more information than a single subject and can eliminate ambiguities that can occur with observing only one aspect. However, multiple subjects usually require *multiple payloads*, which will invariably, drive up the space mission cost and complexity. Thus, a *principal trade* is between a low-cost mission with a single subject and single payload vs. a more expensive mission that achieves higher performance by using several payloads to sense several different subjects related to the same objective.

For FireSat II, we tentatively select the heat of the forest fire as the subject of the mission, keeping in mind that this may change as the design evolves. Of course, we should make these trades as rapidly as possible because they strongly affect how the mission is done.

We should then document the subject selection and review it frequently during the program's early stages, looking for other possible methods and subjects. Looking for *alternative subjects* is perhaps the single most important way to drive down the cost of space missions.

15.3 Payload Design

Payload definition and sizing determine many of the capabilities, as well as, the limitations of the mission. The payload determines what the mission can achieve, while the size of the payload, along with any special structural, thermal, control, communications, or pointing restrictions, influence the design of the remainder of the spacecraft support systems [Pillet et al. 2005].

Let's start by assuming that the mission objectives are defined and the critical requirements are understood. This section will concentrate on the trade space of possible payloads and making an informed selection among them. This process is a useful guide for moving from a blank piece of paper to a preliminary set of payloads. Iterating on the process produces a more detailed definition and more useful set of payloads that can meet the mission objectives at minimum cost and risk.

As shown in Table 15-6, the process begins with an understanding of mission requirements described in Secs. 3.5, 4.4, 5.5, and Chap. 6. The mission requirements have a major effect on all aspects of space vehicle design, but it is frequently necessary to treat the components and subsystems separately for preliminary design and sizing. We begin with the payload because it is the critical mission element bounding spacecraft performance. Chapters 14 and 18 through 22 treat the remainder of the spacecraft systems and trade-offs involved in the overall spacecraft design.

Once the mission requirements are understood, we must determine the level of detail required to satisfy different aspects of the mission. For FireSat II, varying levels of detail are required if the task is to identify the existence of a fire, assess the damage caused by fires, or characterize the combustibles in a fire. Additionally, the timeliness demands placed on the mission could be vastly different depending on whether the data is to support long-term scientific analysis or real-time ground activity.

We summarize the basic steps in this process below and discuss them in more detail in Chap. 16 for Communications Payloads, and Chap. 17 for Observation Payloads.

1. Select Payload Objectives. These objectives will, of course, be strongly related to the mission objectives defined in Sec. 3.3 and will also depend on the overall mission concept, requirements, and constraints from Chaps. 3 through 6. However, unlike the mission objectives which are a broad statement of what the mission must do to be useful, the payload objectives are more specific statements of what the payload must do (i.e., what is its output or fundamental function). For FireSat II, this is specific performance objectives in terms of identifying fires. For the space manufacturing example in Table 15-6, called WaferSat, the payload objective is a definition of the end product to be manufactured.

2. Conduct Subject Trades. The subject is what the payload interacts with or observes. As discussed in detail in Sec. 15.2, a key part of the subject trade is determining what the subject is or should be. For a mobile communications system, it is the handheld user receivers. In that case, the subject trade is to determine how much capability to put in the user unit and how much to put on the satellite. For FireSat II, we may get very different results if we define the subject as the IR radiation produced by the fire or as the smoke or visible flickering which the fire produces. In addition to defining the subject, we need to determine the performance thresholds to which the system must operate. For FireSat II, what temperature dif-

Table 15-6. Process for Defining Space Payloads. See text for discussion. See Chap. 16 for a discussion of communications payloads, including SCS.

Process Step	Product	FireSat II (Remote Sensing) Example	Space Manufacturing Example	Where Discussed
1. Use mission objectives, concept, requirements, and constraints to select payload objectives	Payload performance objectives	Identify smoldering and flaming fires	Manufacture ultra-pure silicon wafers	Secs. 3.3, 3.5, 4.4, 5.5 and Chap. 6
2. Conduct subject trades	Subject definition and performance thresholds	Distinguish smoldering fires that are 3 K warmer than the background from flaming fires that are 10 K warmer than the background	Less than 1 ppb impurities over 50 cm square wafers	Sec. 15.2
3. Develop the payload operations concept	End-to-end concept for all mission phases and operating modes	Determine how end users will receive and act on fire detection data	Define user method to specify product needs, recover and use materials	Secs. 4.2, 15.3 Chap. 29
4. Determine required payload capability to meet mission objectives [identify key characteristics of interest]	Required payload capability	12-bit quantization of radiometric intensity in the 3–5 mm wavelength	Throughput of 5,000 wafers/day on orbit	Sec. 15.3
5. Identify candidate payloads	Initial list of potential payloads	Specifications for Sensors #1 and #2	Specifications for Factories #1 and #2	Sec. 15.3
6. Estimate candidate payload capabilities and characteristics [mission output, performance, size, mass, and power]	Assessment of each candidate payload	Sensor #1 meets the sensitivity requirement but requires a data rate of 10 Mbps. Sensor #2 can only identify flaming fires that are 10 K warmer than the background but requires a data rate of only 1.5 Mbps	Factory #1 produces 6,000 wafers/day, weighs 80 kg, and uses 2 kW Factory #2 produces 4,000 wafers/day (some of which will have >1 ppb impurities), weighs 100 kg, and uses 500 W	Sec. 15.3
7. Evaluate candidate payloads and select a baseline	Preliminary payload definition	Spacecraft and ground architecture based on 1.5 Mbps data rate. Adjust mission requirement to identify flaming fires only (not smoldering)	Select #1 with 1,000 wafers/day margin to be sold to reduce cost	Sec. 15.3 Chaps. 16, 17
8. Assess life-cycle cost and operability of the payload and mission	Revised payload performance requirements constrained by cost or architecture limitations	FireSat II spacecraft with acceptable mission performance and cost	Payload repackaging to accommodate launch as an Ariane secondary payload on ASAP ring	Sec. 15.3 Chaps. 5, 11
9. Identify and negotiate payload-derived requirements	Derived requirements for related subsystems	Data handling subsystem requirement to accommodate payload data rate of 1.5 Mbps	ACS system to provide 140 continuous min of jitter less than ± 1 nm	Sec. 15.3
10. Document and iterate	Baseline payload design	Baseline FireSat II payload	Baseline WaferSat payload	

ferences must we detect? For WaferSat, how pure must the resulting material be? For mobile communications, how much rain attenuation must we be able to accommodate? These will be iterative trades as we begin to define the payload instruments and can intelligently evaluate cost vs. performance.

3. Develop the Payload Operations Concept. Ultimately, the data or product produced by the payload must get to the user in an appropriate form or format. How will the end user of FireSat II data receive and act on the satellite data? How will the manufacturer recover the WaferSat materials and define what is to be done on the next flight? Payload operations will have a major impact on the cost of both the spacecraft and mission operations. As discussed in Chap. 29, payload operations may be done by the same facility and personnel that handle the spacecraft or, as with the Space Telescope, payload operations may be an entirely different operations activity.

4. Determine the Required Payload Capability. What is the throughput and performance required of the payload equipment to meet the performance thresholds defined in Step 2? For FireSat II, what are the specifications on the equipment needed to meet the temperature, resolution, or geolocation requirements? For WaferSat, how many wafers of what size will it produce? For mobile communications, how many phone calls or television channels must it handle simultaneously?

5. Identify Candidate Payloads. Here we identify the possible payloads and their specifications. For simple missions there will be a single payload instrument. For most missions, there will be multiple instruments or units which frequently must work together to meet mission requirements. Different complements of equipment may break the tasks down in different ways and may even work with different aspects of the subject. Thus, a system designed to identify the source of solar storms may have an imager and a spectrometer or a magnetometer and an instrument to map small temperature fluctuations on the photosphere or in the solar wind.

6. Estimate Candidate Payload Characteristics. Here we need to determine the performance characteristics, the cost, and the impact on the spacecraft bus and ground system so that we can understand the cost vs. performance for each of the viable candidate systems. Payloads will differ in their performance and cost, but also in weight, power, pointing, data rate, thermal, structural support, orbit, commanding, and processing requirements. We must know all of these impacts to conduct meaningful trades.

7. Evaluate Candidates and Select a Baseline. We examine the alternatives and make a preliminary selection of the payload combination that will best meet our cost and performance objectives. In selecting a baseline, we must decide which elements of performance are worth how much money. The payload baseline is strongly related to the mission baseline and cannot be defined in isolation of the rest of the parts of the mission and what it will be able do for the end user.

8. Assess Life-cycle Cost and Operability. Ultimately, we want to determine mission utility as a function of cost. This process was described in detail in Chap. 5. Typically it will not be a simple cost vs. level of performance characterization. Rather, it is a complex trade that requires substantial interaction with potential users and with whatever organization is funding the activity. It may become necessary at this point to relax or prioritize some of the mission requirements in order to meet cost and schedule objectives. For FireSat II, we may decide that only one type of fire or one geographic region will be addressed. For WaferSat, we may reduce the purity, the size of the wafers, or the throughput.

9. Define Payload-derived Requirements. In this step we provide a detailed definition of the impact of the selected payloads on the requirements for the rest of the system (i.e., the spacecraft bus, the ground segment, and mission operations). FireSat II will have power, pointing, geolocation, and data rate requirements. WaferSat may care very little about pointing and geolocation, but will have requirements on the spacecraft cleanliness levels and jitter control. These, in turn, may levy secondary requirements such as storage for on-board commands or thermal stability for pointing and jitter control.

10. Document and Iterate. Although this point is emphasized throughout the book, we stress again the need to document what we have decided, and why. The "why" is critical to allowing the system trades to proceed at a future time. We can make preliminary decisions for a wide variety of reasons, but we must understand these reasons in order to intelligently continue to do payload and system trades. Like all of the space mission analysis and design processes, payload definition is iterative. We will come back to the process many times as we learn more about the consequences of preliminary choices.

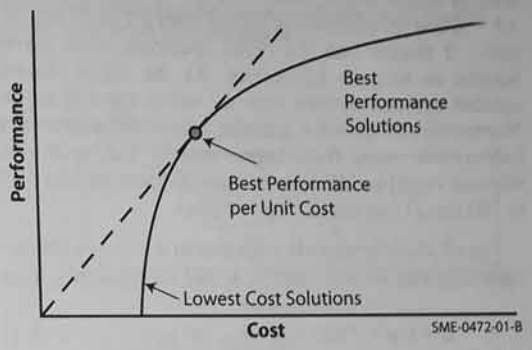


Fig. 15-2. Performance vs. Cost. The tangent point is the highest performance per unit cost.

Figure 15-2 illustrates the conceptual process of payload sizing. At the bottom end of the curve, we need to spend a minimum amount of money to achieve any performance at all. Near minimum performance, a small amount of additional expense will substantially increase performance. At the top end of the curve, we can spend a lot of money for very small improvement. The overall payload performance per unit cost follows a straight line through the origin and whatever point on the performance vs. cost curve we are working at. Therefore, the maximum performance per unit cost occurs where a straight line through the origin is tangent to the curve. There are good reasons for operating at any region along the curve in Fig. 15-2. To design a good payload, we must decide where along the curve our particular mission should be. At the high end, we obtain the best available performance. This would be appropriate for some military or science missions, such as the Space Telescope or Chandra X-Ray Observatory. Very small satellites such as CubeSats are at the bottom end of the curve. They perform modestly at very low cost. They may also be appropriate for multi-satellite, distributed architecture systems. Large commercial activities such as communications satellites, need the best performance per unit cost.

Table 15-6◀, Fig. 15-2, Eq. 15-0

The key to deciding how to size our payload is to look carefully at the mission objectives, particularly the tacit rules which often imply how well we want to do. Do we need the best performance regardless of cost? Can the mission proceed only on a minimum budget? Is this a long-term, continuing and potentially competitive activity in which performance per unit cost is critical? The answers to these questions will let us correctly size the payload and the mission to meet our mission objectives.

15.4 The Electromagnetic Spectrum

As Fig. 15-4 illustrates, the e*lectromagnetic spectrum* extends across a very broad range that spans at least sixteen orders of magnitude in wavelength or frequency. It includes *gamma rays* and *X-rays*, with extremely short wavelengths measured in angstroms (Å = 10^{-10} m), as well as *visible* and *infrared* (IR) wavelengths of 10^{-7} to 10^{-3} m and the *microwave* region from 0.1 to 30 cm. Finally, it ranges into the *radio* spectrum, with wavelengths as long as kilometers. As the figure shows, satellite systems operate over the entire spectral range. Normal wavelengths for comsats, radars, and microwave radiometers range from approximately 1 m to 1 mm, whereas visual and IR systems operate from around 0.35 to 100 mm (1 micron 10^{-6} m = 1 μm).

For all electromagnetic radiation in a vacuum, the relation between the wavelength, λ, and the frequency, υ, is

$$c = \lambda \upsilon = 2.997\,924\,58 \times 10^8 \text{ m/s} \qquad (15\text{-}1)$$

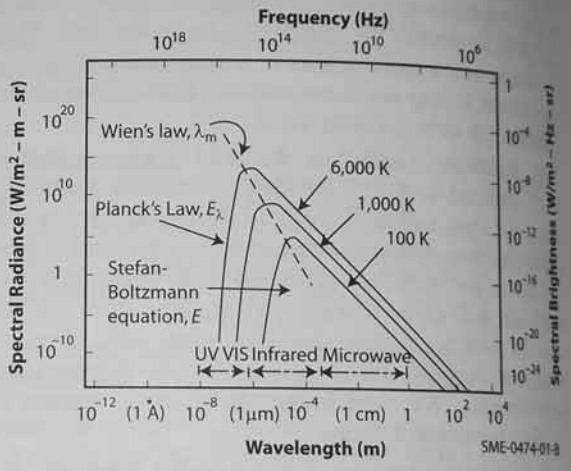
where c is the speed of light. Thus, in terms of frequency, the usable electromagnetic spectrum ranges from radio waves measured in kilohertz (kHz) to gamma rays with frequencies in the tens of exahertz (EHz). (1 kilohertz = 1,000 cycles/s ≈ 300 km wavelength; 1 exahertz = 10^9 GHz = 10^{18} cycles/s ≈ wavelength of 3 Å or 3×10^{-10} m).

At any temperature above absolute zero (0 K), all matter continuously emits electromagnetic radiation. This is called *thermal radiation* or *blackbody radiation*. For a *perfect blackbody*, the rate of total energy emission and the energy distribution by wavelength or frequency is a function only of the temperature, *T*. The spectrum of emitted radiation by a real body is related to the perfect blackbody by a radiation efficiency factor called *emissivity*, which is the ratio at every wavelength of the emission of a real body to that of the ideal or blackbody under identical conditions. For solid opaque objects without extended atmospheres like spacecraft or asteroids, emissivity depends largely on surface characteristics. For planets or stars that have atmospheres, emissivity is determined by how light propagates through the atmosphere of these objects. Even though the emissivity of real objects can be much less than one, in practice, the blackbody energy distribution is a good starting point for analysis. The spectral energy distribution of a blackbody is given by *Planck's Law*:

Table 15-6◀, Fig. 15-3, Eq. 15-4

$$E_\lambda = \frac{2\pi hc^2 \varepsilon}{\lambda^5} \frac{1}{e^{ch/kT\lambda} - 1} \qquad (15\text{-}2)$$

where E_λ is the emitted power density per unit wavelength (also called the spectral irradiance and typically measured in $Wm^{-2}\ \mu m^{-1}$), λ is the wavelength, h is Planck's constant ($6.6260755 \times 10^{-34}$ Ws^2), *T* is the temperature in K, c is the speed of light, ε is the emissivity and k is Boltzmann's constant (1.380658×10^{-23} Ws/K). Figure 15-3 shows typical energy distribution curves for various blackbody temperatures. When E_λ is divided by the solid angle (in steradians) for an extended source emitting in a given direction, the result is L_λ which is called the *spectral radiance* (typical units, $Wm^{-2}\mu m^{-1}sr^{-1}$).



Live Calc

Fig. 15-3. Planck's Blackbody Radiation Curves as a Function of Wavelength and Frequency [Chen, 1985]. Planck's Law defines the shape of the curve over all frequencies, the Stefan-Boltzmann's Law defines the area under the curve (the total energy emitted over all wavelengths), and Wien's displacement law defines the wavelength of maximum radiance.

From Planck's Law we can derive two other important relations. First, we obtain the *Stefan-Boltzmann's Law* by integrating Eq. (15-2) over the complete spectrum, which results in the total *radiant emittance*, W_b:

$$W_b = \varepsilon\, \sigma\, T^4 \qquad (15\text{-}3)$$

where σ is the Stefan-Boltzmann constant, 5.67051×10^{-8} $Wm^{-2}\ K^{-4}$, and W_b is in Wm^{-2}.

Next, we derive *Wien's Displacement Law* by differentiating Eq. (15-2) and setting the result equal to zero. The straight-line result defines the locus of peak spectral radiance vs. temperature, as shown on Fig. 15-4 and defined by

$$\lambda_{max} = 2.898/T \qquad (15\text{-}4)$$

where λ_{max} is in μm and *T* is in K.

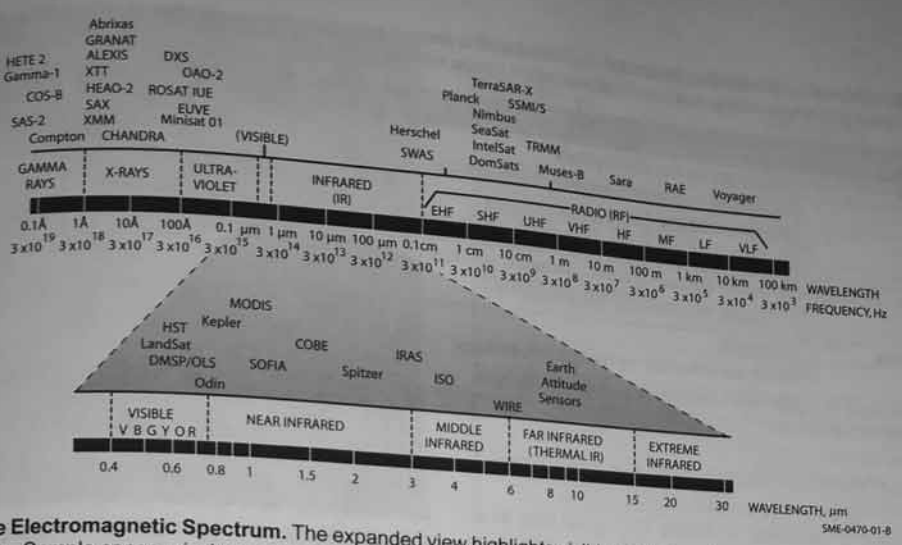


Fig. 15-4. The Electromagnetic Spectrum. The expanded view highlights visible and infrared wavelengths frequently exploited by satellites. Sample space missions across the entire spectrum are listed above the band region.

Remote-sensing instruments are aimed at a target on the Earth's surface or in space. Radars measure the characteristics of reflected, self-generated signals. For other sensors, an object's spectral radiance, or brightness, depends on its equivalent blackbody temperature or brightness temperature. This is the temperature of a perfect radiating body which has the same radiance. Visible wavelength systems can take advantage of the Sun's reflected energy, based on its brightness temperature of about 6,000 K. Of course, without sunlight, visible wavelength images are much less distinct. Laser-based sources of illumination can improve image quality for visible wavelength systems operating with low levels of natural illumination. On the other hand, systems using infrared and passive microwave radiometry measure the thermal emission from the Earth or other objects or scenes of interest. Thus, they can operate day or night, and for long enough wavelengths these systems can operate through clouds and other atmospheric disturbances. Note, however, as shown in Fig. 15-3, that thermal emission in the RF is much weaker than in the infrared spectral region. Well known sources of thermal radiation include the cosmic background radiation with brightness temperature of 2.7 K, room temperature objects at about 300 K and the hottest known stars with brightness temperatures close to 40,000 K. A typical Earth surface temperature is about 300 K, depending on location and season. Noctilucent clouds in the Earth's mesosphere at altitudes of about 80 km can be as cold as 150 K.

As Fig. 15-5 shows, the electromagnetic spectrum has many frequency bands for which the Earth's atmosphere is nearly opaque. We must avoid these bands if we wish to observe ground scenes or communicate to the surface of the Earth. This phenomenon also allows us to sound the atmosphere and measure such interesting data as the thickness and location of cloud layers, water vapor contained in clouds, and other upper-atmospheric phenome-

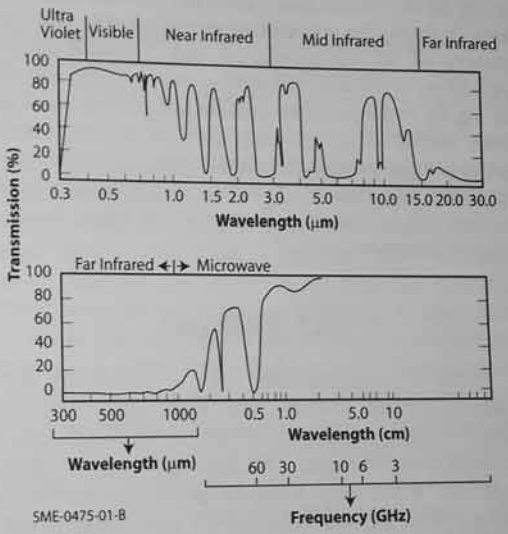


Fig. 15-5. Transmission Characteristics of the Earth's Atmosphere. Transparent regions are referred to as *windows* in the atmosphere.

na using the opaque bands. Clouds, rain, and snow tend to produce noise and thus attenuate signals for both communication and remote-sensing, even in the window bands. However, as the figure shows, for long wavelengths, atmospheric effects are relatively weak, which is one of the important reasons why almost all communication systems operate in the RF and why RF imaging techniques such as Synthetic Aperture Radar (SAR) discussed in Sec. 17.2 are so valuable in providing nearly all weather imaging capability.

When a sensor views an area in space, the radiation that reaches the sensor could come from a number of sources. The energy reflected directly from the target is usually the dominant feature of interest for optical re-

Table 15-6◄, Fig. 15-5, Eq. 15-4◄

Table 15-7. Sources of Radiation. Radiation that reaches the sensor can come from a number of different sources. This table shows all direct and single-scatter radiation that reaches a space sensor.

Source	Path to Sensor	Spectral Band
Sunlight	Scattered by the atmosphere	Visible
	Scattered by the atmosphere, then reflected off the Earth	Visible
	Reflected off the Earth	Visible
	Reflected off the Earth, then scattered by the atmosphere	Visible
Ground	Emission (including fires, volcanos, and man made lighting)	IR & Visible
	Emission scattered by the atmosphere	IR
Atmosphere	Emission	IR
	Emission reflected by the Earth's surface	IR
	Emission scattered by the atmosphere	IR

mote sensing, but other emitted, reflected and scattered energy can complicate the picture (Table 15-7).

The sources of radiation in Table 15-7 give rise to a number of different strategies for distinguishing different phenomena within the atmosphere or on the surface. For a given application, any of the sources of radiation will either be the subject being analyzed or noise to be minimized. Radiative measurements include the full complexity of all the effects on that radiation such as reflection, refraction, absorption, transmission, and scattering by material substances in solid, liquid, and gaseous phases. Distinguishing and identifying features using remote sensing techniques must take all of these variables into account. As Miller and Friedman [1996] advise, "when modeling the real world, allow for some slack to represent reality."

15.5 Examples

Space mission payloads span a remarkably broad range of subject, function, size, mass, power and other characteristics, depending on what is needed to respond to mission requirements. This final section of the chapter examines three very different payloads: the Hubble Space Telescope, the TerreStar-1 communication satellite, and QuakeSat. Both Hubble Space Telescope and TerreStar-1 were designed to put significant capability into space, to provide a unique astronomical observation capability in the case of Hubble and, in the case of TerreStar-1, to enable use of small handheld communication transceivers from the surface of the Earth to geosynchronous orbit. QuakeSat was constrained by cost to simplify the space segment to the greatest degree possible and to build system capability through post processing of data in the ground segment.

Hubble Space Telescope. The Hubble Space Telescope (HST) is one of the most important astronomical observatories in history, because of the breathtaking range of discoveries made by scientists using this unprecedented research tool. Scientific discoveries made with HST range from imaging the first planet outside the solar system to discovering the source of relatively long duration gamma ray bursts first noticed in the 1960s by the DoD Vela satellites. Perhaps, the most fundamental scientific research made with HST involves measuring the expansion rate of the Universe so accurately that the rate of change of that expansion can be assessed with much higher confidence. This work shows the expansion rate of the universe is actually increasing rather than decreasing as had been previously thought and may require a form of energy called dark energy to explain.

The idea of putting a telescope into space was first suggested by Oberth [1923]. The first steps toward the HST began in 1946, when the American astronomer Lyman Spitzer wrote a seminal paper describing the advantages of a space-based astronomical observatory. The two main advantages of an extraterrestrial observatory include: angular resolution limited only by diffraction, rather than by local atmospheric turbulence, as occurs on the surface of the Earth and the capability to observe entire electromagnetic spectrum including portions of infrared and ultraviolet strongly absorbed by atmosphere (Fig. 15-5). Professor Spitzer and others pursued the idea of a large space telescope for decades, until it was finally approved for development in 1978. HST was launched into orbit by the space shuttle in 1990.

HST's mission is to enable scientific discovery and study of astronomical objects from space in ways that cannot be achieved from the Earth's surface. The passive subjects of this mission are every astronomical object ranging from the moon to the most distant known objects, with some emphasis and priority given to the most distant objects. Unlike virtually every other space payload developed and launched to date, HST was designed to be serviceable, so that its suite of scientific instruments could be repaired and updated with the latest technology over a period of decades. Five servicing missions have occurred to date, with the most recent one completed in 2009. Clearly, HST's mission would benefit from the largest and highest quality telescope that could be launched into space within budget and other constraints. Early design thinking pointed to a 3 m aperture telescope. However, in the end, due to budget constraints and the need to fit HST within the space shuttle cargo bay, the HST ended up as a 2.4 m diameter telescope. Likewise, a higher orbit would have benefitted the mis-

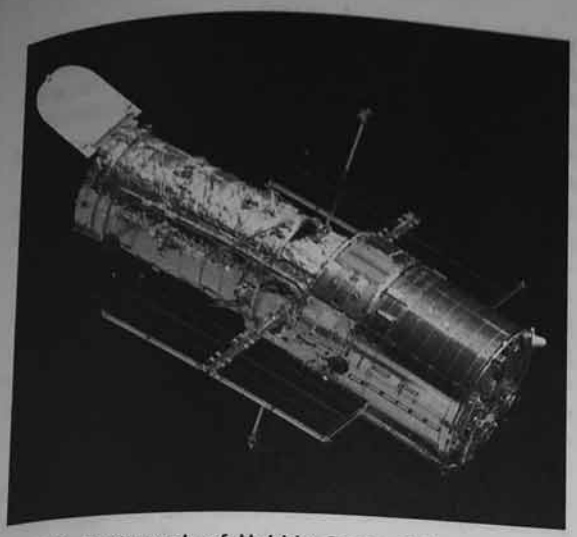


Fig. 15-6. Photograph of Hubble Space Telescope Taken from the Space Shuttle Atlantis at the End of the Final Servicing Mission (STS-125). (Photo courtesy of NASA)

sion objectives by being more distant from the Earth thereby increasing the solid angle available for astronomical observations at any given time and by reducing thermal and scattered light effects of the Earth on the system. However, the important need for astronauts to service the observatory required the HST be within range of the space shuttle.

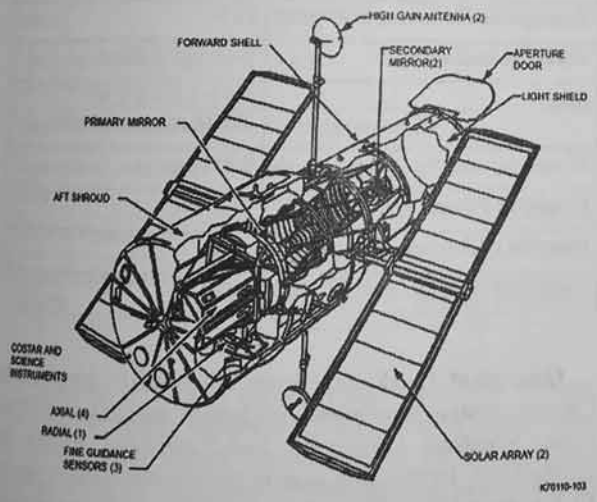


Fig. 15-7. Cutaway View of the Hubble Space Telescope. (Photo courtesy of NASA)

HST orbits at an altitude of slightly less than 600 km. The telescope is a 2.4 m Ritchey-Chretien reflector. Figure 15-6 shows a photograph of the HST captured by the space shuttle Atlantis, at the end of the fifth and almost certainly final servicing mission (STS-125). Key characteristics of the system are summarized in Table 15-8. Figure 15-7 shows a cutaway view of the HST and its instruments.

Table 15-8. Hubble Space Telescope Characteristics.

Launch Date	1990 April 24
Orbit	569 km altitude, 28.5 deg inclination
Primary Mirror Diameter	2.4 m
Telescope Focal Length	57.6 m
Length	13.2 m
Maximum Diameter	4.2 m
Mass	11,110 kg
Required Power	2.8 kW
Current Instrument Suite	NICMOS: near infrared camera/spectrometer COS: Cosmic Origins Spectrograph ACS: optical survey camera WFC3: wide field optical camera STIS: optical spectrometer/camera FGS: three fine guidance sensors
Line of Sight Stability	34 nrad over 24 hours
Cost at Launch	$1.5B (1990$)
Expected Mission End of Life	2014

HST was launched with a flawed primary mirror. Shortly after launch, HST images revealed a serious problem with the optical system. Images appeared sharper than ground-based images, but the best image quality obtained was much lower than the expected diffraction limited performance. Point sources spread out over more than one arcsec, instead of the specified and expected 0.1 arcsec. Analysis showed the primary mirror had been ground to a wrong shape resulting in a large halo in the point spread function (PSF), despite a detailed and extended pre-launch test program. The mirror was ground very precisely to the wrong shape, because a key piece of test equipment had been assembled incorrectly. This piece of equipment called a null corrector is intended to make a non-spherical mirror like the HST primary look spherical in testing. This fabrication flaw reduced effectiveness of the telescope for observations of very faint objects and for high contrast imaging. Spectroscopy and observations of bright objects were not seriously affected. Following a detailed investigation, a "Corrective Optics Space Telescope Axial Replacement" (COSTAR), essentially two mirrors in the light path with one figured to correct the spherical aberration, was designed and installed in the HST during the first servicing mission with Endeavor (STS-61) in 1993. COSTAR fixed the problem and enabled many of the key discoveries made by HST. However, to fit COSTAR into HST, one instrument had to be removed. Astronomers selected the High Speed Photometer. Subsequent HST instruments were designed to compensate for the distortion caused by the primary mirror so that COSTAR was no longer

Table 15-8, Fig. 15-7, Eq. 15-4◀

needed. It was returned to Earth on May 24, 2009 at the end of the last servicing mission.

The intended serviceability of HST along with highly skilled work by the engineers and astronauts who designed and implemented the COSTAR repair made it possible to rescue the HST mission. Normally, space mission hardware is not repairable in flight, although software fixes are frequently implemented to improve performance in-flight. (Some would argue that it would have been less expensive to build a better, more modern replacement telescope and launch it on a less expensive expendable.) Recent trends with development of fractionated spacecraft with replaceable modules and robotic servicing spacecraft to increase mission lifetime of large communication satellites suggest the lessons learned by HST may find more widespread implementation in future space missions.

TerreStar-1. At the time of its launch in 2009, TerreStar-1 was the largest satellite ever put into geosynchronous transfer orbit and the largest commercial communications satellite ever built. TerreStar-1 carries S-band (~2 GHz carrier frequency) transponders for providing mobile voice and data communications across North America. An interesting feature of this system is that it is enhanced by an Ancillary Terrestrial Component (ATC) to improve coverage performance in areas where the satellite signal could be blocked by buildings, dense forests or other surface features. One of the defining characteristics of the satellite is an 18 m reflector that was unfurled after launch. The reflector is used to generate hundreds of spot beams across the coverage area to enable use of small handheld phones for two-way communication. Since its launch, Terrestar-1 has been surpassed in size by the SkyTerra satellite, launched in 2010. Table 15-9 summarizes key characteristics of TerreStar-1.

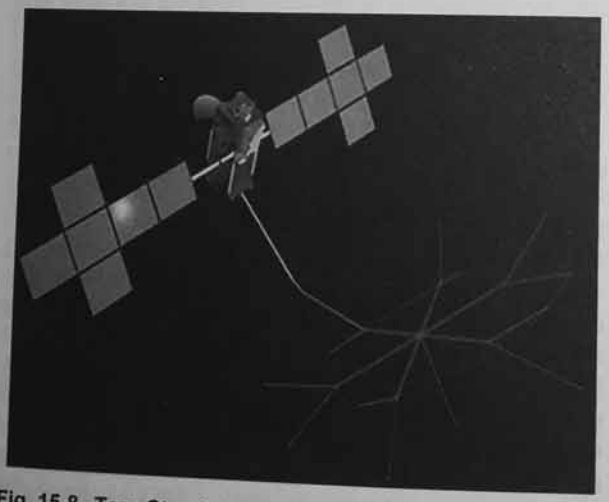


Fig. 15-8. TerreStar-1 Satellite Showing the Unfurled 18m Reflector. (Graphic courtesy of Space Systems/Loral)

TerreStar-1's mission is to provide two-way communication across a broad region for small handheld phones that will operate seamlessly with a ground-based 2G/3G cellular phone network or with the satellite in remote areas and during times of natural disaster or other interruptions in ground-based network service. The active subject in this mission is small, highly mobile communication terminals with users that aren't terribly concerned with whether their needs are being met by a ground-based network or a satellite-based network. The mission need to maintain communication in remote areas and during times of crisis, drove the design to include a highly reliable space segment. The need to make the ground-based communication terminals small and relatively low power, drove the space payload design to include a large aperture reflector on board to generate the large number of spot beams. From Eq. (17-6), we can estimate the size of each spot as being ~700 km in diameter at nadir, assuming diffraction limited performance for the reflector. By steering these spots actively throughout the coverage area, the system can minimize the total amount of power required to maintain contact between the satellite and the ground-based terminals while maintaining the relatively low data rate communication required for this system. The satellite generates hundreds of spot beams covering the Continental US, Canada, Alaska, Hawaii, Puerto Rico, and the US Virgin Islands.

Table 15-9. TerreStar-1 Characteristics.

Launch Date	2009 July 1
Orbit	Geostationary (35,786 km altitude) at 111 W longitude
Deployable Reflector Diameter	18 m
Communication Band	S-band (~2 GHz)
Size	7.6 × 3.6 × 2.8 m^3 (32.44 m span in orbit)
Mass	6,910 kg
Required Power at End of Life	14.2 kW
Number of Beams	~500
Expected Mission End of Life	2024

QuakeSat. QuakeSat (flown 2003–2010) was a triple Cubesat system developed by QuakeFinder/Stellar Solutions, Stanford University and Lockheed Martin in 2002–2003. Its mission was to detect, record and report ELF radio emission by seismic activity occurring in conjunction with earthquakes and to help establish whether such signals could be used to predict earthquakes. The intent, as described by Long et al. [2002] was to follow up previous reports of ELF detection by the Russian satellite, Cosmos 1809, which detected ELF signals immediately after a magnitude 6+ earthquake in Spitak, Armenia in 1989 and a French satellite, Aureol 3, that also reported similar results in a worldwide survey of ELF frequencies over active earthquake regions. Early trade studies showed that this mission could be implemented at very low cost using a Cubesat-based approach.

The key element in the QuakeSat payload was an AC magnetometer sensitive to low frequency magnetic field variations in the 0.5–1,000 Hz frequency range. The magnetometer was mounted at the end of a 70 cm long boom, as shown in Fig. 15-9. To provide an indication of spacecraft orientation, an infrared sensor was pointed along the direction of the boom to detect either the bright thermal emission from Earth or the cold darkness of deep space.

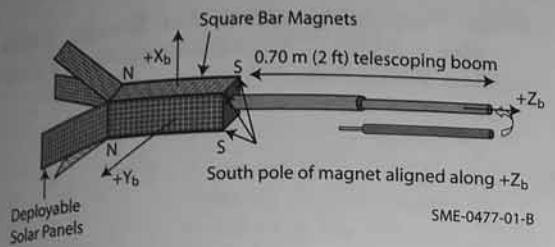


Fig. 15-9. QuakeSat Satellite Showing ELF Sensor on Deployable Boom. (Graphic courtesy of QuakeFinder/Stellar Solutions, Stanford University, and Lockheed Martin)

Table 15-10. QuakeSat Characteristics.

Launch Date	2003 June 30
Orbit	840 km altitude, polar sun synchronous orbit at terminator
Size	36 × 11 cm^2 (stowed) and 152 × 81 cm^2 (deployed)
Mass	4.5 kg
Required Power at Beginning of Life	14 W
Key Payload Element	AC magnetometer
Spacecraft Attitude Alignment	Within ~10 deg of local magnetic field line
Cost at Launch	~$1M
Expected Mission End of Life	6 months after launch (operated successfully for at least 11 months)

Flagg et al. [2004] reported that QuakeSat detected the type of signals expected for the mission within about 2 months after launch. Interpretation of the data was complicated by the complex propagation path taken by the signal along Earth magnetic field lines and by being in a terminator orbit, whose crossing time coincides with turbulent activity in the ionosphere. In parallel with this space-based measurement system, the QuakeSat team made similar measurements using a ground-based network. Results from these measurements found that the signals of interest appear to be short duration pulses lasting only 1–10 sec that occur with a duty cycle of approximately 5–10%. Thus, a LEO satellite like a single QuakeSat would be within measurement range of specific earthquake areas for only about 2–15 minutes per day, so chances of being at the right place when pulsations occur are relatively small. Nevertheless, an advantage of a low cost Cubesat approach (Sec. 25.4) is that it would be relatively inexpensive to deploy a large constellation of these satellites, if the mission appears promising. Work continues in this area.

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16 Communications Payloads

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16.1 Space Mission Communications Architectures
Space Mission Communications Architectures; Constellation Definition and Network Topology; Terrestrial Terminals; Spectrum Allocation; Communications Links; Communications Payload Architecture Development; Satellite Communication Missions; Concluding Remarks

16.2 Communication Link Analysis
Overview of Physical Principles of Satellite Communications Links; Link Equations and Link Budgets; Communications Satellite Link Budget Example

16.3 Communications Payload Design
Preliminary Antenna Design; Communications Payload Block Diagram Design; Communications Payload Sizing

16.4 Sample Missions
SCS; FireSat II

16.1 Space Mission Communications Architectures

All spacecraft must include a communications function, whether associated with telemetry, tracking, and control (TT&C) in support of satellite operations (Chap. 21), transmission of data generated by observation and scientific payloads (Chap. 17), or as a primary mission objective (Chap. 16). By any measure, communications applications are dominant among all satellite missions, with support for terrestrial communications being most significant. This chapter discusses architecture and key design concepts and considerations for RF communications payloads dedicated to support terrestrial communications. Table 16-1 describes the process flow for developing space mission communications architectures and detailed designs.

Before a design team can begin to develop a communications payload architecture, there must exist a higher-level *space mission communications architecture* to define the context within which the communications payload architecture must function, and to define connectivity, interfaces, and requirements associated with other elements of the overall communications system. Since these higher-level connectivities, interfaces, and requirements determine the feasibility of communications payload architectures and guide their optimization, we begin with a brief overview of space mission communications missions and associated architectures.

We then follow with the *communications payload architecture* which describes the configuration of payload elements that is used to receive transmitted RF signals, and to filter, amplify, direct, and re-transmit them to other receivers.

The *communications link design* is the detailed link assessment, where we incorporate the ground and satellite terminal characteristics, with propagation details to establish a solution which provides an acceptable positive

Table 16-1. The Communications Payload Design and Sizing Process. The link budget (Step 3) is a deterministic process that allows us to determine that each communications link will work successfully under the assigned conditions.

Step	Comment	Where Discussed
1. Define Space Mission Communications Architecture	Defines mission objectives and requirements	Sec. 16.1.1
2. Define Communications Payload Architecture	Defines the configuration of payload elements	Sec. 16.1.6
3. Perform Link Analyses	Required for all links	Sec. 16.2
4. Define Communications Payload Design	This is the hardware implementation of each link	Sec. 16.3
5. Estimate Payload Mass, Power, and Thermal Demands	Key sizing parameters	Sec. 16.3
6. Document and Iterate	Throughout the Architecture and Design Process	Secs. 16.1, 16.2, 16.3

Table 16-1, Fig. 16-0, Eq. 16-0

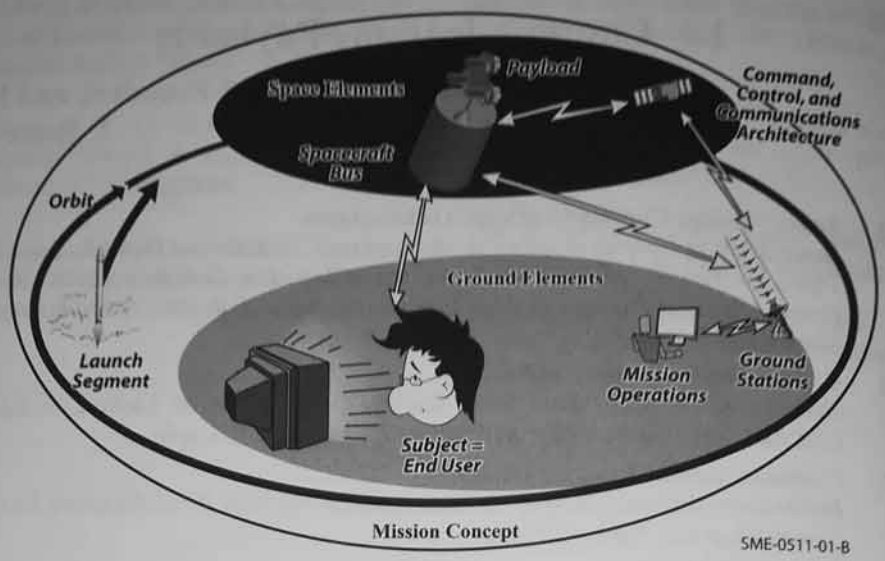


Fig. 16-1. Space Mission Communications Architectures Are Typically Divided Into Space Elements and Ground Elements. Although there is no universally accepted definition for the boundary between the space and ground elements, the full system architecture comprises a network of one or more satellites, ground terminals, control facilities, and user terminals (in some applications) that are interconnected by communications links.

margin to the link performance and thereby consistent with the architectural description. The *communications payload design* is the detailed implementation of the architecture, comprising the specification of components and the detailed description of the way that those components are integrated on the satellite to satisfy system performance requirements, again in a manner that is consistent with the architectural description.

Following a sequence analogous to that developed in Chap. 4 for the overall *space mission architecture*, this section describes a systems engineering process flow for synthesizing and refining *communications payload architectures*. Development of the corresponding *communications link design* is presented in Sec. 16.2 and *communication payload design* is presented in Sec. 16.3.

16.1.1 Space Mission Communications Architectures

Space mission communications architectures are typically divided into two segments, a space segment and a ground segment, as illustrated in Fig. 16-1, that between them comprise a network of satellites, ground terminals, control facilities, and user terminals (in some applications) that are interconnected by communications links.

Table 16-2 describes the elements of the space mission communications architecture that the communications payload architecture and design must support. (See Chap. 15.) In short, the space mission communications architecture defines the topology for the overall communications network and the data rates and protocols in each direction on the communications link between each node in the network of system elements (nodes being

Table 16-2. Space Mission Communications Architecture Elements. These elements differ somewhat from those in Table 4-13 because for communications payloads the "subject" often is the end user.

	Element	Description/Comment	Where Discussed
1	***Constellation Definition and Network Topology***	Includes orbit type and the number of satellites that operate together within the system as well as physical-layer connections between nodes of the satellite communications system, satellite to ground-terminal links, satellite to end-user links, and satellite-to-satellite links (cross-links)	Chap. 10, Sec. 16.1.2
2	***User terminal characteristics and the number and distribution of user terminals***	Individual user terminals and ground terminals User terminals are not present in all space mission architectures or even in all space communications system architectures. Where they do exist, the user-terminal characteristics and distribution are often abbreviated in terms of a description of the coverage areas, and the corresponding *G/T* and *EIRP* contours (*G/T* and *EIRP* are discussed in Sec. 16.3). Ground terminals typically used as principal interfaces to terrestrial wide-area networks	Secs. 16.1.3, 16.3
3	***Spectrum Allocation and Reuse***	Operating bands (typically RF) of the electromagnetic spectrum used in each direction of each communications link in the system, including any spectral reuse that may be required to satisfy capacity and geographic coverage requirements	Sec. 16.1.4
4	***Traffic Characteristics***	Data rates in each direction for each link or type of link, and any switching that may be required between different nodes of the communications network. FSS, MSS, and BSS for example, exhibit very different traffic characteristics	Sec. 16.1.7

defined as satellites, user terminals, and ground terminals). That is, the space mission communications architecture provides all of the information that is needed to understand the number of communications links that each satellite's communications payload must support, the characteristics of those links, whether switching may be required between different beams, and whether the satellite beams that support the communications links can be fixed or whether it may be necessary to modify either the shape or the direction of satellite beams while on-orbit.

16.1.2 Constellation Definition and Network Topology

The orbit type and the number of satellites in the constellation have a fundamental impact on the design of any space mission. For a communications mission, orbit type and constellation size are interrelated. As an example, whereas a single satellite (free flyer) in *geosynchronous* (or *geostationary*) orbit can provide continuous availability to a fixed region of the earth, multiple satellites must be used in a coordinated manner to provide continuous availability for systems based on constellations that employ other orbit types. (See Chap. 10.)

Orbit type also plays an important role in the definition of the satellite communications payload. Orbit diameter plays a principal role (in combination with user-terminal and ground-terminal characteristics and the geographic distribution of user terminals and ground terminals that must be served) in determining the power and sensitivity that will be required for the communications payload to maintain terrestrial communications links, for example. And whereas satellites in geosynchronous orbit may be able to use fixed antennas for terrestrial communications links and satellite cross-links, systems based on satellites in other types of orbit must either use tracking antennas or lower-gain antennas that have more limited communications capacity.

The network topology of the space communications network is defined by the number and locations of the various nodes: satellites, user terminals, and ground terminals and by the flexibility that is required for establishing connections between nodes. Like the orbit type, this topology, combined with the required traffic characteristics plays a fundamental role in shaping the requirements for the communications payload architecture. In fact, just as satellite constellations are classified according to orbit type (LEO, MEO, GEO, polar, or inclined), they too are classified according to the topology and flexibility of the network required for a given communications mission. Table 16-3 describes several examples of orbit-type and space communications network topology, and summarizes key advantages and disadvantages associated with each.

Table 16-3. Comparison of Five Sample Space Mission Communications Architectures, with Focus on the Relationship Between Orbit Type and the Space Communications Network Topology.

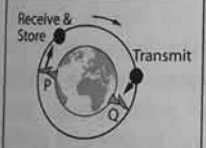
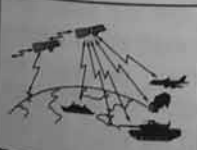
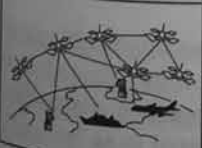
	Orbit Type	Advantages	Disadvantages
	***A. Low-Altitude Single Satellite, Store & Forward***	• Low-cost launch • Low-cost satellite • Polar coverage with inclined orbit	• Long message access time and transmission delay (up to several hours)
	***B. Geostationary Orbit***	• No switching between satellites • Ground station antenna tracking often not required	• High-cost launch • High-cost satellite • Need for stationkeeping • Propagation delay • No coverage of polar regions
	***C. Molniya Orbit***	• Provides coverage of polar region • Low-cost launch per satellite	• Requires several satellites for continuous coverage of one hemisphere • Need for ground station antenna pointing and satellite handover • Network control more complex • Need for stationkeeping
	***D. Geostationary Orbit with Crosslink***	• Communication over greater distance without intermediate ground-station relay • Reduced propagation delay • No ground stations in foreign territory: – Increased security – Reduced cost	• Higher satellite complexity and cost • Need for stationkeeping • Relay satellite and launch costs • No coverage of polar regions
	***E. Low-Altitude Multiple Satellites with Crosslinks***	• Highly survivable—multiple paths • Reduced jamming susceptibility due to limited Earth view area • Reduced transmitter power due to low altitude • Low-cost launch per satellite • Polar coverage with inclined orbit	• Complex link acquisition ground station (antenna pointing, frequency, time) • Complex dynamic network control • Many satellite required for high link availability

Table 16-3, Fig. 16-1◀, Eq. 16-0

16.1.3 Terrestrial Terminals

In this context, the term *ground terminal* is equivalent to Earth station, ground station, gateway, and Earth terminal, and is used in connection with the principal terrestrial nodes within the communications system. Ground terminals are distinguished from user terminals by virtue of their typically much larger RF sensitivity and much greater peak radiated power flux density, and in their frequent role as trunk-line interfaces to land-line or terrestrial RF wide-area networks [Elbert, 2000].

In contrast, user terminals, which may be land mobile, airborne, or ship borne, typically have much lower sensitivity than ground terminals, operate at much lower power, and serve a single user or a relatively small number of users on a local-area network. Both ground terminals and *user terminals* share in common the fact that they each include one or more antennas, a transmitter, a receiver, and control equipment required to communicate with the satellite.

16.1.4 Spectrum Allocation

To ensure that all users of the RF spectrum can operate without interfering with or being interfered by other users, *spectral allocation* is governed both by national and international regulatory bodies. To ensure the interoperability of both terrestrial and space-based systems, including commercial, scientific, military, and personal uses, these regulatory bodies have defined very detailed specifications that control the *frequency ranges* (bands) and maximum *power flux densities* that must be used for each application, as well as coordination that is required to take place between users in overlapping or adjacent geospatial domains. Because these regulatory requirements and coordination with other users place strict limits on the operating bands and radiated power flux density, spectral allocation has a direct impact on the architecture of satellite communications payloads.

Within the US, for example, the Federal Communications Commission (FCC) regulates spectrum allocation for commercial applications, while the National Telecommunications and Information Administration (NTIA) coordinates and regulates government users, including military, intelligence, meteorological, scientific, and other missions. International standards, including regulations that govern spectrum allocation*, power flux density limits, and coordination requirements, are defined by the Radio Regulations (RR) of the International Telecommunications Union (ITU), an agency chartered by the United Nations to standardize and coordinate the operation of telecommunications systems and services.

Although cognizant regulatory authorities define permitted uses for each portion of the RF spectrum, coordination with existing users often imposes further restrictions on the operating bands available for a new application. And while the *operating band* specified for new applications must clearly be compatible with other parts of the space mission *communications architecture* (e.g., existing user terminals), further constraints must usually be observed in order to maintain interference within prescribed limits near the boundaries between operating bands, and at geospatial boundaries that are reserved for existing users or are governed by other regulatory entities. Given the complexity of regulatory compliance and coordination with other users, spectral allocation can be one of the fundamental factors that determines whether a satellite communications architecture will be designed to operate at L, S, C, X, K_u, or K_a band, and the feasibility of specific constellation types, orbits, and other factors that are fundamental to both system and satellite payload architectures.

Figure 16-2 shows how the NTIA allocates the RF spectrum across various applications. As a specific example of the way that these allocations can affect the architecture of a satellite communications payload, Fig. 16-2 highlights the 500 MHz wide K_u-band transmit and receive operating bands (11.7 to 12.2 GHz and 14.0 to 14.5 GHz, respectively) that are typical for many commercial communications satellite applications. The NTIA defines non-commercial uses of these bands as including communications satellites and Earth stations operated by both Federal civilian and military agencies for voice, data, and video signals using commercial geostationary satellites, and for radio astronomy research conducted by the National Science Foundation.

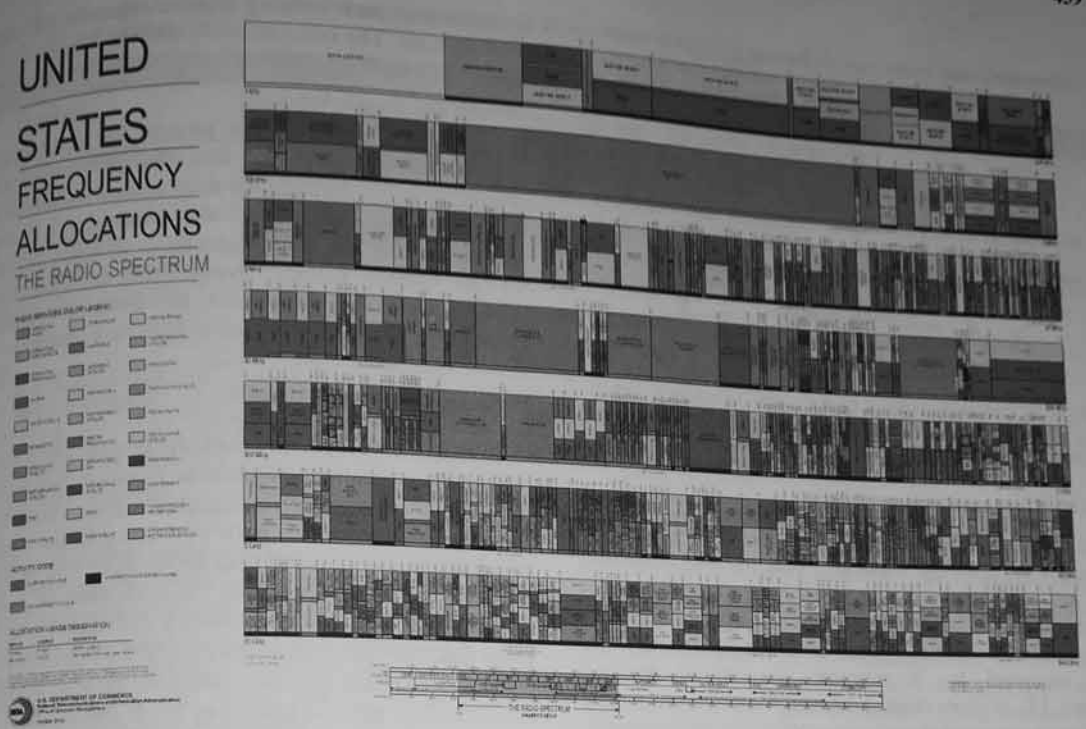
Spectrum Reuse

The *capacity* of a *communications link* is directly related to the *spectral bandwidth* available to support the link. Given the tremendous constraints to bandwidth allocation presented by regulatory prescriptions and coordination with other users, the total bandwidth available to a new application is frequently far smaller than what is necessary to support the traffic capacities required with a single beam for each geographic coverage area that must be served. This limitation can be overcome, however, by dividing the allocated spectrum into non-overlapping bands and designing the satellite communications payload so that each service area is covered by an array of beams, each of which occupies a band that is different than the bands occupied by adjacent beams, as illustrated in Fig. 16-3. This strategy for satisfying capacity needs within limits of spectral constraints is known as *frequency reuse*, because each of the spectral sub-bands can be reused many times, provided that beams operating at the same frequency are sufficiently isolated from one another by virtue of the shape (or roll-off) and direction of each beam.

Using this technique, the effective bandwidth available can be multiplied to achieve a reuse factor that is given by the ratio of the number of beams to the number of unique frequency bands (also referred to as "colors").

* It should be noted that the ITU is also charged with allocating satellite orbits (orbital slots).

Table 16-3◀, Fig. 16-1◀, Eq. 16-0



Get Pic

Fig. 16-2. Radio Spectrum Frequency Allocations, as Defined by the United States National Telecommunications and Information Administration (NTIA), Showing the Detailed and Complex Assignments of Bandwidth Across Applications [NTIA, 2010].

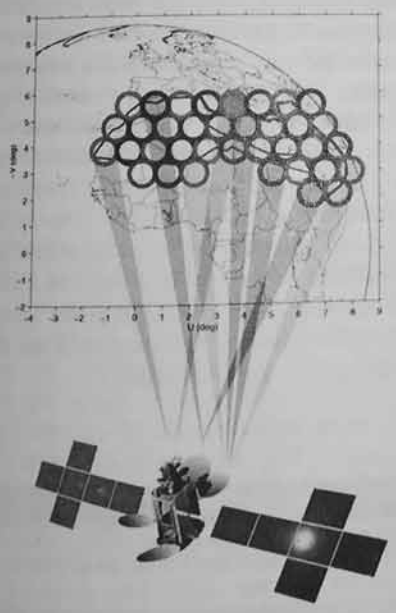


Fig. 16-3. By Dividing Available Bandwidth into Non-overlapping Sub-bands, Each Sub-band Can be "Reused" Over Non-adjacent Geographic Areas.

Likewise, *polarization reuse* can be used to further multiply the total usable bandwidth by taking advantage of the isolation (typically in the range of about 30 dB) that can be achieved between orthogonal polarizations (whether linear or circular).

In Fig. 16-3 we can see the benefit of polarization and frequency reuse. If there was only polarization reuse with *linear horizontal* and *vertical polarization* applied to one single ubiquitous coverage area, then the effective bandwidth would be 2 × 500 MHz = 1 GHz divided into a total of 24 (36 MHz) transponders. If however, we also employ frequency reuse, with 4 color spot beam coverage as shown in Fig. 16-3, we achieve sufficient isolation between beams of the same color. Then the effective bandwidth for n = 36, m = 4 is 2 × 500 × 36/4 = 9 GHz or an equivalent of 225 36 MHz transponders. In other words, we have achieved an order of magnitude improvement in capacity. The drawback to this is the increased architecture complexity of the communications payload.

Although frequency and polarization reuse can be used effectively to increase the capacity of space mission communications systems, these system-level architectural elements drive fundamental aspects of the corresponding communications payload architectures by requiring the implementation of multi-beam arrays or phased arrays. Defining channelization, bandwidth allocations, decisions as to the use of analog vs. digital payload alternatives, are also among the many attributes that can be driven by the decision to implement frequency and polarization reuse.

Table 16-3◀, Fig. 16-3, Eq. 16-0

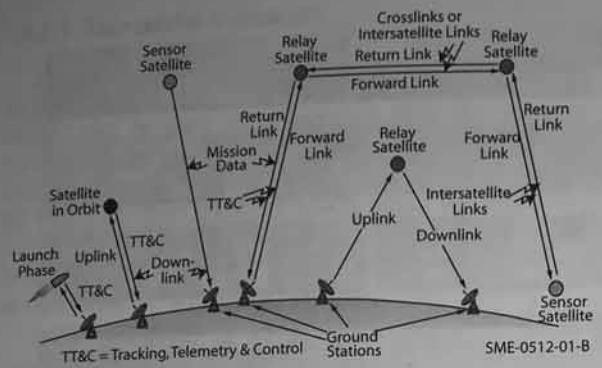


Fig. 16-4. Communications Links Carry Telemetry, Tracking, and Command (TT&C) Data or Mission Data Between Nodes of the Space Mission Communications System.

16.1.5 Communications Links

Communications links allow a satellite system to function by carrying *telemetry, tracking and command* (TT&C) data (Chap. 21) or mission data between nodes of the space mission communications system. In many cases, this involves transmission of the sensor data that are collected by a separate observation payload (Chap. 17). In other cases, however, the entire purpose of the space mission is to provide satellite-based communications, where the data on the communications links and the mission data are one and the same. (Fig. 16-4).

In this chapter, we focus on the architecture and design of spacecraft communications payloads that support the exchange of mission data between system elements via *radio frequency* (RF) communications links. *Link analysis* is fundamental to the development of both communications payload architectures and detailed designs. A detailed discussion of link analysis is presented in Sec. 16.2.

16.1.6 Communications Payload Architecture Development

Following a sequence analogous to that developed in Chap. 4 for the overall space mission architecture, this section describes a systems engineering process flow for synthesizing and refining communications payload architectures. The corresponding process for developing a communications payload design is described in Sec. 16.3.

Table 16-4 describes the process for developing a *communications payload architecture*. The process comprises five steps: defining the architectural requirements, identifying the architectural elements subject to trade and identifying the main alternatives that are available for each tradable element, synthesizing candidate architectures, assessing performance, and down-selecting and iterating concepts until a final architecture emerges.

Definition of Requirements

Like any development effort, the process for developing a communications payload architecture begins with defining an appropriate set of requirements. The specific set of requirements that is flowed down to the satellite communications payload will differ from one development to the next, depending on a number of factors, including technical and regulatory considerations, program risk, business considerations, and other factors. An important example of a regulatory factor is in the specification of the specific operating bands that must be used for each coverage area (Sec. 16.1.1). While it is common for the operating bands for each beam to be specified on the basis of existing ground segment hardware that is to be used or on the basis of licenses that have been applied for or already received, there are cases in which payload architectures may be developed with little or no practical limit as to specific operating bands. This may occur, for example, in the development of new satellite communications systems that do not require the use of specific existing user terminals. In the absence of such constraints, it may be possible to better optimize overall system architecture and performance. Rather than having ground segment or regulatory considerations impose an *a-priori* constraint on system and payload architectures, payload driven requirements might guide specifications that are flowed to gateway and user terminal developers and spectrum license applications.

Table 16-4. Communications Payload Architecture Development Process

	Step	Description/Comment
1	***Define architectural requirements***	Specify the attributes that must be exhibited in order for the communications payload architecture to be viable and how candidate architectures will be assessed to rank alternatives
2	***Identify elements subject to trade and identify the main alternatives for each tradable element***	List those payload elements that can be supported by more than one alternative. Identify the design degrees of freedom, focusing on the alternatives that have the greatest potential to materially affect measures of effectiveness
3	***Synthesize candidate architectures***	Identify element alternatives and combinations of element alternatives that work well and that do not work well
4	***Assess compliance***	Evaluate the performance of each candidate architecture in terms of the requirements
5	***Ranking, selecting, iterating, and finalizing architectural concepts***	Rank candidate architectures based on measures of effectiveness, assess sensitivity of highest-rated alternatives, and iterate architectural concepts until a final architecture can be selected

Table 16-5. Partial Listing of Typical Requirements Flowed Down From the Communications System to the Communications Payload Architecture.

	Requirement	Representative Requirements
1	*Coverage*	Definition of uplink and downlink coverage areas and gateway locations Specification *G/T* and *EIRP* required for each coverage, including gateways
2	*Coverage Flexibility*	Description of coverage areas and gateway locations that are to be fixed, gateway locations and user beam coverages that may be moved after launch, and user-beam coverages whose shape must be capable of being changed while on-orbit
3	*Connectivity*	Definition of the mapping required between uplink beams/channels and downlink beams/channels
4	*Connectivity Flexibility*	Specification of switching and routing that the communications payload must accommodate
5	*Operating Bands*	Intended uplink and downlink operating bands for both communications signals and for bus and payload telemetry, tracking, and control Approximate channel sizes and counts (in some cases)
6	*Size, Mass, Power, and Thermal Budgets*	Aperture suite must be compatible with a specific bus configuration and launch fairing Maximum allowable mass, DC power, and thermal dissipation
7	*Lead Time (Development Risk)*	Target date or calendar months from contract to launch or on-orbit delivery
8	*Payload Cost/Price*	Target cost or price for payload development, production, integration, and test

Table 16-5 provides a list of requirements that are typically involved in the definition of a communications payload architecture. Because these requirements will be used to assess compliance and rank performance in steps 4 and 5 of the process, it is important that requirements are stated in a way that allows them to be objectively evaluated for each of the candidate architectures.

Identification of Tradable Elements and Principal Alternatives

Once the payload architectural requirements have been defined, the next step in the architecture development process involves identifying the elements of the communications payload that are subject to trade, and the main alternatives for each *tradable element*. The satellite communications payload architecture comprises the four elements described in Table 16-6: antennas, spectrum allocation, repeater type, and payload topology. In most instances, antennas, repeater type, and payload topology will be tradable, though there is sometimes at least some flexibility with respect to spectrum allocation. For each of the tradable elements, many alternatives are available. Note that although some of these alternatives are represented at a very high level in Table 16-6, the actual array of alternatives from which the design team has to choose spans many dimensions within each architectural element, and cannot possibly be represented in such a simple tabulation. Of the universe of alternatives within each tradable architectural element, only a subset will have potential applicability for a particular payload, and these are the building blocks from which the payload architecture will be constructed.

Synthesis of Candidate Architectures

Based on the main alternatives that have been identified for each tradable element, synthesis of the communications payload architecture begins by considering the alternatives (a) in different combinations, and (b) with different connectivity between components. As will be true for the detailed design, synthesis of candidate architectures typically begins by considering the orbital slot that the satellite will occupy, the geographic areas that must be serviced, the operating bands for uplinks from and downlinks to each coverage area, the characteristics of the ground terminals and user terminals with which the satellite must establish RF connections (frequently referred to as communications or *Comm links*), and the data rates that must be supported on each link.

Ordinarily, the *orbital slot* and the required bandwidth allocations will have been specified as part of the space mission communications architecture, as will the characteristics of the ground terminals and user terminals. In these cases, it is common for the terminal characteristics to have been translated into (A) the power flux density at the spacecraft that will be received from terrestrial terminals and (B) corresponding power flux density that must be transmitted by the communications payload (expressed as *equivalent isotropic radiated power,* or *EIRP* (Sec. 16.2).

Based on an understanding of the alternatives available in terms of *Low-Noise Amplifiers* and typical input signal losses, *link analyses* (Sec. 16.2) are performed to obtain an initial estimate as to the *antenna directivity* that will be required to reliably detect and distinguish uplinked signals that reach the satellite from thermal noise and other sources of interference. A similar link analysis is performed for each downlink from the satellite, as well, but in this case with the objective of estimating the characteristics of the transmit antennas that will be required for each beam.

The link analyses yield estimates of the directivity that will be required for both receive and transmit antennas. Another outcome of the link analysis is an initial estimate of the bandwidth that will be required for each link. From the directivity, the design team can estimate the size that will be required for each antenna. The bandwidth requirements give insight as to whether the spectral allocations are sufficient to avoid the cost and complexity of *frequency reuse*. While the required con-

Table 16-5, Fig. 16-4◀, Eq. 16-0

Table 16-6. Tradable Elements of the Communications Payload Architecture and Principal Alternatives for Each.

	Element Types	Description	Principal Alternatives
1	*Antennas*	Both transmit and receive antennas for both user links, gateways, and inter-satellite cross-links	• Single element vs. array • Element Geometry (e.g., horn, reflector, helix) • Antenna geometry (e.g., direct radiating, Newtonian, Gregorian, Cassegrain) • Physical vs. electronic beam forming • Fixed vs. mechanical steering vs. electronic steering • Articulation for antennas stowed during launch
2	*Spectrum Allocation*	Spectrum allocated for the communications mission within each beam or coverage area	• L, S, C, X, K_u, K_a • Specific operating bands to be used for each transmit and each receive beam.
3	*Repeater Type*	The subsystem that performs the following functions: receiving, frequency conversion, filtering, analog-to-digital (A/D) and digital-to-analog (D/A) conversion, switching/routing, and amplification	• Transponder: Receives, filters, routes, amplifies, and retransmits analog RF signals • Analog Processor: Supports flexible bandwidth reallocation by filtering received signals into channels and using switch matrices for routing • Digital Channelizer: Filters received analog signal into narrow channels that are digitized to allow highly flexible, reconfigurable bandwidth routing before being converted back to high-power analog RF signals suitable for re-transmission • Router: Performs A/D conversion on received analog signal to support extraction of routing information for demodulated signals. Regenerates and remodulates high-power analog signals for re-transmission. • Digital Processor: Uses information extracted from demodulated signals to control payload configuration or information handling, as, for example, in providing automated phase and amplitude control for phased array bandwidth and spatial coverage adjustments
4	*Payload Topology*	Highest-level concept describing (a) the signal-processing sequence, and (b) the overall strategy for directing the flow of analog and digital signals through the repeater	• Few-to-few (typical for fixed coverages linking gateways and VSATs that serve geographically dispersed terrestrial WANs and LANs) • Many-to-few/Concentrating (typical for broadband satellite services (BSS) and mobile satellite service (MSS) applications) • Few-to-many/Diverging (typical for digital audio radio services (DARS) • Many-to-many (typical for satellite-based broadband IP access) • Intersatellite links

nectivity and connection flexibility between beams determine the nature of switching that will be required, the question of whether beam steering (either mechanical or electrical) needs to be part of the communications payload architecture dictates whether beam forming is necessary. In this manner, by taking consideration of each of the requirements, as well as each of the alternatives available for tradable elements of the architecture, the design team synthesizes one or more candidate architectures to be considered.

Assessment of Compliance

Once candidate architectures have been developed, an assessment must be performed against each. This assessment is performed with respect to the requirements that were identified or flowed down as the first step of Table 16-4. Despite the fact that many requirements will have been identified, even at this early stage of development, assessment of compliance at the first several iterations is ordinarily quite fast because some aspects of the candidate architectures (e.g., switching flexibility) will be compliant by design. While others, such as *EIRP*, can be readily determined on the basis of relatively simple link budget calculations, there are some (such as the ability of available satellite buses to accommodate the number and sizes of apertures) that are likely to yield to the initial judgments of members of the design team, based on experience in previous development efforts.

Table 16-6, Fig. 16-4◀, Eq. 16-0

Ranking, Selecting, Iterating and Finalizing Candidate Architectures

Like the design process, development of architectures is an intrinsically iterative process. At each iteration, candidate architectures are ranked according to their compliance assessment. Equally important as the selection process, the *compliance assessment* also provides insight into the sensitivity of each alternative architecture with respect to pertinent measures of performance. This insight provides valuable guidance as to refinements, optimizations, and elaborations that should be considered, identifying requirements that cannot be satisfied, for additional information that is needed in order for a final architecture to be selected. In this manner, the design team proceeds through successive iterations until a final architecture is identified or it becomes clear that compliance to the requirements cannot be achieved. In the latter case, the design team may solicit or propose relaxation of one or more requirements, at which point the process may resume.

To complete its work in the case that a preferred architecture does emerge the design team must elaborate the preferred architecture at a level sufficient to (A) obtain the concurrence of the program sponsor, and (B) ensure that the documentation of the preferred architecture is sufficiently detailed and unambiguous as to provide a basis for the detailed design.

System-Level Architecture Trades: Partitioning of Performance and Functionality Between Space and Ground Segments

The satellite communications payload is a classic example of an *active* or *controllable subject* (Chap. 4), in the sense that system architects have broad flexibility in partitioning performance and functionality between ground-based and space-based elements. One of the most fundamental examples of this is found in the partitioning of power and aperture sizes between the satellite and ground segment terminals (gateways and user terminals).

Although it might be technically feasible to design a system that uses essentially the same amplifiers and antennas on the satellite, gateways, and user terminals, this never happens in practice. The reason for this is that launch vehicle constraints on mass and size create tremendous incentive to minimize the size, mass, and total power consumed by the satellite payload. On the other hand, in comparison with the satellite, gateways are virtually unconstrained in any of these dimensions. Likewise, user terminals (which may be as small as a mobile phone or a pager), frequently have even more severe size, mass, and power constraints than the satellite communications payload. As a result, the design of satellite communications systems always biases amplifier power and aperture size allocations away from the satellite and toward the gateway, while the nature of the mission will determine the direction and the degree to which aperture size and power are partitioned between the satellite and ground terminals.

In a similar sense, system architectures allow certain functional elements of the satellite communications system to be moved between the space and ground segments. Whereas routing of bit streams is typically performed within the ground segment, some military missions in particular may lead to the development of space-based routing capabilities, such as in applications like command and control of remotely controlled weapons systems, for example, that can have severe latency constraints. Likewise, for satellites that use antenna arrays with electronic steering or beam forming, there are potential advantages in moving beam forming hardware and electronics from the satellite (where these functions have typically been implemented) to ground-based systems where these subsystems can be both implemented at lower cost and more readily maintained over the course of mission life.

This partitioning of performance and functionality typically takes place at the system level. Although these types of trades are not typically open to trade at the level of the communications payload architecture, it is nevertheless worthwhile for payload designers to recognize that such trades exist so that alternatives can be presented when initially stated requirements cannot be satisfied.

Table 16-7. Comparison of Principal Types of Commercial Space Communications Missions.

	Fixed Satellite Service (FSS)	Broadcast Satellite Service (BSS)	Mobile Satellite Service (MSS)
Coverage Areas	Typically large, geographically separated	May be large, but often subdivided into smaller regional coverages, depending on market-driven differences in required capacity and content	Service typically provided over large coverage areas by patterns of overlapping spot beams
Terminals	Usually gateway ground terminals or VSATs	Satellite uplinks from ground terminals and broadcasts directly to small fixed or mobile user terminals	Vehicle-mounted terminals (e.g., providing voice/internet access to aircraft or train passengers), and small hand-held terminals (e.g., satellite-enabled mobile phones) or laptop computers
Traffic	Typically symmetric voice and data flows between coverage areas Supports broadcast to VSATs	Asymmetric, with low to moderate data rates to individual user terminals, and little or no data returning from users	Nominally symmetric low bandwidth voice and data to and from individual user terminals

16.1.7 Satellite Communications Missions

Space Mission applications are dominated by satellite communications missions. This section describes the *principal classifications* used for commercial space communications missions, and key ways in which differences between these missions affect the payload communications architectures. Table 16-7 summarizes key attributes that differentiate the three principal categories of commercial satellite communications missions.

Fixed Satellite Services (FSS)

For many years, *Fixed Satellite Service* (FSS) missions have been the backbone of the commercial satellite industry. Although many permutations exist, FSS satellite communications are ideal for providing fixed-bandwidth, broadcast or telephonic connectivity between large geographically separated coverage areas. And while it is increasingly common for FSS satellites to incorporate a limited amount of beam switching capability, the coverage areas served by FSS systems are typically fixed throughout the mission lifetime of each satellite in the system. Because they provide essentially fixed connectivity and capacity, FSS satellite payloads can be implemented by relatively simple, low cost satellite designs that incorporate fixed, dedicated antennas, and analog repeaters built from very mature component technology, including dedicated high-power amplifiers, analog signal filtering, and mechanical switching. Although business considerations continually push toward larger, more powerful, and higher capacity solutions, most FSS satellite payloads can ordinarily be packaged on low to moderate power satellite buses.

Table 16-7, Fig. 16-4◀, Eq. 16-0

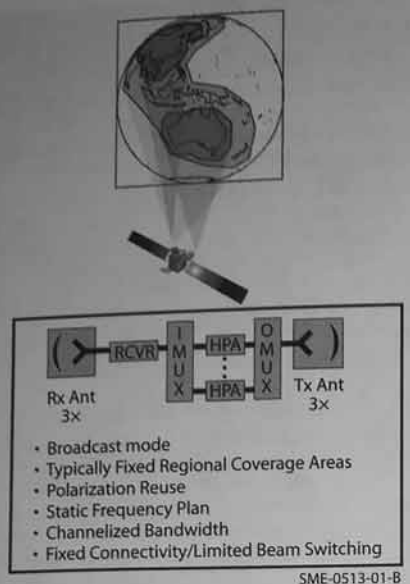


Fig. 16-5. An FSS Satellite Providing Fixed Connectivity Between Ground Terminals in Large Coverage Areas. Depending upon mission requirements, the communications payload may be configured to provide connectivity between terminals within any single coverage area or between any coverage area and one or both of the others. The communications payload architecture, as described in the simplified payload block diagram, is directly related to the FSS mission through the number and type of antennas, the absence of beam switching, and the very straightforward filtering and amplification chain.

FSS systems are capable of supporting broad beam coverage of large geographic areas, providing point-to-point, point-to-multipoint (broadcast), and multipoint-to-point connectivity to ground terminals ranging from very small aperture terminals (VSATs with apertures less than 1 m in diameter) to large gateways with aperture sizes of 10 m or more.

Figure 16-5 illustrates a hypothetical FSS satellite and the kernel of the corresponding payload architecture, as described by a highly simplified "payload block diagram." While the FSS satellite depicted in the figure may be a free-flyer, or may be part of a broader constellation of satellites, this particular satellite provides service to three coverage areas whose operating frequencies are coordinated such that the two smaller coverage areas are able to overlap with the largest. Depending upon mission requirements, the communications payload may be configured to provide connectivity between terminals within single coverage areas or (more commonly) between any coverage area and one or both of the others. Communication may be asymmetric or unidirectional (as in broadcast applications), or may be highly symmetric (as in telephony). A heterogeneous mixture of ground terminals may be served within each region.*

The payload architecture, as described by the simplified payload block diagram, is directly related to the key elements of this FSS payload through the inclusion of three fixed (non-steering) *transmit (Tx)* and three fixed *receive (Rx) antennas*†, each with shaping appropriate to support the required coverage areas, a very straightforward filtering and amplification chain, and fixed connectivity between coverage areas (no switching).

Broadcast Satellite Services (BSS)

Although the FSS satellite communications systems can support broadcast mission elements and functionality, another category of satellite communications systems is more highly optimized specifically for particular broadcast applications. As a category, *broadcast satellite services* (BSS) are growing in commercial significance, and include specialized variants: *direct-to-home* (DTH) video broadcasts to small fixed user terminals, digital audio radio service to small fixed and mobile user terminals (digital radio receivers), and *digital mobile broadcast* (DMB) of video and voice to handheld or vehicle-mount terminals.

Whereas the data flows in FSS networks are typically symmetric, with similar amounts of traffic flowing in each direction between coverage areas, the data flows in BSS systems are highly asymmetric by definition, with large capacity in the "forward" direction (toward end users), and little or no capacity in the "return" direction (from end users). This difference alone drives a fundamental difference in satellite architecture in terms of payload topology. And whereas the terrestrial terminals in FSS systems tend to be collectors that serve as aggregators of data from networks of data generators and consumers, BSS systems are designed to broadcast directly to individual users. This difference has direct implications for payload architecture because user terminals are invariably smaller and less sensitive than the gateway and VSATs that are the basis for FSS systems. As a result, the BSS satellites must transmit substantially greater power flux densities in order to communicate at a given rate with its user terminals than an FSS satellite must transmit to communicate at the same rate with a VSAT or gateway ground terminal. This transmit power differential drives BSS satellites toward the use of large, high-gain antennas and super high-power amplifier combining networks, integrated on a high-power satellite bus, as is illustrated in Fig. 16-6.

Mobile Satellite Services (MSS)

The last main category of commercial satellite communications mission is referred to as *mobile satellite services* (MSS). As in the case of the BSS mission, MSS

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* It should be noted that whereas the example provided in Fig. 16-5 represents a mission in which data originating from terrestrial users is transmitted to the satellite, and filtered, amplified, and retransmitted to other terrestrial users, most observation satellites include a communications payload designed to provide similar functionality. In this case, however, the observation payload and associated signal-processing electronics generate the signals that are to be transmitted to Earth; typically to fixed ground terminals, instead of receiving communications uplinks from the Earth.

† These will often be combined into single antennas that support both transmit and receive functions.

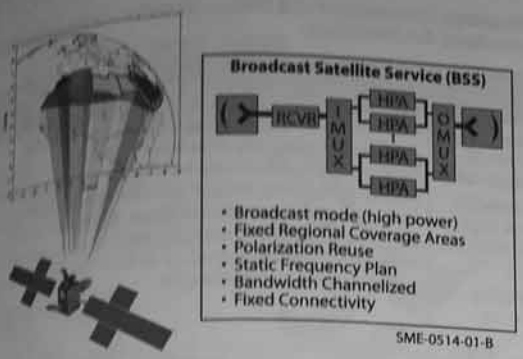


Fig. 16-6. A BSS Satellite Providing Fixed Broadcast Communication Directly to Small Fixed and Mobile User Terminals Within Regional-beam Coverage Areas Tailored for Market-driven Capacity and Content. Asymmetric capacity requirements and service to disadvantaged terminals tend to drive payload architectures for these missions toward asymmetric topologies and high-power combining networks.

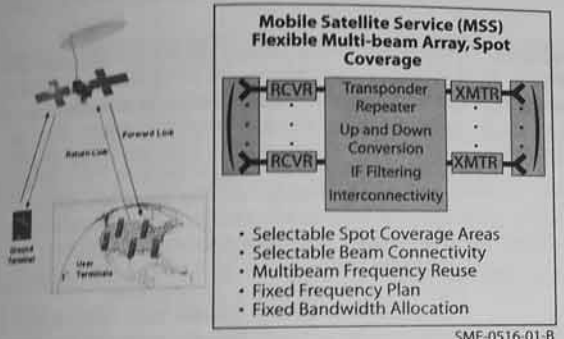


Fig. 16-7. An MSS Satellite Providing Two-way Communication Directly to Disadvantaged Mobile User Terminals Over a Wide Geographic Area. Service to disadvantaged terminals requires very high gain, which is achieved by overlapping highly directive beams where adjacent beam spots occupy different frequency bands. Corresponding military applications (COTM) may also require beam steering, beam forming, and reallocation of bandwidth between different beams. All of these mission requirements result in communications payload architectures that differ dramatically from those used to support FSS and BSS missions.

missions are optimized for direct communications with small individual user terminals. In contrast, MSS missions are optimized for individualized low data-rate user links rather than broadcast of common data at higher rates to many users, and typically require the ability to accommodate similar forward and return traffic flows to each user terminal. As a result, a typical MSS mission therefore involves two-way (full duplexed) communication between the satellite communications payload and mobile phones or laptop computers equipped with the appropriate antennas and electronics. Given these form factors, user terminals are limited in both their sensitivity to signals transmitted from orbiting satellites, and in the signals that they are able to transmit back to the satellite.

Both the need for nominally symmetric two-way traffic, and the transmit/receive limitations of the user terminals drive the communications payload architectures for MSS missions in dramatically different directions than those that are typical for FSS and BSS missions. As a consequence, MSS satellites are designed to have high sensitivity for uplinked return-direction signals from the mobile user terminals, and must also be able to radiate with high power flux density in the forward direction toward the mobile user terminals. For MSS systems based on geosynchronous orbits, these considerations typically drive satellite communications payload designs toward the use of multiple high-gain spot beams whenever the service areas that must be covered are larger than a few hundred miles in diameter.

Although uncommon in purely commercial MSS applications, it is increasingly common for corresponding military missions (referred to as *Comm on the Move*, or COTM) to also require additional capabilities and flexibility beyond those described above for commercial MSS systems. Among these are the ability to support flexible bandwidth allocation between beams, and various processing functions, such as those related to electronic beam forming and the detection and suppression of jamming threats. Figure 16-7 illustrates how both typical commercial MSS requirements and corresponding requirements for military COTM applications can result in the development of payload architectures that are dramatically different than those for FSS and BSS communications systems.

Military Satellite Communications Systems

Although this chapter is presented mainly in the context of commercial communications applications, it should be noted that similar principles apply to non-commercial satellite communications missions, as well. In particular, while military satellite communications systems share important similarities with those of their commercial counterparts, there are important differences between the two that frequently result in fundamentally different architectures and designs. Table 16-8 provides a high level comparison between typical commercial and military satellite communications mission requirements, and how these differences may influence system or payload architectures, despite similar core FSS-like or MSS-like missions.

16.1.8 Concluding Remarks

This section describes the principal classifications used for commercial space communications missions, and key ways in which differences between these missions affect the payload communications architectures. Table 16-7 summarizes key attributes that differentiate the three principal categories of commercial satellite communications missions.

Section 16.1.6 described the process by which communications payload architectures are developed, including the way that the space communications architecture and

Table 16-8. Qualitative Comparison Between Key Factors that Differentiate Commercial and Military Satellite Communications Missions, and Corresponding Consequences for Satellite Communications Architectures.

	Commercial	Military	Typical Consequence to Satellite Communications Architecture
Coverage Areas	May be large or small, but typically stable over extended time horizons that can be predicted at the time initial architectures are being conceived, years in advance of launch	• Global or nearly global access may be required for a small number of users at small to moderate data rates • System may be required to provide large capacity (many users at high data rates) to theaters of operation on short notice (days to months) • Premium is placed upon flexibility, availability, and robustness over maximizing total capacity	Whereas commercial communications satellite payloads are usually able to make extensive use of fixed apertures and fixed bandwidth allocation, with limited beam switching and beam shape flexibility, military satellite communications payloads must often be designed to support steering or tracking to allow beams to be moved, or beamforming to allow beam shapes to be changed
Capacity	• Required capacity tends to be stable or slowly changing over mission lifetime • Often strong incentives to maximize total payload capacity (Gbps), with little or no need for coverage flexibility	• May require only low capacity in most places at most times, but the ability to augment communications capacity to much higher levels on short notice (hours/days) • Peak capacity is typically far below the peak capacity delivered by commercial satellite systems	Commercial satellite applications tend to be optimized for maximum capacity, while corresponding military applications tend to be optimized to robustly provide lower capacity on short notice as theaters of operation are defined and evolve
Reliability, Availability, and Robustness	• Business considerations motivate maximum reliability and availability to maximize return on investment • Operating environment is typically assumed to be permissive/ uncontested	• Maximum reliability and availability are motivated by the desire to maximize military readiness and speed • Contested operating environment must often be assumed	Commercial and military satellite communications missions may be driven to similarly high levels of reliability and availability, but may achieve those objectives based on different motivations, through different solutions, and in response to dramatically different threat and risk assessments. For example, whereas commercial satellite communications systems may emphasize component-level redundancy and ground terminal diversity to minimize weather-related availability reduction, military systems may add component protection/hardening, satellite level redundancy, and ground terminal diversity to provide additional protection against adversarial threats

the requirements of the overall space communications mission affect the communications payload architecture. Section 16.1.7 described how fundamental differences between different types of communcation missions can drive differences in communications payload architectures and designs.

Communications payload design, the subject of Sec. 16.3, is the process of defining the specific and detailed implementation of the communications architecture. Communications links were defined in Sec. 16.1.5. Although a certain amount of link analysis is required in support of communications payload architecture development, communications payload design can not even begin without link analysis, and link analyses remain as a core element of the entire communications payload design process. For this reason, Sec. 16.2 provides a detailed discussion of link analyses and link budgets, which are tools used to systematically perform link analyses.

Although the discussions in both 16.2 and 16.3 are generic, concepts are illustrated through examples based on FSS applications. Additional sample missions are provided in web-based supplements (Sec. 16.4) in which examples are provided for both the SCS and FireSat II missions, with the SCS examples illustrating analysis of alternative missions and solutions for situations in which the original mission requirements could not be satisfied by currently available technology.

A further web supplement covers special topics (Sec. 16.5), including extensions of the concepts described here to other payload types and topologies, including regenerative payloads, processing payloads, routing payloads, ground-based beam forming, hosted payloads, and non-commercial payloads.

16.2 Communication Link Analysis

An RF communication link consists of a transmitter and receiver separated by a free space distance. The transmitter antenna radiates an electromagnetic wave into free space, and the receiver antenna collects the portion of radiated energy which falls on it.

Figure 16-8 shows a satellite communications system which transmits information from a single ground terminal to a number of users, and receives information at the ground terminal from a number of users. The system includes uplink from Earth to satellite, and downlinks

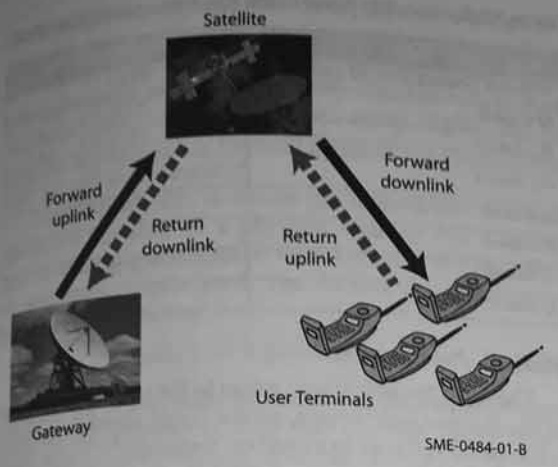


Fig. 16-8. Space Mission Communications System Uplinks and Downlinks. Forward communications links carry information toward user terminals, while Return communications links carry information away from user terminals. Both Forward and Return links include uplinks and downlinks.

from satellite to Earth. In addition to differentiating uplinks from downlinks, a further distinction is made between transmissions that are directed toward end users *(forward links)* and those that are directed away from end users *(return links)*. A space mission communications system may also contain *intersatellite links* (ISLs) that transmit information from one satellite to another, a feature that is used to relax signals between user and/or ground terminals that are beyond the line-of-sight perspective from one another from a single satellite.

The link consists of three parts: transmitter, the propagating electromagnetic signal, and a receiver. The link equation is expressed in terms of received power, transmitted power, the factor by which the transmit and receive antennas amplify signals (gain), and total losses:

$$P_{Rx} = P_{Tx} + G_{Tx} + G_{Rx} - L_{Tx+Rx} \qquad (16\text{-}1)$$

where P_{Rx} is the power at the receiver, after being collected by the receive antenna and passing through the input section of the receiver; P_{Tx} is the power radiated from the transmitter amplifier; G_{Tx} is the gain of the transmit antenna; G_{Rx} is the gain of the receive antenna; and L_{Tx+Rx} represents all losses, including those associated with the transmitter, the receiver, free space dispersion, and other losses, including atmospheric effects.

Losses include antenna pointing loss, propagation losses in the atmosphere, free space loss, and line losses at the transmitter and receiver.

Regardless of whether the link under consideration is a forward or a return link, link analysis must be performed on an end-to-end link between user terminal and ground terminal that represents the combination of two individual links: the uplink to the satellite and the downlink from the satellite. The principal result of the calculation is the ratio of received power to noise. To interpret the result, this ratio is compared to detection thresholds that are tabulated for appropriately selected operating parameters (Sec. 16.2.1.5). The difference between the power-to-noise ratio and the tabulated detection threshbold is a parameter referred to as the *link margin*. When this value is positive, the received signal can be detected, and the link is said to "close." When it is negative, however, the signal cannot be detected, and the link does not close. To ensure the robustness of a given link, the design team typically designs to a link margin greater than zero, with the magnitude of the positive margin usually being dependent upon the mission criticality of the link.

The *link equation* and the *link analyses* that use it are fundamental to the design of communications system of all types. Despite the simple appearance of Eq. (16-1), however, many physical processes and concepts are represented by each term of the equation. Before the details of the link equation can be adequately described or understood, it is therefore necessary to develop a basic foundation in regard to the following topics:

- Sec. 16.2.1.1 Electromagnetic Waves
- Sec. 16.2.1.2 Polarization
- Sec. 16.2.1.3 Decibels
- Sec. 16.2.1.4 Antennas
- Sec. 16.2.1.5 Modulation, Coding, and Spectral Efficiency
- Sec. 16.2.1.6 Bit Energy to Noise Ratio (E_b/N_o) and Coding Gain
- Sec. 16.2.1.7 Atmospheric Effects

Once these topics have been introduced, we will be prepared to examine in Sec. 16.2.2, the details of the link equation and link analysis in greater detail, and to introduce the *link budget* (step 3 of Table 16-1), a tool for performing link analyses that allows us to optimize design parameters and verify that each communications link will work successfully under the required conditions. Because of the importance of link analyses to the design of communications systems, this section closes with a detailed example of a link analysis and corresponding link budget in the context of an FSS communications link (Sec. 16.2.3).

16.2.1 Overview of Physical Principles of Satellite Communications Links

Fundamental to satellite communications is the propagation and use of electromagnetic waves in establishing a communication link. This section examines the fundamentals of electromagnetic waves and their characteristics for communications.

16.2.1.1 Electromagnetic Waves

The communications links used in space mission communications systems are based on the propagation of electromagnetic radiation through the Earth's atmosphere and the space beyond the atmosphere. Although

Table 16-8◀, Fig. 16-8, Eq. 16-1

the vast majority of satellite communications links are based on radio frequency (RF) electromagnetic waves, a small but important and growing fraction is based on signals that are generated in other parts of the electromagnetic spectrum. The discussion in this chapter is presented in the context of RF communications links.

Electromagnetic radiation is generated by accelerating electric charges and by changing electric currents. A current oscillating at a frequency, f, generates electromagnetic radiation in the form of propagating waves whose frequency match that of the source. For the purposes of this chapter, we can consider only the electric component of these waves, which can be described by the following relations.

$$e = |\mathbf{E}|\sin(\omega t) \qquad (16\text{-}2)$$
$$\omega = 2\pi f$$
$$\phi = \omega t$$
$$\lambda = v/f = c/f$$

where

- e ≡ instantaneous magnitude of the electric field vector, **E** (V/m),
- E ≡ amplitude of the electric field vector
- ω ≡ angular rotation of **E** (rad/s)
- ϕ ≡ phase angle of **E** (rad)
- t ≡ time (s)
- f ≡ frequency of the electromagnetic oscillations (Hz)
- λ ≡ wavelength (m)
- c = speed of light (2.99792458×10^8 m/s)

Figure 16-9 illustrates an electromagnetic wave with the electric and magnetic field vectors orthogonal both to one another and to the direction of propagation.

By convention, the portion of the electromagnetic spectrum used for satellite communications is divided into a number of ranges (or bands). Letter designations for these bands are listed in Table 16-9.

As was discussed in Sec. 16.1.1, regulatory contraints governing satellite operating bands impose a significant constraint on both the architecture and design of communications satellite payloads, with regulatory authority for these assignments being shared between a combination of national and international regulatory bodies.

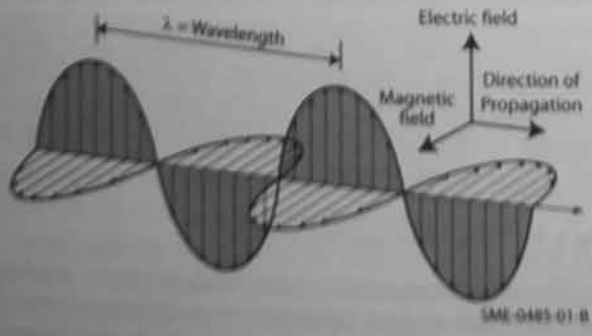


Fig. 16-9. Electromagnetic Wave.

Table 16-9, Fig. 16-9, Eq. 16-2

Table 16-9. Satellite Band in the Electromagnetic Spectrum

Band	Frequency (GHz)
V-band (intersatellite links)	40 to 75
Q-Band	30 to 40
K_a-band	18 to 30
K_u-band	11 to 18
X-band	7 to 11
C-band	4 to 7
S-band	2 to 4
L-band	1 to 2

16.2.1.2 Polarization

The term "polarization" refers to the spatial alignment of the electric field vector, with a single electromagnetic wave being referred to as either "vertically" (V) or "horizontally" (H) polarized. Although the designation is ultimately at the discretion of the user, polarization assignment is typically based on the orientation of the electric field with respect to the Earth's surface.

Circular or elliptical polarization can be obtained through the superposition of two electromagnetic waves having the same frequency that propagate in the same direction and have electric fields that are perpendicular to one another. In this case, the linearly polarized constituent waves combine through vector addition to form a single resultant wave, whose direction of propagation is the same as that of the constituent waves, but whose electric field rotates in a manner that is dependent upon the phase relationship between them. So-called "circular" polarization results if the constituent waves are of equal amplitude, while unequal amplitudes result in "elliptical" polarization. In each case, the polarization sense is *right-handed* (RHCP) if the electric field vector rotates clockwise as viewed in the direction of propagation, and *left-handed* (LHCP) otherwise.

The concept of "polarization" is important because antennas can be designed to respond with high specificity to electromagnetic radiation of one polarization sense, while strongly rejecting orthogonally polarized signals. As a result, by designating both frequency and polarization, polarization becomes a tool that communications engineers can use to effectively double the spectrum available for use. Therefore, the process of bandwidth allocation discussed in Sec. 16.1 takes both frequency and polarization into account.

16.2.1.3 Decibels

Communications engineering is a discipline that involves physical processes (such as amplification and attenuation) and physical quantities (such as power) that typically change by many orders of magnitude within even a small part of the physical system. As an example, the power flux density at a point 100 meters from a ground terminal aperture radiating information toward a geosynchronous satellite may be more than one hundred billion (100,000,000,000) times greater than the portion of the signal that ultimately reaches the receive antenna

of the satellite. Although it often appears to be a significant and unnecessary complication to those who are first introduced to the technique, communications engineers use logarithms to both compactly express values that might otherwise extend to many, many digits, and to quickly and efficiently perform many of the calculations that are routinely required.

The decibel (dB) is a base ten logarithmic unit that describes the ratio of a physical quantity (such as the power of an electromagnetic wave) relative to a specified or implied reference level. The decibel value of a number, P is computed as

$$P(\text{dB}) = 10 \log(P/P_{ref}) \tag{16-3}$$

where P_{Ref} is the value to which the decibel expressed quantity is referenced. Note that because decibels express the value of interest in terms of a ratio to a reference level or value, the reference value is critical to interpreting a value that is expressed in decibels.

As an example, 2 Watts (2,000 mW), can be expressed in decibel form as either

$$10\log_{10}(2 \text{ W}/1 \text{ W}) = 3 \text{ dBW} \tag{16-4}$$

or as

$$10\log_{10}(2{,}000 \text{ mW}/1 \text{ mW}) = 33 \text{ dBmW} \tag{16-5}$$

Although 3 dBW and 33 dBmW refer to the same physical quantity, the notation "dBW" tells the engineer that the value "3" is to be referenced to 1 W, whereas the notation "dBmW" indicates that the value "33" is to be referenced to 1 mW. The only time that a reference value is *not* provided is when the value being expressed represents a pure ratio (which is dimensionless) because pure ratios are obtained by dividing two values that each have the same units. Thus in our earlier comparison of the power received at a geosynchronous satellite, as compared with the power near a terrestrial transmitter, we would express the pure ratio as $1/10^{11}$. Thus, if the power near the transmitter were 10 μW, we might express the ratio in decibel form as

$$10\log_{10}(10^{-11} \text{ μW}/1 \text{ μW}) = -110 \text{ dB} \tag{16-6}$$

In addition to the advantages of compact notation, certain calculations are simplified once values have been expressed in decibels by virtue of the fact that the values being manipulated are proportional to the base-ten logarithms of the quantities they represent. As a result, we can multiply two quantities by adding their decibel equivalent. Likewise, once two numbers are expressed as decibels, relative to the same reference, we can find their ratio to one another by subtracting the denominator from the numerator. Similarly, exponentiation is replaced by multiplication (positive exponents) and division (negative exponents). Because communications engineering involves large amplification and attenuation factors, and frequently involves calculations with numbers that may differ by many orders of magnitude, decibel math can be simpler, faster, and less error prone, allowing simple addition and subtraction of small numbers to replace multiplication and division of numbers that extend to many digits. Once one becomes accustomed to this type of arithmetic, remembering the decibel equivalents of several common factors listed in Table 16-10 allows rapid calculations and unit conversions.

As examples, the decibel equivalent of a 134.9 W amplifier output power relative to a 1 W reference is 21.3 dBW. If this value needs to be used in a calculation in which other values are expressed in mW, we can express the amplifier's output power as either 134,900 mW, or as 21.3 dBW + 30 = 51.3 dBmW. Likewise, if we are interested in using an amplifier that has only half this power, we can compute that power as either 67.45 W or as 21.3 dBW – 3 db = 18.3 dBW. If the output of the original 139 W amplifier is attenuated by a factor of 40, we can use paper and pencil to find that the attenuated signal will be 134.9 W / 40 ≈ 3.37 W, or we can compute the value quickly by subtracting 16 dB (the decibel equivalents of two factors of two and one factor of ten) from 21.3 dBW to find the attenuated power level of 5.3 dBW.

Table 16-10. Decibel Equivalents Useful for Frequently Used Multiplication and Division, Including Unit Conversions.

Factor	Multiply by Adding/Divide by Subtracting Decibel Values
1	0
2	3
10	10
20	13
100	20
1,000	30

16.2.1.4 Antennas

Antennas are used to launch an electromagnetic wave into space or receive an electromagnetic wave, and to amplify the transmitted or received signals that travel in particular directions relative to the antenna. There are many types of antennas, including simple dipoles, direct radiating horns, reflector antennas, and direct radiating or reflected arrays. Many factors, including the operating frequency, coverage/pattern requirements, and the amplification factor delivered by the antenna determine which antennas are likely to be useful for a given application.

Antenna Gain

Although antennas may be as simple as a single dipole, they frequently include a number of components that operate together as precise mechanical assemblies of feeds, reflectors, and subreflectors. The total amplification factor of an antenna is given by its *gain*, G, a value that includes both the focusing effect of the antenna (directivity, D), which is a measure of the antenna's ability to concentrate energy in a given direction, and various losses (including ohmic transmission-line losses, L for example).

$$G = D - L \tag{16-7}$$

Table 16-10, Fig. 16-9◀, Eq. 16-7

Gain is commonly calculated from the frequency, physical size of the antenna aperture, and the antenna efficiency, according to the formula:

$$G = \eta\left(\frac{4\pi}{\lambda^2}\right)A \qquad (16\text{-}8)$$

where G is the gain, η is the dimensionless antenna efficiency, as a function of all losses, including ohmic losses, aperture efficiency, and spillover losses, A is the antenna area (m^2), and λ is the signal wavelength (m).

For an antenna with a circularly symmetric radiating aperture, such as a parabolic reflector antenna of diameter, D, this can be expressed in terms of the signal wavelength or frequency, f, as

$$G = \eta\left(\frac{\pi D}{\lambda}\right)^2 = \eta\left(\pi D\frac{f}{c}\right)^2 \qquad (16\text{-}9)$$

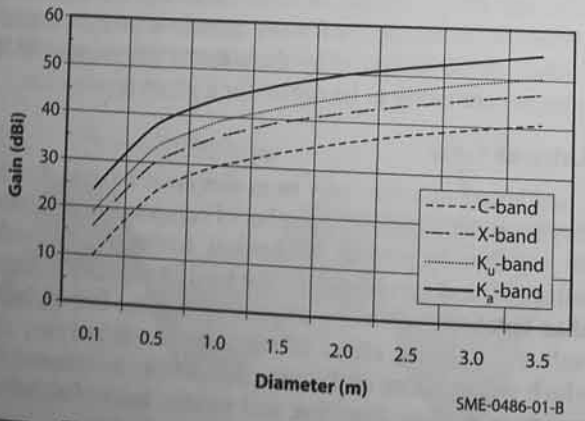
where, c denotes the speed of light.

Using the decibel-based arithmetic described in the last section, we can express the gain (dB) in terms of aperture size, antenna efficiency, and signal frequency as

$$G = 20.4 + 20\log(f) + 20\log(D) + 10\log(\eta) \qquad (16\text{-}10)$$

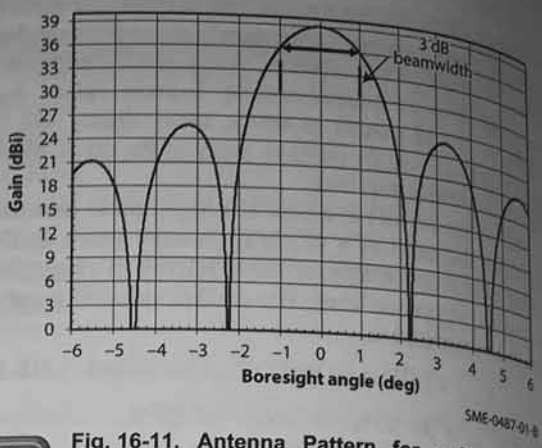
where, here, we have expressed f in GHz, combining this conversion with the constants π and c in to the constant 20.4. The relationships given in Eqs. (16-7) through (16-10) can be applied to antennas in a variety of contexts that are pertinent to satellite communications payload architecture and design, including calculations involving point-to-point links between ground terminals and the satellite, and highly directive user terminals. They cannot be applied to antennas whose shaping has been defined to yield irregular coverage contours.

Figure 16-10 illustrates the variation of antenna gain as a function of signal frequency and aperture size, assuming an antenna efficiency of 55%, and with the dBi notation indicating that the directivity in this instance has been normalized to that of an isotropic radiator.



Live Calc **Fig. 16-10. Gain vs. Aperture Diameter, Plotted for Operating Frequencies Common to Satellite Communications Systems.**

Table 16-10◀, Fig. 16-11, Eq. 16-11



Live Calc **Fig. 16-11. Antenna Pattern for a Hypothetical 0.75 m Parabolic Reflector with Uniform Illumination, and Operated at 14 GHz.**

The pattern of antenna gain as a function of angle is usually described in relationship to its peak, or *boresight* value. The half-power beamwidth is the angle with respect to boresight, within which the gain is within 3 dB (50%) of the peak. An antenna having a circularly symmetric aperture is approximated by the formula

$$\Theta = 21/(fD) \qquad (16\text{-}11)$$

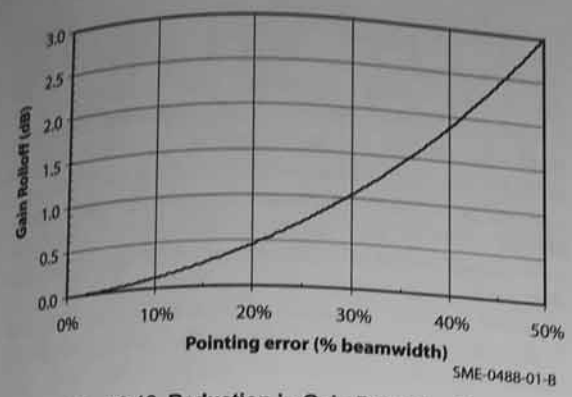
where Θ is the half-power beamwidth (deg), f is the frequency (GHz), and D is the antenna diameter (m).

Figure 16-11 shows the variation, or *pattern*, of such an antenna's gain as a function of the angle from boresight for a uniformly illuminated, 0.75 m diameter paraboloid operating at 14 GHz with 65% efficiency and 39 dBi peak gain. 72% of the total power from this antenna is radiated in directions that fall within the 2 deg half-power beamwidth, while 90% is radiated into the main lobe at angles between boresight and the first null, which appears at ~2.2 deg from the boresight.

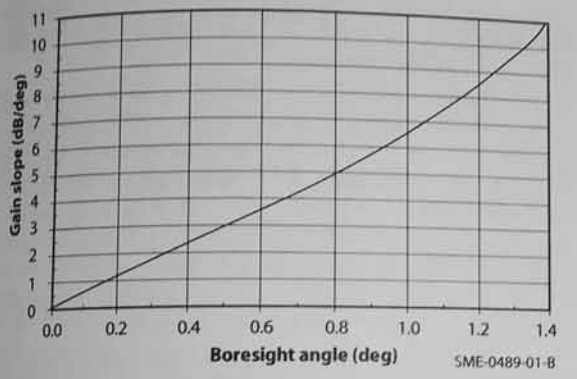
Pointing Errors

All antennas have pointing errors. These errors may be relatively static, as in the case of fixed misalignment, or variable, as in the case of wind driven pointing errors that can often be significant for large ground terminal apertures. One of the most important sources of pointing error in satellite antennas is related to thermal distortion that results from variable heating of different antenna and spacecraft structures, which changes as the orientation of the satellite and its antennas change with respect to the Sun, Earth, and other satellite structures, like solar panels and other antennas. When analyzing a communications link, it is important to recognize that the antennas on both ends of the link contribute to pointing errors.

Figure 16-12 illustrates the beam peak pointing loss (rolloff in gain) as a function of pointing error, expressed as a fraction of the 3 dB beamwidth of an antenna with a uniformly illuminated circular aperture. Though wide variations are possible, typical design points allow for a maximum pointing error that is 10% of the half-power beamwidth (e.g., 0.2 deg for 2 deg beam). Figure 16-12



Live Calc **Fig. 16-12. Reduction in Gain Resulting from Pointing Loss, Expressed as a Function of 3 dB beamwidth.**



Live Calc **Fig. 16-13. Absolute Value of Gain Slope vs. Boresight Angle for the Antenna Whose Pattern is Presented in Fig. 16-11.**

shows that at this limit, losses on the link associated with this aperture alone will amount to approximately 0.12 dB. At 20% of the beamwidth, pointing losses will grow to 0.45 dB.

In general, pointing loss is computed as the product of the *pointing error*, measured in deg, and the *gain slope*, measured in dB/deg. Because beam antenna patterns are typically flattest near the peak, pointing induced losses are usually smallest near the peak. In practice, it is therefore most important to consider pointing errors near the *edge of coverage* (EOC), where the gain is reduced and where gain slopes can be greatest. Figure 16-13 shows the gain slope corresponding to the pattern shown in Fig. 16-11. If for this antenna, the EOC point were taken to be 3 dB down from the peak, then the corresponding EOC gain slope would be 6.5 dB per deg, yielding a 1.3 dB EOC pointing loss for a 0.2 deg pointing error. One conventional practice for addressing pointing error is to use reflector shaping to design antennas in a manner that minimizes the variation in directivity across a coverage area. This has two effects: (A) making more efficient use of power, by reducing peak gains and increasing EOC gains, and (B) reducing gain slope within the coverage area. By designing antennas to have EOC gain margin that is greater than or equal to the predicted pointing losses, requirements will be satisfied as long as actual pointing is within the tolerances that have been predicted.

Shaped Beams

In many space communications missions, satellite antennas are required to cover an irregular area, such as the continental United States (CONUS), which subtends an angle approximately 5 deg × 3 deg as viewed from geosynchronous orbit. At 14 GHz, the antenna required to produce such a beam would be about 0.3 m × 0.5 m. If such an area were to be covered by a simple parabolic reflector, it would be necessary to adjust the pointing and the operating point of the system so that the gain at the edges of the specified coverage, and about 3 dB below the peak of the beam, would be compliant to the minimum requirements of the application. As a result, the gain over most of the coverage area would be greatly in excess of the requirements, while coverage would also be provided in areas where it is not likely to be useful (ocean areas) or may be either unwanted or even prohibited (Canada, Mexico, and Caribbean islands). Shaped reflector antennas allow the limited transmitted power and receiver sensitive resource of the spacecraft to be used more optimally by producing flatter patterns that conform more closely to the contours of the desired coverage area. In so doing, these antennas also maximize opportunities to coordinate the operations of new satellite communications systems with existing systems.

The directivity of an antenna whose shaping has been tailored can be estimated by considering the solid angle subtended within the coverage area. Given that the directivity of an antenna represents its ability to concentrate energy, and is often referenced to the directivity of an isotropic antenna, we can estimate the directivity of a shaped antenna by comparing the solid angle subtended within the coverage to the solid angle of an isotropic antenna (4π steradians = 41,253 deg^2). Because the directivity of an isotropic radiator is, by definition, 0 dBi, the maximum directivity achievable for an antenna that covers a smaller solid angle is given by

$$D_{Ideal} = 10 \log_{10}(41{,}253/A_\theta) \qquad (16\text{-}12)$$

where D_{Ideal} is in dB and coverage area, A_θ is measured in deg^2. Using the decibel-based arithmetic described earlier in this chapter, Eq. 16-12 can be recast as

$$D_{Ideal} = 46.15 - 10 \log(A_\theta) \qquad (16\text{-}13)$$

This provides a first level approximation for the link equation, the subject of Sec. 16.2.2. As an example, an ideal antenna that concentrates all of its radiated power into a spot beam whose area is 1.0 deg^2 would have directivity of 46.15 dBi.

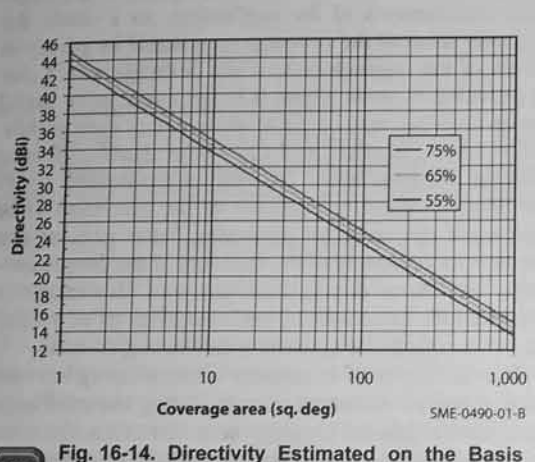
For a satellite in geosynchronous orbit at 35,786 km altitude above mean sea level, the Earth has an apparent diameter of 17.2 deg, corresponding to a solid angle of about 232 deg^2. Using Eq. (16-13), we find that an ideal Earth coverage antenna would have a directivity of 22.5 dB.

Table 16-10◀, Fig. 16-13, Eq. 16-13

To estimate the directivity of an aperture designed to provide coverage over CONUS, one would substitute the 13.3 deg^2 CONUS land area into Eq. (16-13) to obtain an estimate of the ideal directivity for such an antenna as approximately 34.9 dBi.

The ideal directivity calculations do not include antenna radiated efficiency, which is a measure of the power being radiated outside the desired coverage area. Antenna efficiencies typically range between 55% to 75%, corresponding to a –2.6 to –1.25 dB adjustment to the ideal directivity estimated from Eq. (16-13). Equation (16-14) modifies Eq. (16-13) to account correctly for antenna efficiency, where D_{Ideal} is in dBi. Equation (16-14) is plotted in Fig. 16-14 for several values of η.

$$D_{Ideal} = 10\log(41{,}253/A_\theta) + 10\log(\eta) \qquad (16\text{-}14)$$



Live Calc

Fig. 16-14. Directivity Estimated on the Basis of Coverage Area for Several Values of Antenna Efficiency.

Antenna Noise Temperature and Noise from Feed Losses

Equally important as accounting for signal power, the link analysis must also account for all significant sources of noise. In the broadest sense, noise includes all in-band sources of electromagnetic energy received other than those that are desired to support the communications mission. These include *black body thermal noise* radiated from the environment, noise generated from within the link, and non-thermal sources of interference from outside the system, including other RF transmitters that are not part of the space mission communications system being designed. Although they represent very different processes, the first two of these are usually considered together, while outside sources of interference are accounted for separately.

The first noise component that we consider is thermal radiation that originates from objects within the field of view of the receive antenna. Although perfect black bodies do not exist in nature, each object within the antenna's field of view radiates electromagnetic radiation with power spectral density that is approximated by that of a black body.

Table 16-10◀, Fig. 16-15, Eq. 16-16

$$N_o = kT \qquad (16\text{-}15)$$

where N_o is in W/Hz, k = 1.3806504×10^{-23} J/K = –228.6 dBJ/K is the Boltzmann constant, and T is the object's effective *brightness temperature* (K), the temperature that would be required of a black body in equilibrium with its environment to exhibit a similar radiation spectrum.

Despite the apparent simplicity of Eq. (16-15), generating a detailed and accurate prediction of the noise power received by an antenna is complicated by the fact that the noise power received must be weighted by both the solid angle of the objects within its field of view and by the antenna's pattern of directivity, yielding a value of the *antenna noise temperature*, T_A, given by

$$T_A = \frac{1}{4\pi}\iint T(\theta,\varphi)D(\theta,\varphi)\sin(\theta)\,d\theta\,d\varphi \qquad (16\text{-}16)$$

where $T(\theta, \varphi)$ is the brightness temperature of each point within the antenna's field of view, and $D(\theta, \varphi)$ is the antenna's directivity pattern.

In practice, although Eq. (16-16) is likely to be used in support of the detailed design to predict the *noise spectral density* for the specific antenna patterns, satellite location, and orientation, early link analyses for satellite receive antennas are likely to be based on more subjective estimates of the antenna noise temperature, based on a general understanding of the antenna's intended coverage and simpler models of the Earth's brightness temperature, such as that shown in Fig. 16-15.

Likewise, initial downlink analyses, which involve terrestrial receivers will be estimated, based on typical brightness temperature for the sky that ranges from approximately 5 K at zenith to about 150 K near the horizon at K_u band, and brightness temperature of approximately 300 K for ground that may occupy a portion of an antenna's field of view (including side lobes) under some circumstances.

The concept of *noise temperature* is used to characterize the noise performance of RF components. For antennas, the noise temperature comprises two components: (A) noise that originates from the thermally radiated energy of objects that appear within the antenna's field of view, and (B) noise that originates from the temperature of the antenna feed structure and transmission losses within the antenna and feed assembly itself.

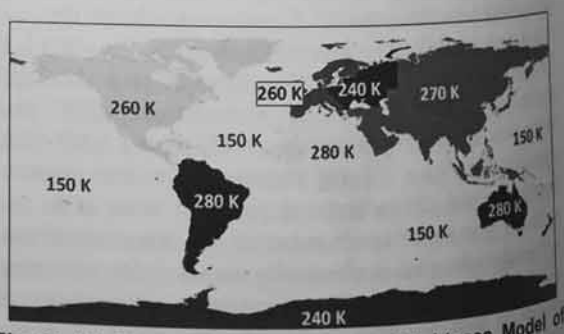


Fig. 16-15. ESA/EUTELSAT K_u-band Brightness Model of the Earth.

The second component of noise received by the antenna is associated with RF losses from within the antenna feed network itself, as well as losses in other passive components between the antenna feed and the receiver. While these losses will eventually be determined either by computer simulation or by direct measurement, estimates are used to support initial link analyses. In either case, it is typical for the noise power associated with antenna feed and other input losses to be represented as a simple resistive element that injects a constant *noise power spectral density*, $N_o = kT_L$, where T_L is the effective thermodynamic temperature of the modeled resistance. In Sec. 16.2.2.4, this noise source and thermally radiated noise received from objects within the field of view of the antenna will be combined with receiver noise to define an effective *system temperature*, T_S, that can be used to compactly represent the noise received from all three of these sources.

16.2.1.5 Modulation, Coding, and Spectral Efficiency

To carry information content between transmitter and receiver, the *electromagnetic waves* that comprise the operating bands of the communications system (carriers) are modified to encode data. Carrier *modulation* is performed by varying the carriers' frequency, amplitude, or phase. *Demodulation* is the process by which signals are recovered from modulated carriers. The modulation scheme and commonly used error-correction coding schemes play an important role in determining the power required to transmit a given amount of information on a communications link, and together with the characteristics of the filters used to minimize interference between adjacent carriers, determine the *spectral efficiency* (expressed in bps/Hz) with which information can be transmitted on that link. Both of these factors are fundamentally important parts of the link analyses that must be performed in support of both the communications architecture development and the detailed communications payload design.

Signals in modern communications systems are typically transmitted in digital format, at data rates measured in *bits/second* (b/s or bps). Modulators and demodulators operate on *symbols*, with the modulation scheme determining the number of bits that are encoded per symbol. Modulation techniques that function by varying carrier phase, known as *phase-shift keying* techniques [Ford, 2007], have distinct advantages over amplitude and frequency modulation techniques, both in terms of *bit energy* (E_b) and spectral efficiency (bps/Hz), and are therefore virtually ubiquitous in modern satellite communications applications. The most basic of these modulation schemes, *binary phase-shift keying*, (BPSK), encodes information onto carriers through the use of two phase states per symbol, thereby encoding 1 bit of information per symbol. Higher order modulations are also possible and in use, with *quadrature phase-shift keying* (QPSK), 8-PSK, and 16-PSK using 4, 8, and 16 phase states per symbol, respectively, corresponding to 2, 3, and 4 bits per symbol.

Although higher order modulation schemes improve spectral efficiency by encoding more bits per symbol than lower order schemes, this efficiency comes at a cost. With *m* representing the number of bits per symbol, the phase difference between allowable symbol states is inversely proportional to 2^m. As a consequence, implementing higher order modulation schemes increases the susceptibility of communications links to phase-noise-induced errors. In addition to requiring more careful attention to phase noise and phase stability in the communications payload, implementation of higher order modulation schemes requires signals to be transmitted with higher values of bit energy. Based on this, the tradeoff between spectral efficiency and total payload power consumption is an important factor for designers and operators of satellite communications systems.

Despite the fact that digital encoding is used on a communications link, a series of analog processes are involved in the formation of the digital bit stream. While the desire to maximize the total information carrying capacity motivates the designer to minimize bit energy, the *bit error rate* (BER) increases as bit energy becomes closer to that of the noise associated with these processes. *Forward Error Correction* (FEC) is used to mitigate the effects of increasing BER without requiring the implementation of more complicated communications channels and handshaking between transmitter and receiver. By encoding redundant bits onto the communications link, FEC algorithms allow the receiver a limited ability to detect and correct errors without the need for a return-channel to confirm that data have been successfully transmitted. The *code rate* of an FEC algorithm is defined as

$$\rho = n/(n+r) \tag{16-17}$$

where *r* represents the number of redundant bits added by the FEC algorithm for every *n* unique information bits transmitted on the link. Although increasing the coding rate improves the ability of the receiver to correct bit errors, thereby allowing bit energy to be reduced below the level that could otherwise be used to achieve a given BER, bandwidth use per information bit increases in direct proportion to the ratio *r*/*n*. Selection of the code rate for a particular application involves a balance between considerations for power consumption and spectral efficiency, with typical code-rate values ranging between approximately ½ and 9/10.

In practice, filters are used to minimize interference between carriers that occupy adjacent frequency bands. To optimize overall performance, a *rolloff factor* is specified for these filters, with the rolloff factor having the effect of creating small unused spectral bands between channels. Together, the modulation scheme, code rate, and rolloff factor determine the spectral efficiency, Γ, of a communications link as

$$\Gamma = m/[\rho(1+\alpha)] \tag{16-18}$$

where *m* represents the number of bits per symbol for the encoding scheme, ρ is the FEC code rate, and α is the filter rolloff factor. Table 16-11 gives examples of spectral efficiency (exclusive of filter rolloff) for several combinations of modulation and code rate.

Table 16-10◀, Fig. 16-15◀, Eq. 16-18

Table 16-11. Spectral Efficiency (exclusive of filter rolloff) for Several Combinations of Modulation and Code Rate.

Modulation Method	Bits per Symbol	Code Rate	Spectral Efficiency
BPSK	1	0.5	0.5
QPSK	2	0.5	1.0
QPSK	2	0.8	1.6
8-PSK	3	0.8	2.4
16-PSK	4	0.8	3.2

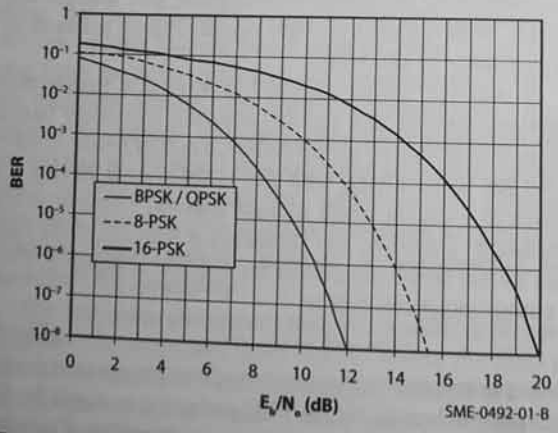
16.2.1.6 Bit Energy to Noise Ratio (E_b/N_o) and Coding Gain

The figure of merit used in link budgets is the bit error rate (BER), the principal performance metric that is a measure of quantization error in digital communications links. As was discussed qualitatively in Sec. 16.2.1.5, the probability of quantization errors increases as bit energy is reduced and becomes closer to the noise level that characterizes the communications link. In particular, provided that the noise source itself can be appropriately characterized, it is possible to predict the *BER* of a communications link as a function of the bit-energy to noise-spectral-density ratio, denoted as E_b/N_o, for a given modulation scheme and code rate. BER can be approximated by

$$BER \approx \frac{1}{m}\operatorname{erfc}\left(\sqrt{\frac{mE_b}{N_o}}\sin\left(\frac{\pi}{M}\right)\right) \qquad (16\text{-}19)$$

where N_o is the noise power spectral density, and the M-ary PSK modulation scheme encodes m bits per symbol, where $M = 2^m$. Figure 16-16 plots this relationship for BPSK/QPSK (same curve), 8-PSK, and 16-PSK modulation.

Figure 16-16 shows that for fixed bit energy, quantization error increases as the number of bits per symbol increases, and that to maintain a constant BER as the symbol rate increases, it is necessary to increase bit energy.

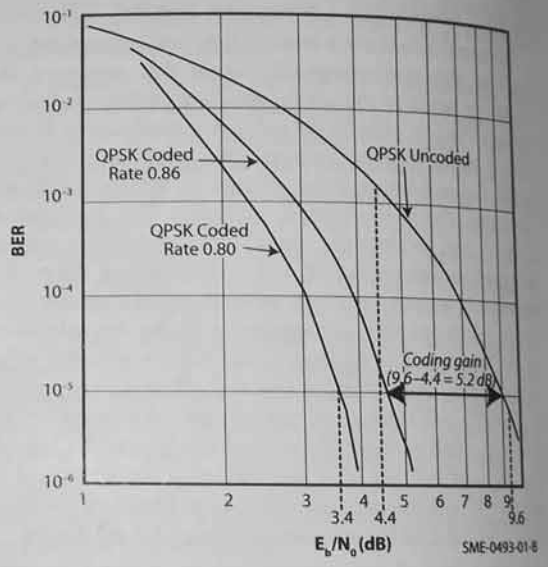


Live Calc

Fig. 16-16. Predicted BER Curves as a Function of E_b/N_o

The effect of FEC coding upon BER is illustrated in Fig. 16-17 for QPSK modulation with three code rates: $\rho = 1$ (no forward error correction), 0.86 (6/7), and 0.80 (4/5). *Coding gain* is the reduction in E_b/N_o from the value required for an uncoded link to the values that can be achieved by increasing redundancy (decreasing ρ). As was described in Sec. 16.2.1.5, for a fixed bandwidth allocation, this comes at the cost of reduced throughput on the communications link.

In the example illustrated in Fig. 16-17, an uncoded QPSK signal requires $E_b/N_o = 9.6$ dB to achieve a BER of 10^{-5}. With $\rho = 0.86$, E_b/N_o can be reduced to 4.4 dB, and with $\rho = 0.80$, it can be further reduced to 3.4 dB, for coding gains of 5.2 and 6.2 dB, respectively. Bit error rates vary widely with application, with voice applications allowing BER values in the range of 10^{-3}, and data links often requiring values in the range of 10^{-11}.



Live Calc

Fig. 16-17. Predicted BER as a Function of E_b/N_o

16.2.1.7 Atmospheric Effects

Gases and rain in the atmosphere attenuate electromagnetic radiation. Iononospheric effects are not significant at frequencies above 1 GHz.

Attenuation by Atmospheric Gases

Figure 16-18 depicts the total attenuation resulting from atmospheric gases as a function of frequency corresponding to the ITU "mean annual global reference atmosphere," with attenuation corresponding to a vertical path from sea level to the top of the atmosphere [ITU, 2011]. For elevation angles between 5 deg and 90 deg, attenuation increases by the cosecant of the elevation angle.

Also shown in Fig. 16-18 are contribution to total atmospheric attenuation due to dry air alone, and due to water vapor in the atmosphere. These curves clearly

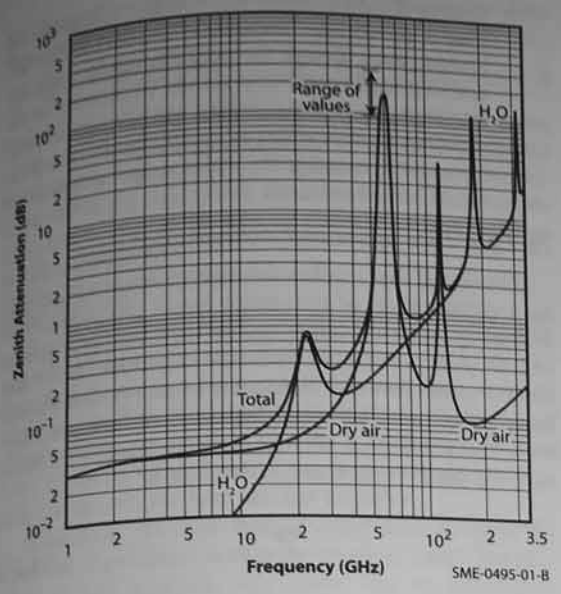


Fig. 16-18. Total Attenuation of Electromagnetic Radiation Versus Frequency due to Atmospheric Gases, for a Vertical Path from Sea Level to the Top of the Atmosphere (approximately 15 to 17 km at the equator)

show peaks due to water vapor at 22 GHz, and molecular oxygen at approximately 60 GHz. One interesting consequence of the enormous attenuation due to molecular oxygen is that terrestrial transmitters and sources operating near 60 GHz are very effectively isolated from those in orbit. Both the ITU and the US FCC have exploited this by designating bands in the range of 60 GHz for use as intersatellite links.

Precipitation Losses

Precipitation in the form of rain and wet snow attenuates RF signals through both absorption and scattering (depolarization). Precipitation-induced losses are dependent upon the product of a rate-dependent attenuation coefficient and the path length that the RF signals are required to traverse through rainfall regions within the atmosphere.

Because losses are a function of the concentration of water in the atmosphere, precipitation-induced losses are heavily dependent upon geographic and seasonal variations in weather. To provide a means for predicting rain precipitation effects upon satellite communications, the ITU [2002] has created a geographic database of rainfall rates, as well as algorithms for determining the frequency with which rainfall rates in excess of a user-specified value are likely to be exceeded. Using these resources, link analyses are often performed on the basis of availability objectives for the link being analyzed.

For a specific transmitter/receiver geometry (satellite and terrestrial terminal location), this is done by using the ITU model "*PROPA*" to determine a precipitation rate that will not be exceeded with frequency greater than the target availability. With this precipitation rate known, the attenuation due to precipitation can be computed. By accounting for this attenuation throughout the course of the link analysis, precipitation-driven link availability can be achieved so long as the overall link budget closes.

Scintillation

Scintillation refers to variations in signal amplitude, phase, and angle-of-arrival result from refractive index variations within the troposphere and the ionosphere, and occur over short time frames; typically less than a second. Within the troposphere, refractive index changes are driven by weather, and are frequency independent. Ionospheric refractive index changes are frequency dependent, and result from localized variations in charged particle densities, which can vary rapidly in the various layers of plasma that comprise the ionosphere.

Although a budget is allocated in link budgets to accommodate signal variations due to scintillation, scintillation does not ordinarily represent significant constraint to the design of satellite communication links above 1 GHz. For example, though peak-to-peak variations as large as 1 dB were very infrequently observed in association with heavy rainfall events, K_a-band data from the 20.2 and 27.5 GHz beacons of the ACTS satellite at K_a-band showed scintillation values typically below 0.2 dB. Although scintillation amplitudes may be as much as 5 dB or more at frequencies below about 1 GHz, they are typically no more than a few tenths of a dB for L-band signals, and smaller still at higher frequencies.

16.2.2 Link Equations and Link Budgets

As was mentioned earlier, link analyses are fundamentally important during both the synthesis and assessment of architectural alternatives, and during the development of the detailed communications payload design. Using Eq. (16-1) as a starting point, the analysis proceeds by accounting for increases and decreases in power through each part of the link from the transmitter and its antenna, and through space to the receiver antenna and electronics where the signal is measured and decoded. In this manner, the link analysis is treated much as a budget, in which positive allocations are required from specific identified sources, and negative allocations are made to accommodate expected losses. Because of the high cost and complexity associated with launching and operating a communications satellite, there are enormous pressures to ensure that the communications payload is highly optimized to deliver the best possible combination of performance and reliability. Given this, each component of the *link budget* is scrutinized to ensure that it is realistic and achievable, without being overly conservative [Gordon and Morgan, 1993]. In the end, positive allocations (power, and power amplification) must balance with negative allocations (attenuation, other losses, and margin) for the budget to close.

Table 16-11◀, Fig. 16-18, Eq. 16-19◀

Because noise is added at various stages through both the uplink and the downlink on payloads, link budgets must be computed for these payloads in both the uplink to the satellite and the downlink to a terrestrial terminal, and for both forward and return links. The principal metric in this budget is the margin, or excess above the minimum threshold requirement for the composite link, in received signal-to-noise power ratio at the receiver, E_b/N_o. The link is said to "close" if the margin to the requirements is ≥ 0. These requirements are determined and widely tabulated according to the modulation scheme and code rate that are to be used for the link. In practice, it is common to design to a positive margin target, referred to as implementation margin, whose specific value will vary depending upon various elements of perceived risk, as a means of accommodating normal variations in unit performance and unforeseen factors associated with production, integration, and test.

In the subsections that follow, we first consider budget entries that account for signal power (Secs. 16.2.2.1 through 16.2.2.3), after which we describe how noise factors into the link budget (Secs. 16.2.2.4 and 16.2.2.5). We conclude the discussion of the link budget by showing how signal power and noise factors are combined to compute the carrier-to-noise ratio (Sec. 16.2.2.6), and how margin is computed for the overall link that comprises both uplink and downlink. We end in Sec. 16.2.2.7 with a sample budget computed for an FSS satellite communications link.

16.2.2.1 Transmitted Power and Transmitter Output Losses

A transmitter receives baseband data, modulates it onto a higher-frequency carrier, amplifies it, and broadcasts it via an antenna. So far as the link analysis is concerned, the principal product of the transmitter is a measure of the power radiated in the direction of the peak of the transmit antenna directivity pattern: *equivalent isotropic radiated power*, *EIRP*. The *EIRP* (dBW), is equal to the power of the transmitter (dBW) minus losses associated with output transmission lines and other hardware between the transmitter and antenna (dB), plus the gain of the transmit antenna (dBi). The result corresponds to the power that an isotropic radiator with no output losses would be required to have in order to radiate with power flux density equivalent to that at the beam peak of the transmit antenna.

$$EIRP = P_{Tx} + G_{Tx} - L_{output+Tx\ antenna} \qquad (16\text{-}20)$$

Two things are worth noting at this point:

- The value of the transmitter power used in Eq. 16-20 is the *RF output power of the high-power amplifier*, which is related to the *DC bus power* required to operate the high-power amplifier through the amplifier's efficiency factor, which varies from amplifier to amplifier, and as a function of the amplifier's operating parameters
- It is common to carry the nominal maximum RF output power (or "nameplate" power) in the link budget, even though the amplifier might not be operated at that power in practice. To account for the intended operating point, the *output backoff* (OBO) value is carried as an additional loss term to represent the factor by which output power is reduced below nominal

16.2.2.2 Free-Space and Atmospheric Losses

The signal that reaches the receive antenna is reduced by a combination of free-space losses and atmospheric scattering and attenuation losses. So-called propagation or *free-space loss* (L_S) is by far the largest loss factor, and simply represents the reduction in power flux density that results from geometric dispersion of the electromagnetic radiation as it travels through space. Free-space loss, L_S, can be expressed in decibels as

$$L_s = \left(\frac{4\pi r}{\lambda}\right)^2 \qquad (16\text{-}21)$$

where r is distance between the transmit and receive antennas, and λ is the wavelength of the RF carrier. Free space loss L_S (dB) can be be expressed as

$$L_S = 92.45 + 20\log_{10}(r) + 20\log_{10}(f) \qquad (16\text{-}22)$$

where r is the distance (km), and f is the carrier frequency (GHz).

The signal that reaches the receive antenna is further reduced by atmospheric losses. In practice, these are best estimated either by using geographic databases and models of precipitation rates and attenuation factors published by the ITU, or corresponding rain-rate maps and tabulations, each of which is provided with detailed use instructions and guidelines.

16.2.2.3 Received Power

The accounting for changes in power that occur at the receiver is similar and complementary to that described in Sec. 16.2.2.1 for changes in power that occur within the signal path of the transmitter. In this case, however, the signal is first increased by the gain of the receive antenna, then reduced by losses (principally transmission-line) before the signal reaches the receive amplifier.

One final factor that must be accounted for in predicting the power available at the receiver is pointing loss, which is the result of pointing errors at both the transmit and the receive antennas. Pointing loss can be estimated based on a bottoms-up estimate of maximum pointing errors for both transmit and receive antennas, combined with knowledge of the predicted antenna patterns, or by setting a loss budget that is consistent with previous experience. If estimated on the basis of estimated pointing errors, pointing loss is determined by finding the directivity of the transmit or receive antenna at the boresight angle corresponding to the maximum predicted pointing error, and subtracting this from the antenna's peak directivity.

Table 16-11◀, Fig. 16-18◀, Eq. 16-22

The total power at the receive amplifier can therefore be expressed as

$$C = EIRP + G_r - L_S - L_{atm+prec} - L_{in} - L_{Tx+Rx} \quad (16\text{-}23)$$

where *EIRP* and L_S have the meanings and units described previously, $L_{atm+prec}$ represents atmospheric and precipitation losses (dB), G_r is the gain of the receive antenna (dBi), L_{in} represents input losses to the receiver (dB), and L_{Tx+Rx} is the sum of transmitter and receiver pointing losses (dB).

16.2.2.4 Noise

The link budget must account for all significant sources of noise in the link. In Sec. 16.2.1.4, we described the *noise-power spectral density* and absolute noise power that originate from thermal radiation of objects within the receive antenna's field of view, and from RF losses within the antenna feed and passive components between the feed and the receiver.

In addition, the link budget must also account for noise that is generated within the receiver itself. For purposes of link analysis, this noise can be adequately characterized in terms of the *noise figure*, which is defined for a cascaded chain of one or more units as the ratio of the signal-to-noise at the input of the device or chain to the signal-to-noise at the output.

Noise figures are specified for all active units in the payload, and representative values are therefore readily available for link analyses that may be performed at any stage of the development process. It is important to note that **noise figures are always quoted at a reference temperature of $T_0 = 290$K.**

For the receiver's input signal chain, we can determine the *effective system noise temperature*, T_S, as a composite of the antenna and input-loss noise temperatures as

$$T_S = T_A(L) + (1-L)T_L + (F-1)T_o \quad (16\text{-}24)$$

where T_A and T_L are the antenna noise temperature and the effective temperature of the feed and other passive input components, respectively, L is the loss through the antenna feed and passive components upstream of the receiver, expressed as a ratio of the output power to input power, and F is the receiver noise figure. [Note that F is usually quoted in dB, and must be converted to its non-logarithmic representation for use in Eq. (16-24).]

Using the effective system noise temperature given by Eq. (16-24), we can compactly represent all three of the significant sources of noise within the link other than RF interferers, which are accounted for separately. Thus, the noise power spectral density at the receiver is simply computed as $N_o = kT_S$ [W/Hz]. Likewise, by multiplying this by the *noise bandwidth**, B_N, of the receiver or the receiver prefilter we can determine the expected value of the measurable in-band noise

$$N = kT_S B_N \quad (16\text{-}25)$$

which can be expressed in terms of decibel values as

$$N = -228.6 + T_S + B_N \quad (16\text{-}26)$$

where N is in dBW, −228.6 is the Boltzman constant in dBJ/K, T_S is in dBK and B_N is in dB-Hz.

16.2.2.5 Receiver G/T

The ratio of the receive-antenna gain to the noise temperature of the receiver, T_S, obtained from Eq. (16-24) is referred to as *G/T*, and is used in link analyses to compute the signal-to-noise ratio. As such, *G/T* is also the principal figure of merit used to characterize receiver sensitivity, just as *EIRP* is the principal figure of merit used to characterize the transmitter. Like *EIRP*, *G/T* is typically expressed in decibels, and is computed as

$$\frac{G}{T} = G_R - T \quad (16\text{-}27)$$

where *G/T* has units of dB/K, and G_R and T have already been expressed in dBi and dBK, respectively.

16.2.2.6 Receiver C/No

The received signal power, C, from Eq. (16-23), and the noise power, N, from Eq. (16-26) come together in the *carrier-to-noise* ratio, which can be computed by combining these equations to obtain the result

$$C/N = (EIRP + G_R - L_{Comb.}) - (-228.6 + T_S + B_N) \quad (16\text{-}28)$$

where all losses have been combined into the single term, $L_{Comb.}$.

Combining with Eq. (16-27), this reduces to

$$C/N = EIRP + G/T - L_{Comb.} - B_N + 228.6 \quad (16\text{-}29)$$

To obtain the bandwidth-independent ratio of carrier power to noise power spectral density, we multiply the carrier-to-noise ratio by the bandwidth (equivalent to adding B_N to Eq. (16-29), which has all terms expressed in decibels):

$$C/N_o = EIRP + G/T - L_{Comb.} + 228.6 \quad (16\text{-}30)$$

16.2.2.7 Available E_b/N_o and Composite Link Performance

The final step of the link analysis, which involves determining and interpreting the values of E_b/N_o for both the uplink and the downlink, varies slightly, depending upon whether the communications payload is to be implemented as a *transponder* or as a *regenerative* payload. The reason for this is that whereas *transponding payloads* amplify and re-transmit both the signal and the noise that are uplinked to them, regenerative payloads demodulate the signals carried on the uplink, stripping away uplinked noise before remodulating and transmitting signals onto the downlink.

Table 16-11◀, Fig. 16-18◀, Eq. 16-30

* Note that the *noise bandwidth* of a component may be different from the nominal (typically 3 dB) bandwidth quoted or measured for a device because of noise that is admitted to the link within the rolloff region of the applicable filter

Because the transponding payload includes signal-to-noise contributions from both the uplink and downlink beams, the last step of the link analysis requires the uplink and downlink to be treated as a single combined link. For regenerative payloads the analysis is performed independently for uplink and downlink beams. Once E_b/N_o has been computed for the appropriate link, this value is compared against the value required for the modulation scheme and code rate to be used. The difference between the predicted and required values of E_b/N_o is the *link margin*, the principal metric of link quality. This same analysis must be completed for each link that the satellite must support, including both forward (toward end users) and return (originating from end users) links.

Because C represents the total RF power required to carry all of the information in the signal that is to be transmitted, E_b/N_o is obtained from C/N_o by simply dividing by the data rate, R_b, that the carrier is required to support. Performing the calculation with each of the terms expressed in decibels, we compute R_b (dB-Hz) = $10 \log_{10}(R_b \text{ (bps)})$, with which we find

$$E_b/N_o = C/N_o - R_b \qquad (16\text{-}31)$$

For regenerative payloads, we can interpret this result separately for each uplink and downlink by comparing these results with the threshold value required to close the link for the modulation scheme and code rate to be implemented.

For transponding payloads, however, we must first determine a composite value for E_b/N_o by expressing E_b/N_o non-logarithmically for both uplink and downlink, adding these intermediate results, and re-expressing the combined result in decibels so that

$$\left(E_b/N_o\right)_{Composite} = 10\log_{10}\left(10^{-(E_b/N_o)_{uplink}/10} + 10^{-(E_b/N_o)_{downlink}/10}\right) \qquad (16\text{-}32)$$

Because satellite communications payloads always have lower gain antennas than those used at ground terminals, and because ground terminals often are able to implement higher-power amplifiers than those that can be used on the satellite payloads with which they are linked, the composite value of E_b/N_o for forward communications links is always dominated by the downlink component that represents the satellite-to-user-terminal portion of the overall link. Likewise, even if the return-link *EIRP* from user terminals to the satellite is greater than the *EIRP* for the downlink to the ground terminal, the ground terminal receiver will always have much greater sensitivity (as reflected in its G/T) than the corresponding satellite receiver. The composite value of E_b/N_o for return links is therefore dominated by the uplink, which again represents the user terminal-to-satellite portion of the overall link. Thus in each direction, it is the combined performance limits of the satellite and user terminals that constrain link performance most.

Table 16-11◀, Fig. 16-18◀, Eq. 16-33

To interpret the value of E_b/N_o for each link that has been analyzed, these results must be compared with the threshold value required for the coding scheme that is to be implemented. Table 16-12 lists key parameters for a variety of modulation and code-rate combinations that are supported by the Digital Video Broadcasting-Satellite, Second Generation (DVB-S2) standard.

Link margin, M_L is given by

$$M_L = (E_b/N_o)_{\text{Predicted}} - (E_b/N_o)_{\text{Required}} \qquad (16\text{-}33)$$

The link closes when the link margin is zero, which from Eq. 16-33 is clearly the condition for which $(E_b/N_o)_{\text{Predicted}}$ is precisely equal to $(E_b/N_o)_{\text{Required}}$. The link does not close if link margin is negative. To ensure that the system will function as intended despite risk factors such as adverse stackup of normal variations in unit performance and unforeseen factors associated with production, integration, and testing, the design team typically designs to a link margin greater than zero, with the magnitude of the positive margin usually being dependent upon the mission criticality of the link.

16.2.3 Communications Satellite Link Budget Example

We conclude by providing a detailed description of the link budget for a representative transponder-based FSS satellite to be placed in geosynchronous orbit and operated at K_u-band to provide service to users in the continental United States (CONUS). The spectrum allocation for this example is 14.0 to 14.5 GHz for uplinks and 11.7 to 12.2 GHz for downlinks. The communications payload will have 24 channels, with 12 on each polarization. Each channel has a 36 MHz bandwidth, and adjacent channels are separated by 5 MHz guard bands so that all channels fit within the 500 MHz band that has been allocated for this satellite. Table 16-13 shows the link budget in the forward direction for this communications link for two user terminal profiles:

1. "Small Terminal": 0.50 Mbps data rate and 1.3 m user terminal aperture
2. "Large Terminal": 44 Mbps data rate and 2.5 m user terminal aperture

Wherever applicable, Table 16-13 provides references to equation numbers and data sources from earlier in the chapter. Entries that are annotated as "Given" are those that are subject to regulatory constraints, requirements of the design, or constraints dictated by components likely to be selected in the design, and are assumed to be outside the control of the link designer. These include:

- Uplink and downlink frequencies (specified by ITU and national authorities, such as the US Federal Communications Commission)
- Channel bandwidth (design requirement, in this case to provide compatibility with existing services)
- Ground-terminal output losses (typical for existing facilities)

Table 16-12. Symbol Rate, Spectral Efficiency, and E_b/N_o Required for Link Closure for Various Combinations of Modulation and Code Rate that are Supported by the Digital Video Broadcasting-Satellite, Second Generation (DVB-S2) Standard.

Modulation	Bits/ Symbol	Code Rate	Spectral Efficiency	E_b/N_o
QPSK	2	1/4	0.49	0.75
QPSK	2	1/3	0.66	0.59
QPSK	2	2/5	0.79	0.73
QPSK	2	1/2	0.99	1.05
QPSK	2	3/5	1.19	1.48
QPSK	2	2/3	1.32	1.89
QPSK	2	3/4	1.49	2.31
QPSK	2	4/5	1.59	2.67
QPSK	2	5/6	1.65	2.99
QPSK	2	8/9	1.77	3.73
QPSK	2	9/10	1.79	3.89
8PSK	3	3/5	1.8	3.00
8PSK	3	2/3	2.0	3.65
8PSK	3	3/4	2.2	4.43
8PSK	3	5/6	2.5	5.41
8PSK	3	8/9	2.6	6.46
8PSK	3	9/10	2.7	6.7
16APSK	4	2/3	2.6	4.76
16APSK	4	3/4	3.0	5.49
16APSK	4	4/5	3.2	6.03
16APSK	4	5/6	3.3	6.42
16APSK	4	8/9	3.5	7.42
16APSK	4	9/10	3.6	7.61

- Satellite line loss (typical value for satellite payloads of this type)
- System noise temperature (defined by coverage area, input losses and likely component selection)
- Implementation Margin = 3 dB (conservative for initial scoping analyses)

Entries labeled as "Inputs" are parameters that can be optimized and traded by the link designer to achieve link closure with the required margin. These include:

- User data rate (not common for this to be a tradable parameter)
- Modulation and coding (tradeoff of bandwidth and capacity)
- Ground-terminal aperture size and power
- Satellite antenna directivity (subject to satisfying coverage requirements)
- Satellite power per channel
- User terminal size and power (often not subject to trade because of requirements to support terminals that are already deployed)

Entries that are designated as "Assume" represent initial estimates, and will be validated and optimized as the payload design evolves and becomes more mature. These include:

- Amplifier backoff
- Input section losses between the satellite antenna and receiver
- Antenna efficiency

The user-terminal size and the data rata, modulation, and coding are adjusted so that the end-to-end links close with the required link margin Satellite *EIRP* and *G/T* requirements are computed for the coverage areas that have been specified by the customer, based on ground-terminal *EIRP*, user-terminal *G/T*, and desired data rate, modulation and coding, and 3 dB margin.

With support from antenna design engineers a single antenna with a 3.5 m × 2.4 m shaped reflector and a 0.7 m diameter subreflector is defined to support both receive (up) and transmit (down) links on orthogonal linear polarizations.

Forward Uplink Sample Link Budget Description

After considering several alternative trades between free design parameters, the budget closes with a set of parameters that appear to be realistic and achievable. In the final configuration, the ground-terminal is specified to have a 10 m antenna and one 200 W high-power-amplifier for each 36 MHz uplink channel.

In the case of the "Small Terminal" scenario, the uplink *EIRP* for each ground-terminal channel is shared among multiple users, whereas the uplink *EIRP* is dedicated to single terminals in the "Large Terminal" scenario.

The ground terminal backoff is 4 dB for the small terminal, multiple carrier case and 2 dB for the large terminal, single carrier case. Ground-terminal output losses are fixed at 2 dB.

At 14 GHz, the free-space loss (L_S) between the ground terminal and the geosynchronous satellite at the 38,000 km gateway-to-satellite range is 207 dB. Atmospheric loss for the specified ground terminal location and required link availability is calculated to be 10 dB. The satellite receive antenna meets the customer specified *G/T* requirement of 6 dB/K. The predicted value of E_b/N_o on this link is approximately 20 dB, a value that is sufficiently large as to contribute very little to the value of E_b/N_o for the overall link.

Forward Downlink Sample Link Budget Description

The *EIRP* is calculated for each scenario based on the 34.1 dBi *edge-of-coverage* (EOC) transmit gain for the antenna described above, assuming that each channel will be served by a 150 W high power amplifier.

At 11.7 GHz, the space loss for the GEO satellite at the maximum satellite-to-user range of 39,000 km is 205.9 dB*. Atmospheric loss for the user-terminal location considered in this case is 7 dB.

Table 16-12, Fig. 16-18◀, Eq. 16-33◀

Table 16-13. Sample FSS Link Budget, Forward Direction. Note that equation numbers and data sources are referenced in the right-hand column.

FSS Forward Link Cases	Units	Small user	Large user	Comments and References	
Uplink frequency	GHz	14.00	14.00	K_u-band uplink	Given
Gateway terminal type		Tracking	Tracking		
Diameter	m	10	10		Input
Beamwidth	deg	0.15	0.15		Eq. 16-11
Gain	dBi	60.7	60.7	55% efficiency	Eq. 16-10
Transmit power	W	200	200		Input
Backoff and line loss	dB	–6.0	–4.0	2 dB line loss + backoff	Assume
***EIRP*, gateway**	dBW	77.7	79.7	Gateway	Eq. 16-20
EIRP per user		62.3	79.7	*EIRP* / no. users	Calculation
Propagation range	km	38,000	38,000	Gateway to satellite	Given
Space loss	dB	–207.0	–207.0		Eq. 16-22
Atmospheric losses	dB	–10.0	–10.0	99.97% availability	
Net path loss	dB	–217.0	–217.0		Calculation
Satellite antenna, type		3.5 × 2.4 m, shaped		Gregorian	
Coverage area	deg^2	**13.30**	13.30	CONUS	Given
Antenna efficiency	%	70%	70%		Assume
Gain	dBi	33.4	33.4	Derived from coverage	Eq. 16-14
Line loss on satellite	dB	–2.0	–2.0		Assume
Received carrier power, *C*	dBW	–123.3	–105.9		Eq. 16-23
Sys. noise temp	dB-K	27.2	27.2		Derive
G/T	dB/K	6.2	6.2		Eq. 16-27
Receiver C/N_o	dB-Hz	78.1	95.5	k = –228.6	Eq. 16-30
Data rate per user	dB-Hz	57.0	76.4	Single user data rate	Calculation
Available E_b/N_o, uplink	dB	21.1	19.1		Eq. 16-31
Downlink frequency	GHz	11.70	11.70	K_u-band downlink	Given
Satellite antenna, type		3.5 × 2.4-m, shaped		Gregorian	
Coverage area	deg^2	13.30	13.30	CONUS	Given
Antenna efficiency	%	70%	70%		Assume
Antenna gain	dBi	33.4	33.4	Derived from coverage	Eq. 16-14
Satellite *TX* power	W	150	150	Per channel	Input
Backoff and line loss	dB	–5.5	–4.5	2.5 dB line loss	Assume
EIRP per channel	dBW	49.6	50.6		Eq. 16-20
EIRP per user	dBW	34.2	50.6	*EIRP* / no. users	Calculation
Propagation range	km	39,000	39,000	Satellite to users	Given
Space Loss	dB	–205.6	–205.6		Eq. 16-22
Atmospheric losses	dB	–7.0	–7.0	99.97% availability	
Net path loss	dB	–212.6	–212.6		Calculation
User terminal type		Small user	Large user		
Diameter	m	**1.2**	**2.3**		Input
Beamwidth	deg	1.5	0.8		Eq. 16-11
Gain, *G*	dBi	40.8	46.4	55% efficiency	Eq. 16-10
Line loss	dB	–2	–2		Assume
Receive carrier power, *C*	dBW	–139.7	–117.6	Single carrier	Eq. 16-23
System noise temperature, *T*	dB-K	27.0	27.0		Derive
G/T	dB/K	13.8	19.4		Eq. 16-27
Receiver C/N_o	dB-Hz	61.9	84.0	k = –228.6	Eq. 16-30
Data rate per user	dB-Hz	57.0	76.4	Single user data rate	Calculation
Available Eb/No, downlink	dB	4.9	7.6		Eq. 16-31
End-to-end E_b/N_o	dB	4.8	7.3	Uplink plus downlink	Eq. 16-32
Modem implementation loss	dB	–1.2	–1.2	QPSK	Table 16-11
Required E_b/N_o	dB	0.6	3.0	DVB-S2 waveform	Table 16-12
Link Margin	dB	**3.0**	**3.1**		Given
Channel bandwidth	MHz	36	36		Given
Number of channels		24	24	500 MHz, 2 polarizations	Sec. 16.2.1.5
No of users/ channel		35	1		Input
Single user data rate	Mbps	0.500	44		Input
Code rate		0.33	0.83	DVB-S2 waveform	Table 16-12
Single user bandwidth	MHz	1.015	35.52	QPSK, factor 1.34×	Sec.16.2.1.5
Bandwidth used / channel	MHz	35.5	35.5		Sec. 16.2.1.5
Total capacity	**Mbps**	**420**	**1,056**		

For the "Small Terminal" scenario, QPSK modulation and code rate 1/3 are assumed. From Table 16-13, we see that this set of operating parameters will require $E_b/N_o \geq 0.6$ dB for closure on the combined link. For the "Large Terminal" scenario, QPSK modulation and code rate 5/6 are assumed, requiring $E_b/N_o \geq 3.0$ dB for closure on the combined link. In each case the tradable parameters have been varied so that these links close with the required implementation margin.

Link Capacity

The *link capacity* shown at the bottom of Table 16-13 for each scenario is calculated as the number of channels (24) multiplied by the number of user terminals and the maximum required single user data rate. Link capacities of 420 and 1,056 megabits per second (Mbps) are computed for the "Small Terminal" and "Large Terminal" scenarios, respectively.

* Note that the free-space loss for the downlink in this example is smaller than that for the uplink, despite the larger propagation range for the downlink. This is the result of the frequency dependence of the free-space loss, as described by Eqs. (16-21) and (16-22).

16.3 Communications Payload Design

Communications payload design is the process of developing a specific and detailed description as to how the communications payload architecture (Sec. 16.1) is to be implemented. Link analysis (Sec. 16.2) is used in the development of the communications payload architecture, but assumes an even more significant role in the payload design process. Link budgets are maintained and updated throughout the design process to continually refine predicted performance, verify compliance to system-level requirements, and to support analyses for troubleshooting and optimization of subsystems and lower-level assemblies.

Table 16-14 describes the typical process flow for the design of space mission communications payloads. As

Table 16-14. Typical Process Flow for Communication Payload Design.

	Step	Description	Principal Assumptions/Inputs	Outcome
3.1	***Communication Payload Connectivity and Aperture Design***	Define payload connectivity and detailed antenna implementation to meet the initial mission/link requirements, and also the spacecraft/mission constraints	• Payload connectivity requirements, orbital nodes, beam to beam connectivity • Aperture sizing, link characteristics, and frequency reuse requirements • To meet the mission and link requirements • To meet the spacecraft constraints for size, mass, power, thermal	• Payload implementation and subsystem definition • Antenna implementation and subsystem definition
3.2	***Communication Payload Block Diagram Design***	Define detailed frequency channelization and corresponding amplification, filtering, frequency conversion, switching, routing between receive and transmit apertures	• Understanding of typical unit performance specification ranges and limits • Connectivity, connectivity flexibility, and reliability from top-level requirements flow-down	• Detailed frequency channelization • Payload block diagram illustrating component functions, counts, and reference designation, connectivity between components, channel numbering, switching, and redundancy provisions
3.3	***Communications Payload Sizing, with Detail Equipment List, Size, Mass, Power, Thermal***	Estimate the payload size, mass, power, and thermal requirements, and verify that they are within bus and launch-vehicle allocated budgets	• Aperture specifications from previous step • Link parameters for all other beams: Operating bands, terminal *G/T* and *EIRP* values, modulation, coding, data rates, availability, and expected receiver noise and interferences • Required connectivity flexibility (beam-to-beam routing/switching) • Bus and launch vehicle constraints: payload size, mass, power, and thermal	• Total aperture count and configuration concept • Total payload power requirement • Payload switching / routing concept • Redundancy concept for critical units • Total payload mass and volume • Verification that payload can be accommodated by bus and launch vehicle
3.4	***Communication Payload Detail Design and Requirements Flowdown***	Determine component specifications and select components Finalize and document the design	• Detailed frequency plan • Payload block diagram • Payload equipment list • Provides a component capabilities driven feedback into the design for iterative optimization • Redundancy requirements • Environmental requirements • Update to size, mass, power, thermal, within risk/uncertainty margin allocated	• Detailed specification for all components and parts to be used in the communications payload, with detail sufficient to support both procurement and performance assessments • Finalized frequency plan and payload block diagram • Specifications for all parts, units, and higher-level assemblies • Mechanical drawings • Final design approval / design freeze

Table 16-14, Fig. 16-18◀, Eq. 16-33◀

was described in Sec. 16.1, customer mission requirements flow into the communications payload architecture. By describing interfaces to other parts of the space communications system (links and aperture counts) and an overall concept for the types of payload elements that are to be used (e.g., a transparent payload with fixed antennas having shaped reflectors) and the way that those components are to be configured (e.g., specification of uplink and downlink switching), the architecture imposes constraints on the design that can be translated into specific requirements that will both bound and guide the direction of the detailed design process, as described in Table 16-14.

The communications payload design process begins with the development of preliminary antenna designs, a process that begins with link analyses for each beam to define the *G/T* and *EIRP* that each antenna is required to deliver. With fundamental aspects of the antenna concepts having been defined as part of the architecture, computer simulations are used to develop preliminary design concepts and performance predictions. The key objectives of the first step of the process are to develop preliminary antenna designs to a level of maturity required to (A) understand the approximate size, weight, and mechanical configuration of each antenna, including attachment to the bus and any articulation that may be required, (B) predict key measures of antenna performance with high fidelity so that the payload design can be developed in parallel with development of more detailed antenna designs, and (C) demonstrate that the preliminary mechanical designs and RF performance predictions are compatible with mission requirements.

In the second step of the process, payload block diagrams are developed, building on the topology described by the payload architecture to elaborate all details of the channelization and routing of RF signals between the receive and transmit apertures, including amplification, filtering, frequency conversion, and switching. The payload block diagram specifies not only the types of components that will be required, but also describes the number of each component required for the complete payload. The way that the components are connected to one another, and key characteristics about the components, (such as part identification codes, nominal power, operating band, and channel assignments of certain active and passive components), will be useful throughout the remainder of the design process and into later testing and troubleshooting. Development of the payload block diagram requires initial budgets to be created for numerous aspects of payload performance. These budgets, analogous in many respects to the link budgets described in Sec. 16.2, are used to determine initial component specifications, evaluate design alternatives as each aspect of the design matures, and as the basis for testing that must be performed throughout the process of satellite payload integration.

From the payload block diagram, the design team has sufficient information in component counts and top level component specifications to estimate the total mass, power requirements, and thermal dissipation requirements that will be required for the payload, confirm that these are within the design envelope of the satellite bus, and begin developing concepts as to mechanical layouts and configurations. With this third step of the process, the detailed physical configuration of both the communications payload and the satellite begin to emerge on a large scale, as do challenges that inevitably force certain aspects of the design to be iterated.

The final step in the communications payload design process, comprises finalizing and documenting component specifications and detailed designs. Outputs from this step include finalized versions of the payload block diagram, equipment lists, specifications for individual parts, units, and higher level assemblies, mechanical and electrical drawings and assembly instructions, and end-user documentation that describes all aspects of routine payload operations, operating constraints, and procedures for diagnosing and recovering from faults and performance anomalies.

16.3.1 Preliminary Antenna Design

The architecture includes a definition of the payload antennas that are to be implemented in terms of the basic geometry, whether frequency reuse is required, and whether beam forming or beam steering are required. Table 16-15 describes the principal degrees of freedom in the aperture selection process, gives examples of some of the main alternatives that are available, and describes which types of application requirements typically play a role in identifying candidates from among the available alternatives. From this, it is clear that the details of the antenna design process can vary dramatically from one application to the next.

The preliminary antenna design is further complicated because it involves optimization in multiple dimensions, and is subject to combinations of constraints that can be traded against one another. For example, while receiver sensitivity and the output power from high power amplifiers can each be traded against antenna gain to achieve link closure, geographic coverage is reduced as gain increases. Likewise, although the wide separation between uplink and downlink frequencies means that the optimal design for transmit and receive antennas that serve the same coverage area should be very different from one another, size and mass limitations create a strong motivation to use single apertures to support both transmit and receive functions whenever possible. The preliminary antenna design process must work through these trades at a level sufficient to allow the approximate size, weight, and mechanical configuration of each antenna to be understood, and at a level sufficient that key measures of antenna performance can be predicted with high fidelity so that the payload design can be developed while detailed antenna designs are being completed. In order for the preliminary antenna design to be accepted, the design team must be able to demonstrate that the preliminary mechanical designs and RF performance predictions are compatible with mission requirements.

Table 16-14◀, Fig. 16-18◀, Eq. 16-33◀

Table 16-15. Principal Antenna-Specification Degrees of Freedom, and Typical Alternatives and Associated Requirements or Considerations.

Antenna Attribute	Available Alternatives	Related or Relevant Requirements
Antenna Geometry	• Direct radiating (dipole, horn, spiral) • Reflector (aligned/offset, Gregorian, Cassegrain) – Non-parabolic reflector shaping – Multiple reflectors • Lens	Coverage area and directivity, frequency plan vs. capacity (need for frequency reuse), frequency reuse beam-to-beam isolation, coverage keep-out zones, bus/launch vehicle packaging, mass, size, electronic beam forming/steering, thermal distortion, pointing accuracy and precision
Feed	• Feed/element type • Single vs. multiple elements • Transmit only, receive only, or transmit/receive • Polarization (linear, circular)	Operating band and bandwidth, aperture efficiency, electronic beam forming/steering, frequency reuse), bus/launch vehicle packaging, beam-to-beam isolation, transmit receive isolation, ground terminal polarization
Coverage Flexibility	• Mechanical steering (gimbaled assembly, steerable feeds, steerable or selectable reflectors) • Electronic steering and beam forming (amplitude or phase)	Coverage flexibility, reconfiguration/tracking speed, bus/launch vehicle packaging, reliability, intended service life, cost/complexity, discrete vs. continuously variable coverage specification, required pointing/tracking accuracy and precision
Articulation for Apertures Stowed During Launch	• Deployable rigid reflectors • Unfurlable reflectors	Bus/launch vehicle packaging, pointing accuracy, precision, and stability, reliability, coverage flexibility

Regardless of the antenna definition specified by the payload architecture, the preliminary antenna design process begins with link analyses for each link that the communications payload must support, using information flowed down from mission requirements and the architecture definition. The information flowed down into the initial link analyses includes details of link geometry (satellite locations, coverage areas), RF band allocations, key characteristics of the user and ground terminals with which the communications payload must establish RF links, required data rates, and link availability. With these inputs, link budgets are developed that define minimum *G/T* values that the payload receivers must achieve, and *EIRP* that must be delivered at the *payload transmit antennas*. From the initial link analysis results and the antenna concept defined by the payload architecture, combined with an understanding of available alternatives and key performance parameters for receivers and high power amplifiers, the design team assesses the appropriate partition between receiver sensitivity and receive antenna gain, and between amplifier power and transmit antenna gain.

Although it is routine for apertures to be optimized in some way for the requirements of many, if not all of a satellite payload user beams, there is always a strong motivation to reuse successful, flight-proven designs to the greatest extent possible. Once the design team has identified the antenna configuration, detailed and typically intensive, computer-based numerical simulations are conducted to predict all aspects of antenna performance required to support the development of payload budgets, and to demonstrate compliance to system-level requirements. These simulations will typically yield very accurate and precise pattern predictions, which will be useful for higher fidelity modeling of link performance, and will serve as the basis for subsequent steps in the payload design process.

Parabolic reflector antennas were used for the high-gain applications discussed in Sec. 16.2. Such antennas are best suited for applications that require peak gain greater than 20 dB, and beamwidth less than 15 deg, requirements that are typical for satellite ground terminals and satellite reflectors.

Lighter and simpler antennas [Ford, 2009] tend to be appropriate for lower gain, wider beam applications. For example, an Earth-coverage antenna for a geosynchronous satellite might be designed with a beamwidth of about 18 deg. For this requirement, a simple direct radiating horn antenna would likely be selected for virtually any operating band, whether C-band, X-band, K_u-band, or K_a-band.

Figure 16-19 illustrates the configuration of a typical communications satellite that incorporates both high-gain parabolic reflectors and low gain direct radiating Earth-coverage horn antennas.

Mission applications that require beam scanning may be supported with either mechanically or electrically steered beams, with important trades to be made between beam agility (scan rate), and payload cost, weight, and size. *Phased-array* antennas offer significant potential performance advantages over mechanically steered arrays, offering virtually instantaneous repointing and beam forming, but at considerable penalties in terms of cost and complexity.

As Table 16-15 suggests, the range of antenna alternatives available to the communications payload design team is enormous, and it is only possible here to convey a sense of the key trades and considerations that are made in the communications payload design process. Table 16-16 summarizes the attributes of several antenna types that can provide additional insight into these trades.

Table 16-15, Fig. 16-18◀, Eq. 16-33◀